

卷三 · 诊断与生物标志物

肌酐、SDMA、UPC、血压怎么判读，以及 2026 年出现的新型标志物（胱抑素 C/B、KIM-1、GPX4、尿液蛋白组学、肾功能方程）。核心结论：肌酐正常不等于肾没事。

#	文献（中文标题 / 原文标题 / 要点）	来源	页
1	SDMA与肌酐在猫CKD肾功能评估中的对比 Symmetric dimethylarginine (SDMA) has been shown to be an accurate and precise biomarker for calculating 血清对称性二甲基精氨酸和肌酐浓度作为肾功能的比较 对称二甲基精氨酸（SDMA）已被证明是计算人体估算肾小球滤过率（GFR）的准确且精确的生物标志物，同时也是比血清肌酐浓度（sCr）更敏感的评估肾功能障碍的生物标志物。本回顾性研究的目的是报告在猫体内测量血清SDMA浓度用于在通过常规sCr测量诊断之前检测慢性肾病（CKD）的实用性。慢性肾病（CKD）猫（n = 21）包括持续氮质血症≥3个月的猫（n = 15）、非氮质血症但GFR比正常猫的中位数GFR下降>30%的猫（n = 4），以及非氮质血症但患有草酸钙肾结石的猫（n = 2）。健康老年猫（n = 21）选自同一群体。对氨基二甲基精氨酸（... SDMA肌酐与GFR相关性第三方研究 Glomerular filtration rate (GFR) is the gold standard in assessing renal function but is impractical. Serum creatinine 猫在正常和肾功能下降情况下的浓度与肾小球滤过率（GFR）。肾小球滤过率（GFR）是评估肾功能的金标准，但操作不便。血清肌酐（sCr）在识别早期慢性肾病（CKD）方面敏感性有限，而对称二甲基精氨酸（SDMA）已被商业化为更准确的生物标志物。在猫中比较SDMA和sCr与GFR的研究有限。为进一步调查SDMA在非氮质血症和氮质血症猫中的诊断性能。研究对象为49只客户拥有的猫：17只CKD猫，15只糖尿病（DM）猫，17只健康猫。回顾性研究，利用来自有记录的sCr和GFR结果的猫剩余血样进行SDMA分析。使用相关系数、敏感性、特异性和受试者工作特征曲线评估了SDMA和血清肌酐(sCr)的诊断性能。与...	—	13
3	纵向评估SDMA与肌酐不一致作为GFR标志物 Longitudinal evaluation of discordant symmetric dimethylarginine and creatinine as markers of glomerular filtration rate in radioiodine-treated hypert 目的 本研究的目的是确定在接受放射性碘（RAI）治疗的成熟和老年甲状腺功能亢进猫中，血清SDMA（SDMA）和肌酐浓度与GFR（GFR）不一致的频率及临床意义。研究的次要目标是比较两种SDMA上参考区间（RIs; >14 和 ≥18 µg/dl）及肌酐（≥175 µmol/l）在检测GFR下降方面的敏感性和特异性，并评估两种生物标志物的个体内变异性。方法 对27只接受低剂量RAI治疗的客户拥有的甲状腺功能亢进猫进行了前瞻性纵向研究。在基础时、RAI治疗后1个月、6个月和12个月，对猫的甲状腺状况、GFR（GFR）、S... 猫CKD进展从病理生理到新兴生物标志物 Understanding the Progression of Chronic Kidney Disease in Cats: From Pathophysiology to Emerging Biomarkers 猫慢性肾病（CKD）是老年猫死亡的主要原因，其特征是肾功能的进行性和不可逆性丧失。尽管患病率很高，但由于肾单位的代偿机制以及传统生物标志物的敏感性有限，早期诊断仍然具有挑战性，从而产生了一个诊断空白，需要探索新的生物标志物以实现更早的检测。本文综述了该疾病的复杂病理生理学，包括肾素-血管紧张素-醛固酮系统激活、肾小管间质纤维化和矿物质代谢紊乱。通过分析最新的科学文献，本文评估了当前的诊断现状以及新兴生物标志物的临床相关性。证据表明，SDMA和FGF-23-23可改善早期代谢和滤过变化的检测，而尿液生物标志物如胱抑素...	Journal of Feline Medicine and Sur 2026	25
5	甲亢治疗后猫SDMA浓度的变化 Changes in serum symmetric dimethylarginine concentrations after treatment of feline hyperthyroidism with antithyroid medications. 目的 本研究的目的是报告猫甲状腺功能亢进症在使用抗甲状腺药物治疗后血清肌酐和对称性二甲基精氨酸（SDMA）浓度的变化，并比较治疗前血清肌酐和SDMA水平在治疗后出现或未出现氮质血症猫之间的差异。方法 在这项回顾性研究中，确定了在治疗后1个月（T1）和/或2-9个月（T2）恢复至甲状腺功能正常（总甲状腺素[TT4]浓度7-40nmol/l）的甲亢猫，并根据血清肌酐浓度定义的肾功能状态进行分组。比较使用非参数统计方法进行，并使用Spearman相关系数评估相关性。数据以中位数（最小值-最大值）表示。结果 共纳入19例...	Journal of feline medicine and sur 2026	21
6	血清胱抑素C与尿胱抑素B提升猫CKD诊断精度 Beyond traditional biomarkers: do serum cystatin C and urinary cystatin B improve diagnostic precision in feline and canine renal disease? A systemati 慢性肾病（CKD）（CKD）是猫和狗中最常见的肾病，尤其在老年动物中，而急性肾损伤（AKI）则是一种急性发作的疾病，伴随高发病率和死亡率。血清肌酐和尿素被广泛用于肾功能评估；然而，它们在早期发现肾功能障碍方面的敏感性较低。在此背景下，血清胱抑素C（sCysC）和尿液胱抑素B（uCysB）等替代生物标志物被作为潜在的诊断工具进行了研	Frontiers in Veterinary Science 2026	14

究。本研究按照PRISMA指南进行了荟萃分析，以评估这些生物标志物在患有CKD的猫和AKI的狗中的表现。纳入了三项关于sCysC的研究和两项关于uCysB的研究，比较了患病猫狗与健康对照...

7	尿KIM-1作为非氮质血症猫尿道梗阻肾小管损伤标志 Urinary KIM-1 as a Marker of Renal Tubular Injury Associated with Urethral Obstruction in Non-Azotemic Cats	Animals : an open access journal f 2026	23
8	猫CKD尿谷胱甘肽过氧化物酶4评价 Evaluation of urine glutathione peroxidase 4 in cats with chronic kidney disease. 引言铁死亡是一种独特的受调控细胞死亡形式，其特征是依赖铁的脂质过氧化，这种过氧化会破坏细胞膜并导致细胞死亡。谷胱甘肽过氧化物酶4（GPX4）是唯一能够在细胞内还原脂质过氧化产物的酶，是该过程的关键调控因子。目的GPX4在猫慢性肾病（CKD）（CKD）中的作用此前尚未研究。本研究旨在确定尿液中GPX4水平是否与猫的CKD严重程度相关，并评估其作为疾病进展生物标志物的潜力。方法使用商业猫ELISA试剂盒测定尿液中GPX4水平。计算尿GPX4与肌酐比值（UGCR）。研究纳入了十五只健康猫、61只慢性肾病（CKD）猫和六只...	Frontiers in veterinary science 2025	11
9	猫肾功能方程分类器_概念验证 Evaluation of equation as a classifier for kidney function in domestic cats: a proof-of-concept study. 在人体医学中，估算GFR（eGFR）常用于评估肾功能。而在猫中，目前尚无可靠的eGFR公式。为了验证一个包含四种代谢物的人类eGFR方程是否能有效区分肾功能不同的猫，我们回顾性地纳入了两组猫群。第一组包括34只群体猫，第二组包括29只宠物猫。在猫血清中定量了代谢物标志物，并使用改良方程（eRFfel）估算肾功能。对于eRFfel作为二分类器的受试者工作特征曲线下面积（ROC AUC），第一组和第二组分别为0.9714和0.9958；作为三分类器时，第一组和第二组分别为0.9731和0.9643。队列1中，eRFfe...	npj aging 2026	23
10	猫肾盂肾炎血清淀粉样蛋白A与尿KIM-1 Increased serum amyloid A and urinary kidney injury marker-1 concentrations in cats with pyelonephritis: a diagnostic accuracy study. 目的：研究血清淀粉样蛋白A、嗜中性粒细胞:淋巴细胞比率、单核细胞:淋巴细胞比率在诊断猫肾盂肾炎中的鉴别潜力，并评估肾小管损伤。材料与方法：诊断准确性研究，使用前瞻性入组猫和储存样本。根据诊断将猫分类为肾盂肾炎、输尿管梗阻、慢性肾病、亚临床菌尿、膀胱炎或健康。比较不同组别的血清淀粉样蛋白A、嗜中性粒细胞:淋巴细胞比率、单核细胞:淋巴细胞比率及尿液肾损伤标志物-1。利用受试者工作特征曲线分析评估炎症生物标志物的表现。血清淀粉样蛋白A和尿液肾损伤标志物-1在一部分肾盂肾炎猫中连续测量。结果纳入了71只猫的样本（肾盂肾炎 ...	The Journal of small animal practi 2026	40
11	早晚期猫CKD血清非靶向代谢组学 Page 1/32 Untargeted metabolomic proling of serum from client-owned cats with early and late-stage chronic kidney disease 第1页/共32页 客户拥有的猫在早期和晚期慢性肾脏疾病（CKD）中的血清非靶向代谢组分析 Nora Jean Nealon 美国俄亥俄州立大学兽医学院 Stacie Summers 美国俄勒冈州立大学卡尔森兽医学院 Jessica Quimby 美国俄亥俄州立大学兽医学院 Jenessa Winston 美国俄亥俄州立大学兽医学院 发布日期: 2023年11月9日 DOI: https://doi.org/10.21203/rs.3.rs-3538472/v1 许可: ☐ ☐ 本作品已根据知识共享署名 4.0 国际许可协议获得许可。阅读全文许可 其他声明: 未报告竞争利益。正式版本: 本文预印本的一个...	—	32
12	氮质血症前肠道来源尿毒素改变 Alterations in gut-derived uremic toxins before the onset of azotemic chronic kidney disease in cats 标准文章 小动物内科 肾脏学/泌尿学 肠源性尿毒素在发病前的变化 猫氮质血症性慢性肾病（CKD） Laurens Van Mulders 1,2 Ellen Vanden Broecke 1,2 Ellen De Paepe 2 Femke Mortier 1 Lynn Vanhaecke 2,3 Sylvie Daminet 1 1 根特大学兽医学院, 小动物系, Merelbeke, 2 兽医学院, 转化生理学、感染学系 公共卫生, 整合代谢组学实验室, 根特大学, 梅雷尔贝克, 3贝尔法斯特女王大学, 学院 生物科学, 全球食品安全研究所, 贝尔法斯特, 英国 Sylvie Daminet...	—	14
13	猫血浆FGF-23浓度与生存时间 Fibroblast growth factor-23 (FGF-23) and parathyroid hormone (PTH) are commonly increased in cats with azotemic 猫慢性肾病（CKD）中血浆成纤维细胞生长因子（FGF-23）-23浓度与生存时间的关系 成纤维细胞生长因子（FGF-23）-23（FGF-23）和甲状旁腺激素（PTH）在患有氮质血症性慢性肾病（CKD）的猫中通常升高。这两者都是人类患者生存时间的预测指标，但这些关系在猫中尚未被研究。为了研究在猫CKD诊断时血浆FGF-23和PTH浓度与生存时间及疾病在12个月内进展的关系。本研究纳入214只氮质血症猫（≥9岁），均由客户拥有。回顾性研究：评估氮质血症CKD诊断时的生化和尿液变量，包括血浆FGF-23和PTH浓度，通过Cox回归分析其对生存时间（全因死亡率）的预测作用，并通过逻辑回归分析其对12...	—	14
14	猫FGF-23血浓度scoping review	—	25

This study undertook a scoping review of research on blood fibroblast growth factor 23 (FGF-23) concentrations in
猫血液成纤维细胞生长因子 (FGF-23) 23浓度在有无慢性肾病 (CKD) 猫中的情况: 一项
范围性综述 本研究对健康非氮质血症猫和慢性肾病 (CKD) 猫的血成纤维细胞生长因子
(FGF-23) 23 (FGF-23) 浓度研究进行了范围性综述, 以描述现有文献的数量和性质, 确
定已发表的研究是否为在临床实践中使用FGF-23作为生物标志物提供了充分证据, 并识别知
识中存在的任何空白。研究设计和执行范围性综述时采用了PRISMA扩展范围性综述指南。
利用在线数据库, 识别了截止2022年12月前发表的健康猫和CKD猫血液FGF-23浓度的观察
性和临床研究。提取了研究和人群特征以及FGF-23浓度的描述性数据。 共审查...

15	2019猫CKD硫酸吡哆酚与FGF-23关联研究 The association of indoxyl sulfate with fibroblast growth factor-23 in cats with chronic kidney disease - PMC 吡哆酚硫酸盐 (IS) 与慢性肾病 (CKD) 猫纤维母细胞生长因子-23的关联 叶育伦 1, 周志忠 1,2, 李雅珍 3,4,☐ 已有报道显示, 吡哆酚硫酸盐 (IS) 不仅会随着肾功能损害的严重程度而增加, 还可能是与慢性肾病 (CKD) 人类中与纤维母细胞生长因子-23 (FGF-23) 相关的骨异常有关的因素。目前尚不清楚这种IS与FGF-23之间的相关性是否同样存在于CKD猫中。IS的积累与CKD猫的FGF-23分泌有关。回顾性评估了20只临床健康猫和73只CKD猫病例。血浆中IS和FGF-23的浓度分别通过高效液相色谱和ELISA测定。进展定义为血清肌酐浓度在3个月内增加0.5 mg/dL。随着肾功能下降...	—	26
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Comparison of Serum Concentrations of Symmetric Dimethylarginine and Creatinine as Kidney Function Biomarkers in Cats with Chronic Kidney Disease

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Abstract

Background

Symmetric dimethylarginine (SDMA) has been shown to be an accurate and precise biomarker for calculating estimated glomerular filtration rate (GFR) in humans, as well as a more sensitive biomarker than serum creatinine concentration (sCr) for assessing renal dysfunction.

Objectives

The purpose of this retrospective study was to report on the utility of measuring serum SDMA concentrations in cats for detection of chronic kidney disease (CKD) before diagnosis by conventional measurement of sCr.

Animals

Chronic kidney disease cats ($n = 21$) included those persistently azotemic for ≥ 3 months ($n = 15$), nonazotemic cats with GFR $>30\%$ decreased from median GFR of normal cats ($n = 4$), and nonazotemic cats with calcium oxalate kidney stones ($n = 2$). Healthy geriatric cats ($n = 21$) were selected from the same colony.

Methods

Symmetric dimethylarginine concentrations (liquid chromatography-mass spectroscopy) and sCr (enzymatic colorimetry) were determined retrospectively from historical data or banked serum samples in azotemic cats or at the time GFR (iohexol clearance) was measured in nonazotemic cats.

Results

Serum SDMA ($r = -0.79$) and sCr ($r = -0.77$) concentrations were significantly correlated to GFR (both $P < .0001$). Symmetric dimethylarginine became increased before sCr in 17/21 cats (mean, 17.0 months; range, 1.5–48 months). Serum SDMA had higher sensitivity (100%) compared with sCr (17%), but lower specificity (91% versus 100%) and positive predictive value (86% versus 100%).

Conclusion and Clinical Importance

Using serum SDMA as a biomarker for CKD allows earlier detection of CKD in cats compared with sCr, which may be desirable for initiating renoprotective interventions that slow progression of CKD.

Abbreviations

asymmetric dimethylarginine

blood urea nitrogen

chronic kidney disease

dimethylarginine dimethylaminohydrolase

glomerular filtration rate

International Renal Interest Society

monomethylarginine

nitric oxide synthase

negative predictive value

positive predictive value

serum creatinine concentration

symmetric dimethylarginine

urine protein : creatinine

urine specific gravity

Chronic kidney disease (CKD) is a major cause of morbidity in cats, especially as they age.^{1, 2} Although irreversible, the rate of progression in cats is not predictable, and CKD in many cats can be successfully managed for months or even years.^{1, 2} Early recognition of CKD is desirable in order to know when to initiate renoprotective interventions that may slow its progression (eg, dietary modifications).^{2, 3} However, early decreases in glomerular filtration rate (GFR) are not recognized by commonly used diagnostic tests. CKD usually is diagnosed on the basis of later indicators including impaired urine concentrating ability and azotemia (increased serum creatinine [sCr] and blood urea nitrogen [BUN] concentrations). Cats with CKD are staged according to guidelines developed by the International Renal Interest Society (IRIS) and accepted by the American and European Societies of Veterinary Nephrology and Urology.⁴ The 4-tier IRIS CKD staging system is based on sCr concentration, magnitude of proteinuria as measured by urine protein : creatinine (UPC) ratio, and blood pressure.⁵

Determination of GFR is a valuable tool for estimating renal function and staging kidney disease in dogs and cats.⁶ Renal clearance of inulin is the gold standard for measuring GFR.⁶ If a marker is eliminated solely by the kidneys via glomerular filtration, and is neither reabsorbed nor secreted by the tubules, then renal clearance is equal to plasma clearance and provides an accurate estimate of GFR.⁶ Measuring iohexol plasma clearance appears to be the simplest and most accurate method for determining GFR in clinical practice.⁶ Because measuring GFR is technically cumbersome and expensive, sCr concentration remains the standard surrogate for GFR.⁵ However, sCr has limitations as a marker of kidney function, most notably insensitivity because it remains in the normal reference interval (with a flat slope over much of the GFR range) until GFR is decreased approximately 75%.⁷ In addition, other nonrenal factors can influence sCr, including endogenous production by muscle such that muscle mass, breed, and sex may influence sCr concentration. Although serial evaluation of sCr in the same cat does increase its sensitivity for detecting progressive changes in GFR,⁵ baseline concentrations often are unavailable because they are not commonly monitored in clinical practice. Thus, there is a need for a better biomarker for earlier detection of CKD.

Symmetric dimethylarginine (SDMA) has been shown to be an accurate and precise biomarker for calculating estimated GFR in humans,⁸ as well as a more sensitive biomarker than sCr for assessing renal dysfunction.⁹ A meta-analysis of 18 studies involving human patients showed that 1/SDMA concentrations correlated highly with inulin clearance ($r=0.85$), and SDMA concentrations correlated highly with sCr concentrations ($r=0.75$).¹⁰ In another study using human subjects, a close correlation of GFR (measured by an iodothalamate clearance technique) with sCr and SDMA was reported ($r=-0.84$ and -0.89 , respectively).¹¹ Plasma SDMA concentrations also have been shown to be increased in cats with CKD and to correlate with plasma Cr concentration ($r=0.74$).¹²

In brief, SDMA is a byproduct of protein methylation. There are 3 main species of methylated arginine: monomethylarginine (MMA), asymmetric dimethylarginine (ADMA), and SDMA.¹³ Subsequent protein degradation of methylated proteins yields individual methylated arginine amino acids. Symmetric dimethylarginine is excreted primarily ($\geq 90\%$) by renal clearance.^{14, 15} Because SDMA is eliminated by the kidneys, plasma concentrations are affected by changes in GFR.

The purpose of this retrospective study was to report on the utility of measuring serum SDMA concentrations to detect CKD in cats before diagnosis by conventional single-point sCr measurements. Our goal was to measure serum SDMA concentrations in all CKD cats before the time they became azotemic, provided banked serum samples were available for measurement, in order to compare the sensitivity of sCr and serum SDMA as biomarkers for early detection of renal dysfunction.

Materials and Methods

Animals and Study Design

All study protocols were reviewed and approved by the Institutional Animal Care and Use Committee, Hill's Pet Nutrition, Inc., Topeka, KS. Each cat had an annual physical examination, CBC, serum biochemical analyses, urinalysis, and urine culture if indicated by the urinalysis results. In addition, after 2010, serum was frozen at -70°C and banked for retrospective analyses. Cats were housed individually or in groups and allowed exercise in indoor runs. Cats had access to natural light that varied with seasonal changes. All cats were provided with regular opportunities to exercise, with access to toys. All cats were owned by the commercial funders of this research or their affiliates, who gave permission for them to be included in this study.

All cats had been used historically for palatability studies. Many different types of commercial and non-commercial foods of varying nutrient compositions were fed, including dry and canned cat foods. All foods met the requirements established by the Association of American Feed Control Officials (AAFCO) for complete and balanced pet foods for adult cats.

Cats with CKD ($n=21$) came from the colony of over 400 domestic short hair cats. Cats in the colony ranged in age from 1 to 19 years (mean, 7 years) with approximately 25% of cats >10 years. The cats ($n=21$) with CKD included cats that were persistently azotemic over an extended period, typically for ≥ 3 months ($n=15$). CKD was diagnosed in nonazotemic cats (with stable sCr concentrations) based on GFR testing by iothexol clearance ($n=4$). Nonazotemic cats with calcium oxalate kidney stones also were considered as having CKD ($n=2$). Cats with CKD had no evidence of other confounding disease, notwithstanding history and physical examination findings (changes in urine volume or changes in kidney size or shape) that were consistent with CKD. Mean age of cats with CKD was 14.3 years (range,

8.0–18.5 years). There were 15 ovariohysterectomized females and 6 neutered males. Mean body weight was 4.0 kg (range, 2.6–5.7 kg).

Acute kidney disease may have been present in some cats with CKD and gone undiagnosed. For example, because an unremarkable urine sediment does not definitively rule out infection, especially in cats with CKD, cats with occult pyelonephritis may have gone undiagnosed. In cats with chronic urolithiasis, stones could have moved, leading to partial ureteral obstruction and acute kidney injury.

A similar number of healthy geriatric cats ($n = 21$) was selected from the same colony. Criteria for inclusion were age >10 years, requirement of 3 normal GFR tests, 3 normal sCr concentrations, and 3 urine specific gravity (USG) measurements >1.040 over a 6-month period. In addition, these cats lacked historical or physical evidence of confounding disease at the time of inclusion, and had banked serum samples available for determination of SDMA concentrations. Mean age of healthy geriatric cats was 11.7 years (range, 10.2–13.1 years). There were 12 ovariohysterectomized females and 9 neutered males. Mean body weight was 4.5 kg (range, 3.1–6.3 kg).

Retrospective data was used to document sCr concentrations in cats with CKD. Serum SDMA concentrations were determined from serum stored in serum banks. sCr and SDMA concentrations also were measured from blood collected prospectively in CKD cats.

Analyses

Serum Biomarkers

Serum Cr and BUN concentrations were determined by enzymatic colorimetric methods.¹ Reference ranges for sCr (0.7–2.1 mg/dL) and BUN (15.0–31.0 mg/dL) in other adult cats were previously established for this in-house laboratory.

Urinalysis

Urine specific gravity was determined using a refractometer. Urine creatinine concentration was used as an internal reference and measured with the same assay as sCr. Urine protein concentrations were determined using urine supernatant. In brief, urine was pre-incubated in an alkaline solution containing EDTA, which denatures protein and eliminates interference from magnesium ions. Benzethonium chloride then was added, producing turbidity, which is assessed by a turbidimetric method.¹ Urine protein : creatinine ratio results are reported as mg/dL protein : mg/dL creatinine.

Glomerular Filtration Rate

Serum concentrations of iohexol were measured by a commercial laboratory² using an inductively coupled plasma-atomic emission spectroscopy (ICP-AES) method.¹⁶ In brief, a single IV injection of iohexol, an iodinated radiographic contrast agent (300 mg I/kg), was administered and 3 serum samples were collected at 2, 3, and 4 hours after injection. GFR was estimated from calculations made using a one-compartment model for serum iohexol clearance. Median (range) GFR for normal healthy cats in this study determined from 3 iohexol clearance tests per cat over a 6-month period was 1.94 mL/min/kg (range, 1.34–3.79 mL/min/kg).

Symmetric Dimethylarginine

Symmetric dimethylarginine concentrations were determined using liquid chromatography-mass spectroscopy (LC-MS). In brief, 50 μ L of serum from cats, or 50 μ L of standard, were transferred to centrifuge tubes and diluted with 50 μ L of internal standard solution³ in water. Next, 300 μ L of high performance liquid chromatography (HPLC)-grade acetonitrile⁴ was added to extract the SDMA. The samples were centrifuged and the supernatant was decanted and tested by LC-MS for SDMA concentration.

The LC-MS procedure was performed using an API 4000 coupled with Shimadzu Nexera HPLC system.⁵ Ten μ L of serum extract was injected into the HPLC equipped with a C18 column⁶ (5 μ m, 4.6 $\times$ 30 mm). The SDMA was eluted using 0.1% formic acid⁴ and 0.5 mM perfluoroheptanoic acid⁷ in water (Mobile Phase A) and acetonitrile with 0.1% formic acid (Mobile Phase B) with a flow rate of 1 mL/min and column temperature of 25°C. Under these conditions, the retention times for SDMA and internal standard D-7-ADMA are both 1.72 minutes. The effluent of HPLC was directed to the electrospray ionization source of the mass spectrometer, which produced single charged adduct ions $[M+H]^+$. The $[M+H]^+$ ions of SDMA and D-7-ADMA were analyzed in the multiple reaction monitoring (MRM) mode of the mass spectrometer. Fragmentation was accomplished using collision energy of 20 V. The observed MRM transitions for SDMA and D-7-ADMA were 203.1 to 172.1 and 210.2 to 46.1 m/z, respectively.

A standard curve for SDMA quantitation was prepared using calibrator solutions. First, a solution of SDMA stock standard was prepared (1 mg/mL) in water. The stock standard was diluted 50-fold in water to obtain working standard solutions (20 μ g/mL). Next, 100 μ L of working standard was spiked into 1,900 μ L of charcoal-stripped feline serum to obtain 100 μ g/dL SDMA. Serial dilutions of SDMA were similarly prepared in charcoal-stripped feline serum ranging from 100 to 1.56 μ g/dL of SDMA. The calibrator samples were analyzed using the same protocol described for samples. The standard curve, prepared following the LC-MS protocol for SDMA detection, was linear with a correlation coefficient >0.999 . The intra-assay ($n = 5$) and inter-assay ($n = 25$) coefficients of variation for SDMA concentrations were 2.2 and 2.5%, respectively. The normal reference limit for SDMA in other healthy cats previously was determined to be <14 μ g/dL.⁸

Statistical Analysis

Statistical analyses were performed using Statistical Analysis Software version 9.2.⁹ Data were assessed for normality by the Shapiro–Wilk test. To investigate the relationships between SDMA and GFR, sCr and GFR, and SDMA and sCr, correlation coefficients were measured between these response variables. To accomplish this, SDMA and sCr were plotted against GFR, and equations were derived from linear, exponential, logarithmic, or quadratic plots. Multiple equations were assessed to better approximate changes at the edges of the plots. These analyses were performed in Excel and verified using a regression model in SAS. Significance was accepted as $P < .05$. Using derived equations, the GFR that corresponded to the upper limit of the normal reference interval for SDMA (14 μ g/dL) and sCr (2.1 mg/dL) concentrations, was calculated. The sensitivity, specificity, positive predictive value (PPV), and negative predictive values (NPV) also were calculated for SDMA and sCr concentrations to detect CKD.

All healthy geriatric cats had 3 evaluations for sCr, BUN, and SDMA concentrations, USG, and GFR over a 6-month period. These measurements were used to calculate standard deviation and coefficient of variation for these variables. The data for GFR measurements were not normally distributed. To describe the variability of GFR in healthy geriatric cats, median, range, and percentiles were determined. The lower 2.5 percentile was used to establish the lower limit of

normal. All cats with GFR below the lower 2.5 percentile were considered to have abnormal renal function whereas cats above this threshold were considered to have normal renal function.

Healthy geriatric cats then were compared with cats with CKD at the time SDMA concentrations were first increased ($>14 \mu\text{g/dL}$) and at the time sCr concentrations were first increased ($>2.1 \text{ mg/dL}$). Comparisons included age; body weight; sCr, BUN, and SDMA concentrations; USG; UPC ratios; and, GFR measurements. In addition, the estimated time that serum SDMA concentrations were increased before sCr concentrations were increased was calculated. Data are reported as mean (range) unless otherwise indicated. A $P < .05$ was considered statistically significant.

Results

Concurrent serum SDMA and sCr concentrations were plotted for CKD cats and healthy geriatric cats (1 time point only) using a scatter plot (Fig ¹; $r = 0.72$; $P < .0001$). In cats with CKD, repeated measures (2 sets of data) within the same animal were included (ie, at the time when SDMA first increased above normal and at the time when sCr was first increased above normal; Fig ¹; Table ¹). There were no situations in which serum SDMA concentration was within the normal reference interval and serum sCr concentration was increased above the normal reference interval (ie, no points in the upper left quadrant). All healthy geriatric cats had serum SDMA and sCr concentrations in the normal reference interval (lower left quadrant). As CKD cats became azotemic, serum SDMA and sCr data points moved into the upper right quadrant (see Fig ¹).

Table 1

Demographic data for healthy cats and cats with CKD at time SDMA first elevated and at time when sCr first elevated

	Healthy Cats	Cats with CKD Data When SDMA First Detected as $>14 \mu\text{g/mL}$	Cats with CKD Data When sCr First Detected as $>2.1 \text{ mg/dL}$		
N	21	21	P Value	21	P Value
Age (years)	11.7 (10.2–13.1)	12.8 (4.9–17.2)	.08	14.3 (8.0–18.5)	$<.001$
Body weight (kg)	4.5 (3.1–6.3)	4.1 (2.6–5.7)	$>.10$	4.0 (2.6–5.7)	$>.10$
Creatinine concentration (mg/dL)	1.2 (0.7–1.6)	1.6 (1.0–2.5)	.001	2.6 (1.2–5.8) ^a	$<.001$
Blood urea nitrogen concentration (mg/dL)	19.7 (16.7–26.0)	28.6 (19.8–41.6)	$<.001$	46.0 (28.7–88.9)	$<.001$
Urine specific gravity	1.059 (1.039–1.078)	1.030 (1.009–1.056)	$<.001$	1.020 (1.009–1.047)	$<.001$
Urine protein to creatinine ratio	0.2 (0.1–0.3)	0.2 (0.1–0.4)	$>.10$	0.3 (0.1–1.2)	$>.10$

	Healthy Cats	Cats with CKD Data When SDMA First Detected as >14 µg/mL	Cats with CKD Data When sCr First Detected as >2.1 mg/dL		
N	21	21	P Value	21	P Value
Glomerular filtration rate (mL/min/kg)	1.94 (1.34–3.79) ^b	1.17 (0.87–1.45)	<.001	0.98 (0.49–1.45)	<.001
SDMA (µg/dL)	9.9 (7.3–12.4)	19.9 (14.0–45.0)	<.001	32.1 (15.0–115.0)	<.001
Approximate time SDMA increased before creatinine (months)	0.0 (0.0–0.0)			17.0 (1.5–48.0) ^c	<.001

Data are presented as mean (range).

CKD, chronic kidney disease; SDMA, symmetric dimethylarginine; sCr, serum creatinine.

a

Six of 21 cats had sCr < 2.1 mg/dL. In 4 cats, the diagnosis of CKD was based on GFR being <1.36 mL/min/kg. In 2 cats with calcium oxalate kidney stones, sCr never increased above the normal reference interval, although serum SDMA concentrations were >14 µg/mL in both cats.

b

Median (range) GFR for normal healthy cats in this study determined from 3 iohexol clearance tests per cat over a 6-month period.

c

Serum SDMA concentrations increased before sCr in 17/21 cats. In 4/21 cats, banked serum samples were not available to measure SDMA concentrations before the time cats developed azotemia.

Figure 1

Relationship between serum symmetric dimethylarginine (SDMA; µg/dL) and serum creatinine (sCr; mg/dL) concentrations in 21 healthy geriatric cats (mean age, 11.7 years; range, 10.2–13.1 years; diamonds) and 21 cats with chronic kidney disease (CKD; mean age, 14.3 years; range, 8.0–18.5 years; squares). There is a positive linear relationship between serum SDMA and sCr concentrations ($r = 0.72$). No cats with sCr concentrations above the normal reference interval (>2.1 mg/dL) had normal serum SDMA concentrations (<14 µg/dL).

The median (range) of GFR measurements in healthy geriatric cats was 1.94 mL/min/kg (1.34–3.79 mL/min/kg). The lower 2.5 percentile, which was used to establish the lower limit of normal, was determined to be 1.36 mL/min/kg. All cats with GFR < 1.36 mL/min/kg (which corresponds to a >30% decrease below the median GFR) were considered to have CKD whereas cats above this threshold were considered to have normal renal function.

Using serum from data banks to measure SDMA concentrations, serum SDMA concentrations were shown to increase above the normal reference interval before sCr concentrations increased above the normal reference interval in 17/21

cats by a mean of 17.0 months (range, 1.5–48.0 months). A representative example of the relationship between serum SDMA concentrations and sCr concentrations across time is shown for 1 cat in a time versus analytes concentration bar graph (Fig ²). In 4/21 cats, banked serum samples were not available to measure SDMA concentrations before the time cats developed azotemia. For the latter, the time that serum SDMA was increased before an increase in sCr was recorded as 0 months. In 4/21 cats with CKD, sCr concentration did not increase above the normal reference interval and the diagnosis of CKD was based on GFR being >30% below median GFR for healthy cats. In these 4 cats, the estimated time that serum SDMA concentration was increased before sCr increased was recorded as months until death in 3 cats (17, 28.5, and 48 months), and 41 months in 1 cat that was still alive. In 2/21 cats (both cats with calcium oxalate kidney stones), sCr never increased above the normal reference interval, although serum SDMA concentrations were increased above normal in both cats. One of the cats was euthanized because of anorexia of 2 weeks' duration with a sCr of 2.03 mg/dL; serum SDMA had been increased for 16 months (same duration as the nephrolithiasis diagnosis). The other cat was alive with a sCr of 1.74 mg/dL; serum SDMA has been increased for 9 months. Nephrolithiasis had been diagnosed 14 months before. Three azotemic cats also had calcium oxalate nephrolithiasis; nephrolithiasis had been diagnosed 13 and 17 months before death in 2 cats, but was not diagnosed before necropsy in the third cat.

Figure 2

Representative cat (01ASU3; born September 2001; neutered male) with serum symmetric dimethylarginine (SDMA; black bars) and sCr (gray bars) concentrations indicated across time. Glomerular filtration rate was measured in March 2011 and found to be 40% below the mean of 1.94 mL/min/kg for the feline population of the reference laboratory. Serum SDMA was increased in April 2012 (16 µg/dL). The cat became azotemic in November 2012 (sCr, 2.12 mg/dL), approximately 8 months after serum SDMA was increased. The cat died in January 2013.

The best fit equation for the relationship between SDMA concentration and GFR was a quadratic relationship: $[GFR \text{ in mL/min/kg}] = 0.0031 [\text{SDMA concentration in } \mu\text{g/dL}]^2 - 0.1703 [\text{SDMA concentration in } \mu\text{g/dL}] + 3.2444$ ($r = -0.79$; $P < .0001$). Setting the upper limit of the normal reference interval for SDMA at 14 µg/dL corresponded to a GFR of 1.47 mL/min/kg, which represents an approximately 24% decrease from median GFR of healthy cats. Using a 30% decrease from median GFR as the gold standard for confirmed decrease in renal function, sensitivity (100%), specificity (91%), PPV (86%), and NPV (100%) were calculated for serum SDMA concentration. There were 2 false positives, whereby SDMA was >14 µg/dL (18 and 19 µg/dL), but GFR was not <1.36 mL/min/kg. Both cats had GFR of 1.45 mL/min/kg (25% decrease below the median).

The best fit equation for the relationship between sCr concentrations and GFR was an exponential relationship: $[GFR \text{ in mL/min/kg}] = 4.5501e^{-0.84 [\text{sCr concentration in mg/dL}]}$ ($r = -0.77$; $P < .0001$). Setting the upper limit of the normal reference interval for sCr at 2.1 mg/dL corresponded to a GFR of 0.78 mL/min/kg, which represents approximately 60% decrease from the median. Using a 30% decrease from median GFR as the gold standard for confirmed decrease in renal function, the sensitivity (17%), specificity (100%), PPV (100%), and NPV (70%) were calculated for sCr concentration.

In healthy geriatric cats, the standard deviation and coefficient of variation for other variables measured over a 6-month period were sCr (0.23 mg/dL; 19.85%), BUN (2.61 mg/dL; 12.99%), SDMA 1.54 µg/dL; 15.53%), and USG (0.01; 0.87%), respectively. Compared with healthy geriatric cats, cats with CKD at the time when serum SDMA was

first detected as being $>14 \mu\text{g/dL}$ (Table ¹) were of similar age ($P=.08$) and body weight ($P>.10$), but had higher sCr, BUN, and SDMA concentrations (all $P\leq.001$), more dilute urine based on USG ($P<.001$), and lower GFR ($P<.001$). The UPC ratios did not differ ($P>.10$). Cats with CKD at the time when sCr was first detected as being $>2.1 \text{ mg/dL}$ were older ($P<.001$), but still of similar body weight and with similar UPC ratios (both $P>.10$). Otherwise, cats with CKD continued to have higher sCr, BUN, and SDMA concentrations, more dilute urine based on USG, and lower GFR (all $P<.001$) compared with healthy geriatric cats.

Discussion

Using retrospective serum samples, we were able to show that serum SDMA increases before sCr in cats with CKD. Serum samples had been banked as part of annual examinations or as part of protocols for other studies. Thus, exact interval data were not available. Nonetheless, serum SDMA concentrations increased above the normal reference interval before sCr concentrations increased above the normal reference interval by a mean of 14.6 months (range, 0–48 months) in 21 cats diagnosed with CKD. Cats with CKD under consideration were IRIS Stage 1 (sCr $<1.6 \text{ mg/dL}$) or Stage 2 (sCr $1.6\text{--}2.8 \text{ mg/dL}$) and nonproteinuric (UPC ratio ≤ 0.2) or borderline proteinuric (UPC ratio $0.2\text{--}0.4$) at the time of diagnosis. All healthy geriatric cats had serum SDMA and sCr concentrations in the normal reference interval. Thus, serum SDMA represents a promising biomarker for early detection of CKD in cats with compromised renal function. Early detection is desirable because it may be beneficial to initiate dietary or other interventions even earlier in order to slow progression of CKD, as was shown for cats with IRIS stages 2 and 3 CKD.²

A significant positive correlation ($r=0.72$) was present between serum SDMA and sCr concentrations, similar to what has been reported previously in cats with CKD.¹² These results support that SDMA is excreted primarily by the kidneys. Furthermore, our evidence suggests that nonazotemic cats with SDMA concentrations $>14 \mu\text{g/dL}$ eventually progress to azotemic CKD.

Serum SDMA is a relatively new biomarker that allows early detection of CKD in a population of nonazotemic cats. Serum SDMA concentration was $>14 \mu\text{g/dL}$ in all nonazotemic cats with GFR $<1.36 \text{ mL/min/kg}$ (median for healthy cats, 1.94 mL/min/kg). We chose GFR $<1.36 \text{ mL/min/kg}$ (30% decrease from median of healthy cats) as consistent with a diagnosis of CKD based on the lower 2.5 percentile of GFR measurements in healthy geriatric cats being 1.36 mL/min/kg . The normal reference interval for SDMA of $<14 \mu\text{g/dL}$ corresponded to a GFR of 1.47 mL/min/kg (approximately 24% decrease from median). Thus, 2 cats with CKD had SDMA concentrations $\geq 14 \mu\text{g/dL}$, but GFR measured by iohexol clearance was not $<1.36 \text{ mL/min/kg}$. Therefore, SDMA concentration indicated a false positive for decreased GFR in these 2 cats. Both cats had GFR of 1.45 mL/min/kg (25% decrease below median GFR). Our cut-off ($<1.36 \text{ mL/min/kg}$) for the diagnosis of CKD lowered the sensitivity of the SDMA assay to 91%.

Serum SDMA is a byproduct of intranuclear protein methylation. Although phosphorylation is the best understood post-translational modification of proteins, arginine methylation also is important.¹³ Arginine methylation occurs when the nitrogens of arginine within polypeptides are post-translationally modified to contain methyl groups. There are 3 main species of methylated arginine: MMA, ADMA, and SDMA.¹³ The methyl group is transferred from *S*-adenosylmethionine to a nitrogen of arginine by 1 of several protein arginine methyltransferases (PRMT).¹³ Subsequent protein degradation of methylated proteins by hydrolysis yields individual methylated arginine amino acids, which are biologically active. ADMA functions as an endogenous nitric oxide synthase (NOS) inhibitor,⁸ preventing production of nitric oxide from *L*-arginine. Although SDMA is not a direct inhibitor of NOS, it can compete

with arginine for transport across membranes,⁸ and thus, indirectly decreases nitric oxide synthesis by limiting l-arginine supply.^{8, 17} Asymmetric dimethylarginine is degraded primarily to dimethylamine and citrulline ($\geq 80\%$) after metabolism by dimethylarginine dimethylaminohydrolase (DDAH) whereas only a small amount of intact ADMA ($\leq 20\%$) is eliminated by the kidneys.^{12, 14} Symmetric dimethylarginine is eliminated primarily ($\geq 90\%$) by renal clearance.^{14, 15} The presence of dimethylarginines in the urine has been recognized for over 40 years.¹⁸ Because SDMA is excreted by the kidneys, plasma concentrations are affected by changes in GFR. Asymmetric dimethylarginine accumulates in patients with renal dysfunction, although it is less abundant than SDMA, and its accumulation may be related to renal parenchymal damage, resulting in decreased DDAH expression and activity, rather than to decreased glomerular filtration of ADMA.¹⁴ In healthy human subjects from the Framingham offspring cohort, dietary components were not associated with plasma SDMA concentration.^{19, 20} Symmetric dimethylarginine concentrations were positively correlated with age and sCr concentration whereas body mass index, effective GRF, and diastolic blood pressure were inversely related to SDMA concentration.²¹

Glomerular filtration rate is directly related to functional renal mass. Early detection of a decrease in GFR may be important for initiating dietary or medical interventions in cats with CKD when sCr concentrations are still within the normal reference range (≤ 2.1 mg/dL).² However, GFR testing remains an underused tool for diagnosis and management of kidney disease because it is difficult to measure, expensive, and requires multiple blood samples or a 24-hour urine collection.⁶ Instead, serial sCr concentrations commonly are used to assess renal function because sCr is easily measured and less expensive. Yet, cats with IRIS stage 1 and 2 CKD may have normal sCr because sCr does not increase above the reference range until approximately 75% of nephrons are nonfunctioning. Other limitations of using sCr as a biomarker for CKD include increased concentrations that may occur with some types of assays, and there is additional sCr secretion into the tubules in male dogs. Because creatinine is a non-enzymatic breakdown product of phosphocreatine in muscle, daily production of creatinine in the body is determined largely by muscle mass, such that total lean body weight may influence sCr concentration as well. Overall, it is less than ideal as an early biomarker of CKD.⁶

Normal GFR is poorly defined in dogs and cats such that recommendations are to use the reference range established by the laboratory measuring the injected GFR marker. This helps minimize the broad reference range because of variations in protocols.⁶ Markers used to measure GFR should be freely filtered through the glomerulus, not bound to plasma proteins, neither reabsorbed nor secreted by renal tubules, nontoxic, and cannot themselves alter GFR. Inulin is considered the gold standard for measuring GFR.^{6, 16} In cats, however, plasma clearance of iohexol using 2–4 plasma samples provides an accurate estimate of GFR.¹⁶ Using plasma clearance of iohexol to estimate GFR in healthy cats, we then selected the lower 2.5 percentile to establish the lower limit of normal GFR.

Serum SDMA concentrations varied inversely with GFR ($r = -0.79$), increasing in cats as GFR decreased. Because SDMA is thought to be excreted exclusively by the kidneys, SDMA has been proposed as a marker of renal function in humans.^{10, 14, 21} Symmetric dimethylarginine recently was reported in human volunteers to provide an accurate and precise estimate of GFR and to serve as a more sensitive biomarker of renal dysfunction than sCr.⁹

Assuming pre-renal (volume responsive) causes of decreased GFR and causes of acute kidney injury have been eliminated, an increased serum SDMA concentration (positive test) as a single-point measurement or screening test is moderately effective for confirming CKD (PPV = 86%), although further monitoring is needed. Increased SDMA,

however, correctly identified all positive CKD cases (100% sensitivity; also called the true positive rate, which is the ratio of true positives divided by true positives plus false negatives). As a screening test, a negative result is excellent at reassuring the owner that the patient does not have CKD (NPV = 100%), but the initial screening test correctly identifies only 91% of cats that do not have CKD (91% specificity; also called the true negative rate, which is the ratio of true negatives divided by false positives plus true negatives). In comparison, using sCr as a screening test, an increased sCr (positive test) is excellent for confirming CKD (PPV = 100%). It does, however, only correctly identify 17% of all cases (sensitivity). This is consistent with the conclusion of others that plasma Cr is not sufficiently sensitive for early diagnosis of renal dysfunction in cats.²² As a screening test, a negative result is poor at reassuring the owner that the patient does not have CKD (NPV = 70%), although at the initial screening the test correctly identifies 100% of those that do not have CKD (specificity). Because nutritional intervention using appropriately formulated renal diets for cats with CKD is relatively safe, it could be argued that treating cats for CKD based on using SDMA as a biomarker, because of its higher sensitivity, is more logical than missing CKD cats by reliance on sCr, which has higher specificity for CKD.

There is considerable interest in finding new biomarkers to track the progression of renal disease in cats (eg, SDMA, ADMA, and fibroblast growth factor 23)^{12, 23, 24} and dogs (eg, homocysteine, cystatin C, and clusterin).^{25, 26} Because plasma SDMA and ADMA concentrations were shown to increase in cats with CKD and to correlate with plasma Cr, they were suggested as possible markers for renal disease although no correlation to GFR was studied in cats.¹² In humans, SDMA is more abundant than ADMA in patients with CKD and the correlation between ADMA concentrations and GFR is considerably weaker than that between SDMA and GFR.¹⁴ Fibroblast growth factor 23 concentration in cats was shown to negatively correlate with GFR ($r = -0.472$) and to increase in the early nonazotemic stage of CKD.²⁴ However, as in humans, independent factors such as phosphate, parathyroid hormone, and calcium concentrations may be more important predictors of plasma fibroblast growth factor 23 concentration when GFR is markedly decreased.^{23, 27}

Feeding a renal diet to cats with IRIS CKD stage 2 or higher is considered the current standard of care with strong evidence supporting this recommendation.⁵ Dietary modifications include decreased protein, phosphorus, and sodium content; increased water soluble vitamins and fiber content; increased caloric density; and additional n-3 fatty acids, antioxidants, and potassium.⁵ The kidney diet evaluated in 1 study was superior to an adult maintenance diet in minimizing uremic episodes and kidney-related deaths in cats with spontaneous stage 2 or 3 CKD.² The use of a biomarker to help identify cats with CKD earlier in the disease course, corresponding to sCr in the range of 1.6–1.9 mg/dL, may provide additional options for slowing progressive loss of kidney function in order to ameliorate clinical and biochemical consequences of CKD, while maintaining adequate nutrition.⁵

In summary, this study demonstrated the utility of using SDMA as an early indicator of compromised renal function in cats with CKD. Future studies are needed to demonstrate that early intervention improves the outcome of CKD in cats.

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Plasma symmetric dimethylarginine and creatinine concentrations and glomerular filtration rate in cats with normal and decreased renal function

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Abstract

Abstract

Background

Glomerular filtration rate (GFR) is the gold standard in assessing renal function but is impractical. Serum creatinine (sCr) has limited sensitivity in identifying early chronic kidney disease (CKD), whereas symmetric dimethylarginine (SDMA) has been commercialized as more accurate biomarker. Studies comparing SDMA and sCr with GFR in cats are limited.

Objectives

To further investigate the diagnostic performance of SDMA in nonazotemic and azotemic cats.

Animals

Forty-nine client-owned cats: 17 cats with CKD, 15 cats with diabetes mellitus (DM), and 17 healthy cats.

Methods

Retrospective study using spare blood samples from cats with documented sCr and GFR results for SDMA analysis. Diagnostic performances of SDMA and sCr were evaluated using correlation coefficients, sensitivities, specificities, and receiver operator characteristic curves.

Results

Compared to healthy cats and cats with DM, CKD cats had significantly higher SDMA_{plasma} (26.7 ± 9.9 µg/dL) and sCr (249.7 ± 71.6 µmol/L [2.8 ± 0.8 mg/dL]; both $P < .001$) values. SDMA_{plasma} ($\tau_B = -0.57$; $P < .001$) and sCr ($\tau_B = -0.56$; $P < .001$) were significantly correlated with GFR. SDMA_{plasma} ($\tau_B = 0.52$; $P < .001$) had a significant relationship with sCr. SDMA_{plasma} and sCr had similar sensitivity (76%-94% and 71%-88%, respectively) in detecting reduced renal function. Creatinine had higher specificity (94%-96%) than SDMA_{plasma} (75%-76%) ($P < .05$).

Conclusion and Clinical Importance

In this study of azotemic and nonazotemic cats, SDMA was a reliable marker to identify decreased GFR. However, superiority of SDMA over sCr could not be confirmed.

Abbreviations

area under the curve

chronic kidney disease

diabetes mellitus

glomerular filtration rate

International Renal Interest Society

liquid chromatography-mass spectrometry

reference interval

receiver operator characteristic

serum creatinine

symmetric dimethylarginine

urine-specific gravity

1

INTRODUCTION

Chronic kidney disease (CKD) is characterized by loss of structure and function of 1 or both kidneys over a time span of 3 months or longer.¹ Between 1.6% and 20% of all cats develop CKD and the disease mostly affects elderly cats, the prevalence of CKD reaching up to 80% in the population of geriatric cats.^{2, 3, 4} It is hoped that early treatment will delay the natural progressive course of the disease, making early diagnosis important and regular health screening of senior (11-14 years) and geriatric (≥ 15 years) cats essential.^{5, 6, 7}

Assessing early renal function loss by measuring glomerular filtration rate (GFR) is the gold standard but is impractical.⁸ Chronic kidney disease is routinely diagnosed using a combination of thorough physical examination, extensive laboratory testing (including complete blood counts, serum biochemistry profile, and urinalyses), diagnostic imaging, and blood pressure measurement. Compatible clinical signs, persistent azotemia, and decreased urine concentrating ability (urine-specific gravity [USG] < 1.035) indicate onset of azotemic CKD.⁶ Presence of renal azotemia reflects irreversible loss of 50% to 75% of functional nephrons, correlating to 50% to 60% decrease in kidney function.^{9, 10} This discrepancy is attributable to compensatory hyperfiltration of remaining nephrons. With the goal of maintaining the homeostasis, reduction of nephron mass leads to increase in perfusion and filtration of surviving nephrons. In the short term, the compensatory hyperfiltration is beneficial as the GFR is partly maintained, but in the long term this might be detrimental as it promotes intraglomerular hypertension and ultimately progression of CKD.¹¹ Although GFR decline can already be substantial in early stages of CKD, serum creatinine (sCr) will not necessarily exceed the upper limit of the reference interval (RI).¹² Additionally, extra-renal factors may interfere with the results of routine biomarkers, sCr being influenced by diet, age, sex, and muscle mass.^{13, 14, 15} Moreover, sCr has high interindividual variability, hence the RI is usually wide.¹⁶ A consequence of the high interindividual variability is

that sCr is not a sensitive biomarker to identify early CKD using population-based RI.¹⁶ These limitations create an urgent need for sensitive markers to identify impaired renal function before azotemia arises.

Symmetric dimethylarginine (SDMA) is a by-product of protein methylation that enters the circulation after proteolysis.¹³ Its excretion is mainly renal (>90%), via glomerular filtration and active secretion.¹⁷ Tubular reabsorption is absent.¹⁴ In small animals, SDMA is not affected by muscle mass^{18, 19} or sex,¹⁸ creating an advantage over sCr. It has been claimed that SDMA has superior sensitivity to detect renal dysfunction compared to sCr. This is mainly based on a retrospective study containing 42 CKD and healthy geriatric cats in which SDMA (cut-off 14 µg/dL) showed a higher sensitivity than sCr (cut-off value 185 µmol/L [2.1 mg/dL]) to identify a >30% decrease in GFR.²⁰ The correlation between SDMA and GFR ($r = -0.79$; $P < .001$) proved equivalent to the correlation between sCr and GFR ($r = -0.77$; $P < .001$), indicating that both renal markers are closely related to the GFR.²⁰ This finding was also confirmed by other studies which obtained a coefficient of determination between SDMA and GFR ($R^2 = 0.82$; $P < .001$), similar to the relationship between sCr and GFR ($R^2 = 0.81$; $P < .001$).¹³

Since research on the diagnostic performance of SDMA in cats is limited, this study aimed to verify whether SDMA has added value over sCr to detect impaired GFR. Our first objective was to compare the strength of the correlation between the biomarkers and GFR. The second goal was to evaluate sensitivities, specificities, and optimal cut-off values for SDMA and sCr.

2

MATERIAL AND METHODS

2.1

Study sample and design

This retrospective study was performed at the Small Animal Department of the Faculty of Veterinary Medicine. Frozen blood samples (-80°C) from adult, privately owned cats that had undergone GFR estimation and general health screening as part of previously published prospective studies^{21, 22} were used. The cats had been recruited in a 5-year period between 2009 and 2014.

Based on the results of an extensive physical examination and routine laboratory analysis, all animals had been categorized into 1 of 3 predefined groups: CKD, diabetes mellitus (DM), and the healthy control group. The presence of CKD had been determined by a sCr value higher than the RI ($>161.8\text{ }\mu\text{mol/L}$ [$>1.83\text{ mg/dL}$]) in combination with USG <1.035 (renal azotemia) together with a compatible history and associated clinical signs.²³ The diagnosis of DM had been based on the combination of hyperglycemia, glucosuria, increased serum fructosamine concentration, and representative clinical complaints. Cats within the healthy control group did not show any significant abnormalities on physical examination, blood examination (complete blood count, serum biochemistry profile, and total thyroxine concentration), and urinalysis (urinary protein : creatinine ratio and bacterial culture included).²¹ These cats were mainly recruited among staff and students of the faculty of veterinary medicine for participation in the mentioned prospective research projects.^{21, 22}

Signalment data, USG, total thyroxine values, sCr concentration, and GFR results determined by clearance of exo-iohexol were retrospectively retrieved from the medical file of all cats. Symmetric dimethylarginine concentrations were retrospectively measured in spare plasma samples. Animals with unknown sCr or exo-iohexol GFR results and cats with insufficient plasma samples were excluded from the study. Recently, a nonsignificant relationship was found between SDMA and GFR in a sample of hyperthyroid cats, possibly because both production and metabolism of SDMA may be altered due to thyroxine changes in the blood, independent of the GFR.²⁴ In the present study, hyperthyroidism was diagnosed based on thyroid gland palpation, compatible clinical signs and a serum total thyroxine concentration (with RI: 14.19-45.15 nmol/L and measured in the majority of the study cats). Hyperthyroid cats were excluded from the current study.

2.2

Analyses

2.2.1

Glomerular filtration rate

Glomerular filtration rate (GFR) had been measured using a combined plasma exogenous creatinine Iohexol clearance test (PEC-ICT).^{21, 22, 25} Briefly, 64.7 mg/kg exo-iohexol and 40 mg/kg creatinine had been injected intravenously. Ethylenediaminetetraacetic acid (EDTA) plasma samples had been collected before and 5, 15, 30, 60, 90, 120, 180, 360, and 600 minutes after injection. For our study, only values of the exo-iohexol clearance were used. The plasma levels of exo-iohexol had been analyzed using high-performance liquid chromatography with ultraviolet detection.²³ Pharmacokinetic analyses had been performed using WinNonlin (WinNonlin Version 4.0.1, Scientific Consulting Inc, Apex, NC). The plasma data had been subjected to noncompartmental analysis using a statistical moment approach. The area under the curve (AUC) of plasma concentration-vs-time had been calculated using the trapezoidal rule with extrapolation to infinity, as described by Watson et al.²⁶ Plasma exo-iohexol clearance had been determined by dividing dose administered by AUC and indexed to bodyweight (mL/[min kg]).

Evaluation of sensitivity and specificity of indirect markers, SDMA and sCr, was done by 2 different GFR cut-off values using the same clearance technique as described by Paepe et al.²³ A borderline GFR cut-off value, indicating mildly impaired renal function, was set at 1.7 mL/(min kg).²³ The low GFR cut-off value indicating CKD was set at 1.2 mL/(min kg).²³

2.2.2

Serum creatinine

The sCr concentration had been determined using a modified Jaffe method with an RI of 64.5 to 161.8 μ mol/L (0.73-1.83 mg/dL).²⁷

2.2.3

Symmetric dimethylarginine

Residual plasma samples (−80°C) from the clearance test were thawed to 20°C and sent in batches to IDEXX Laboratories GmbH, Leipzig, Germany for analysis. Quantification of SDMA concentration was performed using the validated immunoassay IDEXX SDMA Test (IDEXX Laboratories Inc, Westbrook, Maine). The upper limit of the RI was 14 µg/dL, meaning that values >14 µg/dL were indicative of impaired renal function. Symmetric dimethylarginine analysis was performed preferably on the basal (T0) sample that was collected just before the start of the GFR measurement. When plasma samples of T0 were missing, samples obtained 5 minutes after injection (T5) of the ex-iohexol marker were used. Storage time of the samples was at least 3 years and a maximum of 8 years. Plasma samples that underwent 1 freeze-thaw cycle were only used in cases where there was no alternative.

2.3

Statistical analysis

SAS (Statistical Analysis Software Version 9.4, SAS Institute Inc, Cary, North Carolina) was used for all statistical analyses using a global significance level of 5%.

An ANOVA *F*-test was used to check for the presence of a group effect on each variable (SDMA_{plasma}, GFR, and sCr) and was followed by Dunnett's multiple comparisons where diseased animals (DM/CKD) were compared pairwise with healthy controls. *P* values were adjusted for multiple comparisons. Kendall's Tau correlation coefficients (τ_B) were calculated to investigate the relationship between GFR and SDMA_{plasma} and sCr. Subsequently, we calculated the specificities and sensitivities (and 95% confidence interval) for SDMA_{plasma} and sCr at their predefined cut-off values and at 2 different levels of GFR impairment since the renal clearance is the gold standard for CKD diagnosis. With the 2 different threshold values for GFR in mind, we calculated sensitivities and specificities corresponding to a wide range of alternative cut-off values of SDMA and sCr. Sensitivities were plotted against 1-specificities ultimately resulting in 4 receiver operator characteristic (ROC) curves. For each plot, the AUC was calculated (with the 95% confidence interval included²⁸) as an evaluation of the diagnostic accuracy or distinctiveness of the biomarker to detect a decreased GFR.

3

RESULTS

3.1

Study sample and descriptive statistics

A total of 49 cats were included in this study. Seventeen animals were diagnosed with CKD. Based on the current International Renal Interest Society (IRIS) guidelines,²⁹ 11 had IRIS stage 2 (sCr: 140-250 µmol/L [1.6-2.8 mg/dL]) and 6 had stage 3 CKD (sCr: 251-440 µmol/L [2.9-5.0 mg/dL]). Fifteen cats had DM and 17 cats were considered healthy control animals. The majority of the animals with DM were not sufficiently controlled for DM despite therapy, had fructosamine levels > 600 µmol/L and presence of glucosuria (blood glucose > 15 mmol/L [270 mg/dL]).

Further characteristics of the study sample are shown in Table ¹.

TABLE 1

Overview of the different breeds and sexes within the CKD group, DM, and healthy control group presented in absolute numbers and (percentage ratio between brackets)^{21 22},

		CKD n = 17	DM n = 15	Control n = 17	Total n = 49
Breed	Eur SH	n = 11 (65%)	n = 12 (80%)	n = 17 (100%)	n = 40 (82%)
Other breeds	British SH n = 2 Siamese n = 1 Persian n = 1 Ragdoll n = 1 Burmese n = 1	Eastern SH n = 1 British SH n = 1 Burmese n = 1		British SH n = 3 Burmese n = 2 Persian n = 1 Ragdoll n = 1 Siamese n = 1 Eastern SH n = 1	
Sex	M	n = 0	n = 1 (6.5%)	n = 0	n = 1 (2%)
F	n = 1 (6%)	n = 1 (6.5%)	n = 0	n = 2 (45%)	
Mc	n = 10 (59%)	n = 12 (80%)	n = 5 (29.5%)	n = 27 (55%)	
Fc	n = 6 (35%)	n = 1 (6.5%)	n = 12 (70.5%)	n = 19 (39%)	

Abbreviations: British SH, British Shorthair; CKD, chronic kidney disease; DM, diabetes mellitus; Eastern SH, Eastern Shorthair; Eur SH, European Shorthair; F, female; FC, female castrated; M, male; MC, male castrated.

Spare plasma samples were available for all cats. Samples of T0 were missing in 4 cats (all healthy cats), but spare plasma collected at T5 during the GFR procedure were available for SDMA analysis in these animals. Five plasma samples underwent 1 freeze-thaw cycle before SDMA measurement.

As shown in Table ², a GFR < borderline cut-off value of 1.7 mL/(min kg) was present in 21/49 cats. Sixteen of them belonged to the CKD group while 2 were diagnosed with DM, and the remaining 3 cats belonged to the healthy group. Fifteen of these animals obtained a GFR < the low cut-off value of 1.2 mL/(min kg). One of these was recruited as a healthy cat and had sCr of 104 µmol/L and USG of 1.038. All other cats with a GFR < 1.2 mL/(min kg) belonged to the CKD group.

TABLE 2

Classification of the cat sample of 49 cats based on estimation of the GFR by exo-iohexol clearance test

Group	GFR (mL/[min kg])		
<1.2	1.2 ≤ GFR < 1.7	≥1.7	
Chronic kidney disease	15	1	1
Diabetes mellitus	0	2	13
Healthy	1	2	14
Total	16	5	28

Note: Within each group (CKD, DM, and healthy control group), the number of cats with a normal renal clearance ($\text{GFR} \geq 1.7 \text{ mL}/[\text{min kg}]$), mild renal impairment ($\text{GFR} \geq 1.2 \text{ mL}/[\text{min kg}]$ and $<1.7 \text{ mL}/[\text{min kg}]$), and severe renal dysfunction ($\text{GFR} < 1.2 \text{ mL}/[\text{min kg}]$) are displayed.

Abbreviations: CKD, chronic kidney disease; DM, diabetes mellitus; GFR, glomerular filtration rate.

Mean results of exo-iohexol GFR, sCr, SDMA_{plasma}, and USG for the complete study sample and the different subgroups were available and presented in Table ³. As total thyroxine serum values were missing in 7/49 cats (subgroup CKD: n = 2; DM: n = 3; control: n = 3), mean values were calculated based on the available results.

TABLE 3

Overview of the mean values and (SDs between brackets) for GFR, sCr, SDMA_{plasma}, USG, bodyweight and age for 17 CKD cats, 15 DM cats, 17 healthy cats, and the complete study sample of 49 cats ^{21 22},

	CKD n = 17	DM n = 15	Control n = 17	Total n = 49
Exo-iohexol GFR	0.9 (0.4) ^a	2.1 (0.6)	2.1 (0.5) ^a	1.7 (0.8)
sCr	249.7 (71.6) ^b [2.8 (0.8)]	111.6 (24.8) [1.3 (0.3)]	104.1 (25.9) ^b [1.2 (0.3)]	156.9 (82.4) [1.8 (0.9)]
SDMA _{plasma}	26.7 (9.9) ^c	12 (2.4)	12.5 (4.6) ^c	17.3 (9.5)
USG	1.020 (0.009)	1.036 (0.010)	1.045 (0.009)	1.033 (0.014)
TT4	24.3 (10.1)	14.7 (8.2)	30.2 (6.3)	23.7 (10.3)
Bodyweight	4.2 (1.2)	4.9 (1.2)	4.5 (1.3)	4.5 (1.2)
Age	10.3 (4.9)	9.5 (2.9)	10.5 (3.1)	10.1 (3.7)

Note: Mean values and (SDs between brackets) for TT4 were available for 15 CKD cats, 12 DM cats, 15 healthy cats, and a total of 42 study cats. ^{21 22} GFR in mL/(min kg), sCr in $\mu\text{mol}/\text{L}$ and $[\text{mg}/\text{dL}]$, SDMA_{plasma} in $\mu\text{g}/\text{dL}$, TT4 in nmol/L, bodyweight in kg, and age in years were available for the 49 cats. Means sharing the same letter differed significantly from each other ($P < .05$).

Abbreviations: CKD, chronic kidney disease; DM, diabetes mellitus; GFR, glomerular filtration rate; sCr, serum creatinine; SDMA, symmetric dimethylarginine; TT4, total thyroxine; USG, urine-specific gravity.

SDMA_{plasma}, sCr, and GFR differed significantly among the groups ($P < .001$ for each variable). Cats with CKD possessed significantly lower GFR results and significantly higher sCr and SDMA concentrations compared to healthy individuals (all $P < .001$). No significant differences in GFR, sCr, and SDMA were found between DM and healthy animals.

3.2

Correlation between kidney function tests

Kendall's Tau correlation coefficients (τ_B) revealed for the 49 cats that the correlation between SDMA_{plasma} and GFR ($\tau_B = -0.57$; $P < .001$) was moderate; however, the correlation between sCr and GFR was of the same magnitude ($\tau_B = -0.56$; $P < .001$). High concentrations of both biomarkers were associated with reduced filtration rate. The correlation between the 2 renal biomarkers SDMA_{plasma} and sCr was also moderate ($\tau_B = 0.52$; $P < .001$).

3.3

Relationship between SDMA and sCr

SDMA_{plasma} and sCr concentrations of the 49 animals were plotted against each other in Figure ¹. Concordant results between both renal biomarkers were confirmed in 39/49 cats. 24/49 had SDMA and sCr results within RI in quadrant I and 15/49 had SDMA and sCr results above the upper reference limit in quadrant IV. Discordant results were present in 10 animals. Five healthy control cats and 3 DM cats had normal sCr concentrations but increased SDMA values (quadrant II). However, 6 of them had a normal GFR (>1.7 mL/[min kg]). Within reference SDMA_{plasma} and increased sCr levels were represented in quadrant III and contained 2 cats that were earlier assigned to the CKD group. The first cat had a GFR measurement <1.7 mL/(min kg), while the renal clearance of the second was <1.2 mL/(min kg), which indicated that in this cat SDMA obviously failed to identify CKD in contrast to sCr.

FIGURE 1

Scatter plot illustrating the relationship between SDMA (on the x-axis) and sCr (on the y-axis) concentrations measured in 17 healthy animals, 17 CKD cats, and 15 cats with DM. Chronic kidney disease animals are visualized as blue triangles; DM and control cats obtain the green circle and orange rhombus shape, respectively. Horizontal and vertical lines refer to the threshold values of sCr (161.8 μ mol/L [1.83 mg/dL]) and SDMA (14 μ g/dL), respectively. Quadrant I and IV represent the cats with concordant results between SDMA and sCr. Quadrant II represents the cats with increased SDMA and normal sCr values. Quadrant III represents the cats with normal SDMA and increased sCr values. The graph demonstrates the positive relationship between both kidney markers ($\tau_B = 0.52$; $P < .001$). CKD, chronic kidney disease; DM, diabetes mellitus; sCr, serum creatinine; SDMA, symmetric dimethylarginine

3.4

Diagnostic value of sCr and SDMA as renal biomarkers

Sensitivities and specificities for SDMA_{plasma} and sCr to detect GFR exceeding the borderline (1.7 mL/[min kg]) and low (1.2 mL/[min kg]) GFR cut-off values are shown in Table ⁴.

TABLE 4

Sensitivity and specificity for SDMA and sCr with upper reference limits 14 μ g/dL and 161.8 μ mol/L (1.83 mg/dL), respectively, and upper reference limits 18 μ g/dL and 155.6 μ mol/L (1.76 mg/dL), respectively

		Cut-off GFR 1.7 mL/(min kg)	Cut-off GFR 1.2 mL/(min kg)		
Cut-off	Sensitivity	Specificity	Sensitivity	Specificity	
SDMA _{plasma}	14 μ g/dL	76.2 (52.8-91.8)	75 (55.1-89.3)	93.7 (69.8-99.8)	75.7 (57.7-88.9)

		Cut-off GFR 1.7 mL/(min kg)	Cut-off GFR 1.2 mL/(min kg)		
Cut-off	Sensitivity	Specificity	Sensitivity	Specificity	
	18 µg/dL	71.4 (47.8-88.7)	96.4 (81.6-99.9)	87.5 (61.6-98.5)	93.9 (79.8-99.3)
sCr	161.8 µmol/L (1.83 mg/dL)	71.4 (47.8-88.7)	96.4(81.6-99.9)	87.5 (61.7-98.4)	93.9 (79.8-99.3)
	155.6 µmol/L (1.76 mg/dL)	76.2 (52.8-91.8)	92.9 (76.5-99.1)	93.8 (69.8-99.8)	90.9 (75.7-98.1)

Note: 95% CIs are added between the brackets. Test positive and negative results are objectively evaluated by means of exceeding the borderline GFR cut-off and the low GFR cut-off.

Abbreviations: GFR, glomerular filtration rate; sCr, serum creatinine; SDMA, symmetric dimethylarginine.

We determined the threshold concentrations for SDMA and sCr that resulted in the ideal combination of sensitivity and specificity to optimize the diagnostic force for both biomarkers. To identify any cat with a GFR result <1.7 mL/(min kg), a SDMA cut-off value of 18 µg/dL and sCr threshold value of 155.6 µmol/L (1.76 mg/dL) was more appropriate. The associated sensitivities and specificities are also presented in Table ⁴.

Receiver operating characteristic curves (Figure ²) illustrate that the AUC of SDMA_{plasma} and sCr was 0.86 (95% CI = 0.79-0.93) and 0.90 (95% CI = 0.84-0.96), respectively, to detect mild kidney dysfunction (GFR < borderline GFR cut-off of 1.7 mL/[min kg]). To detect obvious kidney dysfunction (GFR < low GFR cut-off of 1.2 mL/[min kg]), SDMA_{plasma} had an AUC of 0.95 (95% CI = 0.91-0.99) while sCr achieved an AUC of 0.93 (95% CI = 0.89-0.98) (Figure ³).

FIGURE 2

Receiver operator characteristic curves of SDMA (left) and sCr (right) showing the diagnostic ability of the biomarkers to detect GFR < borderline GFR cut-off (1.7 mL/[min kg]). On the x-axis, 1-specificity is shown for a range of possible cut-off values. The y-axis demonstrates the sensitivity of the renal biomarker for a range of possible cut-off values. The figure demonstrates a similar AUC of 0.86 (with a 95% CI of 0.79-0.93) and 0.90 (with a 95% CI of 0.84-0.96) for SDMA_{plasma} and sCr, respectively. AUC, area under the curve; GFR, glomerular filtration rate; sCr, serum creatinine; SDMA, symmetric dimethylarginine

FIGURE 3

Receiver operator characteristic curves of SDMA (left) and sCr (right) showing the diagnostic ability of the biomarkers to detect GFR < low GFR cut-off (1.2 mL/[min kg]). On the x-axis, 1-specificity is shown for a range of possible cut-off values. The y-axis demonstrates the sensitivity of the renal biomarker for a range of possible cut-off values. The figure demonstrates a similar AUC of 0.95 (with a 95% CI of 0.91-0.99) and 0.93 (with a 95%CI of 0.89-0.98) for SDMA_{plasma} and sCr, respectively. AUC, area under the curve; GFR, glomerular filtration rate; sCr, serum creatinine; SDMA, symmetric dimethylarginine

DISCUSSION

The present study aimed to investigate the benefit of SDMA as an indirect renal biomarker in 49 nonazotemic and azotemic cats. By using an exo-iohexol clearance test, renal function was objectively established and diagnostic results of SDMA and sCr were evaluated in light of this gold standard. Using retrospective plasma samples, we recognized that both biomarkers were equally correlated with the GFR and we confirmed a mild and clinically nonrelevant difference in sensitivity between SDMA and sCr in the detection of renal function loss. In conclusion, the diagnostic results of both tests were comparable.

Unlike sCr, SDMA is only slightly subjected to extra renal influence.^{18, 19} Based on the claim that SDMA has superior sensitivity than sCr to detect renal dysfunction, a stronger correlation between SDMA and GFR compared to the correlation between sCr and GFR could be anticipated. In contrast, we observed both SDMA_{plasma} and sCr were equally correlated with the renal clearance. Both biomarkers are worthy as a surrogate marker for GFR measurement to objectively evaluate renal function. Plasma SDMA increased with decreasing renal clearance but both variables were not perfectly correlated. Variations in SDMA results cannot be completely attributed to changes in the filtration capacity of the kidney and additional factors in the metabolism or elimination of the SDMA molecule must be considered.

Until now, the influence of systemic conditions or treatment on SDMA concentration has been poorly studied in small animals.³⁰ Recent research with hyperthyroid cats showed a poor correlation between SDMA and GFR values, probably due to changes in protein metabolism by the hyperthyroid state of the cats.²⁴ Therefore, we excluded hyperthyroidism based on a number of diagnostic tools including the determination of serum total thyroxine. For a limited number of cats, thyroxine values were missing but based on their young age, the absence of compatible clinical signs, and a normal thyroid palpation; hyperthyroidism was an unlikely diagnosis. In the present study, a large number of animals with normal GFR suffered from DM so an extra-renal influence of this endocrine disorder on plasma SDMA could not be ruled out. Nevertheless, we could not prove significant difference in SDMA blood concentration between the healthy control group and nonazotemic cats with DM. The influence of (insufficiently controlled) DM on SDMA concentration with significantly lower SDMA blood levels in DM cats (without comorbidities) compared to healthy control animals has been reported, probably due to hyperfiltration and osmotic diuresis.³¹ Since GFR was not measured in that study, these findings are hard to interpret. Further investigation to determine if DM can influence SDMA concentrations in cats is warranted.

The scatter plot illustrating all SDMA and sCr results indicated several discordant results. The GFR values indicated that the majority of cats with conflictingly high SDMA and normal sCr concentrations showed a false positive SDMA result, reflected in a somewhat lower specificity of SDMA (76%) compared to sCr (94%-96%). Using the upper limit of the RI (161.8 $\mu\text{mol/L}$ [1.83 mg/dL] for sCr and 14 $\mu\text{g/dL}$ for SDMA), sCr only generated a minimum number of false positive test results ($n = 1$) while SDMA incorrectly suspected more healthy cats with deteriorated kidney function ($n = 7$). As expected, sensitivity and specificity of both kidney biomarkers was partly influenced by the GFR cut-off value. The borderline GFR cut-off value generally led to fewer false positive test results for SDMA and sCr compared to the low GFR cut-off value. Increased sCr despite normal SDMA results was recorded in 2 cats. Both cats had a decreased GFR value indicating false negative SDMA results. However, the perceived differences in sensitivity between sCr and SDMA in the present study were not clinically relevant. Both SDMA and sCr correctly identified

cats in advanced stages of renal function loss ($<1.2 \text{ mL}/[\text{min kg}]$). Overall, it is important to note that the mild differences in sensitivity and specificity of sCr and SDMA are probably attributable to their RI as both biomarkers show equal performance in the ROC curve analysis.

According to our data, SDMA offered little added diagnostic value compared to the long-implemented sCr. The accuracy in detecting a $\text{GFR} < 1.7 \text{ mL}/(\text{min kg})$ and a $\text{GFR} < 1.2 \text{ mL}/(\text{min kg})$ was investigated, and diagnostic performance of both markers improved as renal impairment progressed. However, SDMA is commercially promoted as a highly sensitive diagnostic tool providing extra value in detecting cats with early renal function loss who are missed with the use of the traditional renal marker sCr.^{10, 32}

Our findings are not completely in line with previous research due to multiple reasons. Across studies the SDMA cut-off value stays fixed, but the sCr RI widely varies due to interlaboratory differences in samples selected for the establishment of the RI. This affects sensitivity of sCr. The higher the upper reference limit, the more false negative test results are generated which will ultimately result in a lower sensitivity for this biomarker. Our RI of sCr was determined by Ghys et al.²⁷ and had the important advantage of being laboratory-specific, but compared to other studies, the upper reference limit for sCr was rather low. In addition, analytical variation can arise by the usage of different laboratory quantification techniques for measuring sCr (enzymatic vs colorimetric) as well as GFR (marker and sampling strategy). This makes comparison between studies challenging. The cross-sectional retrospective nature of our research did not allow us to follow-up the animals over time. This led us to the additional disadvantage that we did not possess sufficient data of cats going through IRIS stage I before the presence of azotemia so it was not possible to assess the biomarkers for this purpose.

Using ROC curves, we evaluated the threshold values for SDMA ($14 \mu\text{g}/\text{dL}$) and sCr ($161.8 \mu\text{mol}/\text{L}$ [$1.83 \text{ mg}/\text{dL}$]), which were not ideal for our data set containing CKD and non-CKD cats. A cut-off for SDMA yielding a more optimal combination of sensitivity and specificity was $18 \mu\text{g}/\text{dL}$. Symmetric dimethylarginine with the predetermined cut-off $14 \mu\text{g}/\text{dL}$ generated many false positives. Since sensitivity and specificity are strongly intertwined, a cut-off of $18 \mu\text{g}/\text{dL}$ inherently led to a slight loss in sensitivity, but this disadvantage was limited and in favor of a higher specificity. After the CKD diagnosis is confirmed in a clinically stable, hydrated animal (with exclusion of pre- or postrenal problems), recently updated IRIS guidelines allow staging of the chronic disease by at least 2 measurements of SDMA and sCr in a fasted animal. Symmetric dimethylarginine repeatedly exceeding $18 \mu\text{g}/\text{dL}$ (even in combination with normal sCr levels) suggests that the animal suffers from at least CKD stage 2.²⁹ Since our study did not include serial monitoring of SDMA and sCr, a direct comparison of our results with the IRIS staging system is not indicated.

For sCr, a slightly lower threshold value of $155.6 \mu\text{mol}/\text{L}$ ($1.76 \text{ mg}/\text{dL}$) was preferred. Since the specificity of sCr was high in our study sample of cats, it was desirable to improve sensitivity to the disadvantage of a minor loss in specificity.

4.1

Limitations

Although the present study is retrospective in nature, the majority of our data (sCr and GFR) originated from cats recruited in the context of previous prospective research studies.^{21, 22} All the cats included were subjected to the same

protocol: a general physical examination, blood and urine testing, and a GFR clearance test. This minimized the limitations of the retrospective nature of the present study. The most important inclusion criterion for our study was the availability of sufficient residual blood samples in order to perform SDMA analysis, implying that the 49 cats were not randomly selected. This was disadvantageous in obtaining a representative study sample.

We aimed to test the accuracy of SDMA and sCr in a sample of nonazotemic and azotemic cats. Performance of a kidney marker is better with advanced renal failure (IRIS stage III/IV) while the challenge and added value of a biomarker is mainly based on its capacity to identify animals with minimal GFR loss.³³ Five cats out of all achieved a renal clearance between 1.2 and 1.7 mL/(min kg), indicating the presence of mild renal impairment. Furthermore, the small number of cats with early CKD could explain the reason for no added diagnostic value of SDMA that could be established in our data set and further studies are needed.

The allocation of the cats to the different subgroups was based on physical examination and routine laboratory analysis. But when the GFR result was taken into account, 3 cats initially considered as healthy and 2 cats with DM were found to have mildly impaired renal function. On the other hand, 1 cat of the CKD group had a GFR value >1.7 mL/(min kg).

Residual frozen blood samples stored for maximum 8 years were used for SDMA analysis. Short-term stability of the molecule in blood has already been reported.³⁴ The effect of long-term preservation on accurate analysis of SDMA is a relevant issue that has not been clarified. In addition, several plasma samples (n = 5) were subjected to 1 freeze-thaw cycle before SDMA analysis. Multiple freeze-thaw cycles do not generate significant changes in the SDMA concentration.³⁴ Furthermore, IDEXX claims the molecule remains stable for several years if the sample is frozen.³⁰

Symmetric dimethylarginine concentrations were quantified using the immunoassay “SDMA IDEXX test” and not the gold standard liquid chromatography-mass spectrometry (LC-MS). The SDMA IDEXX test uses glucose-6-phosphate dehydrogenase conjugate and monoclonal anti-SDMA antibodies, is less expensive and time-consuming than the LC-MS, and is currently widely used in commercial veterinary laboratories. This means that the results of our study have the advantage of being clinically applicable to veterinary practice. The IDEXX SDMA test is an accurate technique. Within the 10 to 45 µg/dL range, the maximum measurement error is estimated to be 1 to 3 µg/dL.³⁵ In addition, the test is not sensitive to lipemia, icterus, or mild-to-moderate hemolysis, which usually arises during blood sampling.³²

5

CONCLUSIONS

In this retrospective study, SDMA behaved as an accurate biomarker for detecting impaired renal function defined by exo-iohexol clearance in cats. However, in the present sample of adult nonazotemic and azotemic cats, we could not prove that SDMA offered prominent added value compared to the conventional sCr biomarker. The diagnostic value of both molecules was approximately equivalent and improved in the advanced phases of renal dysfunction.

CONFLICT OF INTEREST DECLARATION

Authors declare no conflict of interest.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

Authors declare no IACUC or other approval was needed.

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

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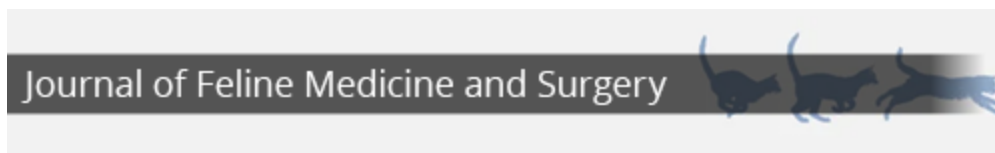
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Longitudinal evaluation of discordant symmetric dimethylarginine and creatinine as markers of glomerular filtration rate in radioiodine-treated hyperthyroid cats

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Abstract

Objectives

The aim of the present study was to determine the frequency and clinical significance of discordant serum symmetric dimethylarginine (SDMA) and creatinine concentrations relative to glomerular filtration rate (GFR) in mature and senior hyperthyroid cats treated with radioactive iodine (RAI). Secondary aims of the study were to compare the sensitivity and specificity of two SDMA upper reference intervals (RIs; >14 and $\geq$ 18 μ g/dl) and creatinine ($\geq$ 175 μ mol/l) for detecting decreased GFR and to evaluate the intra-individual variability of both biomarkers.

Methods

A prospective longitudinal study of 27 client-owned hyperthyroid cats receiving low-dose RAI therapy was conducted. At baseline, 1, 6 and 12 months after RAI, cats were evaluated for thyroid status, GFR, SDMA and creatinine concentrations. Discordance was defined as one renal biomarker exceeding its RI with the other within/below the RI, using laboratory and age-specific SDMA RIs. In euthyroid cats, sensitivity, specificity and predictive values of SDMA/creatinine for decreased GFR and intra-individual variability were determined.

Results

Discordance between SDMA and creatinine was in the range of 10–75% and was highest with the SDMA limit $>14 \mu\text{g/dl}$. SDMA-GFR discordance exceeded creatinine-GFR discordance at all time points and was highest at baseline, when all cats were hyperthyroid. In cats with euthyroid status at 12 months, SDMA $>14 \mu\text{g/dl}$ was more sensitive (83%) but less specific (50%) for decreased GFR than SDMA $\geq 18 \mu\text{g/dl}$ (sensitivity 50%, specificity 71%) and creatinine (sensitivity 50%, specificity 85%). Intra-individual variability was similar for both biomarkers ($\sim 15\%$).

Conclusions and relevance

Discordant SDMA and creatinine concentrations are frequent in older hyperthyroid cats and are influenced by thyroid status and SDMA RIs, complicating interpretation. The SDMA limit $\geq 18 \mu\text{g/dl}$ improves specificity for decreased GFR and is more appropriate for senior cats. SDMA concentrations in the range of $15\text{--}18 \mu\text{g/dl}$ are less likely to reflect true GFR reduction when creatinine is normal; therefore, non-renal influences such as hyperthyroidism should be considered and ongoing monitoring of renal parameters is recommended.

Keywords: Kidney, chronic kidney disease, renal, biomarkers

Plain language summary

A long-term assessment of two different indirect blood indicators of kidney function in cats undergoing radioiodine treatment for hyperthyroidism, to determine any discrepancies between the two and how these relate to direct measurements of kidney function

This study looked at how two common blood tests for kidney function, symmetric dimethylarginine (SDMA) and creatinine, behave in older cats treated with radioiodine for hyperthyroidism. Vets often use these tests to check how well the kidneys are functioning, but the

two markers do not always agree. The researchers wanted to understand how often this happens and which test is more reliable when compared with a more direct measurement of kidney function. A total of 27 hyperthyroid cats were followed for 1 year. Before treatment with radioiodine and again at 1, 6 and 12 months afterwards, the cats had their thyroid levels, kidney glomerular filtration rate (GFR), SDMA and creatinine measured. The team looked for cases where one kidney marker (either SDMA or creatinine) was high but the other was normal or low, and they assessed how well each marker detected a true decrease in kidney filtration function by comparing them to GFR. The researchers found that SDMA and creatinine results did not agree in 10–75% of cats, especially when using a lower SDMA cut-off ($>14\text{ }\mu\text{g/dl}$). SDMA did not agree with the direct measure of kidney function, GFR, more frequently than creatinine, particularly when the cats were still hyperthyroid. At 12 months after treatment, in the cats with normal thyroid levels, the higher SDMA cut-off ($\geq 18\text{ }\mu\text{g/dl}$) was more accurate and produced fewer false-positive results. Overall, the study shows that discrepancies in SDMA and creatinine results are common in older hyperthyroid cats, and thyroid disease itself can affect SDMA. SDMA values of $15\text{--}18\text{ }\mu\text{g/dl}$ may not truly indicate kidney disease if creatinine is normal. For senior cats, using the higher SDMA cut-off improves accuracy. When results do not match, vets should consider non-kidney factors, like thyroid status, and continue monitoring over time.

Introduction

Chronic kidney disease (CKD) and hyperthyroidism are common comorbidities in mature (7–10 years) and senior (>10 years) cats.^{1–3} Restoration of euthyroidism after treatment of hyperthyroidism can result in unmasking decreased renal filtration function associated with CKD as the hyperthyroid state increases glomerular filtration rate (GFR).⁴ Reduced renal filtration function is most sensitively detected by decreased GFR, which has been reported in 29% of hyperthyroid cats in the 12 months after radioactive iodine (RAI) treatment;⁵ therefore monitoring of renal function is recommended.^{4,5} In the clinical setting, direct measurement of GFR is rarely performed because of the requirement for multiple samples to be collected over several hours, costs of analysis and lack of standardisation in interpretation. Therefore, routine evaluation of renal filtration function is indirectly measured using serum concentrations of the renal biomarkers symmetric dimethylarginine (SDMA) and creatinine.^{6,7} Detection of persistent increased serum concentrations of these biomarkers, in the absence of any prerenal or postrenal causes, is fundamental for the diagnosis and staging of CKD as outlined by the International Renal Interest Society (IRIS).⁷

The reference interval (RI) for the commercially available immunoassay for the measurement of serum SDMA classifies serum SDMA $>14\text{ }\mu\text{g/dl}$ as increased and consistent with decreased GFR.⁸

However, a recent study investigating age-specific RIs in healthy mature and senior cats determined that an upper reference limit of $\geq 18 \mu\text{g/dl}$ for serum SDMA concentrations may be more appropriate and that the upper limit of $>14 \mu\text{g/dl}$ may overestimate renal dysfunction.⁹ The suggested age-specific RI of $\geq 18 \mu\text{g/dl}$ is also in line with the SDMA concentrations stated in the IRIS recommendations for staging of IRIS stage 2 CKD in cats.⁷ Therefore, in the older population of cats that develop hyperthyroidism, applying the upper reference limit of $\geq 18 \mu\text{g/dl}$ may be appropriate for monitoring renal function.

Hyperthyroidism can affect biomarkers of renal filtration function due to glomerular hyperfiltration, and loss of lean muscle mass in hyperthyroid cats may decrease endogenous creatinine production and thus circulating creatinine concentrations.¹⁰ A physiological influence of hyperthyroidism on SDMA metabolism is also possible, and previous studies suggest that thyroid status likely influences SDMA concentrations in cats.^{11,12} In human patients with hyperthyroidism, increased SDMA concentrations have been reported in the absence of concurrent renal disease.^{13,14} These findings suggest that production or metabolism of SDMA may be altered by thyroid dysfunction independent of altered GFR.

In the clinical setting, when investigating potential decreased renal function, both SDMA and creatinine are frequently available in biochemistry panels. There have been few studies performed on the clinical application of SDMA to guide decision-making.¹⁵ Therefore, in the clinical scenario whereby SDMA and creatinine results are discordant (ie, one is increased above the RI while the other is not), the clinician faces a conundrum in the interpretation of GFR. The primary aim of this study was to describe the frequency of discordant SDMA and creatinine concentrations in relation to GFR, utilising both the laboratory RI greater than $14 \mu\text{g/dl}$ and the proposed age-specific RI ($\geq 18 \mu\text{g/dl}$)⁹ for SDMA, over a 12-month follow-up period in a population of mature and senior hyperthyroid cats treated with RAI and to determine the effect of hyperthyroidism on discordance. Secondary aims of the study were to determine the sensitivity, specificity and predictive values of both upper limits of SDMA (>14 and $\geq 18 \mu\text{g/dl}$) and creatinine in detecting decreased GFR and the intra-individual variation of markers in euthyroid cats.

Materials and methods

Study population

A total of 27 client-owned hyperthyroid cats presenting to a referral hospital for low-dose (3 mCi; 111 MBq) RAI treatment had been recruited prospectively into a study and the cohort followed longitudinally for a 12-month follow-up period; thyroid function (total thyroxine [TT4] and

thyroid stimulating hormone [TSH]), serum creatinine concentration and GFR data from these cats have been published previously.⁵ Cats had been evaluated at the time of recruitment into the study for baseline measurements (T0) and at 1- (T1), 6- (T2) and 12-month (T3) time points after RAI treatment. Each assessment comprised a full physical examination, systolic blood pressure via Doppler technique, and blood and urine sampling. The study was approved by the University of Bristol Animal Welfare and Ethics Review Body (VIN/12/025).

Measurement of thyroid status

Serum TT4 had been measured using a chemiluminescent immunoassay at a commercial reference laboratory (Langford Diagnostic Laboratories, Langford Vets, University of Bristol, UK)¹⁶ while serum TSH was measured using a validated canine assay at a commercial reference laboratory (Royal Veterinary College Diagnostic Laboratory Services, Hatfield, UK).¹⁷

Each cat was classified into one of four possible thyroid statuses at each of the four time points:

- Hypothyroid: TT4 below 15 nmol/l (<1.2 µg/dl), TSH above 0.3 ng/ml
- Euthyroid: TT4 less than or equal to 60 nmol/l ($\leq$ 4.7 µg/dl), TSH less than or equal to $\leq$ 0.3 ng/ml
- Subclinical hypothyroid: TT4 15–60 nmol/l (1.2–4.7 µg/dl), TSH above 0.3 ng/ml
- Hyperthyroid: TT4 above 60 nmol/l (>4.7 µg/dl), TSH below 0.03 ng/ml

This classification resulted in 4/27 cats being reclassified from the statuses published in the original study by Finch et al⁵ due to revision of TSH concentrations for classifications of hypothyroidism.^{5,18}

Measurement of renal filtration function

Glomerular filtration rate was determined using a slope-intercept iohexol clearance method described by Finch et al,¹⁹ and GFR was defined as being decreased if it was below 0.92 ml/min/kg.²⁰ Serum creatinine concentration was measured at Langford Diagnostic Laboratories, with increased creatinine concentration defined as 175 µmol/l or higher ($\geq$ 2.0 mg/dl), the upper limit of the laboratory's RI.

Measurement of SDMA was performed on serum samples that had been stored at –80°C before analysis. SDMA analysis was performed approximately 16 months after data collection was

completed for the original study. SDMA was determined using a validated immunoassay IDEXX SDMA Test (IDEXX Laboratories) with increased SDMA concentration defined as $>14 \mu\text{g/dl}$. An age-specific RI of $\geq 18 \mu\text{g/dl}$ was also evaluated.⁹

Data analysis

Data analysis consisted of descriptive statistics.

Discordant SDMA and creatinine concentrations in relation to GFR

Discordant SDMA and creatinine was defined as being when either of these biomarkers was above the laboratory RI ($>14 \mu\text{g/dl}$ and $\geq 175 \mu\text{mol/l}$, respectively) while the other was within or below the RI. Discordance was also determined by applying the age-specific RI of $\geq 18 \mu\text{g/dl}$ for SDMA. In cats with discordant SDMA and creatinine concentrations, the discordance between SDMA and GFR, and creatinine and GFR was subsequently determined. Discordance with GFR was defined as being when either GFR was below 0.92 ml/min/kg and either creatinine or SDMA were within or below the laboratory RI, or GFR was normal ($\geq 0.92 \text{ ml/min/kg}$) and either biomarker was above the upper limit of the laboratory's RI.

Sensitivity, specificity, and predictive values of serum creatinine and SDMA concentrations in detecting decreased GFR in euthyroid cats

The sensitivity, specificity, negative predictive value (NPV) and positive predictive value (PPV) for the two upper limits of SDMA and creatinine for detecting decreased GFR were determined at the 12-month time point (T3). This time point was selected as it had the greatest proportion of cats with euthyroid status and cats with decreased GFR to include in the analysis. True positive was defined as decreased GFR with increased biomarker (SDMA or creatinine) serum concentrations, false positive as a biomarker above the laboratory RI with normal GFR, true negative as normal GFR with biomarker concentrations (SDMA or creatinine) within the RI and false negative as a biomarker within or below the laboratory RI with decreased GFR. The standard formulae included below were used to determine sensitivity, specificity, NPV and PPV.

$$\text{Sensitivity} = \frac{\text{true positive}}{(\text{true positive} + \text{false negative})}$$

$$\text{Specificity} = \frac{\text{truenegative}}{(\text{truenegative} + \text{falsepositive})}$$

$$\text{NPV} = \frac{\text{truenegative}}{(\text{truenegative} + \text{falsenegative})}$$

$$\text{PPV} = \frac{\text{truepositive}}{(\text{truepositive} + \text{falsepositive})}$$

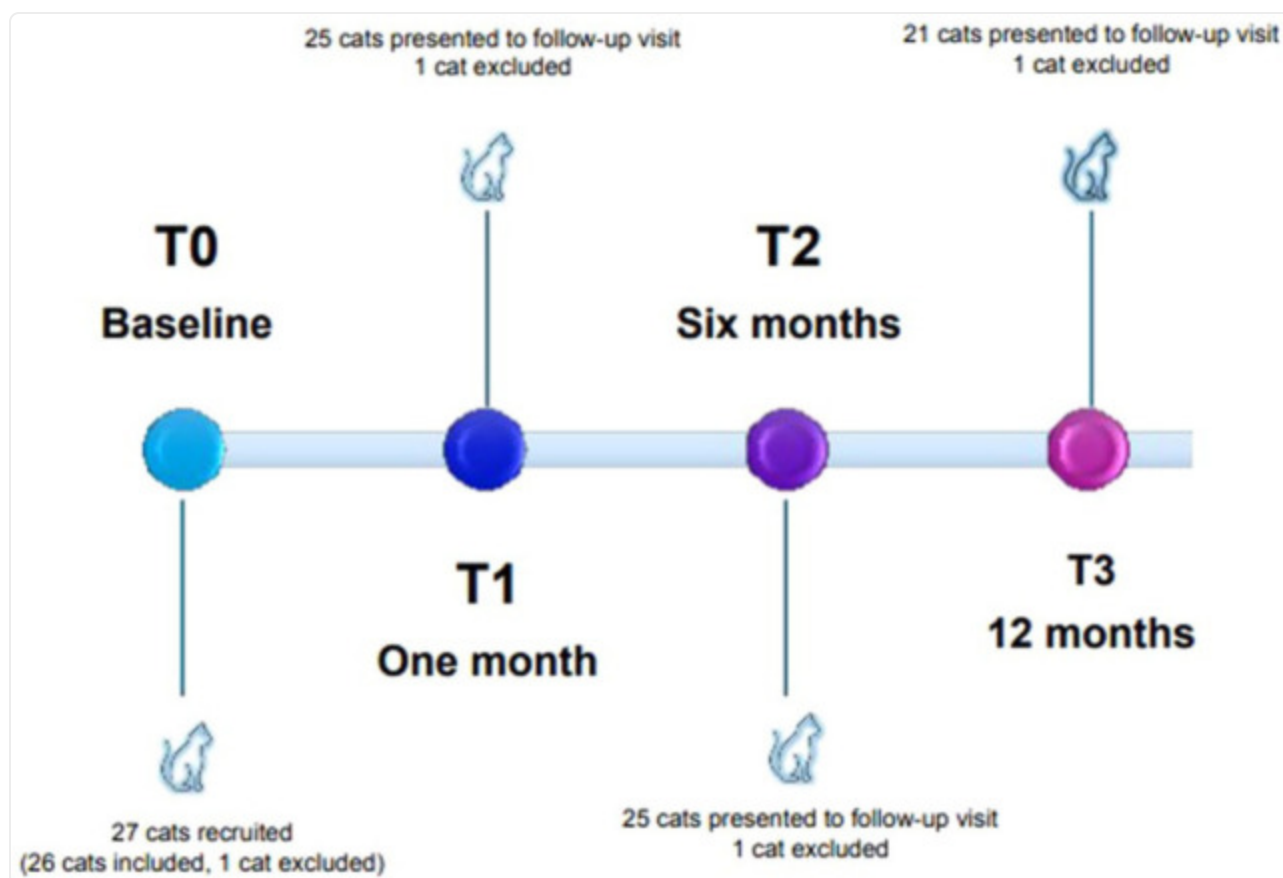
Intra-individual variability in longitudinal measurements of serum SDMA and creatinine in euthyroid cats

The intra-individual variability in longitudinal measurements of serum SDMA and creatinine was evaluated by determining the coefficient of variation percentage, calculated by $(\text{SD}/\text{mean}) \times 100$. At least two SDMA or creatinine measurements were required at time points when cats had euthyroid status.

Results

A total of 27 cats had been recruited to the study; 25 completed the 1- (T1) and 6-month (T2) follow-up visit and 21 completed the 12-month (T3) time points. The median age at recruitment was 11 years (range 7–16). Of the 27 cats, 15 were female neutered, 11 were male neutered and one was female entire. Median (range) TT4, TSH and creatinine concentrations, and GFR have been published previously.⁵ Serum creatinine concentrations had not been measured in one cat at T0 and SDMA had not been measured in one cat at the T1 time point due to insufficient sample volume. GFR measurement was also not available in one cat at the T2 time point and in one cat at the T3 time point. These four cats with incomplete data sets were excluded from analysis at the respective timepoints, leaving 26, 24, 24 and 20 cats for analysis ([Figure 1](#)). A summary of thyroid function status at each time point is provided in [Table 1](#). The median (range) GFR, serum creatinine and SDMA concentrations are presented in [Figure 2](#).

Figure 1.



[Open in a new tab](#)

Study timeline outlining the number of cats analysed at each time point (baseline [T0], 1 [T1], 6 [T2] and 12 months [T3] after radioactive iodine treatment)

Table 1.

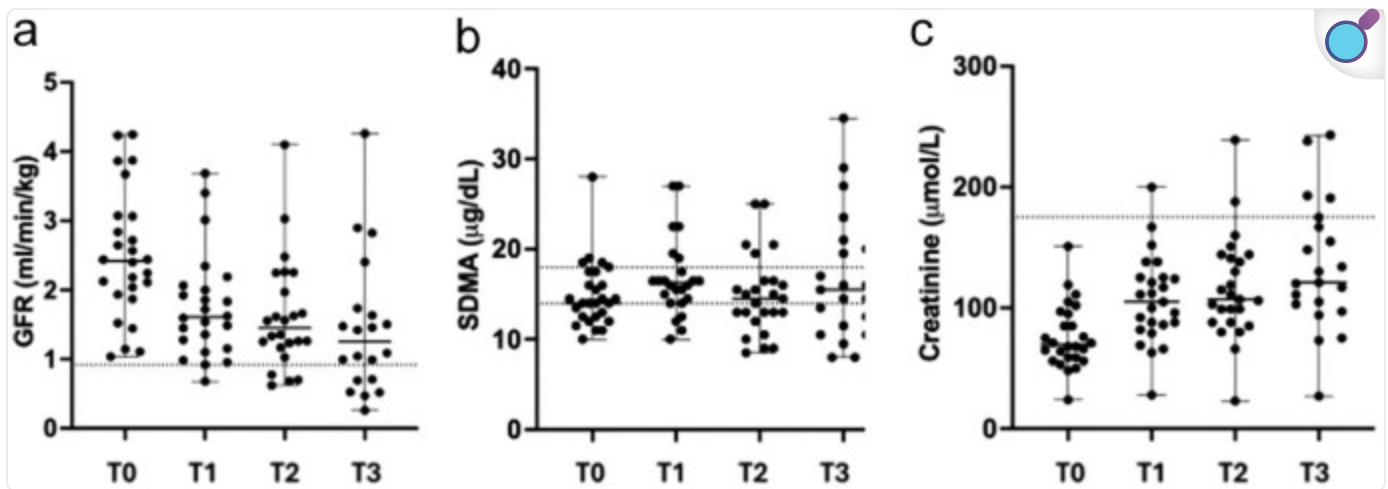
Thyroid status of cats at baseline (T0), 1 (T1), 6 (T2) and 12 months (T3) after radioactive iodine therapy

Time point	T0 (n = 26)	T1 (n = 24)	T2 (n = 24)	T3 (n = 20)
Hyperthyroid	26 (100)	7 (29)	2 (8)	2 (10)
Euthyroid	0 (0)	16 (67)	18 (75)	16 (80)
Subclinical hypothyroid	0 (0)	0 (0)	1 (4)	0 (0)
Hypothyroid	0 (0)	1 (4)	3 (13)	2 (10)

[Open in a new tab](#)

Data are n (%)

Figure 2.



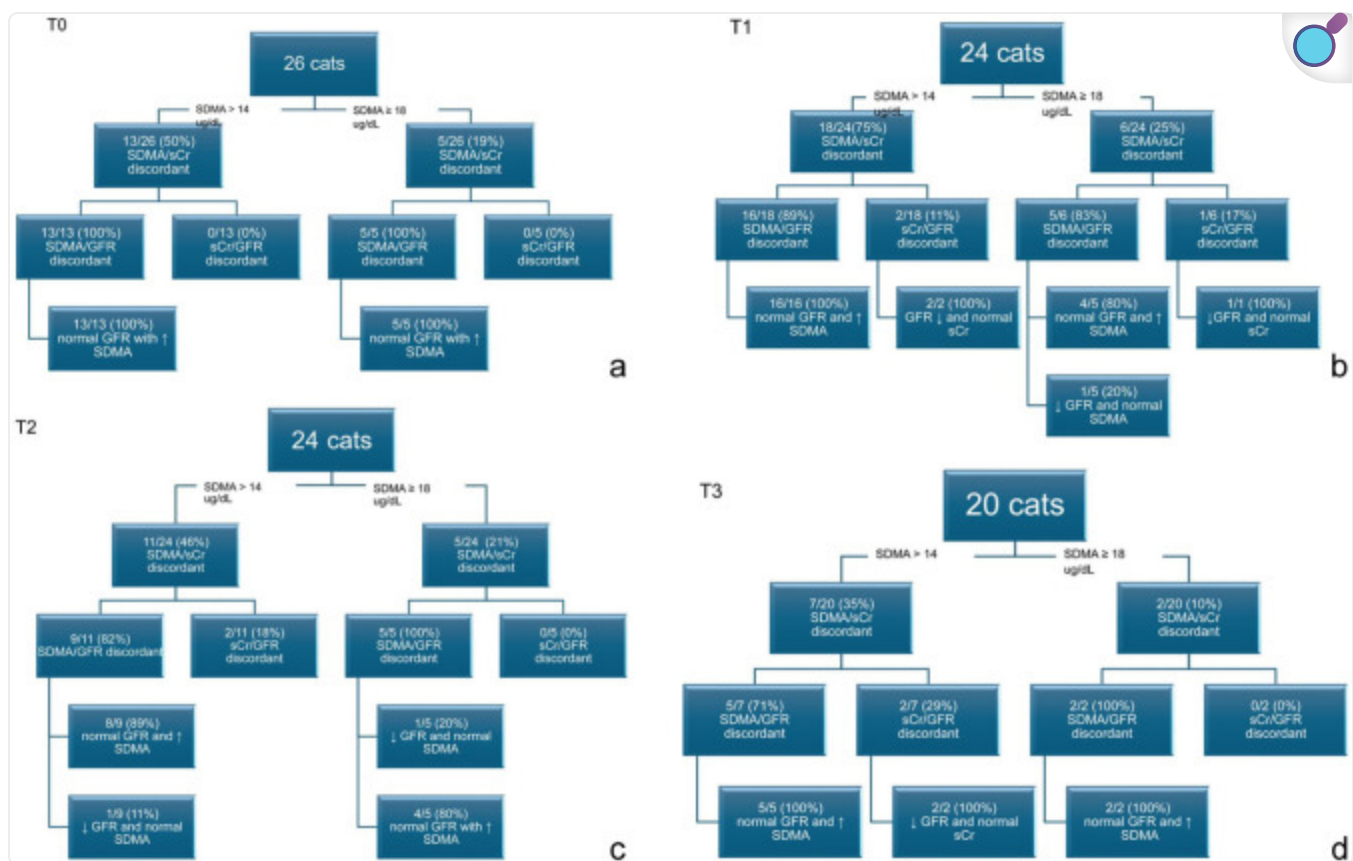
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Measurements of renal filtration function in cats at baseline (T0), 1 (T1), 6 (T2) and 12 months (T3) after radioactive iodine treatment for (a) glomerular filtration rate (GFR), (b) serum symmetric dimethylarginine (SDMA) and (c) serum creatinine. The middle solid line represents the median value and the solid lines above and below this represent the range. The dashed line indicates the value below which GFR was defined as decreased renal filtration function and the upper limit of the reference intervals for serum SDMA and creatinine concentration

Discordant SDMA and creatinine concentrations in relation to GFR

The proportion of discordant results at each time point using both laboratory ($>14 \mu\text{g/dl}$) and age-specific ($\geq 18 \mu\text{g/dl}$) SDMA RIs are presented in [Figure 3a-d](#).

Figure 3.



[Open in a new tab](#)

(a–d) Discordant symmetric dimethylarginine (SDMA) and serum creatinine (Cr) results in cats at (a) baseline (T0), (b) 1 (T1), (c) 6 (T2) and (d) 12 months (T3) after radioactive iodine treatment for both laboratory ($>14 \mu\text{g/dL}$) and age-specific ($\geq 18 \mu\text{g/dL}$) SDMA reference intervals. Of those cats with discordant SDMA and Cr results, SDMA-glomerular filtration rate (GFR) and creatinine-GFR discordance is then presented below at each time point

At T0, all cats were hyperthyroid and none of the cats had decreased GFR or increased serum creatinine. Using the SDMA upper reference limit of $>14 \mu\text{g/dL}$, 13/26 (50%) cats had discordant SDMA and creatinine concentrations (Figure 3a). Of these 13 cats, SDMA and GFR were discordant in all cats (100%). Using the alternative upper reference limit of $\geq 18 \mu\text{g/dL}$, 5/26 (19%) cats had discordant SDMA and creatinine concentrations, of which 100% had discordant SDMA and GFR. At T1, T2 and T3, 18/24 (75%), 11/24 (46%) and 7/20 (35%) cats, respectively, had discordant SDMA

and creatinine concentrations when the upper reference limit of $>14 \mu\text{g/dl}$ and above was applied. Discordance of SDMA and GFR was higher than that of creatinine and GFR at all time points (SDMA vs GFR at T1, T2 and T3 was 16/18 (89%), 9/11 (82%) and 5/7 (71%), respectively, and creatinine vs GFR at T1, T2 and T3 was 2/18 (11%), 2/11 (18%) and 2/7 (29%), respectively) ([Figure 3b-d](#)).

When the upper reference limit of $\geq 18 \mu\text{g/dl}$ was used for SDMA, at T1, T2 and T3, 6/24 (25%), 5/24 (21%) and 2/20 (10%) cats, respectively, had discordant SDMA and creatinine concentrations ([Figure 3b-d](#)). Discordance of SDMA and GFR was higher than that of creatinine and GFR at all time points (SDMA vs GFR at T1, T2 and T3 was 5/6 (83%), 5/5 (100%) and 2/2 (100%), respectively, and creatinine vs GFR at T1, T2 and T3 was 1/6 (17%), 0/5 (0%) and 0/2 (0%), respectively) ([Figure 3b-d](#)).

Concordant results are presented in [S1 in the supplementary material](#) [↗](#).

Sensitivity, specificity and predictive values of serum creatinine and SDMA concentrations in detecting decreased GFR at T3

A total of 17 cats were euthyroid at the T3 time point, of which 16 had complete data available for analysis. Of the euthyroid cats, 6/16 (38%) developed decreased GFR at T3. The sensitivity, specificity, NPV and PPV for detecting decreased GFR was 83%, 50%, 88% and 42% for SDMA ($>14 \mu\text{g/dl}$), 50%, 71%, 77% and 43% for SDMA ($\geq 18 \mu\text{g/dl}$), and 50%, 85%, 80% and 60% for creatinine, respectively ([Table 2](#)).

Table 2.

Sensitivity, specificity, negative predictive value (NPV) and positive predictive value (PPV) of increased creatinine (>175 $\mu\text{mol/l}$) and symmetric dimethylarginine (SDMA) (>14 $\mu\text{g/dl}$ and $\geq 18 \mu\text{g/dl}$) in detecting reduced glomerular filtration rate in cats with euthyroid status at 12 months (T3) after radioactive iodine therapy

	Sensitivity	Specificity	NPV	PPV
SDMA >14 $\mu\text{g/dl}$	83	50	88	42
SDMA $\geq 18 \mu\text{g/dl}$	50	71	77	43
Creatinine	50	85	80	60

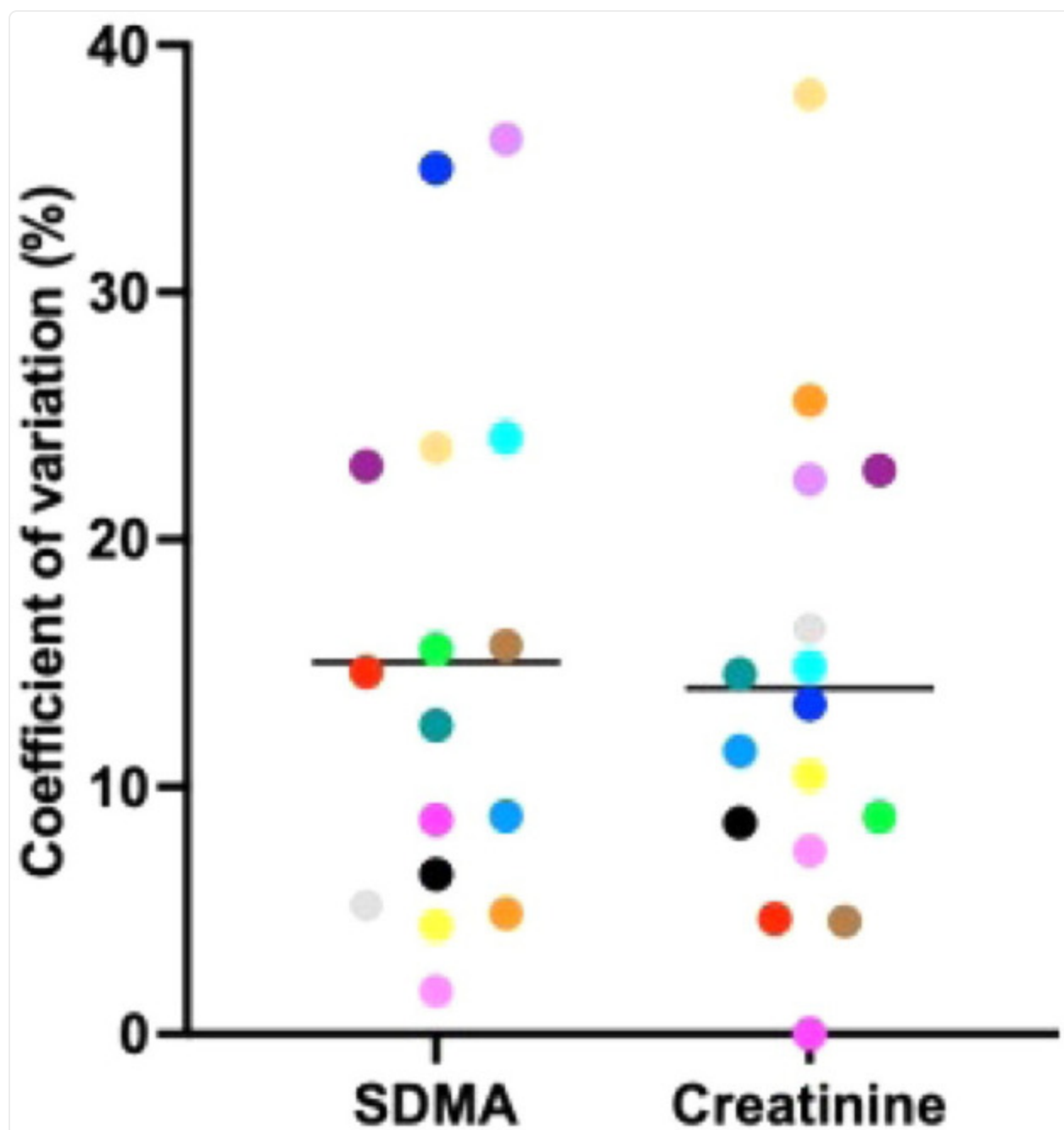
[Open in a new tab](#)

Data are %

Intra-individual variability in longitudinal measurements of serum SDMA and creatinine

The intra-individual variability in longitudinal measurements of serum SDMA and creatinine were evaluated in 16 cats at euthyroid status time points and was similar. The mean coefficient of variation for the 16 cats for SDMA was 15% and for creatinine was 14%. Individual coefficients of variation for each of the 16 cats for SDMA and creatinine are presented in [Figure 4](#).

Figure 4.



[Open in a new tab](#)

Coefficient of variation of longitudinal measurements of symmetric dimethylarginine (SDMA) and creatinine in 16 euthyroid cats. The bold line represents the mean. Each individual cat is represented by the same colour circle in the SDMA and creatinine data

Discussion

The present study identified a high frequency of discordance (10–75% of cats) between creatinine and SDMA concentrations at all time points in this study population of mature and senior cats in hyperthyroid and euthyroid states. There was greater discordance identified when the laboratory upper reference limit of $>14\text{ }\mu\text{g/dl}$ was applied compared with the proposed age-specific reference limit of $\geq 18\text{ }\mu\text{g/dl}$. The highest number of discordant SDMA/creatinine results was observed at T1 and, of the discordant cats, 6/18 (33%) were hyperthyroid. Discordance rates between SDMA and GFR were higher at all time points compared with those of creatinine and GFR. There was 100% discordance between SDMA and GFR vs 0% discordance between creatinine and GFR at the baseline (T0) time point, when 100% of the cats were hyperthyroid, when applying both upper limits of SDMA, suggesting that hyperthyroidism is an important non-renal factor influencing SDMA in cats.

The SDMA upper limit of $>14\text{ }\mu\text{g/dl}$ was more sensitive at detecting decreased GFR compared with the SDMA upper limit of $\geq 18\text{ }\mu\text{g/dl}$ and creatinine. However, the upper limit of $>14\text{ }\mu\text{g/dl}$ for SDMA was less specific (50%) than $\geq 18\text{ }\mu\text{g/dl}$ (71%) and serum creatinine (85%) for the detection of decreased GFR. There are a limited number of studies that have analysed the diagnostic accuracy of SDMA in cats, and reported sensitivities and specificities have varied greatly (between 15% [21](#) and 100% [22](#) and between 75% [23](#) and 98%, [24](#) respectively). Initial studies reported SDMA to be a more sensitive marker for decreased renal function when compared with creatinine, with increases in SDMA concentration first detected 17 months before increases in creatinine concentration in cats developing CKD. [22](#) However, subsequent studies have reported that the relationship between SDMA and GFR and between creatinine and GFR are comparable. [23,25](#) Comparison of these studies is limited by the variability in study design, population analysed and the reference standard for which they determined impaired renal filtration function. [15](#) Unlike creatinine, SDMA is not considered to be affected by muscle mass or sex and may be more reliable in inflammatory states. [26,27](#) However, evidence from studies suggest it may be influenced by non-renal factors, including lymphoma [28,29](#) and diabetes mellitus. [30,31](#)

Importantly, the findings of the present study suggest SDMA may be influenced by hyperthyroidism, which may decrease its specificity in detecting decreased GFR relative to creatinine. When facing the clinical dilemma of discordance between SDMA and creatinine in cats, non-renal factors, such as hyperthyroidism, should be excluded. In addition, ongoing monitoring of renal function, including serial measurement of renal biomarkers and urinalysis, should be performed, as increased SDMA may suggest subsequent development of decreased GFR. Intra-

individual variability of both biomarkers was similar, and therefore increases in both biomarkers of greater than 15% may be of clinical significance in euthyroid cats.

In a previous study of hyperthyroid cats undergoing bilateral thyroidectomy, 35% of samples had discordant SDMA and creatinine concentrations, with SDMA most often increased while creatinine concentration remained within the RI.¹¹ However, the biomarkers were not measured against GFR for comparison. A further study found a lack of correlation between SDMA and creatinine in hyperthyroid cats before RAI treatment but a positive correlation after RAI treatment, with the authors speculating that this was due to the hyperthyroidism-induced metabolic state.¹² SDMA is positively correlated with free T4 in humans and can be increased in humans with hyperthyroidism.^{14,32} Potential mechanisms for this included genetic variation, increased synthesis of arginine N-methyltransferase involved in L-arginine methylation, or decreased SDMA metabolism due to decreased enzymatic hydrolysis.^{14,33} It is clear that SDMA is increased in both cats and humans with hyperthyroidism, despite higher GFR, which should increase renal clearance of SDMA. Whether increased SDMA concentrations in cats with hyperthyroidism is of pathophysiological significance is unclear. Previous studies of hyperthyroid cats have shown that increased SDMA predicts declining renal function.^{21,24} Although such studies can only demonstrate associations, and not causality, one can postulate that SDMA may not only be a predictor of declining renal function but also a mediator. SDMA is a competitive inhibitor of arginine transport and so decreases endothelial nitric oxide (NO) synthesis, probably by limiting L-arginine supply to endothelial NO synthase, and in this way SDMA may play a role in vascular endothelial dysfunction. Indeed, SDMA decreased NO synthesis and increased reactive oxygen species in cultured endothelial cells in a dose-dependent manner.³⁴

The present study identified higher sensitivity of SDMA >14 µg/dl for the detection of decreased GFR than SDMA of ≥18 µg/dl and creatinine in the euthyroid population (83% vs 50% vs 50%), similar to previously reported.²² Serum SDMA >14 µg/dl was less specific (50%) for detecting decreased GFR than both the SDMA upper limit of ≥18 µg/dl and serum creatinine (71% and 85%, respectively), which is considerably lower than the 91% specificity reported by Hall et al.²² Higher sensitivity of SDMA >14 µg/dl, compared with creatinine, for predicting azotaemia in euthyroid cats after RAI,²⁴ and lower specificity of SDMA >14 µg/dl, compared with creatinine, in detecting decreased renal function post-RAI oral therapy,³⁵ have been reported previously.^{24,35} The present study also identified lower PPVs for both SDMA >14 µg/dl and creatinine (42% and 60%, respectively), compared with a previous study (86% and 100%, respectively).²² This may be partly influenced by the lower prevalence of decreased GFR (31%) in our population and smaller population size compared with the previous study and the different upper limit of serum creatinine that was used (≥175 µmol/l in this study compared with 140 µmol/l).²² However, we

consider our study population to be representative of the older feline population in which SDMA and creatinine concentrations would be routinely measured, and therefore consider our findings relevant.

This is the first study reporting the intra-individual biological variability in SDMA over a long (6–12 months) follow-up period in mature and senior cats. The intra-individual variability in SDMA and creatinine was evaluated to understand if this was a potential contributing factor to discordance. We found that variability, as determined by the mean coefficient of variation, was similar in SDMA and creatinine measurements (15% vs 14%) in euthyroid cats. Van Vertloo et al [12](#) observed variability in SDMA concentrations in individual cats across time points, although the coefficient of variation as an objective measurement was not determined. A study evaluating the analytical performance of SDMA in healthy cats aged 1–7 years over a 6-week period reported intra-individual variability to be approximately 19.9%. [36](#) The variability in creatinine concentration was higher in the present study than that reported in a previous study of euthyroid cats (14% vs 8.5%). [37](#) This may reflect the longer follow-up period for repeated measurements in the present study (6–12 months) compared with the previous study (6 months only). Breed can also influence variation, with Birman cats reported to have physiologically higher SDMA and creatinine compared with other breeds. [38](#) No Birman cats were included in this study cohort.

A previous study defined progressive CKD as an increase in plasma creatinine concentration of at least 25% relative to the concentration at diagnosis; [39](#) the variability of the individual creatinine results in the present study (14%) may suggest that progression (ie, decrease in GFR) may be associated with a lower relative increase in creatinine concentration. [39](#) However, analytical variation associated with analyser imprecision should be considered as this will also contribute to variation between replicate measurements. [39,40](#) A previous study identified a lower intra-individual variability and reference change value for creatinine compared with SDMA (enzyme multiplied immunoassay technique) (8.6% vs 19.9% and 24.7% vs 61.7%, respectively), suggesting that more variability can occur in repeated SDMA measurements in an individual cat that may not be clinically significant. [41](#) Previous studies looking at the dispersion of measured serum SDMA, which describes the range of possible values for a test result based on individual biological and analyser variation, identified a wider dispersion for SDMA compared with creatinine. [36,40,42](#)

Limitations of the present study include the small sample size. The gold standard method of liquid chromatography mass spectrometry was not utilised for SDMA measurement; however, the IDEXX laboratory test used is validated and therefore was considered appropriate and most representative of the methodology used for samples submitted from clinical practice. [43](#) The SDMA and creatinine measurements in this study were analysed at different diagnostic laboratories, but

this may also reflect the situation that can occur for samples submitted from clinical practice. In addition, it is possible that muscle mass may have influenced the serum creatinine results, which was not assessed in this study.

Conclusions

There was a high frequency of discordance between serum SDMA and creatinine, and SDMA and GFR, particularly when cats were hyperthyroid. The discordance was lower when the upper RI of $\geq 18 \mu\text{g/dl}$ was applied, indicating that the proposed age-specific RI may be more appropriate for the older cat population. The upper limit of $\geq 18 \mu\text{g/dl}$ for SDMA concentrations was more specific for detecting decreased GFR than when applying the upper limit of $>14 \mu\text{g/dl}$. However, serum SDMA $>14 \mu\text{g/dl}$ was more sensitive for detecting decreased GFR than $\geq 18 \mu\text{g/dl}$, highlighting that cats with SDMA $>14 \mu\text{g/dl}$ may require ongoing monitoring of their renal parameters, including serial measurement of renal biomarkers and urinalysis. The results of the present study suggest that when faced with the clinical dilemma of discordant SDMA and creatinine results in mature and senior cats, a serum SDMA of $\geq 18 \mu\text{g/dl}$ should be considered more significant for the detection of decreased GFR, whereas an SDMA concentration of $15\text{--}18 \mu\text{g/dl}$ has a lower likelihood of indicating a true reduction in GFR, particularly if creatinine is within the RI. Furthermore, we should consider that an increase in SDMA may be influenced by non-renal factors, such as hyperthyroidism.

Supplemental Material

S1a–d:

[sj-pdf-1-jfm-10.1177_1098612X261466163.pdf](#) ^(693.1KB, pdf)

Concordant symmetric dimethylarginine (SDMA) and serum creatinine (sCr) results in cats at baseline (T0), 1 (T1), 6 (T2) and 12 months (T3) after radioactive iodine treatment for both laboratory ($>14 \mu\text{g/dl}$) and age-specific ($\geq 18 \mu\text{g/dl}$) SDMA reference intervals. Of those cats with concordant SDMA and creatinine results, SDMA/sCr and glomerular filtration rate (GFR) concordance and SDMA/sCr and GFR concordance is then presented below at each time point.

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Footnotes

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Author note: Some of the results from this study were presented at the European Congress of Veterinary Internal Medicine Congress 2023 (European Society of Veterinary Nephrology and Urology abstract stream).

Supplementary material: The following file is available as supplementary material:

S1a–d: Concordant symmetric dimethylarginine (SDMA) and serum creatinine (sCr) results in cats at baseline (T0), 1 (T1), 6 (T2) and 12 months (T3) after radioactive iodine treatment for both laboratory (>14 $\mu\text{g/dl}$) and age-specific (≥ 18 $\mu\text{g/dl}$) SDMA reference intervals. Of those cats with concordant SDMA and creatinine results, SDMA/sCr and glomerular filtration rate (GFR) concordance and SDMA/sCr and GFR concordance is then presented below at each time point.

Lorna Hardy declares no conflict of interest. Séverine Tasker, Angie Hibbert and Natalie Finch have received financial remuneration, including for research studies, for providing educational material, speaking at conferences and/or consultancy work; however, none of these activities cause any direct conflict of interest in relation to the research presented here.

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Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals and procedures that differed from established internationally recognised high standards ('best practice') of veterinary clinical care for the individual patient. The study therefore had prior ethical approval from an established (or ad hoc) committee as stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals,

including cadavers, tissues and samples) for all procedure(s) undertaken. No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

S1a–d:

[sj-pdf-1-jfm-10.1177_1098612X261466163.pdf](#)^(693.1KB, pdf)

Concordant symmetric dimethylarginine (SDMA) and serum creatinine (sCr) results in cats at baseline (T0), 1 (T1), 6 (T2) and 12 months (T3) after radioactive iodine treatment for both laboratory ($>14\text{ }\mu\text{g/dl}$) and age-specific ($\geq 18\text{ }\mu\text{g/dl}$) SDMA reference intervals. Of those cats with concordant SDMA and creatinine results, SDMA/sCr and glomerular filtration rate (GFR) concordance and SDMA/sCr and GFR concordance is then presented below at each time point.

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Review

Understanding the Progression of Chronic Kidney Disease in Cats: From Pathophysiology to Emerging Biomarkers

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Simple Summary

Chronic kidney disease is one of the most common medical conditions in older cats, leading to a gradual and permanent loss of kidney function. Unfortunately, by the time a cat shows signs of illness or traditional blood tests detect a problem, a large portion of the kidneys may already be damaged. This review focuses on the mechanisms underlying disease progression and critically examines emerging biomarkers with the potential to detect renal dysfunction at earlier stages than currently available methods. By reviewing the latest scientific research, we highlight how these tools can help veterinarians detect the disease in its silent stages. Earlier identification allows not only earlier intervention but also more accurate monitoring of disease progression. This shift enables veterinarians to move from a reactive approach toward proactive management strategies, which may significantly extend survival time and improve the quality of life of aging feline patients.

Abstract

Feline chronic kidney disease is a leading cause of mortality in geriatric cats, characterized by a progressive and irreversible loss of renal function. Despite its high prevalence, early diagnosis remains challenging due to nephron compensatory mechanisms and the limited sensitivity of traditional biomarkers, creating a diagnostic gap that necessitates the exploration of novel biomarkers for earlier detection. This review examines the complex pathophysiology of the disease, including renin–angiotensin–aldosterone system activation, tubulointerstitial fibrosis, and mineral metabolism disturbances. By analyzing recent scientific literature, this work evaluates current diagnostic landscape and clinical relevance of emerging biomarkers. Evidence indicates that symmetric dimethylarginine and fibroblast growth factor-23 improve detection of early metabolic and filtration changes, while urinary biomarkers like cystatin B and retinol-binding protein provide specific insights into tubular injury. Bridging the diagnostic gap requires a transition from a reactive, azotemia-based framework to a multi-parametric diagnostic approach that integrates novel biomarkers with serial clinical and laboratory monitoring. Although financial constraints and limited availability restrict widespread clinical implementation, incorporating these advances is essential for earlier prognostic stratification and timely therapeutic decision-making. This integrated strategy has the potential to slow disease progression and improve survival and quality of life in cats with chronic kidney disease.



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Keywords: feline chronic kidney disease; renal fibrosis; biomarkers; SDMA; FGF-23; International Renal Interest Society staging; cystatin B; geriatric cats

1. Introduction

1.1. Feline Chronic Kidney Disease (CKD): Epidemiology, Etiology and Risk Factors

Chronic kidney disease (CKD) is a heterogeneous disease syndrome defined by progressive renal damage that encompasses a wide range of renal pathologies and etiologies [1,2]. It is characterized by the presence of either structural or functional abnormalities in one or both kidneys, which have been present for a prolonged period, generally lasting three months or more. It can occur with a range of severity levels, from minor structural lesions in a single kidney to significant nephron loss in both kidneys [3]. Despite extensive characterization, the initiating cause of feline CKD often remains unclear, and the disease is best understood as a multifactorial and progressive continuum rather than a single pathological entity.

This disease is one of the most common in cats and the number of cases has increased in recent years due to better awareness and developments in diagnostic methods [4–6]. Nevertheless, this apparent increase likely reflects improved recognition rather than a true rise in incidence, particularly in aging feline populations.

Data on the exact prevalence of CKD in cats are still scarce [4,5]. Studies show that CKD affects between 1% and 3% of cats in general veterinary practices, but this number can reach up to 50% depending on the population and diagnostic criteria used [4]. The prevalence increases with age, with the highest rates in cats over 15 years old, ranging from 31% to 80% depending on the study [5,7]. In the UK, kidney disease was the most common cause of death in cats older than 5 years, making up 12.1% of cases [1,4]. Notably, CKD is twice as prevalent in geriatric cats as in age-matched canine populations [4], highlighting important species-specific susceptibilities that remain incompletely understood.

The etiology of feline CKD is frequently idiopathic, despite the identification of multiple factors associated with an increased risk of disease development [4]. One of those is breed predisposition, with purebred cats showing a greater susceptibility than non-pedigree cats, such as Domestic Shorthairs [8,9]. Certain breeds, such as the Birman, tend to have higher serum creatinine concentrations, which could lead to misclassification in these studies [10]. The most prevalent breeds affected by CKD include Maine Coon, Abyssinian, Siamese, Russian Blue, and Burmese cats [9,11]. In addition, specific genetic mutations can also contribute to the development of CKD, such as the mutation in the polycystic kidney disease 1 (PKD1) gene, which is highly prevalent in Persian (46%), Scottish Fold (54%), and American Shorthair cats (47%) [12,13]. However, the presence of genetic risk does not fully explain disease onset or progression, reinforcing the multifactorial nature of feline CKD.

Age represents a further critical risk factor for feline CKD. The disease is particularly common in geriatric cats, with the prevalence starting to increase from the age of 5 to 6 years [4,5]. In young cats, CKD development is primarily attributed to hereditary diseases, while in geriatric felines, it is often driven by comorbidities that impair renal function [14]. Demographic variables, such as sex, also modulate CKD risk. Male cats are more susceptible to developing urethral obstruction than females, which can result in kidney injury and subsequently increase the risk of CKD. However, studies have shown that the likelihood of being diagnosed with CKD does not significantly differ between neutered and intact males. Nevertheless, when comparing sexes, neutered males are more likely to be diagnosed with CKD compared to spayed females, suggesting that sex-related

hormonal or behavioral factors may influence disease susceptibility rather than acting as direct causal determinants [8,9,14].

Diet and body condition are also important factors when it comes to feline CKD risk [15]. Cats with a thin body condition or those that are frequently dehydrated tend to have a higher probability of developing the disease [8]. Various studies have investigated how diet influences this risk; for example, feeding cats exclusively commercial dry food has been associated with an increased risk of CKD [11,15]. On the other hand, factors such as free feeding, low fiber intake, high protein, and low potassium levels may increase the risk of developing CKD [10]. These associations, however, remain controversial and may be influenced by differences in study design and dietary recall bias. This is particularly relevant as owners of cats already diagnosed with CKD may retrospectively over-emphasize previous dietary choices, complicating firm causal conclusions.

Specific interventions, such as the administration of antibiotics and non-steroidal anti-inflammatory drugs (NSAIDs) including those used for dental disease, can directly contribute to renal impairment. Similarly, the general anesthesia required for these procedures carries an inherent risk of kidney injury, with both factors collectively increasing the likelihood of developing CKD [8,10]. Regarding preventive medicine, the potential association between vaccination and feline CKD remains a subject of ongoing debate. Rather than a direct causal link to the vaccine product itself, the issue likely reflects a cumulative inflammatory burden on the renal parenchyma. As a highly vascularized organ, the feline kidney is particularly susceptible to immune complex deposition and secondary interstitial inflammation, often acting as a 'bystander victim' of systemic immune stimulation. Consequently, further research into pathophysiological mechanisms, such as renal-specific antibodies, is required to determine if adjusting vaccination protocols by balancing essential protection with the need to minimize unnecessary systemic inflammation can effectively preserve renal function [10].

Beyond demographic and lifestyle-related factors, a variety of underlying diseases and pathological conditions have also been implicated in the development of CKD in cats. Both congenital forms of CKD, including hereditary conditions such as polycystic kidney disease and renal dysplasia, and acquired forms, resulting from infections, toxic insults, or chronic inflammation, contribute to the complex etiology of the disease in feline populations. In this context, a broad range of primary renal disorders, such as amyloidosis, juvenile renal dysplasia, glomerular diseases, polycystic kidney disease, nephro- and ureterolithiasis, and lymphoma, have been identified as potential contributors. Chronic dietary imbalances and bacterial pyelonephritis are likewise recognized as relevant etiologies [4,16].

Other systemic conditions are also frequently seen in cats with CKD and may contribute to its development. Systemic hypertension, cardiovascular disease, and primary hyperaldosteronism are among the most commonly reported [16–18]. High blood pressure is found in approximately 20–65% of cats with CKD and is often linked to proteinuria, though it does not seem to predict survival or cause major kidney damage in treated cats [16,19]. Heart disease, such as hypertrophic cardiomyopathy, is also frequent: around 46.6% of cats with CKD have left ventricular hypertrophy, 59% of cats with hypertrophic cardiomyopathy are azotemic, and 12.7% have CKD. Systolic arterial pressure is also higher in azotemic than in non-azotemic cats with this condition. In addition, primary hyperaldosteronism can lead to mild azotemia, hypokalemia, systemic hypertension, and progressive kidney damage, with histological changes similar to those observed in humans. Although these conditions frequently coexist with CKD, their role as primary risk factors versus parallel comorbidities remains incompletely defined [16,20].

Infectious agents may further contribute to the pathogenesis of CKD. Several viral infections, including feline immunodeficiency virus (FIV), feline leukemia virus (FeLV),

feline infectious peritonitis virus (FIP), feline morbillivirus (FeMV), and foamy virus (FFV), have all been associated with renal damage and may promote the progression of chronic kidney injury [4,5,14]. Bacterial infections, such as leptospirosis and bartonellosis, as well as parasitic diseases like leishmaniosis and heartworm disease, have also been implicated [5,14].

Urinary tract infections, including bacterial pyelonephritis, may also play a contributory role in renal deterioration [4,14]. Another condition increasingly recognized for its potential systemic impact is periodontal disease (PD). Retrospective studies have shown that cats with advanced PD (especially those in stages 3 or 4) have a significantly higher risk of developing CKD [9,10].

1.2. Pathophysiology of Feline Chronic Kidney Disease Progression

CKD in cats progresses through a self-perpetuating cycle of structural and functional alterations in the residual nephrons after injury (Figure 1). Compensatory tools such as increased single-nephron glomerular filtration rate (GFR), glomerular hypertension, and hypertrophy help delay the start of azotemia until more than 75% of renal function is lost. While these adaptive responses are essential for short-term homeostasis, they ultimately become maladaptive, contributing to progressive nephron injury and accelerating disease progression [16].

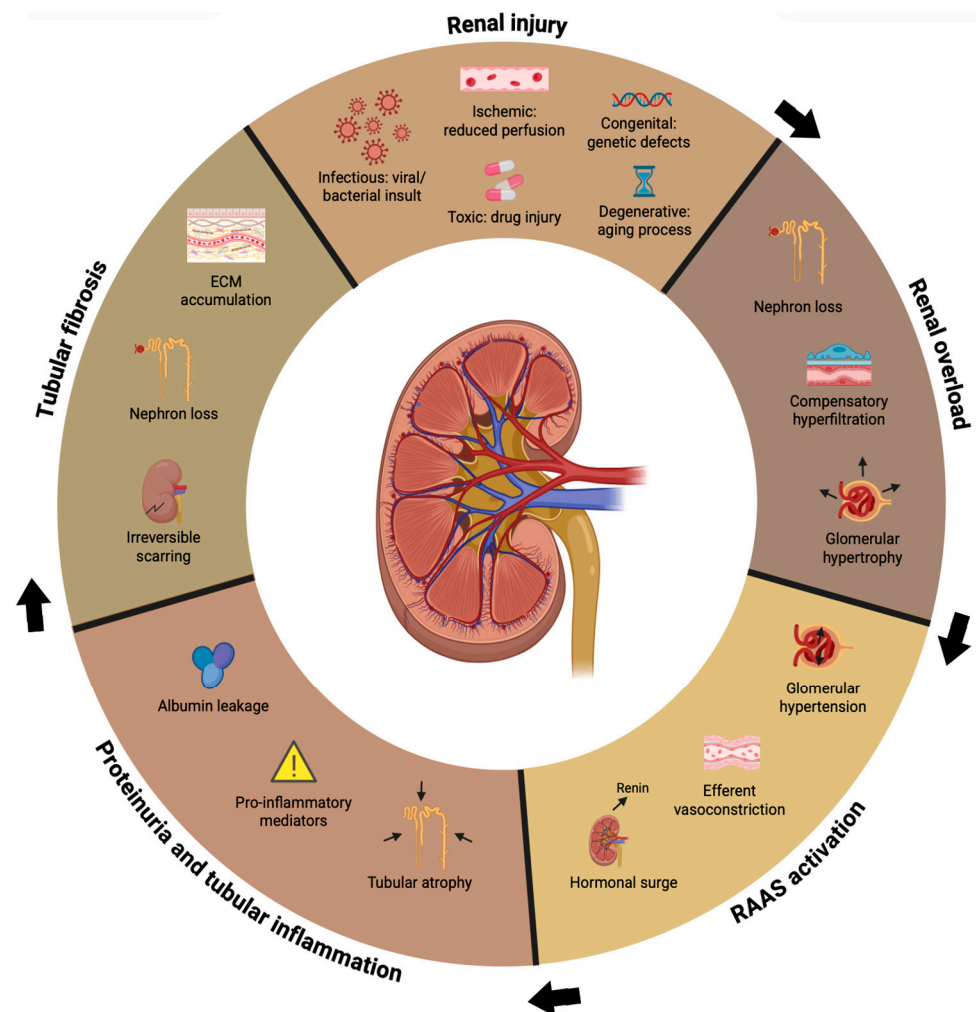


Figure 1. Self-perpetuating cycle of feline CKD progression. The arrows indicate a clockwise sequence, starting with the initial ‘Renal injury’ (top) and progressing through compensatory and inflammatory stages until reaching ‘Tubular fibrosis’ (left). Initial renal injury triggers compensatory mechanisms,

including increased single-nephron glomerular filtration rate (GFR), glomerular hypertension, and activation of the renin–angiotensin–aldosterone system (RAAS). Although initially adaptive, sustained activation of these pathways promotes proteinuria, chronic inflammation, tissue hypoxia, and oxidative stress, ultimately leading to progressive tubulointerstitial fibrosis and irreversible nephron loss. These interconnected processes establish a vicious cycle that drives CKD progression regardless of the initiating renal insult. This fibrosis and loss of functional units represent the final stage of the cycle, which impairs the remaining nephrons and perpetuates the clinical state of kidney failure. ECM, extracellular matrix.

As GFR decreases, uremic toxins accumulate, worsening clinical signs and directly promoting further renal injury [16]. Feline CKD is therefore best conceptualized as a dynamic process driven by the interplay between hemodynamic stress, chronic inflammation, and progressive fibrotic remodeling, rather than a static condition defined solely by biochemical thresholds. This pathophysiological cascade is largely independent of the initiating cause and represents a common final pathway shared across diverse renal insults [4,14,16].

Renal fibrosis plays a significant role in the progression of CKD and represents the principal irreversible lesion responsible for functional decline. In healthy kidneys, the interstitium contains fibroblasts and dendritic cells within an extracellular matrix (ECM) rich in collagens (I, III, VII), fibronectin, and glycoproteins such as tenascin. Following renal injury, inflammatory signaling triggers fibroblast activation, ECM expansion, and tissue scarring. Although fibrosis may initially provide structural support during attempted tissue repair, its persistence and progression ultimately disrupt normal renal architecture and function. Histologically, this process is characterized by excessive ECM accumulation, loss of peritubular capillaries, chronic inflammation, tubular atrophy and dilation, and, in advanced stages, mineralization [14,21]. Myofibroblasts, derived from resident fibroblasts, pericytes, or through epithelial–mesenchymal transition, are key effector cells in renal fibrogenesis. These cells synthesize large amounts of ECM proteins, including collagen IV and fibronectin, and produce profibrotic mediators such as transforming growth factor-beta (TGF- β) [14]. Although the feline kidney retains a limited regenerative capacity, repetitive or severe injury, such as ischemic events or bacterial pyelonephritis, overwhelms reparative mechanisms and results in permanent scarring [21].

Proteinuria further amplifies this fibrotic cascade. Increased glomerular permeability leads to an overload of the megalin and cubulin-mediated endocytosis system, causing tubular epithelial cells to transition from an adaptive response to a pro-inflammatory state. This triggers the release of cytokines and the recruitment of inflammatory cells, ultimately promoting collagen deposition and progressive renal loss [20]. This maladaptive tubular response links glomerular injury to tubulointerstitial inflammation, reinforcing fibrosis even in a species where primary glomerular disease is relatively uncommon. The transition toward a profibrotic phenotype is largely driven by transforming growth factor-beta 1 (TGF- β 1), a master regulator of ECM synthesis and degradation. TGF- β 1 expression is upregulated by multiple stimuli, including renin–angiotensin–aldosterone system (RAAS) activation, oxidative stress, hypoxia, and sustained proteinuria. In cats, urinary TGF- β 1 normalized to creatinine (TGF- β 1:UCrR) has been associated with higher serum creatinine concentrations. However, its clinical utility remains limited due to the lack of direct histopathological correlation and standardized reference intervals, restricting its use primarily to research settings [22]. Another mediator increasingly implicated in renal fibrosis is transglutaminase-2 (TG-2), an enzyme that stabilizes the ECM by cross-linking its protein components, thereby enhancing resistance to proteolytic degradation. TG-2 also facilitates the activation of latent TGF- β 1, further reinforcing fibrotic signaling pathways. In cats with CKD, increased TG-2 activity has been correlated with elevated serum creatinine,

urea, and phosphorus concentrations, suggesting a contributory role in disease severity and highlighting TG-2 as a potential future therapeutic target [21].

1.2.1. Kidney Morphologic Changes

In geriatric cats with CKD, histological lesions are predominantly localized within the tubulo-interstitial compartment, with glomerular sclerotic changes appearing mostly as secondary manifestations. Histopathologically, these lesions are multifocal to segmental and include infiltration of mononuclear cells in the interstitium, tubular degeneration and atrophy, interstitial fibrosis, mineralization of Bowman's capsule and tubular basement membranes, and accumulation of lipids in the interstitium and glomerulosclerosis.

Tubular atrophy, interstitial inflammation, and fibrosis are consistent findings that progressively worsen alongside disease progression. Atrophic cortical tubules typically appear in clusters, exhibiting basement membrane alterations such as thickening and wrinkling or, conversely, thinning. Microscopically, these atrophic tubules are characterized by a marked reduction in epithelial height and luminal dilation. The interstitial space becomes expanded by pale eosinophilic collagen fibers, indicating fibrosis, which displaces the normal parenchyma. This area is often infiltrated by multifocal clusters of mononuclear cells, primarily lymphocytes and plasma cells, signifying chronic active inflammation [4]. Interstitial lipid accumulation, frequently observed in IRIS (International Renal Interest Society) stages 2 to 4, is often associated with granulomatous inflammation; this is likely a consequence of tubular ischemia and subsequent rupture, leading to the release of intraepithelial lipids [23]. Notably, the extent of interstitial fibrosis correlates closely with the severity of azotemia, becoming most prominent in IRIS stage 4 [4].

From a glomerular perspective, affected kidneys often demonstrate compensatory hypertrophy and expansion of the mesangial matrix. While significant nephron loss drives hyperperfusion and potential podocyte damage consistent with focal segmental glomerulosclerosis (FSGS), this secondary manifestation is relatively mild in the feline species. In spontaneous CKD, FSGS typically affects only a limited proportion of glomeruli and capillary tufts. More commonly, glomeruli undergo global sclerosis or obsolescence. Although global sclerosis is a feature of normal aging, its severity is markedly exacerbated in CKD. These sclerotic glomeruli exhibit collapse of the capillary tuft and fibrosis of the urinary space, findings which are more consistent with ischemic glomerular obsolescence than with classic FSGS progression. In advanced stages, these shrunken glomeruli may become indistinguishable from the surrounding fibrotic stroma [4,24].

Structural vascular alterations further complicate the disease profile. Cats with reduced renal mass often exhibit preferential dilation of afferent arterioles, contributing to glomerular hypertension and hyperfiltration in functional nephrons [4]. Furthermore, hypertensive cats frequently display vascular lesions such as hyaline and hyperplastic arteriosclerosis [24]. Ultimately, these structural changes are driven by the activation of myofibroblasts, which can be derived from pericytes, fibroblasts, or via epithelial-mesenchymal transition. These cells deposit substantial quantities of extracellular matrix components, such as collagen and fibronectin, which directly correlate with clinical renal dysfunction [16].

1.2.2. RAAS Activation, Hypoxia, and Oxidative Stress

The RAAS (Figure 2) is a fundamental endocrine mechanism that regulates blood pressure, sodium levels, and extracellular fluid balance [25]. Under physiological conditions, it ensures stable GFR and renal perfusion. When blood pressure or renal perfusion falls, juxtaglomerular cells release renin, an enzyme that catalyzes the conversion of angiotensinogen (ATG) into angiotensin I. This precursor is subsequently converted into angiotensin II by

the angiotensin-converting enzyme (ACE), which is highly active in the endothelial cells of the lungs and within the proximal renal tubules [26,27].

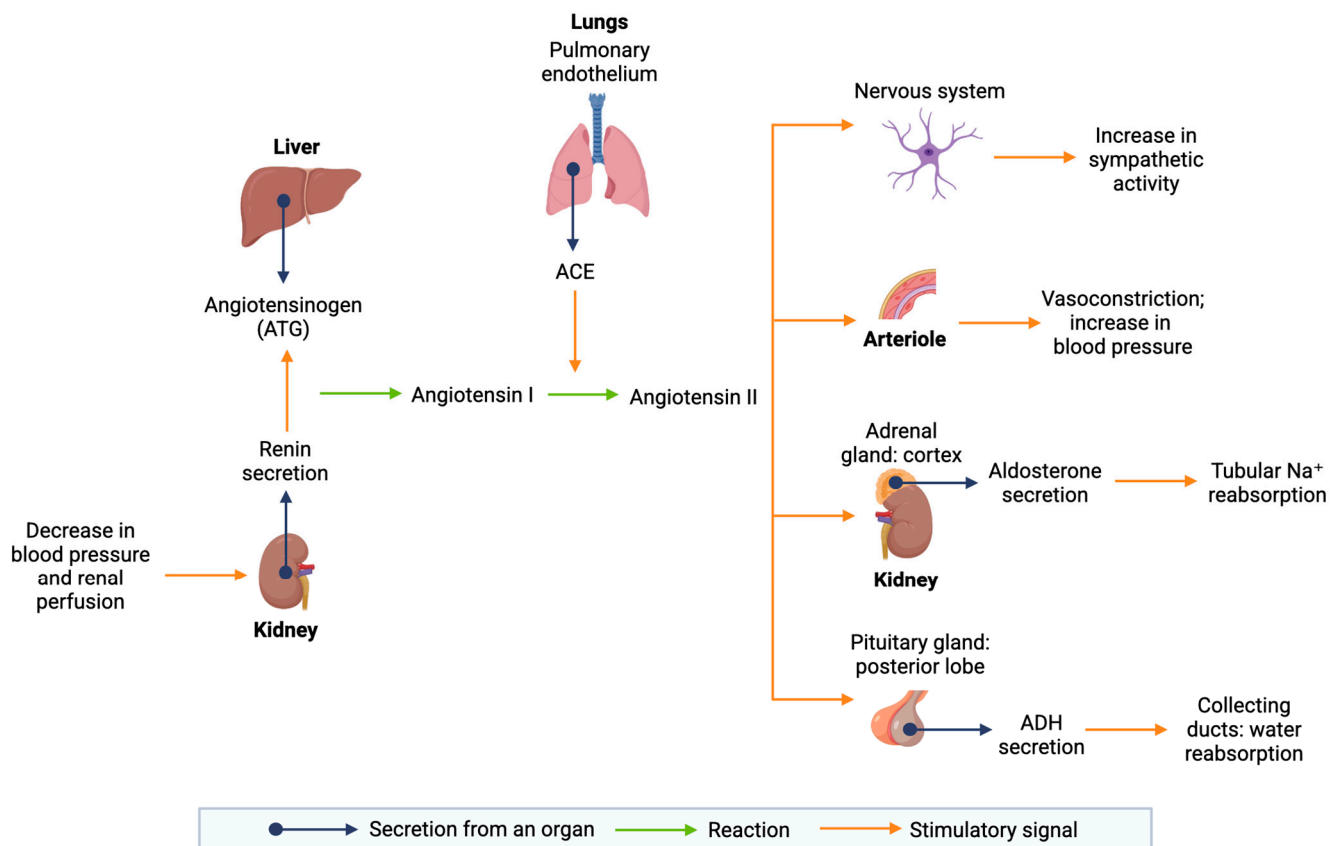


Figure 2. Schematic representation of the renin–angiotensin–aldosterone system (RAAS) and its main physiological effects. Renin is released from the juxtaglomerular cells of the kidney in response to decreased renal perfusion pressure, reduced NaCl delivery to the macula densa, and increased sympathetic stimulation. Renin catalyzes the conversion of angiotensinogen, synthesized by the liver, into angiotensin I. Angiotensin I is subsequently converted into angiotensin II by angiotensin-converting enzyme (ACE), primarily expressed in the pulmonary capillary endothelium. Angiotensin II exerts multiple physiological effects, including arteriolar vasoconstriction with increased blood pressure, stimulation of aldosterone secretion from the adrenal cortex leading to increased sodium reabsorption and potassium excretion in the distal tubules and collecting ducts, stimulation of antidiuretic hormone (ADH) release from the posterior pituitary with enhanced water reabsorption in the collecting ducts, and increased sympathetic activity. This schematic representation summarizes the major components of the RAAS and does not depict all intermediate pathways or feedback mechanisms.

Angiotensin II acts as a potent vasoconstrictor and plays a central role in renal hemodynamics through its interaction with the angiotensin type 1 (AT1) receptor. By inducing vasoconstriction of the efferent arteriole, it increases glomerular capillary pressure to maintain a stable GFR even during systemic hypotension. Furthermore, angiotensin II stimulates aldosterone secretion, promoting sodium and water reabsorption, and triggers the release of antidiuretic hormone (ADH) to enhance fluid retention. While these mechanisms are essential for homeostasis, their chronic activation in feline CKD becomes a primary driver of disease progression [20,26,28].

In cats with CKD, RAAS activation is a prominent feature both systemically and intrarenally. The feline kidney possesses all components necessary for local activation, and intrarenal angiotensin II levels often significantly exceed plasma concentrations, making systemic measurements insufficient to assess the true renal status [14,21,29]. Although initially compensatory, this response becomes maladaptive, contributing to glomerular

hypertension, hyperfiltration, and proteinuria [29]. Other vasoconstrictors, such as thromboxane A2 and endothelin-1 (ET-1), may exacerbate these effects and promote segmental tubular hypoxia, particularly in geriatric cats [4,29].

Beyond its hemodynamic effects, angiotensin II also acts as a potent pro-fibrotic mediator, driving chronic structural damage independently of its effects on blood pressure. By triggering the activation of key growth factors, it promotes the transformation of renal cells into fibroblasts and the progressive accumulation of scar tissue [21]. Additionally, endothelial dysfunction impairs the release of nitric oxide (NO), compromising renal microcirculation and reducing renal oxygenation [17]. Notably, calcitriol serves as a negative regulator of the RAAS by suppressing renin transcription and reducing angiotensinogen expression. Through vitamin D receptor activation (VDRA), it limits fibrosis and inflammation by inhibiting NF- κ B and restoring Klotho expression, which is typically downregulated by angiotensin II [29].

These hemodynamic alterations directly contribute to renal hypoxia, a critical disorder in feline CKD progression. Despite receiving a large blood supply, the renal medulla naturally operates at low oxygen levels. As the disease advances, glomerulosclerosis and RAAS-mediated efferent vasoconstriction disrupt blood flow to the peritubular capillaries, further reducing oxygen delivery to the renal tissue. In response, kidney cells adapt by promoting fibrosis and the epithelial-to-mesenchymal transition (EMT), creating a vicious cycle where worsening hypoxia accelerates tissue damage, a state often compounded by concurrent anemia [14].

Closely linked to hypoxia is oxidative stress, which arises from an imbalance between the production of reactive oxygen species (ROS) and the capacity of antioxidant defense systems. The feline kidney is particularly vulnerable due to its high metabolic rate; in CKD, this susceptibility is exacerbated by compensatory hyperfunction of surviving nephrons, leading to increased cellular oxygen consumption and ROS production. This process is further stimulated by angiotensin II and chronic inflammation. Excess ROS, including superoxide anions and hydrogen peroxide, damage DNA, lipids, and proteins, ultimately inducing apoptosis, necrosis, and progressive fibrotic remodeling [14,30].

Evidence of oxidative stress in feline CKD includes increased GSH:GSSG ratios and elevated glutathione peroxidase activity in advanced IRIS stages [31,32]. Furthermore, proteinuria contributes to oxidative stress, as reabsorbed proteins stimulate tubular cells to produce pro-inflammatory cytokines. Although clinical studies evaluating antioxidant supplementation in cats remain limited, dietary supplementation with vitamins C and E in elderly cats has shown a significant reduction in DNA damage, highlighting the potential role of antioxidant strategies in mitigating CKD progression [14,33].

1.2.3. Proteinuria and Mineral Disorders

Proteinuria is a significant pathological feature of feline CKD, resulting primarily from glomerular hypertension and impaired permselectivity. While the predominant histological lesion in cats is tubulointerstitial, the presence of proteins in the urine has substantial clinical relevance. Filtered proteins, including albumin, immunoglobulin G, and transferrin, trigger pro-inflammatory and profibrotic responses upon reaching the tubular lumen. Tubular epithelial cells respond to this protein load by releasing mediators that promote apoptosis, atrophy, and interstitial fibrosis [4,29]. Additionally, the accumulation of non-esterified fatty acids that dissociate from albumin under intratubular pH conditions can exert direct cytotoxic effects on proximal tubular cells.

In feline medicine, the magnitude of proteinuria is generally modest compared to other species, with median urine protein-to-creatinine ratios (UPC) typically ranging from 0.15 in IRIS stage 2 to 0.65 in stage 4 [14]. Nevertheless, persistent proteinuria is a strong

predictor of disease progression and is associated with the development of azotemia and reduced survival times, particularly in hypertensive cats [14,16,34].

Parallel to these proteinuric changes, the progressive decline in GFR triggers profound disturbances in mineral metabolism, culminating in renal secondary hyperparathyroidism (RHPTH). This condition arises from a maladaptive cycle involving phosphate retention, reduced calcitriol synthesis, and hormonal dysregulation. As renal phosphorus excretion declines, hyperphosphatemia develops and becomes increasingly prevalent in advanced IRIS stages, significantly accelerating renal fibrosis and soft tissue mineralization [3,35].

In the early stages of CKD, serum phosphorus concentrations often remain within reference intervals due to compensatory mechanisms mediated by fibroblast growth factor-23 (FGF-23) and parathyroid hormone (PTH). FGF-23 correlates inversely with glomerular filtration rate and promotes phosphaturia while suppressing calcitriol synthesis [16]. The resulting calcitriol deficiency removes inhibitory feedback on PTH secretion, leading to sustained parathyroid hormone elevation even in the absence of evident hyperphosphatemia [29].

This endocrine imbalance is further exacerbated by reduced activity of renal 1 α -hydroxylase enzyme, impairing the conversion of calcidiol to calcitriol. The combined effects of low calcitriol, hypocalcemia, and phosphate retention stimulate parathyroid gland hyperplasia and persistent PTH secretion. Sustained RHPTH has been associated with increased mortality as it promotes vascular calcification, oxidative stress, and systemic inflammation [3,14]. Evidence shows that dietary phosphate restriction can mitigate these lesions by reducing serum phosphorus, PTH, and FGF-23 concentrations, highlighting the importance of early and sustained control of the mineral axis to slow CKD progression [20].

1.2.4. Anemia, Azotemia, and Uremic Syndrome

As CKD progresses, the cumulative loss of functioning nephrons leads to impairment of the kidney's excretory and endocrine functions, culminating in anemia, azotemia, and the systemic manifestations of uremic syndrome.

Anemia is a common and clinically relevant complication, affecting approximately 30% to 65% of cats as the disease advances [16,30,36,37]. This condition is typically characterized as normocytic, normochromic, and poorly regenerative [16,36,37]. Its development is primarily attributed to insufficient erythropoietin (EPO) production by the peritubular fibroblast type-1 interstitial cells in the renal cortex [16,37].

Beyond EPO deficiency, additional contributory mechanisms include malnutrition, myelofibrosis, metabolic disturbances, and the accumulation of uremic toxins. Iron deficiency, particularly functional iron deficiency, plays a significant role and is often driven by increased hepcidin expression, an acute-phase protein that limits iron availability during the chronic inflammatory state characteristic of CKD [16,37,38]. Furthermore, gastrointestinal blood loss may exacerbate iron depletion and further worsen anemia [38]. The clinical impact of anemia is substantial, as reduced oxygen delivery to remaining functional nephrons, particularly within metabolically active and hypoxic regions, accelerates renal injury [4,36]. In severe cases, anemia has been identified as a predictor of poor prognosis, with reported median survival times of approximately 100 days [38].

Azotemia is defined by the accumulation of nitrogenous waste products, including creatinine and urea, in the bloodstream as a consequence of reduced GFR. The severity of azotemia correlates strongly with survival outcomes in cats with CKD [39]. When accompanied by clinical signs and systemic biochemical derangements, azotemia progresses to uremic syndrome [40]. This condition reflects advanced renal dysfunction and is characterized by fluid overload, electrolyte imbalances, and multisystemic effects. Urea itself exerts direct and indirect toxic effects, particularly on the nervous system [41].

In cats, specific gut-derived toxins, such as indoxyl sulfate, p-cresyl sulfate, and trimethylamine-N-oxide (TMAO), accumulate as renal function deteriorates. These compounds originate from bacterial metabolism of dietary precursors within the intestine, where the typically high-protein feline diet may play a substantial role in toxin production [42]. Indoxyl sulfate, in particular, promotes tubular inflammation and has been implicated in the progression of renal injury. Progressive retention of these toxins contributes to clinical manifestations such as nausea, vomiting, anorexia, lethargy, and altered mentation, ultimately leading to reduced quality of life and increased morbidity and mortality [41].

Collectively, the structural, hemodynamic, metabolic, and endocrine alterations described above illustrate that feline CKD is a dynamic and multifactorial process that evolves long before evident clinical signs or azotemia become apparent. Progressive nephron loss, fibrotic remodeling, mineral imbalance, and systemic consequences such as anemia and uremia develop in parallel, often in a clinically silent manner. This biological complexity underlies one of the major challenges in feline CKD management: the limited ability of conventional diagnostic tools to detect disease in its early and potentially more modifiable stages. In this context, understanding the strengths and limitations of current diagnostic approaches is essential to bridge the gap between pathophysiological progression and timely clinical intervention.

2. Search Strategy

To ensure a comprehensive and objective synthesis of feline CKD progression and biomarkers, a structured literature search was conducted across PubMed, Science Direct, and Google Scholar databases. The search focused on peer-reviewed articles, clinical trials, and consensus guidelines published between 2000 and 2025. Search terms included “feline chronic kidney disease”, “renal pathophysiology”, “tubulointerstitial fibrosis”, “biomarkers”, “SDMA”, “FGF-23”, and “early diagnosis”. Inclusion criteria prioritized longitudinal studies, prospective clinical trials, and the IRIS guidelines. Exclusion criteria were applied to studies with no peer review process, conference abstracts with insufficient data, and studies focusing exclusively on acute kidney injury without relevance to chronic progression. A total of 84 references were selected based on their methodological rigor and clinical relevance to provide a multi-parametric view of the disease.

3. Current Diagnostic Landscape

Feline CKD is typically diagnosed based on compatible clinical signs and the presence of renal azotemia, with urinalysis, particularly urine-specific gravity (USG), being essential for confirmation. While advanced CKD is usually easy to identify through blood and urine testing, these approaches have limited sensitivity for detecting early or non-azotemic disease [43].

Since CKD is more common in older cats, close longitudinal monitoring of their health becomes increasingly important with advancing age. Beginning at approximately seven years of age, it is recommended that cats be examined twice a year, including assessment of body weight, body condition, blood pressure, complete blood count, serum biochemistry, and routine urinalysis [20]. If routine findings are inconclusive but clinical suspicion persists, additional diagnostic procedures, such as imaging or histopathology, may assist in identifying underlying structural changes [3].

3.1. Medical History, Physical Assessment and Clinical Signs

The clinical presentation of CKD in cats varies greatly, reflecting different stages of the disease’s progression. While some animals are diagnosed incidentally during routine screening, others show only mild signs, or present at terminal stages characterized by

emaciation and severe dehydration. Although CKD is typically chronic, sudden clinical worsening is not uncommon. This possibility requires a high index of suspicion, particularly in senior and geriatric cats, although CKD can affect cats of any age [43].

Clinical signs usually appear in more advanced stages, making regular screening essential for high-risk populations [16]. In early stages, CKD rarely causes clinical signs, especially before the onset of azotemia, although progressive weight loss may represent an early indicator [44]. The most common clinical signs are generally nonspecific and include inappetence, polyuria, polydipsia, weight loss, lethargy, halitosis, and vomiting [43,44]. Gastrointestinal manifestations such as anorexia, nausea, melena, and diarrhea are among the most frequent reasons for veterinary consultation, often accompanied by other signs such as muscle wasting, weakness, and urinary incontinence [44].

On physical examination, affected cats may present with a thin body condition, dehydration, periodontal disease, and an unkempt coat. Abnormal renal palpation, revealing small, irregular, or occasionally enlarged kidneys, is a hallmark finding. Additionally, pale mucous membranes may be observed, particularly in cases of non-regenerative anemia [43,45]. A thorough clinical assessment must also account for complications that influence disease progression, such as systemic hypertension, secondary renal hyperparathyroidism, hypokalemia, and metabolic acidosis [16].

A standard diagnostic visit should integrate a detailed history and medication review with a complete physical examination, body weight assessment, and nutritional scoring. This initial clinical assessment provides the necessary context for interpreting laboratory findings, including hematocrit, serum biochemistry, and urinalysis [3].

3.2. IRIS Staging and Substaging

The laboratory diagnosis of CKD primarily relies on identifying persistent azotemia combined with reduced urine specific gravity (USG < 1.035), typically assessed through serum biochemistry and urinalysis. USG provides valuable insight into the kidneys' concentrating ability and, when persistently low, particularly alongside azotemia, supports the diagnosis of renal dysfunction. In these cases, it is important to exclude pre-renal and post-renal causes by evaluating urine concentration, assessing response to fluid therapy, and considering the cat's clinical history and physical examination [7,20,46].

The GFR is considered the most reliable indicator of kidney function. While its direct measurement represents the most accurate method, it is impractical for routine use in companion animals; therefore, GFR is usually estimated using biomarkers such as urea and creatinine, which only begin to rise after approximately 75% of kidney function has been lost. Creatinine is generally preferred, but its nonlinear relationship with GFR means that small early declines in kidney function cause minimal creatinine increases, whereas large rises later may reflect relatively modest decreases in GFR [46]. More recently, symmetric dimethylarginine (SDMA) has emerged as a more sensitive biomarker as it increases earlier than creatinine and is not influenced by muscle mass, allowing earlier detection of CKD, including in non-azotemic cats. SDMA is now included in the IRIS staging guidelines, improving diagnostic accuracy and facilitating timely intervention [46,47].

CKD staging is implemented once the diagnosis has been confirmed, helping guide treatment decisions and clinical monitoring. When performed early, it enables appropriate therapeutic interventions to be initiated, improving patient outcome [43,48]. As the disease progresses, higher stages are generally associated with shorter survival times, highlighting the importance of accurate and consistent staging [49]. IRIS recommendations suggest that staging be based primarily on fasting blood creatinine and SDMA concentrations, ideally both, measured at least twice in a stable, well-hydrated patient. Following staging, cats

are further classified into substages based on the assessment of proteinuria and systemic blood pressure.

Stage 1 CKD in cats is characterized by normal blood creatinine concentrations or normal to mildly increased SDMA levels, alongside other indicators of renal disease, such as impaired urinary concentrating ability, abnormal renal palpation or imaging findings, proteinuria of renal origin, or abnormal biopsy results. Stage 2 involves normal to mildly increased creatinine concentrations with mild renal azotemia and mildly elevated SDMA, often accompanied by minimal or absent clinical signs. Stage 3 is characterized by moderate renal azotemia and may be associated with variable extrarenal clinical manifestations, whereas stage 4 reflects severe renal impairment and is associated with a higher risk of systemic uremic crises [48]. This staging system is intended to be applied during periods of short-term clinical stability, and re-evaluation approximately every six months can assist in monitoring disease progression [50]. Nevertheless, maintaining recommended monitoring frequencies remains a challenge, as owners' constraints often prevent clinicians from reassessing patients as frequently as guidelines suggest [51]. This highlights the need for better veterinarian–caregiver communication to improve adherence and ensure that an early diagnosis helps minimize the caregiver burden [52].

After staging, CKD is further classified through substaging based on the urine protein-to-creatinine (UPC) ratio and systolic blood pressure (SBP) [50]. Ideally, the UPC ratio should be determined once urinary tract inflammation/infection or hemorrhage has been excluded, using species-appropriate methods, and should be measured in all cats with CKD. Substaging should be based on at least two urine samples collected over a two-week period. Cats with persistent borderline proteinuria should be re-evaluated within two months, as sustained proteinuria is strongly associated with disease progression. In advanced stages of CKD (stages 3 and 4), proteinuria may paradoxically decrease as renal function deteriorates [48].

Arterial hypertension is a common complication of CKD in cats and is associated with a poorer renal prognosis [3]. IRIS recommends that blood pressure be measured in all cats with CKD, using multiple readings obtained in a calm environment to ensure accuracy. Final risk classification is based on several SBP measurements, preferably collected during repeat clinic visits. SBP is used to sub-classify cats according to their risk of target organ damage and the presence of associated complications. When target organ damage is not initially evident, confirming persistent hypertension through repeated measurements over one to two weeks is essential [48] (Table 1).

Table 1. IRIS staging and substaging system for feline CKD.

Classification Parameter	Reference Interval	Stage/Substage **	Clinical Significance
I. Primary staging			
Blood creatinine $\mu\text{mol/L}$ mg/dL	≤ 140 < 1.6	Stage 1	Non-azotemic; presence of renal abnormalities
	$140\text{--}250$ $1.6\text{--}2.8$	Stage 2	Mild renal azotemia; clinical signs absent or present
	$251\text{--}440$ $2.9\text{--}5.0$	Stage 3	Moderate renal azotemia; systemic clinical signs may be present
	> 440 > 5.0	Stage 4	Severe renal azotemia; increased risk of uremic syndrome

Table 1. Cont.

Classification Parameter	Reference Interval	Stage/Substage **	Clinical Significance
SDMA μg/dL	<18	Stage 1	Non-azotemic; presence of renal abnormalities
	18–25	Stage 2	Mild renal azotemia; clinical signs absent or present
	26–38	Stage 3	Moderate renal azotemia; systemic clinical signs may be present
	>38	Stage 4	Severe renal azotemia; increased risk of uremic syndrome
II. Substaging based on proteinuria			
UPC value	<0.2	Non-proteinuric	Potential microalbuminuria
	0.2–0.4	Borderline proteinuric	Requires monitoring; potential microalbuminuria
	>0.4	Proteinuric	Strong predictor of disease progression
II. Substaging based on blood pressure			
Systolic blood pressure mmHg	<140	Normotensive	Minimal risk of future lesions in target organs
	140–159	Prehypertensive	Low risk of future lesions in target organs
	160–179	Hypertensive	Moderate risk of future lesions in target organs
	>180	Severely hypertensive	High risk of future lesions in target organs

** Staging is primarily based on fasting blood creatinine and/or symmetric dimethylarginine (SDMA) concentrations. Substaging is subsequently performed according to proteinuria status (urine protein-to-creatinine ratio, UPC) and systolic blood pressure (SBP), allowing standardized clinical classification and assessment of risk for target organ damage. Adapted from International Renal Interest Society (IRIS) guidelines (2023) [48].

3.3. Traditional Laboratory Markers and Their Limitations

The clinical diagnosis and monitoring of feline CKD have traditionally relied on a combination of hematology, serum biochemistry and urinalysis. While these tools are fundamental in clinical practice, they possess inherent limitations, particularly with regard to early detection and susceptibility to extra-renal influences. Anemia is a common finding, making a complete blood count (CBC) essential to assess hemoglobin and total proteins. Although examination of blood smears and parameters such as mean corpuscular volume (MCV), red cell distribution width (RDW), and mean corpuscular hemoglobin concentration (MCHC) assist in differentiating anemia types, these often remain within reference ranges in CKD-associated anemia. Since this anemia is primarily driven by reduced EPO production, reticulocyte counts typically remain low, and these changes often manifest only when the disease is already established.

Regarding renal function assessment, the GFR is the most sensitive indicator, especially useful in early stages when routine biomarkers are still within normal ranges. While inulin clearance remains the gold standard, its clinical impracticality has led to the use of plasma clearance methods, such as iothexol. Iothexol is a safe, non-metabolized marker excreted almost exclusively by glomerular filtration, allowing for the identification of early reductions in renal function [53,54]. The precision of GFR estimates is fundamentally dependent on the mathematical modeling applied. Bicompartamental analysis is considered

the most appropriate pharmacokinetic model as it accounts for the initial distribution phase of the marker between plasma and the extracellular fluid volume (ECFV). In contrast, monocompartmental models assume immediate mixing and distribution, which leads to an underestimation of the area under the curve (AUC) and a subsequent overestimation of GFR. This overestimation is particularly pronounced in uremic patients, where altered elimination kinetics and distribution volumes can shift the time required for the marker to reach equilibrium. Therefore, to balance clinical practicality with diagnostic accuracy, the use of a slope-intercept method with a validated correction formula, such as the one developed specifically for cats ($1.036 \times \text{GFR}_{\text{uncorrected}} - 0.062 \times \text{GFR}_{\text{uncorrected}}^2$), is recommended to align monocompartmental results with the more rigorous bicompartamental gold standard. However, despite its reliability, GFR measurement is often bypassed in routine practice due to the requirement for multiple blood samplings [54].

In routine biochemistry, serum creatinine is the most widely used marker and a central component of the IRIS staging system [55,56]. It serves as a surrogate for GFR; however, the relationship between the two is exponential, leading to a significant diagnostic gap whereby creatinine typically rises only after approximately 75% of nephron loss has occurred [57]. This marker is also heavily influenced by muscle mass, meaning that sarcopenic geriatric cats may maintain deceptively low creatinine levels despite severe renal impairment, while certain breeds, such as the Burmese, may exhibit naturally higher baseline levels [55]. Age further contributes to variability, as kittens up to 8 weeks old can present creatinine concentrations similar to or higher than those of adults [58]. Similarly, blood urea nitrogen (BUN) is considered an insensitive and non-specific marker for early CKD, as it is readily influenced by high-protein diets or gastrointestinal bleeding [56].

The emergence of SDMA has addressed some of these limitations. Produced by all nucleated cells and excreted primarily by the kidneys, SDMA exhibits a more linear relationship with GFR and is not affected by muscle mass, potentially revealing reductions in GFR of up to approximately 40% [47,59,60]. While its inclusion in IRIS guidelines has improved early detection, its specificity remains under investigation, and potential breed-related variations require further study [59].

Progression of CKD also triggers profound disruptions in mineral metabolism. Phosphorus retention drives renal secondary hyperparathyroidism and soft tissue mineralization, yet serum phosphorus concentrations often remain within reference ranges in IRIS stages 1 and 2 due to compensatory mechanisms mediated by FGF-23 and PTH [3,55,61]. Calcium assessment is equally complex and represents a critical component of chronic kidney disease–mineral and bone disorder (CKD-MBD), a systemic syndrome involving mineral, bone, and cardiovascular calcification abnormalities. Ionized calcium (iCa) is the diagnostic gold standard, as total calcium (tCa) is notoriously unreliable in feline CKD due to its susceptibility to hypoalbuminemia and acid–base disturbances, which alter calcium–protein binding. Despite a high specificity of 92%, tCa has only a 30% sensitivity for identifying hypercalcemia in CKD cats. Clinical studies indicate that while normocalcemia is frequent, the prevalence of ionized hypercalcemia can reach up to 30%, whereas ionized hypocalcemia is more often associated with advanced stages and severe secondary hyperparathyroidism [62,63].

Electrolyte disturbances such as hypokalemia further complicate clinical interpretation, as serum potassium concentrations typically transition from a 20–30% prevalence of hypokalemia in IRIS stages 2 and 3 to hyperkalemia in terminal stages due to severe GFR decline [3,64]. This variability, coupled with the fact that blood potassium may not accurately reflect total body stores, as approximately 95% of potassium is intracellular, limits its reliability as a marker of disease stability [55]. Similarly, biochemical markers of acid–base balance, such as bicarbonate and total CO_2 , are often insensitive for early detec-

tion, as compensatory mechanisms typically mask metabolic disturbances until advanced stages. However, metabolic acidosis is a key driver of CKD progression. In cats, impaired organic acid excretion leads to titration acidosis with an increased anion gap. This state promotes protein catabolism, muscle wasting, and bone resorption, while also accelerating tubulointerstitial fibrosis. Furthermore, acidosis triggers an extracellular potassium shift, which may mask total body depletion [65].

Additionally, alterations in lipid markers are frequently overlooked in routine panels, despite histological evidence of renal lipid accumulation in up to 89% of cats with CKD. This high prevalence suggests that dyslipidemia may contribute to early metabolic disturbances that traditional laboratory markers fail to capture while appearing clinically stable [23,66].

Urinalysis remains a central diagnostic pillar; however, its components often fail to reflect the early stages of disease progression [67]. While a USG ≤ 1.035 in conjunction with persistent azotemia is a classic diagnostic criterion, this parameter is easily influenced by non-renal factors such as diet and water intake, and may remain relatively preserved in some cats despite significant parenchymal loss [54,68]. Furthermore, sediment analysis is essential to distinguish renal proteinuria from inflammatory causes, but its reliability is hindered by rapid cellular degeneration and temperature-induced artifacts [69]. Similarly, urine cultures often reveal subclinical bacteriuria that does not correlate with creatinine levels or disease progression, complicating this distinction between clinically relevant infection and incidental findings [70–72].

Finally, assessment of proteinuria is crucial for predicting survival, yet current methods present significant technical limitations. Urine dipstick testing is notoriously unreliable in cats, with a specificity of only 31%, largely due to false-positive results associated with alkaline urine or active sediment. Although the sulfosalicylic acid (SSA) test detects a broader range of proteins, its specificity is even lower, at 25%, and reliance on visual turbidity grading renders it highly subjective [73,74]. The UPC ratio is the standard method for quantification, and even mild elevations (≈ 0.4) are associated with reduced survival. However, UPC testing fails to detect albumin concentrations below the standard detection threshold of 30 mg/dL, creating a diagnostic gap that can only be addressed through microalbuminuria (MA) testing. MA assays detect albumin concentrations as low as 1 mg/dL, offering a substantially more sensitive tool for identifying cats at risk of early renal injury [73,75].

4. Emerging Biomarkers: Bridging the Diagnostic Gap

The search for novel biomarkers is driven by the need to identify renal compromise before irreversible nephron loss occurs. FGF-23 has emerged as a key indicator of early metabolic derangement, as its plasma concentrations rise progressively alongside declining kidney function. It effectively serves as an early warning for phosphate retention; indeed, azotemic cats with hyperphosphatemia exhibit significantly higher FGF-23 levels than normophosphatemic cats at the same disease stage. This observation suggests that FGF-23 may reveal disturbances in mineral metabolism while serum phosphorus concentrations still appear deceptively stable. Nevertheless, reference intervals in geriatric cats are broader than those in humans, potentially reflecting species-specific adaptations or dietary influences that necessitate careful clinical interpretation [76].

Serum cystatin C (sCysC) is also being explored as a GFR marker that remains unaffected by muscle mass, addressing a major flaw of creatinine. Although sCysC is widely used in humans and moderately in canine medicine, its use in cats is still limited by the lack of species-specific assays. Current methods often fail to distinguish clearly between healthy cats and those with early CKD, leading to overlapping results. Furthermore, common

comorbidities such as hyperthyroidism can interfere with sCysC concentrations, reducing its reliability as a standalone diagnostic marker [59].

Shifting the focus toward structural renal integrity, urinary biomarkers of tubular injury provide a more granular assessment than traditional filtration-based markers. Urinary cystatin B (uCysB) and retinol binding protein (RBP) are particularly promising indicators of ongoing tubular injury. Notably, uCysB can detect active tubular damage even when serum creatinine remains stable, effectively narrowing the diagnostic gap for monitoring disease progression [77–79]. Other markers, such as N-acetyl- β -D-glucosaminidase (NAG) and neutrophil gelatinase-associated lipocalin (NGAL), also show a strong correlation with tubular injury. However, their clinical utility remains inconsistent due to high interindividual variability and confounding factors, including urinary tract inflammation and hyperthyroidism, which may artificially elevate biomarker levels and complicate differentiation between primary renal damage and transient systemic stress [59,80].

Table 2 provides a comprehensive overview of feline CKD biomarkers, showing that while emerging tools are essential for bridging the diagnostic gap, they also present specific limitations that necessitate integration with conventional laboratory markers rather than isolated interpretation. Emerging biomarkers therefore complement, rather than replace, traditional diagnostic parameters by enabling earlier detection of renal compromise before irreversible nephron loss occurs.

Table 2. Overview of established and emerging biomarkers for feline CKD.

Marker	Sample	Primary Application	Clinical Limitations /Challenges	Availability	Reference
Gold Standard					
Iohexol	Serum	Precise GFR measurement	Clinically impractical; requires multiple blood samplings	Specialized	[53,54]
iCa	Serum	Mineral status	Requires anaerobic handling; essential to bypass tCa unreliability	Commercial	[62,63]
Traditional					
Creatinine	Serum	GFR surrogate	Low sensitivity; affected by muscle mass, breed and age	Commercial	[55–58]
BUN	Serum	Nitrogenous waste	Easily skewed by high-protein diets or gastrointestinal bleeding	Commercial	[56]
Phosphorus	Serum	Mineral metabolism	Often masked by FGF-23/PTH compensation in early stages	Commercial	[3,55,64]
Potassium	Serum	Electrolyte stability	Serum levels do not reflect total body stores	Commercial	[3,55,64]
tCa	Serum	Mineral balance	High specificity but very low sensitivity	Commercial	[62,63]
USG	Urine	Renal concentrating function	Influenced by non-renal factors; may remain normal in early CKD	Commercial	[54,68]

Table 2. Cont.

Marker	Sample	Primary Application	Clinical Limitations /Challenges	Availability	Reference
UPC ratio	Urine	Proteinuria/substaging	High variability; requires exclusion of inflammation/UTIs	Commercial	[73–75]
Established					
SDMA	Serum	Early/stable GFR assessment	Specificity under investigation; potential breed-related variations	Commercial	[47,59,60]
Emerging					
FGF-23	Plasma/serum	Early indicator of mineral derangement	Broad reference intervals in geriatric cats; requires careful interpretation	Specialized	[76]
sCysC	Serum	Muscle-independent GFR marker	Lacks species-specific tests; overlapping results; confounded by hyperthyroidism	Research only	[59]
uCysB	Urine	Active tubular injury detection	Detects damage while creatinine remains stable; limited clinical availability	Emerging	[77–79]
RBP	Urine	Proximal tubule damage marker	Primarily used in research settings; specific for reabsorption failure	Research only	[77–79]
NGAL/NAG	Urine	Correlation with tubular insult	High individual variability; skewed by UTIs or systemic stress	Research only	[59,80]

Table summarizes the clinical utility and main limitations of currently available biomarkers used in feline CKD. Direct GFR measurement using iohexol clearance remains the reference method but is rarely applied in routine clinical practice. Traditional markers are widely used for staging but may be influenced by extra-renal factors. Established biomarkers such as symmetric dimethylarginine (SDMA), together with emerging markers including fibroblast growth factor 23 (FGF-23) and urinary cystatin B (uCysB), may improve early detection of metabolic and tubular alterations before advanced nephron loss. Availability is categorized as: Commercial (routine veterinary laboratories); Specialized (reference laboratories or specialized equipment); Emerging (recently validated/limited market availability); and Research only (currently restricted to investigative settings). BUN, Blood Urea Nitrogen; iCa, Ionized Calcium; NAG, N-acetyl- β -D-glucosaminidase; NGAL, Neutrophil Gelatinase-Associated Lipocalin; PTH, Parathyroid Hormone; RBP, Retinol-Binding Protein; sCysC, Serum Cystatin C; tCa, Total Calcium; UPC, Urine Protein-to-Creatinine ratio; USG, Urine-Specific Gravity; UTI, Urinary Tract Infection.

Clinical Hierarchy and Practical Application

To assist practitioners in navigating the diagnostic weight of these tools, a clear distinction must be made between routine practice and research-grade markers. While established biomarkers such as SDMA are now integrated into many reference laboratory panels, several emerging proteins, particularly those indicating acute tubular insult, remain primarily within the scope of academic research. Table 3 provides a practical hierarchy of these biomarkers, categorizing them by their current clinical availability and utility in a practice setting.

Table 3. Clinical applicability of emerging biomarkers.

Marker	Clinical Status	Practitioner's Takeaway	Reference
SDMA	Established	Use for early screening; commercially available in most reference laboratories	[47,59,60]
FGF-23	Available	Useful for monitoring mineral disorders; requires specialized referral laboratories	[76]
sCysC	Low Utility	Not recommended for routine use due to lack of feline-specific validation and thyroid interference	[59]
uCysB	Emerging	High potential for detecting active tubular injury, but limited commercial access	[77–79]
RBP	Research	Restricted to academic settings for proximal tubule assessment; reflects reabsorption failure	[77–79]
NAG/NGAL	Research	Valuable for scientific studies; not currently recommended for routine clinical use	[59,80]

This table categorizes biomarkers based on their current clinical availability and diagnostic weight to assist practitioners in differentiating between validated tools and research-grade markers. FGF-23, Fibroblast Growth Factor 23; NAG, N-acetyl- β -D-glucosaminidase; NGAL, Neutrophil Gelatinase-Associated Lipocalin; RBP, Retinol-Binding Protein; sCysC, Serum Cystatin C; SDMA, Symmetric Dimethylarginine; uCysB, Urinary Cystatin B.

5. Clinical Monitoring and Prognostic Value

Regular assessments are fundamental to managing the progressive nature of feline CKD, as the clinical needs of these patients often shift over time. The frequency of evaluation depends largely on the IRIS stage and clinical stability. Cats in stages 1 and 2 typically require monitoring every 4 to 6 months, whereas those in stages 3 and 4 usually necessitate more frequent visits, approximately every 3 to 4 months. In cases of rapid progression, persistent proteinuria, or uncontrolled hypertension, evaluations may be required as often as every 1 to 4 weeks during stabilization phases [3,20].

A standard monitoring visit must integrate a thorough medical history with a complete physical examination, with particular attention to body weight and nutritional status. Serial measurements of BUN, creatinine, and phosphorus are essential for tracking the rate of functional decline. Furthermore, assessment of the mineral axis should rely on iCa and PTH to ensure adequate control of renal secondary hyperparathyroidism, particularly in cats receiving calcitriol therapy. Given that tCa frequently fails to reflect the true ionized status in feline CKD, direct measurement of iCa is essential for accurate clinical decision-making [29,62]. In this context, emerging biomarkers such as FGF-23 may provide additional insight into treatment efficacy and early metabolic disturbances that precede overt hyperphosphatemia [29]. Adjunctive nutritional strategies may also contribute to CKD management, particularly dietary supplementation with omega-3 polyunsaturated fatty acids (EPA/DHA). A recent systematic review in companion animals with neoplastic and non-neoplastic diseases supports consistent anti-inflammatory and lipid-modulating effects of EPA/DHA supplementation, which may be biologically relevant to chronic kidney disease progression [81].

The prognosis for cats with CKD is highly individual and often difficult to predict, as many patients remain stable for years while others experience a more rapid decline [82]. Several modifiable risk factors are directly linked to reduced survival, including poor body condition, anemia, and hyperphosphatemia. Notably, persistent proteinuria and systemic hypertension remain two of the strongest predictors of a poor outcome. Despite these risks, the progression of feline CKD is typically gradual. Veterinary nurses and technicians play a crucial role in long-term disease monitoring, supporting owners and identifying subtle

changes in parameters such as body condition or the UPC ratio that should prompt a more detailed clinical evaluation [3,20,82].

The Clinical Paradox of Early Diagnosis and the Ethics of Analgesia

A critical challenge in the clinical application of early biomarkers is the “utility paradox”: the concern that identifying CKD in its non-azotemic stages may lead to negative therapeutic interventions. A primary risk is the inappropriate withdrawal of necessary medications, particularly NSAIDs for managing comorbidities such as osteoarthritis. Chronic pain has a profound negative impact on feline quality of life and impairs the owner–cat bond, yet its treatment is frequently neglected due to renal concerns [83].

Evidence from landmark studies indicates that the use of NSAIDs in stable, well-monitored CKD cats does not necessarily accelerate the decline of renal function. Specifically, long-term maintenance therapy with meloxicam (at a median dose of 0.02 mg/kg/day) has been demonstrated to be safe for aged cats with stable CKD and may even slow disease progression in some individuals [84]. Therefore, the identification of early renal compromise should be viewed as a mandate for safer, proactive management and monitoring of treatment tolerance, rather than a reason to withhold necessary analgesia. This approach ensures a holistic clinical balance where the management of chronic pain is refined by earlier diagnosis, rather than restricted by it.

6. Conclusions and Future Directions

The management of feline CKD is currently undergoing a transformative shift by moving from reactive stabilization toward a more proactive and personalized approach. The translational potential of novel biomarkers such as uCysB, NGAL, and FGF-23 lies in their ability to detect subclinical renal distress before the threshold of irreversible nephron loss is crossed. Future directions point toward the integration of multi-parametric diagnostic panels that combine filtration markers with indicators of active tubular injury and metabolic derangement. This paradigm shift aims to redefine the diagnostic gap by enabling clinical intervention during the early stages of fibrosis and mineral imbalance, when therapeutic strategies are most likely to slow disease progression effectively. Nonetheless, it is essential to recognize that these emerging markers are not infallible; their clinical utility is still often constrained by biological variability, interference from common comorbidities such as hyperthyroidism, and a lack of standardized species-specific reference intervals. These limitations underscore the need for cautious interpretation and integration alongside established diagnostic parameters.

However, the path toward widespread clinical adoption faces significant practical challenges. High costs and the limited availability of species-specific assays mean that many of these emerging biomarkers remain restricted to research or referral settings. Until these barriers are addressed, traditional diagnostic tools remain indispensable. Urinalysis, in particular, continues to be a cornerstone of feline practice, as its accessibility and high negative predictive value make it a critical tool for excluding complications and monitoring clinical stability. This remains especially relevant in real-world clinical settings where financial constraints restrict access to advanced diagnostic technologies.

In conclusion, understanding the progression of feline CKD requires a holistic view that bridges the gap between molecular pathophysiology and clinical reality. While serum creatinine and SDMA are the current pillars of staging, they represent only static snapshots within a dynamic and evolving pathological process. However, this diagnostic progress must be balanced with clinical reality. Early detection should not lead to unnecessary medical intervention, but rather to a personalized approach that prioritizes the cat’s comfort. In geriatric patients, renal preservation must go hand in hand with pain management and the

reduction in clinic-related stress. Ultimately, the value of these new biomarkers lies not just in improving laboratory numbers, but in ensuring good quality of life and supporting the bond between the owner and the cat throughout the disease. The integration of emerging biomarkers with consistent clinical monitoring focused on body condition, mineral homeostasis, and preservation of remaining renal function offers the most promising strategy for extending both survival and quality of life in aging cats. Ultimately, the goal is to shift feline CKD from a disease primarily managed at advanced stages to one that can be anticipated and intercepted earlier, promoting healthier aging in the feline population.

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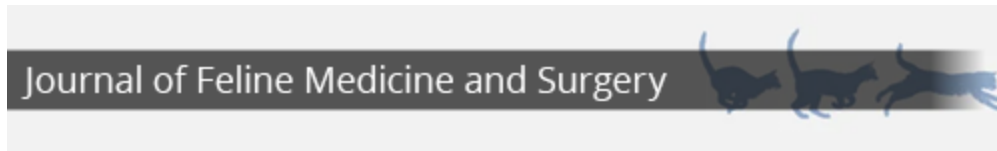
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Changes in serum symmetric dimethylarginine concentrations after treatment of feline hyperthyroidism with antithyroid medications

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Abstract

Objectives

The aim of the present study was to report changes in serum creatinine and symmetric dimethylarginine (SDMA) concentrations after treatment of feline hyperthyroidism with anti thyroid medications and to compare these biomarkers at baseline between cats that were and were not azotaemic after treatment.

Methods

In this retrospective study, hyperthyroid cats that were euthyroid (total thyroxine [TT4] concentration 7–40 nmol/l) at 1 month (T1) and/or 2–9 months (T2) after treatment were identified and grouped by renal status defined by serum creatinine concentrations. Comparisons

were made using non-parametric statistics and the correlations assessed using Spearman's correlation coefficient. Data are presented as median (minimum–maximum).

Results

A total of 19 hyperthyroid cats were included. At baseline, TT4 was negatively correlated with serum concentration of creatinine ($r_s = -0.73$; $P < 0.001$) but not SDMA ($r_s = -0.42$; $P = 0.074$). Serum creatinine concentrations increased significantly at T1 and T2 (137 $\mu\text{mol/l}$ [range 97–241] and 162 $\mu\text{mol/l}$ [range 76–251]) compared with baseline (117 $\mu\text{mol/l}$ [range 62–216]; $P = 0.003$ and $P < 0.001$, respectively), whereas serum SDMA did not change significantly at T1 but did increase by T2 (11 $\mu\text{g/dl}$ [range 8–29] and 13 $\mu\text{g/dl}$ [range 9–24], respectively) compared with baseline (12 $\mu\text{g/dl}$ [range 7–21]; $P = 0.548$ and $P = 0.039$, respectively). There was no significant difference in baseline serum SDMA between cats that were azotaemic after treatment and those that remained non-azotaemic (12 $\mu\text{g/dl}$ [range 7–21], $n = 13$ vs 13 $\mu\text{g/dl}$ [range 11–19], $n = 6$; $P = 0.42$).

Conclusions and relevance

Serum SDMA concentrations are not helpful in predicting post-treatment azotaemia in initially non-azotaemic hyperthyroid cats treated with antithyroid medications and might be influenced by factors other than glomerular filtration rate in hyperthyroidism.

Keywords: Symmetric dimethylarginine, creatinine, chronic kidney disease, hyperthyroidism, methimazole, carbimazole

Plain language summary

Assessment of serum SDMA concentrations in hyperthyroid cats

This study evaluated whether the concentrations of a biomarker of kidney function called symmetric dimethylarginine (SDMA) in the blood change in hyperthyroid cats after treatment. SDMA concentrations did change after treatment but not as quickly as we would expect, which could suggest that SDMA concentrations are affected by factors other than just kidney function.

Introduction

Hyperthyroidism is the most common endocrinopathy in geriatric cats, and although there is significant geographical variance, studies assessing populations of older cats typically indicate a prevalence in the order of 10%.¹⁻⁵ Chronic kidney disease (CKD) is also prevalent in older cats and approximately 10% of untreated hyperthyroid cats have azotaemic CKD, while a further 15% of medically treated hyperthyroid cats become azotaemic after treatment of hyperthyroidism (referred to as 'masked' CKD).⁶ The hypermetabolic state associated with hyperthyroidism results in an increased renal glomerular filtration rate (GFR) and reduced muscle mass, thus decreasing serum creatinine and urea concentrations.⁷⁻¹² These changes therefore complicate the diagnosis of CKD in hyperthyroid cats before treatment. It is established that the presence of pretreatment azotaemia negatively impacts the survival of hyperthyroid cats;⁶ however, the development of azotaemia after treatment in previously non-azotaemic cats is not associated with reduced survival times.¹³ Nevertheless, identifying cats at a higher risk of developing azotaemia after treatment could inform management decisions and enable earlier dietary modification, which can improve long-term survival.¹⁴

Symmetric dimethylarginine (SDMA) is a by-product of L-arginine methylation that is released into the circulation by all nucleated cells after proteolysis. SDMA is excreted predominantly by the kidneys and, as a result, serum SDMA concentrations are inversely correlated with GFR in healthy cats.^{15,16} Serum SDMA concentrations are also not associated with changes in muscle mass in cats.¹⁷ Serum creatinine and SDMA concentrations increase after the restoration of euthyroidism in radioiodine-treated hyperthyroid cats;¹⁸⁻²⁰ however, in some cats, serum SDMA concentrations decrease after radioiodine treatment.^{20,21} In addition, a poor correlation between serum SDMA concentrations and GFR has been demonstrated in hyperthyroid cats, which suggests that factors other than GFR could affect SDMA concentrations.^{21,22} Altered protein metabolism associated with the hyperthyroid state could result in increased SDMA production and release, or hyperthyroidism could be associated with decreased extra-renal degradation and clearance of SDMA.

The effects of radioiodine treatment and bilateral thyroidectomy on serum SDMA concentrations have been reported previously;¹⁸⁻²⁴ however, the effects of methimazole and carbimazole treatment of hyperthyroid cats on serum SDMA concentrations have not yet been characterised. Therefore, investigation of the change in serum SDMA concentrations after treatment of hyperthyroidism with antithyroid medications would be of interest to document whether these differ from those of cats treated with radioiodine.

The aim of this study was to investigate the change in serum creatinine and SDMA concentrations in a population of client-owned hyperthyroid cats treated with methimazole or carbimazole. We

hypothesised that serum SDMA concentrations would increase in parallel with serum creatinine after the restoration of euthyroidism due to the reduction in GFR. We also hypothesised that baseline serum SDMA concentrations would be higher in cats with concurrent or 'masked' CKD.

Materials and methods

Case selection

Ethical approval was granted by the Ethics and Welfare Committee of the Department of Veterinary Medicine, University of Cambridge. The university teaching hospital clinical pathology database was searched for feline serum samples with a total thyroxine (TT4) concentration above the laboratory reference interval (RI) derived in a population of healthy older cats (7–45 nmol/l) [25](#) in the 10-year period between 2014 and 2024. These cases were screened for inclusion based on signalment, age, clinical signs, medications, laboratory testing data, comorbidities and type of treatment. Laboratory testing data were extracted from records from before and after treatment, including serum concentrations of TT4, creatinine, urea and, where available, urine specific gravity (USG).

For inclusion in the study, cats were required to have clinical signs and/or physical examination findings that were consistent with hyperthyroidism, a TT4 above 45 nmol/l (upper limit of laboratory RI) and to have subsequently started medical treatment with oral carbimazole or methimazole. Euthyroidism (defined as serum TT4 7–40 nmol/l) must have been demonstrated at 1 month (T1) and/or at 2–9 months (T2) after initiating treatment. Cats were excluded if they were deemed to be systemically unwell, if follow-up blood work at these time points demonstrated inadequate control of hyperthyroidism or if there was insufficient residual stored serum for SDMA analysis. Azotaemia was defined as a serum creatinine concentration above the laboratory RI of 56–153 $\mu\text{mol/l}$.

Clinicopathological assays

Blood samples were centrifuged within 2 h of collection and residual serum frozen at -80°C within 24 h of initial biochemical analysis. In most instances, serum TT4 and creatinine were measured contemporaneously using a previously validated human TT4 enzyme immunoassay [25](#) and a commercially available spectrophotometric assay for creatinine (OSR6178; Beckman Coulter). Serum SDMA concentrations were measured on residual frozen serum samples up to 10 years after collection using a competitive homogenous immunoassay (IDEXX Laboratories). [26](#)

Classification of hyperthyroid cats

Hyperthyroid cats were divided into three groups based on serum creatinine concentration before and after treatment. Non-azotaemic hyperthyroid cats had a serum creatinine below 153 $\mu\text{mol/l}$ (upper limit of laboratory RI) both before treatment and at the T2 (2–9 months) post-treatment time point. Masked azotaemic hyperthyroid cats had a serum creatinine below 153 $\mu\text{mol/l}$ at baseline but had a serum creatinine above 153 $\mu\text{mol/l}$ at T1 and/or T2. Azotaemic hyperthyroid cats had a serum creatinine above 153 $\mu\text{mol/l}$ at baseline and at T1 and/or T2.

Statistical analysis

Statistical analysis was performed using commercially available software (SPSS Statistics for Windows v27.0; IBM). Results are reported as median (minimum–maximum) unless otherwise specified. The data were assessed for normality by visual histogram assessment and Shapiro–Wilk testing; the data were not normally distributed and non-parametric tests were used. Baseline continuous variables (serum creatinine, urea, TT4 and SDMA concentrations) were compared between groups using the Mann–Whitney U-test or Kruskal–Wallis test. Comparisons within groups at different time points were analysed using the Wilcoxon signed rank test. Correlations between SDMA, TT4 and creatinine were assessed using Spearman's rank correlation coefficient. Statistical significance was defined as $P < 0.05$.

Results

A total of 19 cats were included in the study (see [S1 in the supplementary material](#) [\[4\]](#)). Of them, 14 (74%) were female spayed cats and five (26%) were male castrated cats. All cats were domestic shorthairs or longhairs, and the average age was 14 years (range 8–19). Only five (26%) cats had a recorded USG before initiation of treatment of hyperthyroidism. Comorbidities at the time of diagnosis of hyperthyroidism included hepatic disease ($n = 4$), gingivitis ($n = 3$), steroid-responsive skin disease ($n = 1$), controlled hypertension ($n = 1$), hypertrophic cardiomyopathy (diagnosed or suspected, $n = 3$) and chronic diarrhoea ($n = 1$). Medications at the time of baseline sampling included amlodipine ($n = 1$, Amodip; Ceva Animal Health), benazepril ($n = 1$, Fortekor; Elanco UK AH), furosemide ($n = 2$, generic brands), clindamycin ($n = 1$, Antirobe; Zoetis) and meloxicam ($n = 1$, Metacam; Boehringer Ingelheim). In total, 10 hyperthyroid cats were treated with oral methimazole (either Felimazole [Dechra Veterinary Products] or Thyronorm [Norbrook Laboratories]) and nine cats were treated with oral carbimazole (Vidalta; MSD Animal Health UK).

Baseline biochemical data

Of the 19 hyperthyroid cats in this study, six (32%) were classified as non-azotaemic, nine (47%) were classified as masked azotaemic and four (21%) were classified as azotaemic. Of the masked azotaemic group, three were azotaemic and four were non-azotaemic at T1, and seven were azotaemic at the T2 time point. Two masked azotaemic cats that were azotaemic at T1 had no data at T2 and one of these cats was azotaemic at both T1 and T2. No masked azotaemic cats were azotaemic at T1 but not at T2. In the azotaemic group, three cats were re-evaluated at T1: one cat had a serum creatinine concentration of 148 $\mu\text{mol/l}$, whereas the other two remained azotaemic at T1. All four cats were azotaemic at T2.

When the masked azotaemic and azotaemic groups were combined to form the hyperthyroid-CKD group (representing cats with 'masked' and overt CKD at baseline), there was a tendency towards higher baseline serum creatinine concentrations in the hyperthyroid-CKD group than the hyperthyroid non-azotaemic group (119 $\mu\text{mol/l}$ [range 74–216] vs 97 $\mu\text{mol/l}$ [range 62–118]; $P = 0.058$) ([Table 1](#)). There was no significant difference in baseline serum SDMA concentrations between hyperthyroid-CKD and hyperthyroid non-azotaemic cats (12 $\mu\text{g/dl}$ [range 7–21] vs 13 $\mu\text{g/dl}$ [range 11–19], respectively; $P = 0.42$) ([Table 1](#)). Other clinicopathological comparisons between the hyperthyroid-CKD and hyperthyroid non-azotaemic groups are shown in [Table 1](#).

Table 1.

Selected baseline biochemical parameters in 19 hyperthyroid cats classified by renal function into non-azotaemic (n = 6), masked azotaemic (n = 9) and azotaemic (n = 4) groups

Variable	Hyperthyroid non-azotaemic	Hyperthyroid masked azotaemic	Hyperthyroid azotaemic	Sig [*]	Hyperthyroid-CKD	Sig [†]
Age (years)	15 (14–19)	12 (9–18)	14 (8–15)	0.30	12 (8–18)	0.12
TT4 (nmol/l)	82 (49–90)	80 (49–206) [‡]	53 (47–57) [‡]	0.045	73 (47–206)	0.83
SDMA (µg/dl)	13 (11–19)	10 (7–14) [§]	16 (11–21) [§]	0.049	12 (7–21)	0.42
Creatinine (µmol/l)	97 (62–118) [¶]	99 (74–140) [∞]	169 (157–216) ^{¶,∞}	0.007	119 (74–216)	0.058
Urea (mmol/l)	10.7 (6.5)	11.5 (7–15.4)	12.8 (10.3–18.3)	0.39	11.5 (7–18.3)	0.37

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Data are median (range). Hyperthyroid-chronic kidney disease (CKD) represents the combined masked azotaemic and azotaemic hyperthyroid population (n = 13)

*Comparison of hyperthyroid non-azotaemic, hyperthyroid masked azotaemic and hyperthyroid azotaemic groups using the Kruskal–Wallis test

†Comparison of hyperthyroid-CKD to hyperthyroid non-azotaemic group using the Mann–Whitney U-test

‡Significant difference in total thyroxine (TT4) between masked azotaemic and azotaemic hyperthyroid cats

§Significant difference in symmetric dimethylarginine (SDMA) between masked azotaemic and azotaemic hyperthyroid cats

¶Significant difference in serum creatinine between non-azotaemic and azotaemic hyperthyroid cats

[∞]Significant difference in serum creatinine between masked azotaemic and azotaemic hyperthyroid cats
Sig = significance

Significant differences in baseline serum concentrations of creatinine ($P = 0.007$), SDMA ($P = 0.049$) and TT4 ($P = 0.045$) were evident between the azotaemic, masked azotaemic and non-azotaemic groups, whereas there were no differences in age ($P = 0.30$) or serum urea concentrations ($P = 0.39$) between the groups ([Table 1](#)). Serum creatinine concentrations were significantly higher in the azotaemic group than the non-azotaemic ($P = 0.011$) and the masked azotaemic groups ($P = 0.005$); however, there was no difference in serum creatinine concentrations between the masked azotaemic and non-azotaemic groups ($P = 0.24$) ([Table 1](#)). Serum SDMA concentrations were significantly higher in the azotaemic group than the masked azotaemic group ($P = 0.03$) but were not different between the azotaemic and non-azotaemic groups ($P = 0.33$), nor between the non-azotaemic and masked azotaemic groups ($P = 0.094$) ([Table 1](#)). At baseline, 3/4 cats in the azotaemic group had serum SDMA concentrations above the upper limit of the laboratory RI ($14 \mu\text{g/dl}$), whereas no masked azotaemic cats had serum SDMA concentrations above the laboratory RI. One non-azotaemic cat had serum SDMA concentration above $14 \mu\text{g/dl}$ at baseline. Azotaemic cats had significantly lower serum TT4 concentrations than masked azotaemic cats ($P = 0.02$); however, there was no difference in serum TT4 concentrations between the azotaemic and non-azotaemic groups ($P = 0.056$), nor between the non-azotaemic and masked azotaemic groups ($P = 0.443$).

When all hyperthyroid cats were combined, there was a strong inverse correlation between serum TT4 and creatinine concentrations at baseline ($r = -0.73$; $P < 0.001$); however, there was no significant correlation between serum TT4 and SDMA concentrations ($r = -0.42$; $P = 0.074$). At baseline, there was a moderate positive correlation between serum SDMA and creatinine concentrations ($r = 0.52$; $P = 0.022$).

Post-treatment changes

Follow-up biochemical data were available at T1 for 16 cats and at T2 for 17 cats. T2 was 86 days (range 26–282) after T1 for cats where data from both time points were available. Only two cats had USG recorded at follow-up. Serum TT4 concentrations decreased between baseline and T1 ($n = 16$; $P < 0.001$) ([Table 2](#)) and between baseline and T2 ($n = 17$; $P < 0.001$) ([Table 2](#)), but not between T1 and T2 ($P = 0.753$). When data from all hyperthyroid cats were combined, serum creatinine concentrations increased significantly between baseline and T1 ($n = 15$; $P = 0.003$) ([Figure 1](#)), and

between baseline and T2 (n = 17; *P* <0.001) ([Table 2](#), [Figure 1](#)). Serum creatinine concentrations also increased between T1 and T2 (n = 13; *P* = 0.033).

Table 2.

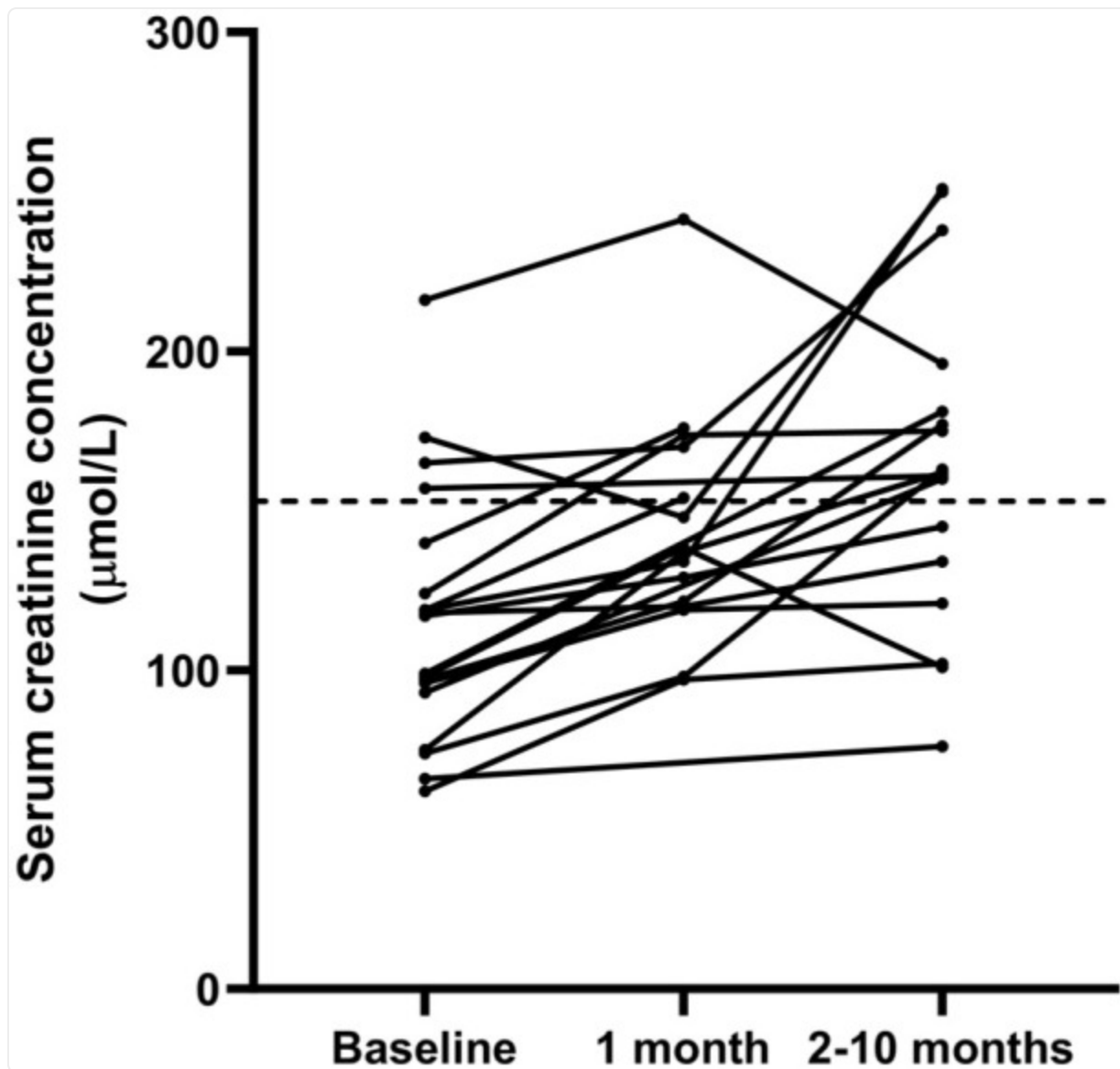
Selected biochemical parameters for all hyperthyroid cats at T0 (baseline, n = 19), T1 (>1 month after treatment, n = 16) and T2 (2–9 months after treatment, n = 17)

Variable	T0	T1	T2
Days after T0	–	32 (21–47)	110 (54–303)
TT4 (nmol/l)	76 (47–206)	20 (7–40)	20 (9–38)
SDMA (µg/dl)	12 (7–21)	11 (8–29)	13 (9–24)
Creatinine (µmol/l)	117 (62–216)	137 (97–241)	162 (76–251)
Urea (mmol/l)	11.5 (6.5–18.3)	12.2 (9.9–19.5)	13.7 (6.8–22.9)
SDMA storage time (years)	–	2.9 (0.6–9.5)	1.9 (0.1–9.2)

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Data are median (range)
SDMA = symmetric dimethylarginine; TT4 = total thyroxine

Figure 1.



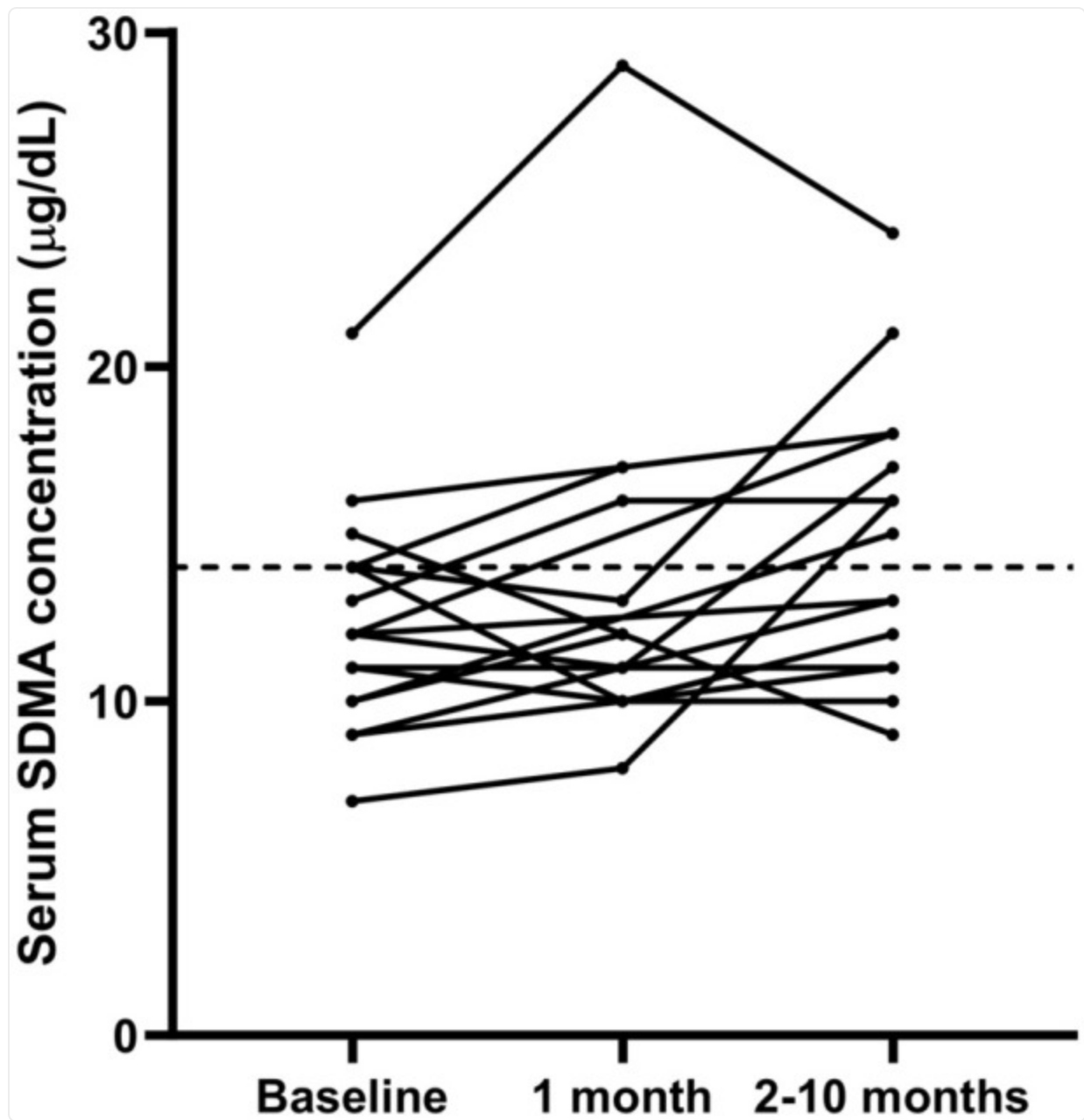
[Open in a new tab](#)

Line graph showing the change in serum creatinine concentrations measured at T0 (before treatment), T1 (~1 month after treatment) and T2 (2–9 months after treatment). Dashed black horizontal line represents the upper limit of the reference interval (153 $\mu\text{mol/l}$)

At T1, one non-azotaemic, one masked azotaemic and one azotaemic cat had serum SDMA concentrations above the laboratory RI. At T2, 2/6 non-azotaemic cats, 3/7 masked azotaemic cats (with available data) and 3/4 azotaemic cats had serum SDMA concentrations above the

laboratory RI. Serum SDMA concentrations did not change between baseline and T1 ($n = 14$; $P = 0.548$) ([Figure 2](#)) but increased significantly between baseline and T2 ($n = 17$; $P = 0.039$) ([Figure 2](#)). Serum SDMA concentrations did not change significantly between T1 and T2 ($n = 12$; $P = 0.084$).

Figure 2.



[Open in a new tab](#)

Line graph showing the change in serum symmetric dimethylarginine (SDMA) concentrations (μg/dl) measured at T0 (before treatment, n = 19), T1 (~1 month after treatment) and T2 (2–9 months after treatment). Dashed black horizontal line represents the upper limit of the reference interval (14 μg/dl)

Discussion

The results of this small retrospective study indicate that both serum creatinine and SDMA concentrations increased after 2–9 months of treatment of hyperthyroidism using antithyroid medications. There was no correlation between pretreatment serum SDMA and TT4 concentrations, whereas serum creatinine concentrations showed a strong negative correlation with serum TT4 concentrations before treatment. Furthermore, neither serum creatinine nor SDMA concentrations were significantly higher in hyperthyroid cats that had concurrent CKD (overt and/or masked) compared with hyperthyroid cats that remained non-azotaemic after treatment.

It was hypothesised that both serum creatinine and SDMA concentrations would increase after restoration of euthyroidism since both correlate with GFR in healthy cats,^{15,16} and GFR would decrease after hyperthyroidism treatment.^{8,9,11} Unexpectedly, serum SDMA concentrations did not increase after approximately 1 month of treatment, whereas serum creatinine concentrations did increase over this time period. Our findings are supported by the results from some studies of radioiodine-treated hyperthyroid cats which reported that serum creatinine, but not SDMA concentrations, increased in the first month after radioiodine treatment despite a significant decrease in GFR,^{21,22} but conflict with the findings of other studies.²⁴ As SDMA concentrations did increase after 2–9 months of treatment in the present study, it could be proposed that SDMA may show a more delayed response to changes in GFR than serum creatinine. However, changes in GFR and SDMA have previously been documented to stabilise 1 month after radioiodine treatment, while changes in serum creatinine associated with increased muscle mass continue for several months.^{9,12,24}

Serum TT4 concentrations were inversely correlated with serum creatinine concentrations but not serum SDMA concentrations in our population. Serum TT4 concentrations are inversely correlated with GFR and serum creatinine concentrations in humans,²⁷ and GFR is also documented to decrease after treatment of hyperthyroidism in cats.^{9,11,28} Therefore, we would expect an inverse correlation between serum TT4 and SDMA concentrations in hyperthyroid cats. Our findings, in conjunction with those of others, support the premise that serum SDMA concentrations are influenced by factors other than GFR in hyperthyroidism.^{20–22} In humans, circulating SDMA concentrations are increased in hyperthyroid patients, and it has been proposed that thyrotoxicosis may upregulate SDMA production.^{29,30} Decreases in serum SDMA have been reported after radioiodine treatment,²¹ and cats that have remained hyperthyroid after bilateral thyroidectomy have a steeper relationship between serum SDMA and serum creatinine.²³

Therefore, it could be proposed that increased SDMA production in hyperthyroidism may reduce the correlation between serum SDMA concentrations and GFR in hyperthyroid cats.

In the present study, serum SDMA concentrations in initially non-azotaemic hyperthyroid cats that developed azotaemia after treatment (masked azotaemic) or the combined group of untreated hyperthyroid cats that were azotaemic at baseline or masked azotaemic (hyperthyroid-CKD group) were not significantly greater than those in untreated hyperthyroid cats that remained non-azotaemic after treatment. These data are in contrast to those from previous studies in which serum SDMA concentrations were higher in masked azotaemic cats compared with non-azotaemic cats,^{[18,19](#)} although serum SDMA concentrations were poorly sensitive in predicting post-treatment azotaemia in these radioiodine-treated cats. Our results may differ because of the smaller sample size of the present study or could reflect the presence of factors other than GFR that affect serum SDMA concentrations in hyperthyroidism. In addition, some cats that were classified as non-azotaemic had elevated serum SDMA concentrations both before and after treatment, which could reflect the presence of concurrent CKD in some cats with low muscle mass (and hence lower serum creatinine concentrations). Furthermore, some cats in the masked azotaemic and azotaemic groups had serum SDMA concentrations within the laboratory RI despite the presence of elevated serum creatinine concentrations. These findings mirror those reported in cats with azotaemic acute kidney injury,^{[31](#)} and might reflect high dispersion (the range of possible results that are feasible for a given sample based on the combination of biological and analytical variation) of serum SDMA concentrations in cats.^{[32](#)} However, correlation between serum SDMA concentrations and direct measurements of GFR would be required to confirm this hypothesis.

In the present study, a relatively high proportion (60%) of initially non-azotaemic hyperthyroid cats became azotaemic after treatment. Approximately 15–51% of initially non-azotaemic cats will become azotaemic after radioiodine, surgical or medical treatment;^{[6,9,19,33,34](#)} however, the high proportion of cats becoming azotaemic after treatment in the present study may reflect bias in the population of cats for which a full biochemistry profile was requested, rather than TT4 alone, after treatment, or could reflect the fact that cats were classified as azotaemic based on documentation of elevated serum creatinine concentrations and without documentation of USG to exclude prerenal causes of azotaemia, such as dehydration. Only 26% of cats in this study had USG measured before treatment and very few at follow-up. Hence, it is possible that some cats classified as masked azotaemic may have had prerenal azotaemia at follow-up – for example, secondary to dehydration or other reversible factors rather than a true renal azotaemia. In addition, the classification of cats according to renal status was based on serum creatinine concentrations; therefore, it is possible that some non-azotaemic cats might have had concurrent

CKD. Direct measurement of GFR would be required to fully classify cats based on renal status; however, this was not possible because of the retrospective nature of the study.

This study had a number of limitations in addition to those already discussed. The three-way subdivision of the population resulted in small group sizes and, as a result, some near-significant results (such as TT4 and SDMA at baseline) may represent real biological differences that could not be demonstrated owing to the study being statistically underpowered. For example, based on the preliminary data in this study, five cats would be needed in each group for the study to be 80% powered to detect an effect size of 6 µg/dl at the 5% significance level, and 21 cats would be needed per group to detect an effect size of 3 µg/dl using the same criteria. SDMA was measured on serum samples that were frozen at -80°C for up to 10 years, and although the stability of SDMA has been demonstrated for up to 24 months,³⁵ the effect of longer term storage on serum SDMA concentrations is unknown; therefore, an impact of prolonged sample storage on the measured SDMA concentrations cannot be excluded. Further studies to establish the long-term stability of serum SDMA concentrations when stored frozen would be useful to help clarify the effect of prolonged sample storage on serum SDMA concentrations. In addition, serum TT4 concentrations alone were used to define thyroid status, and although all cats had serum TT4 concentrations within the RI at all time points, given the lack of concurrent serum thyroid-stimulating hormone measurements and fact that some cats with iatrogenic hypothyroidism can have low-normal serum TT4 concentrations,³⁶ the presence of iatrogenic subclinical hypothyroidism, which might have reduced GFR, could not be fully excluded. Conversely, in the absence of supportive clinical parameters at follow-up, the presence of poorly controlled hyperthyroidism in cats with a TT4 towards the upper end of the RI cannot be excluded, although the RI utilised in this study was derived using a population of non-hyperthyroid and otherwise healthy senior cats.²⁵ As a retrospective study, the decisions on medication, starting doses, dosing intervals and any dose adjustments were made at the discretion of the attending veterinarian and were not standardised. Additional comparisons between once- and twice-daily administration of medication would have produced small subgroups with inadequate statistical power. Finally, in the absence of additional investigations, including abdominal ultrasonography, the presence of other diseases that could have increased serum SDMA concentrations, such as lymphoma, could not be excluded.³⁷

Conclusions

Serum SDMA concentrations were not higher in hyperthyroid cats with concurrent CKD than hyperthyroid cats that remained non-azotaemic after restoration of euthyroidism, and were not useful predictors of post-treatment azotaemia in initially non-azotaemic hyperthyroid cats. Furthermore, serum SDMA concentrations did not increase in parallel with serum creatinine

concentrations after 1 month of treatment with antithyroid medications and were not inversely correlated with TT4 concentrations, which could suggest that factors other than GFR influence serum SDMA concentrations in hyperthyroidism. Larger scale studies measuring serum concentrations of creatinine and SDMA in conjunction with direct measurements of GFR in methimazole/carbimazole-treated hyperthyroid cats are required to further characterise the relationship between GFR and serum SDMA concentrations in this population of hyperthyroid cats.

Supplemental Material

S1

[sj-xlsx-1-jfm-10.1177_1098612X261418859.xlsx](#) (17.2KB, xlsx)

Raw data.

Footnotes

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Supplementary material: The following file is available as supplementary material:

S1: Raw data.

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the use of cadavers. Ethical approval from a committee was therefore not specifically required for publication in *JFMS*. Although not required, where ethical approval was still obtained, it is stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective studies). No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

S1

[sj-xlsx-1-jfm-10.1177_1098612X261418859.xlsx](#) (17.2KB, xlsx)

Raw data.

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Beyond traditional biomarkers: do serum cystatin C and urinary cystatin B improve diagnostic precision in feline and canine renal disease? A systematic review and meta-analysis

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Chronic kidney disease (CKD) is the most common nephropathy in cats and dogs, particularly in elderly animals, whereas acute kidney injury (AKI) represents an abrupt-onset condition associated with high morbidity and mortality. Serum creatinine and urea are widely used for renal function assessment; however, they exhibit low sensitivity for the early detection of renal dysfunction. In this context, alternative biomarkers, such as serum cystatin C (sCysC) and urinary cystatin B (uCysB), have been investigated as potential diagnostic tools. The present study conducted a meta-analysis, following PRISMA guidelines, to evaluate the performance of these biomarkers in cats with CKD and dogs with AKI. Three studies on sCysC and two on uCysB were included, comparing diseased cats and dogs with healthy controls, only one article on uCysB in cats met the inclusion and exclusion criteria and was analyzed qualitatively. The meta-analysis demonstrated a consistent increase in sCysC in cats with CKD, with a large effect size, indicating moderate and stable discriminatory ability. Conversely, uCysB showed a marked elevation in dogs with AKI, with effect sizes exceeding thresholds for strong clinical relevance, suggesting a strong association with tubular injury. However, high heterogeneity was observed among uCysB studies, along with relevant methodological limitations, including predominant comparisons with healthy animals. It is concluded that sCysC has limited utility in current clinical practice, with no proven superiority over creatinine or symmetric dimethylarginine (SDMA), whereas uCysB emerges as a promising biomarker for AKI, although its clinical application still depends on more robust prospective studies.

KEYWORDS

glomerular filtration rate, meta-analysis, nephrology, tubular injury, veterinary medicine

1 Introduction

CKD is the most prevalent nephropathy in dogs and cats, with an estimated prevalence ranging from 1.6 to 20% in the general population and affecting up to 35–50% of cats and dogs over 15 years of age (1, 2). It is a progressive and irreversible condition characterized by the gradual loss of renal function, with a significant negative impact on the quality of life and longevity of affected animals. In this context, the early detection of renal dysfunction is crucial to slowing disease progression, instituting appropriate therapeutic interventions, and improving clinical prognosis (3, 4).

In contrast to the insidious course of CKD, AKI is defined as an abrupt reduction in renal function, which can range from minor increases in serum creatinine to complete loss of renal function, and is associated with a significant prognostic impact in hospitalized and critically ill patients (5). The incidence of AKI has increased alongside the complexity of hospital care, being associated with a higher risk of progression to CKD and increased mortality (5, 6).

Glomerular filtration rate (GFR) is considered the gold standard for assessing renal function, as it directly reflects the kidneys' filtration capacity. However, direct measurement of GFR in animals is laborious, time-consuming, and not readily applicable in clinical practice, making the use of indirect markers indispensable (7). Currently, serum creatinine and urea remain the most widely used biomarkers, but they have significant limitations: both increase only after a loss of more than 75% of functional renal mass and are influenced by extra-renal factors such as dietary intake, drug use, and variations in muscle mass, which are particularly relevant (4). These limitations justify the search for earlier and more reliable biomarkers of renal dysfunction in cats (60).

In this context, CysC emerged as a potentially promising biomarker of renal function. It is a low-molecular-weight protein (approximately 13 kDa), produced at a constant rate by all nucleated cells, freely filtered by the glomeruli, and almost completely reabsorbed and catabolized in the proximal tubules, with no significant tubular secretion or extra-renal elimination (8, 9). These properties theoretically make CysC an ideal endogenous biomarker of GFR (10–12). In humans and dogs, sCysC has demonstrated potentially greater sensitivity than creatinine for the early detection of GFR reduction and for identifying AKI before serum creatinine elevation. Furthermore, detectable urinary cystatin C has been associated with tubulointerstitial injury, reflecting tubular damage (13, 41).

Despite this biotechnological potential, the clinical utility of CysC in cats remains controversial. Previous studies show inconsistent results, with variable correlations between this marker, GFR, and serum creatinine (14, 63). This limitation is compounded by the absence of widely validated commercial assays for the species, restricting its application in routine diagnostics (15).

On the other hand, SDMA, a byproduct of protein methylation eliminated predominantly via renal route, has become established as a sensitive marker of GFR reduction (16). SDMA concentrations are elevated in both AKI and CKD, correlating strongly with serum creatinine in acute conditions (17). However, SDMA lacks specificity regarding the nature of the injury, being unable to distinguish tubular damage from isolated functional loss (16, 17).

More recently, uCysB has emerged as a promising biomarker of AKI in dogs and cats, with higher values being related to worse clinical outcomes (18–20). These findings contrast with the still limited

evidence for CysC and suggest that uCysB may represent a more robust marker of acute tubular injury in this species.

Given this context, the need to differentiate biomarkers aimed at assessing glomerular function from those directed at detecting structural renal damage becomes evident. Thus, the present study aims to critically address the role of CysC in feline CKD and uCysB in AKI in dogs, discussing their clinical applicability, limitations, and potential contribution to the early diagnosis, staging, and prognosis of kidney diseases in this species. uCysB meta-analysis was performed in dogs, while feline evidence was included only qualitatively due to the identification of a single eligible feline study.

2 Materials and methods

2.1 Study design and reporting guidelines

This study was conducted as a systematic review with meta-analysis, in accordance with the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA, 2020). The research questions and eligibility criteria were structured using the PECO strategy (Population, Exposure, Comparison, Outcome).

2.2 Research questions and PECO strategy

For CysC, the PECO strategy was defined as follows: the population (P) consisted of domestic cats; the exposure (E) corresponded to the presence of CKD; the comparison group (C) included clinically healthy cats; and the outcome (O) was the serum concentration of CysC. Based on this design, the following research question was formulated: “Do cats with chronic kidney disease have higher serum CysC concentrations compared to healthy cats?”

For CysB, the PECO strategy was initially designed in a similar manner, however considering dogs the population (P), AKI as the exposure (E), clinically healthy dogs as the comparison (C), uCysB concentration as the outcome (O). The corresponding research question was: “Do dogs with acute kidney injury have higher uCysB concentrations compared to healthy dogs?”

2.3 Information sources and search strategy

A systematic search was performed in the PubMed database, selected for its comprehensiveness and relevance in the biomedical field. For sCysC, the search strategy used the following combination of descriptors and Boolean operators:

(feline OR cats) AND (cystatin C) AND (chronic kidney disease OR CKD).

For uCysB, the following search strategy was employed:

(cystatin B OR uCysB) AND (acute kidney injury) AND (dogs OR canine).

(cystatin B OR uCysB) AND (acute kidney injury) AND (cats OR feline).

No restrictions regarding the year of publication were applied. Only studies published in the English language were considered. The search yielded 20 records for sCysC and 7 records for uCysB. The complete study selection process is presented in the flowchart (Figure 1). Only one study evaluating uCysB in cats met the

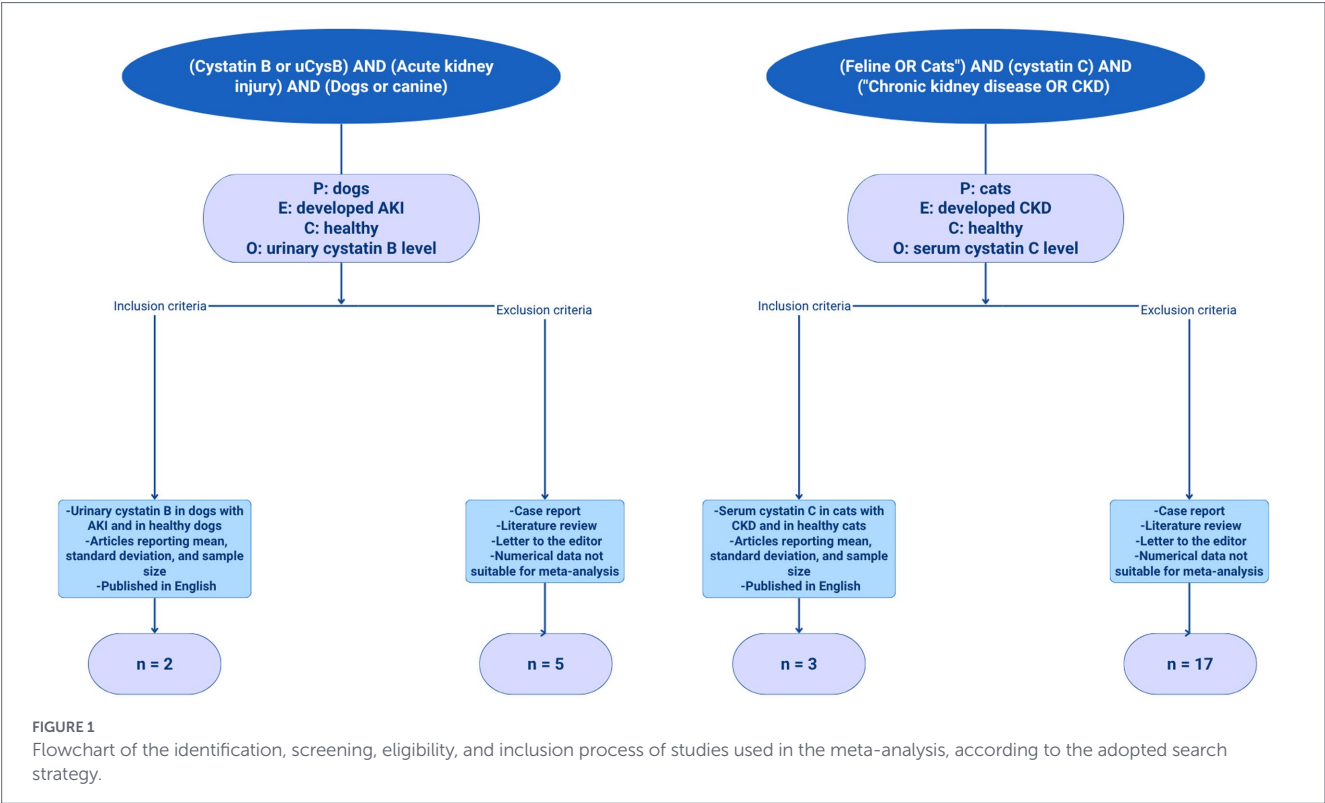


TABLE 1 Articles included in the review, according to the eligibility criteria, presenting the country of origin and the methodology employed for serum cystatin C measurement in cats with chronic kidney disease and healthy controls, as well as the main results obtained.

Study	Country	Methodology	Results
Ghys et al. (14)	Belgium	Nephelometric assay	Cats with CKD had significantly higher serum sCysC concentrations compared to healthy controls.
Ghys et al. (15)	Belgium	Nephelometric assay	sCysC concentrations were, on average, significantly higher in cats with CKD than in healthy controls.
Williams et al. (63)	United Kingdom	Particle-enhanced turbidimetric assay	No significant difference in sCysC concentrations was observed when comparing cats with CKD to healthy control cats.

predefined inclusion and exclusion criteria of this systematic review. Although this study fulfilled the qualitative eligibility requirements.

studies that did not present numerical data compatible with statistical analysis were excluded. Based on these criteria, three studies were included for sCysC and two studies for uCysB, with 17 and 5 studies being excluded.

2.4 Eligibility criteria

Studies were included if they simultaneously met the following criteria:

- (1) Original observational studies;
- (2) Evaluation of sCysC in cats with CKD and AKI or uCysB in dogs with AKI;
- (3) Presence of a control group composed of healthy cats;
- (4) Provision of adequate quantitative data for meta-analysis (mean, standard deviation, and sample size);
- (5) Publication in the English language.

Case reports, narrative or systematic reviews, editorials, letters to the editor, experimental studies not involving CKD or AKI, and

2.5 Study selection process

Study screening was performed independently by two reviewers, initially through the reading of titles and abstracts. Potentially eligible articles were evaluated in full text to confirm inclusion criteria. Disagreements between reviewers were resolved by consensus.

2.6 Data extraction

Data extraction was conducted independently by two reviewers using a standardized electronic spreadsheet. Data regarding study characteristics, sample size, means, standard deviations, and confidence intervals of sCysC and uCysB concentrations in the diseased and control groups were recorded. The extracted data are presented in Tables 1, 2.

TABLE 2 Articles included in the review, according to the eligibility criteria, presenting the country of origin and the methodology employed for uCysB measurement in dogs with AKI and healthy controls, as well as the main results obtained.

Study	Country	Methodology	Results
Gordin et al. (19)	United States	sandwich ELISA	Healthy dogs showed undetectable or low uCysB levels, while cats with acute kidney injury exhibited markedly elevated urinary concentrations.
Gordin et al. (26)	United States	sandwich ELISA	uCysB concentration differed significantly between groups, being higher in the AKI group compared to controls.
Chen et al. (18)	Israel	sandwich ELISA	uCysB progressively increased in cats with AKI.

Only one article on uCysB in cats met the inclusion and exclusion criteria and was therefore analyzed qualitatively.

2.7 Risk of bias assessment

The risk of bias of the included studies was assessed independently by two reviewers using an appropriate tool for observational studies, such as the Newcastle–Ottawa Scale. Any disagreements were resolved by consensus.

2.8 Statistical analysis

Statistical analyses were performed using RStudio software. Differences in biomarker concentrations between the CKD vs. control groups for sCysC, and AKI vs. control groups for uCysB were evaluated using meta-analysis under a random-effects model, considering the clinical and methodological heterogeneity among the studies.

Effect sizes were estimated using Cohen’s d and Glass’s delta, with respective 95% confidence intervals. The pooled standard deviation and statistical power based on Glass’s delta were also calculated. Heterogeneity among studies was assessed using Cochran’s Q test and quantified by the I² index, which expresses the proportion of total variability attributed to heterogeneity among studies rather than chance. As this study involved a secondary synthesis of previously published data, it did not involve direct manipulation of animals and, therefore, did not require submission to an animal research ethics committee.

3 Results

3.1 Difference in means between groups

The comparative meta-analysis included two studies that evaluated the urinary concentration of uCysB in dogs with AKI and three studies that investigated the serum concentration of sCysC CKD in cats, all reporting means, standard deviations, and sample sizes adequate for quantitative synthesis. In both analyses, the initial estimation was conducted using the weighted mean difference, employing a fixed-effect model. The statistical weights assigned to the studies were calculated based on the inverse of the variance, incorporating the standard deviations and sample sizes of the diseased and control groups.

In the analysis of uCysB, the pooled weighted mean difference was 317.65, indicating markedly higher urinary concentrations in dogs with AKI compared to healthy controls. However, the forest plot demonstrated considerable dispersion among

studies, with lack of vertical alignment suggesting low homogeneity. Notably, the study presenting the largest difference in means contributed the least weight due to its higher standard deviation and smaller sample size.

Similarly, the meta-analysis of sCysC demonstrated a consistent increase in serum levels in cats with CKD compared to controls. The overall weighted mean difference was 0.38, also represented by the central diamond in the forest plot. Notably, the study that presented the largest absolute difference in means contributed the least statistical weight (5.5%), due to its reduced sample size and high standard deviation, evidencing greater internal variability. As observed for uCysB, studies with greater variability presented wider confidence intervals, suggesting heterogeneity among the individual results (Figure 2).

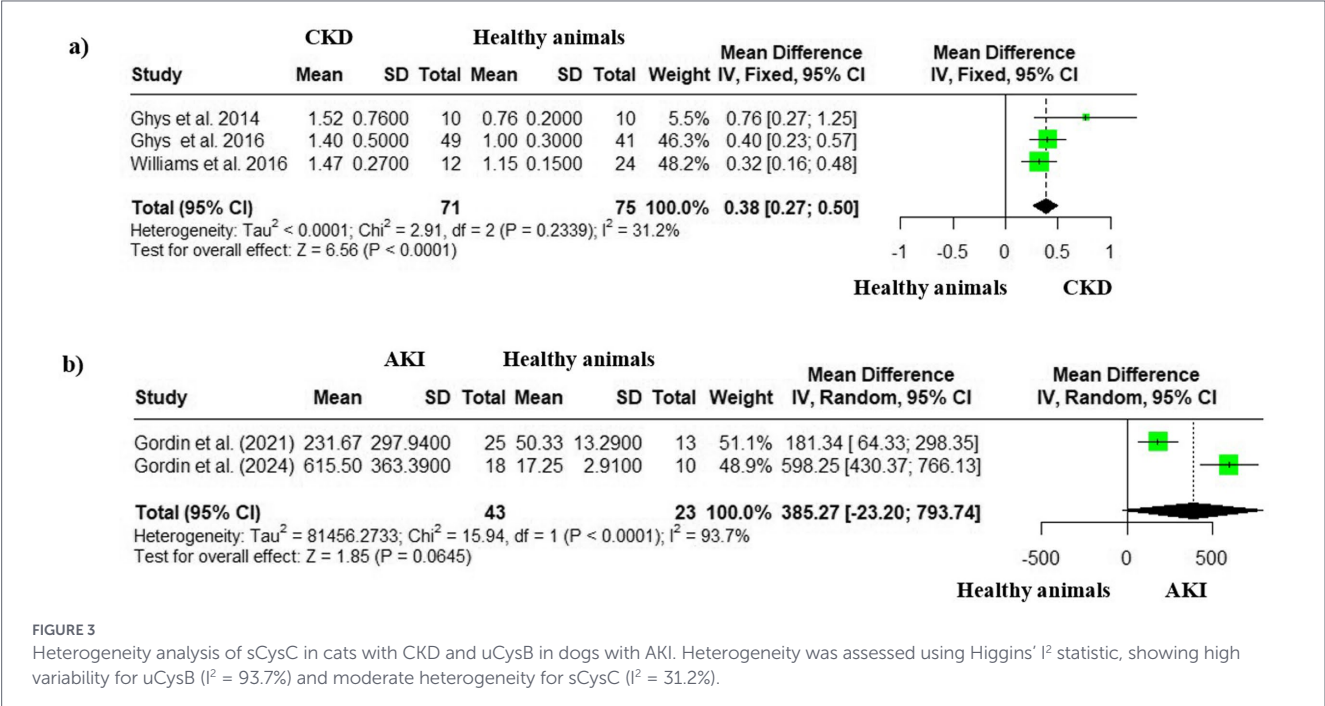
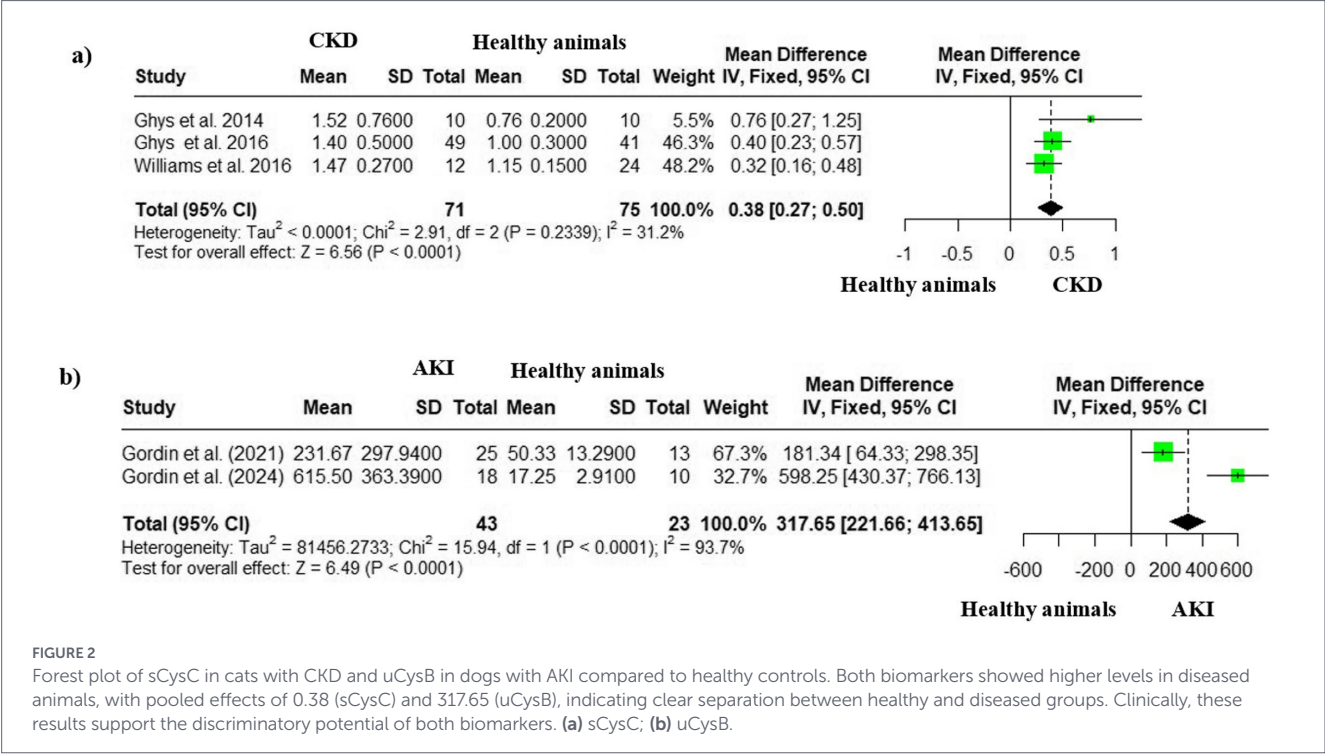
3.2 Heterogeneity and effect models

Heterogeneity among the studies was quantified using Higgins’ I² statistic for both biomarkers, revealing distinct patterns of variability among the included works. In the meta-analysis of uCysB, high heterogeneity was observed (I² = 93.7%), indicating that most of the variability among the results cannot be attributed to sampling error, but likely to uncontrolled factors, such as differences in age, breed, sex, laboratory methods employed, and clinical stages of AKI.

Given this scenario of high heterogeneity, the application of a random-effects model was necessary, aiming to incorporate the variability between studies and provide a more conservative and generalizable estimate. The adoption of this model resulted in a more balanced redistribution of statistical weights, accompanied by an increase in the width of the confidence intervals of the overall estimate, graphically represented by a wider diamond in the forest plot. In this context, the pooled estimate approached the null line (zero), suggesting that, under conditions of high heterogeneity, the behavior of uCysB may vary substantially across populations, and may approximate between control and AKI groups depending on the clinical and demographic characteristics evaluated.

In contrast, the meta-analysis of sCysC presented a distinct pattern. Under the fixed-effect model, partial homogeneity was observed among the studies, with moderate overall heterogeneity (I² = 31.2%). Nevertheless, the complementary application of a random-effects model was chosen, aiming to incorporate residual variability among the studies and produce a more conservative and biologically plausible estimate.

The application of the random-effects model for sCysC resulted in a reduction in the discrepancy between the weights assigned to the studies and a widening of the confidence intervals of the pooled estimate (Figure 3).



3.3 Standardized effect sizes and clinical relevance

To enable comparisons independent of the measurement scale and to assess the clinical relevance of the biomarkers, the meta-analysis results were converted into standardized effect sizes. The effect size expresses the magnitude of the difference between groups and is classified as small, moderate, or large, with at least a moderate to large effect size being expected for diagnostic biomarkers.

In the analysis of CysC as a biomarker of CKD, large and consistent magnitudes were observed across different effect size metrics. Cohen's d indicated a large effect size ($d = 1.14$), while Hedges' g , adjusted for small or unbalanced samples, presented a similar value ($g = 1.12$), confirming the robustness of the observed effect. The application of Glass's Delta (Δ), which exclusively uses the standard deviation of the control group as the denominator, resulted in the largest effect size ($\Delta = 1.63$), suggesting high diagnostic efficiency of sCysC (Figure 4). These findings remained consistent under both the fixed-effect and random-effects models, although, in the latter, a widening

of confidence intervals was observed, reflecting greater expected variability in heterogeneous populations. It is important to note that none of the estimates crossed the null line, reinforcing the consistency of sCysC as a biomarker, regardless of factors such as age, sex, or breed (Figures 4, 5).

For uCysB as a biomarker of AKI, conversion to standardized effect sizes revealed exceeding thresholds for strong clinical relevance. Cohen's *d* demonstrated a very large effect size ($d = 1.19$), consistent with high discriminatory capacity between cats with AKI and healthy dogs. Considering the small number of studies and possible sample imbalances, the effect was adjusted using Hedges' *g*, which maintained an effect size classified as large ($g = 1.16$), although with a lower magnitude than initially estimated. Subsequently, the calculation of Glass's Delta evidenced a substantial increase in effect size ($\Delta = 14.29$), indicating impressive performance of uCysB in distinguishing between groups, especially when evaluated in relation to the variability of the control group (Figure 6).

The introduction of the random-effects model led to additional adjustments in effect sizes for uCysB. Although the values of Cohen's *d*, Hedges' *g*, and Glass's Delta remained high, an increase in confidence intervals was observed, reflecting greater expected population variability (Figure 7).

Together, these results indicate that both biomarkers present effect sizes compatible with clinical relevance. However, while sCysC stands out

for its statistical stability and consistency across models, uCysB, despite exhibiting effects of large magnitude, proves to be more sensitive to population and methodological heterogeneity, an aspect that should be considered in the interpretation and clinical extrapolation of the findings.

3.4 Estimation of the ratio between groups and clinical interpretation

The estimation of population means, derived from weighted means, pooled standard deviations, and standardized effect sizes, allowed inference of the expected values of the biomarkers in the control and exposed groups. For uCysB, the weighted mean in control animals was estimated at 418.65 (95% CI: 42.63 to 794.68), indicating higher values compared to controls, but with substantial variability, as reflected by the wide confidence interval. This suggests that, although the biomarker is elevated in disease, its magnitude may vary significantly depending on clinical context (Figure 8).

Similarly, the analysis of sCysC, the weighted mean in control cats was 0.98 (95% CI: 0.75–1.20). Based on Glass's Delta, the estimated mean in cats with CKD was 1.43, corresponding to an approximate 1.47-fold increase compared to controls. This estimate remained entirely within the confidence interval (95% CI: 1.34–1.54), indicating consistent and predictable elevation in diseased animals (Figure 9).

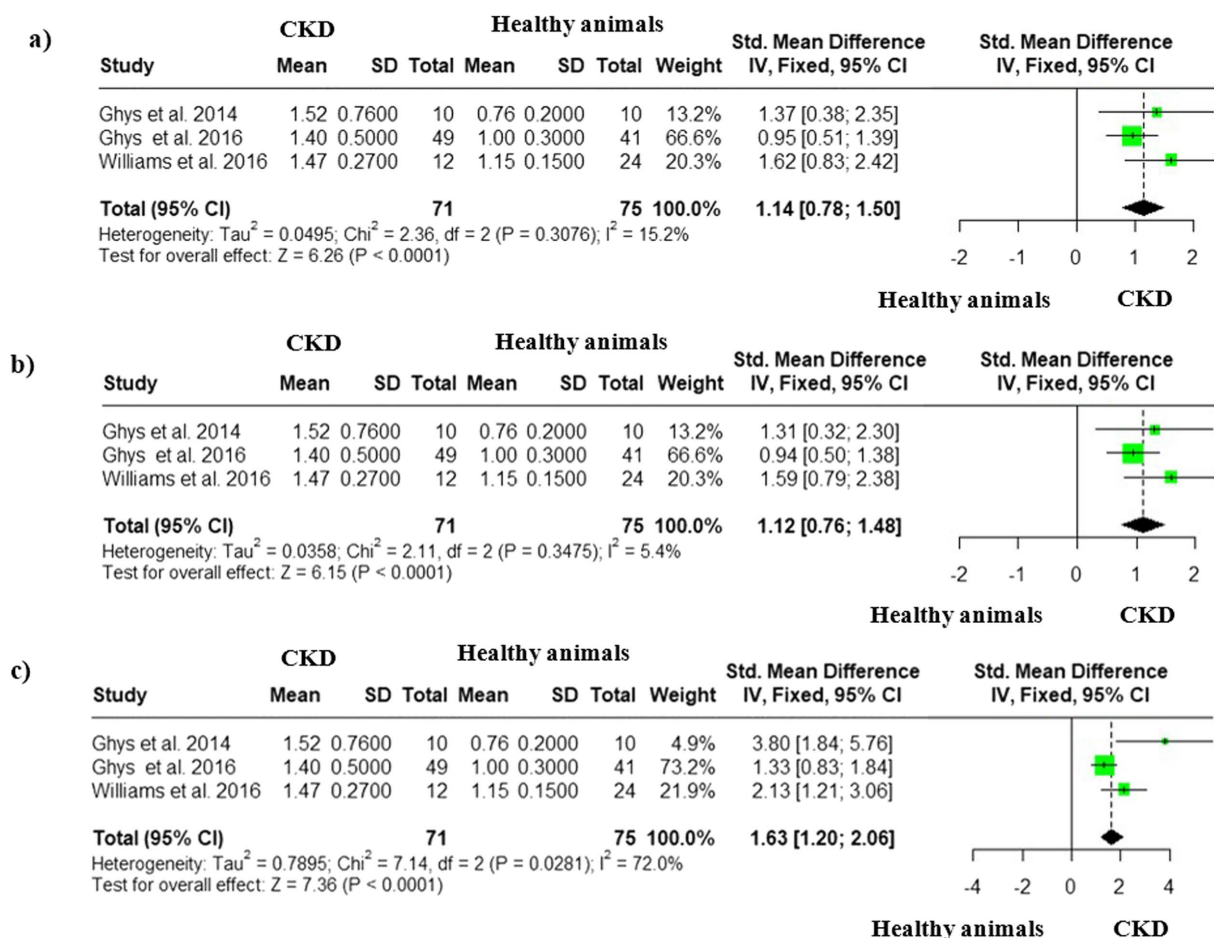
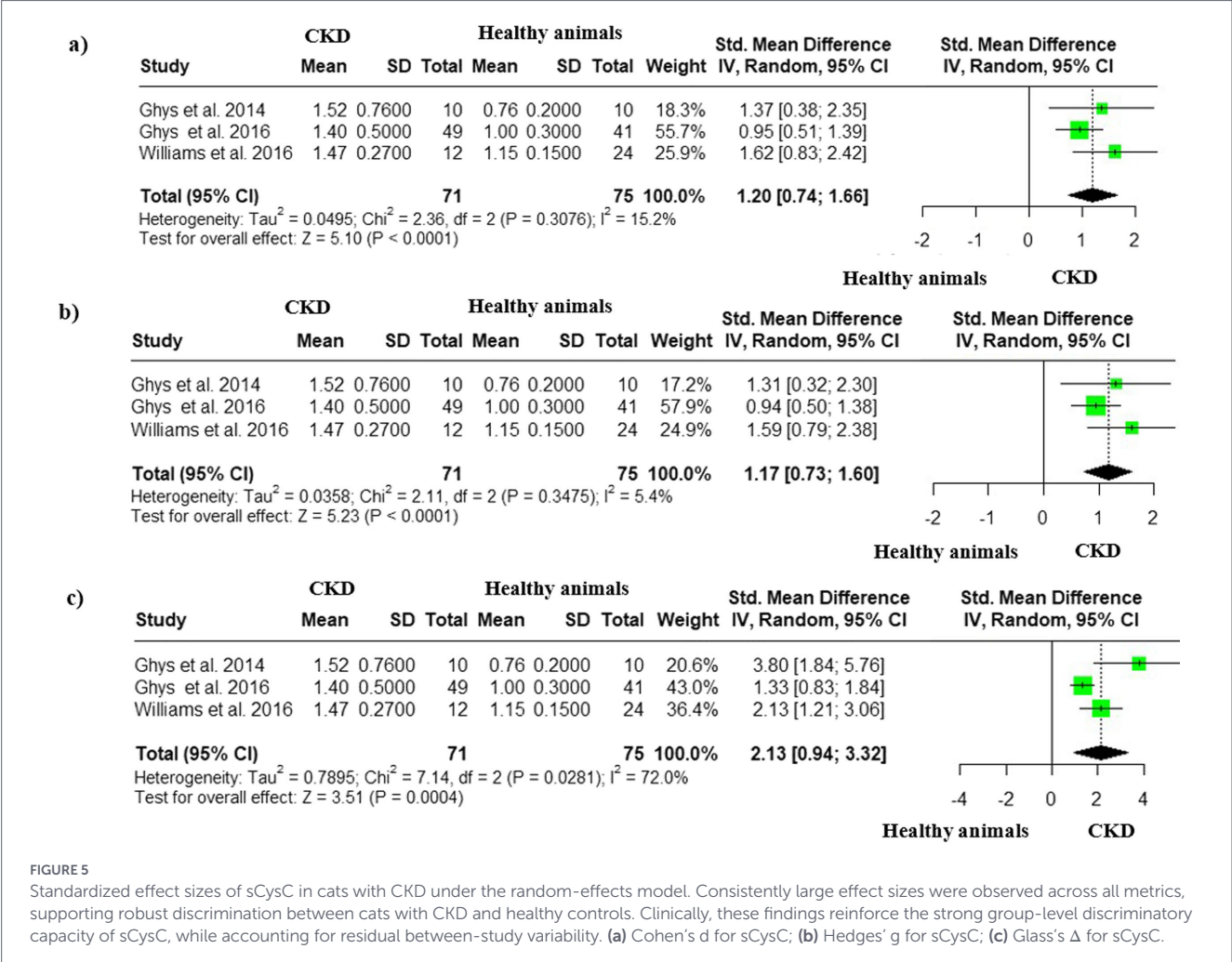


FIGURE 4

Standardized effect size estimates for sCysC in cats with CKD. sCysC showed consistently large effect sizes across all measures, indicating a strong separation between cats with CKD and healthy controls. Clinically, these findings support the robust discriminatory capacity of sCysC at the population level. (a) Cohen's *d* for sCysC; (b) Hedges' *g* for sCysC; (c) Glass's Δ for sCysC.



Together, these population estimates demonstrate that while uCysB presents extremely high absolute and relative increments in dogs with AKI, sCysC shows a more moderate and consistent increase in cats with CKD, reflecting important differences in the biological behavior and clinical potential of these biomarkers in distinct contexts of kidney injury. In practical terms, these magnitudes reflect a strong discriminatory capacity between healthy and diseased animals, suggesting that both sCysC and uCysB are capable of clearly separating populations with and without renal pathology at a group level (Table 3).

3.5 Statistical power

The assessment of statistical power, based primarily on Glass's Delta, demonstrated values close to 90%, even with the small number of included studies. This result indicates a high probability that the observed effects represent real phenomena in the population, reducing the chance of type II errors and conferring robustness to the obtained estimates. In the case of uCysB, such statistical performance was observed even with only two studies, suggesting that, although additional investigations are desirable to refine estimates and reduce uncertainties, new studies are not strictly necessary to confirm the association with dogs AKI (Figure 10).

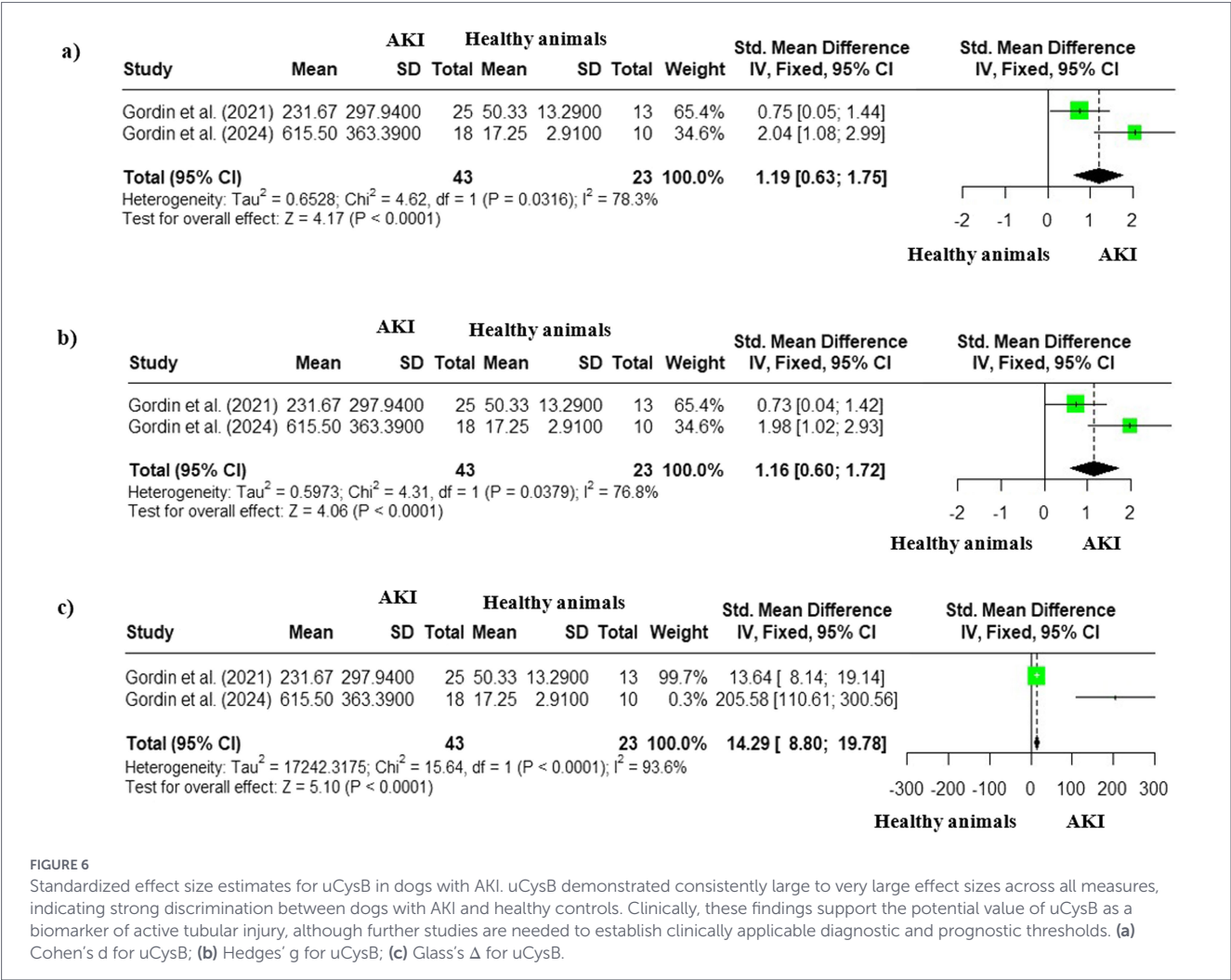
Consistent results were equally observed for sCysC, whose statistical power assessment, also derived from Glass's Delta, indicated

power close to 90%, reinforcing the statistical validity and clinical relevance of the biomarker in discriminating cats with chronic kidney disease. Complementarily, the agreement of the findings with those obtained from Cohen's d reinforces the robustness of the results, demonstrating that the discriminatory capacity of uCysB and sCysC remains consistent regardless of the effect size metric used (Figure 11).

3.6 Urinary cystatin B in feline medicine: qualitative synthesis

In the present systematic review, the available evidence on uCysB in feline medicine was limited to a single study that met the predefined inclusion criteria. Due to this, a quantitative synthesis across outcomes was not feasible, and the findings were therefore summarized qualitatively.

Overall, findings from Chen et al. (18) demonstrated a progressive increase in median uCysB concentrations according to the severity of renal dysfunction, with values of 22 ng/mL in healthy cats, 171 ng/mL in UO, 112 ng/mL in CKD, and 1,052 ng/mL in AKI, with statistically significant differences among groups ($p < 0.001$). These results support a consistent pattern of uCysB elevation associated with renal injury, particularly in acute conditions.



In the association analysis, uCysB showed a moderate correlation with serum creatinine ($r = 0.60$; $p < 0.001$) and SDMA ($r = 0.67$; $p < 0.001$) in the overall population, reinforcing its relationship with conventional renal function biomarkers. However, these correlations were not maintained within the AKI subgroup alone, suggesting clinical and pathophysiological heterogeneity among different presentations of acute kidney injury.

Regarding diagnostic performance, uCysB demonstrated high accuracy for distinguishing AKI from healthy controls, with an area under the ROC curve (AUC) of 0.92 (95% CI: 0.84–1.00), sensitivity of 90%, and specificity of 92% at a cutoff value of 84 ng/mL. When the entire cohort was analyzed, diagnostic accuracy remained moderate to high (AUC = 0.83), although with the expected reduction in performance in more heterogeneous clinical settings. Additionally, uCysB showed relevant prognostic value for 30-day mortality in cats with AKI, with an AUC of 0.77 and significantly higher levels observed in non-survivors.

4 Discussion

This systematic review and meta-analysis evaluated the performance of sCysC and B uCysB as biomarkers of kidney disease in cats

and dogs. The main findings were: (1) sCysC was elevated in cats with CKD, demonstrating large effect sizes and relatively low heterogeneity; (2) uCysB was markedly increased in dogs with AKI, exhibiting very large effect sizes despite substantial between-study heterogeneity; and (3) both biomarkers demonstrated high statistical power, supporting the biological relevance of the observed associations. Collectively, these findings reinforce the potential role of cystatins as complementary biomarkers in veterinary nephrology while highlighting important differences in the current level of evidence available for each marker.

uCysB demonstrated marked elevations in dogs with AKI compared with healthy controls, with effect sizes ranging from large to very large. This finding is consistent with the pathophysiology of AKI, in which structural tubular injury frequently precedes measurable declines in renal function (5, 6, 20). uCysB is an intracellular protein involved in the regulation of proteolytic and apoptotic pathways, and its urinary excretion is believed to reflect active cellular injury within the kidney (21). Consequently, uCysB presents an important conceptual advantage over traditional functional biomarkers, as it may identify ongoing tubular injury during a potentially reversible stage of disease (18, 20, 22, 23, 40).

The available evidence also suggests that uCysB may increase before traditional renal biomarkers such as creatinine and SDMA, supporting its potential utility for the early detection of tubular

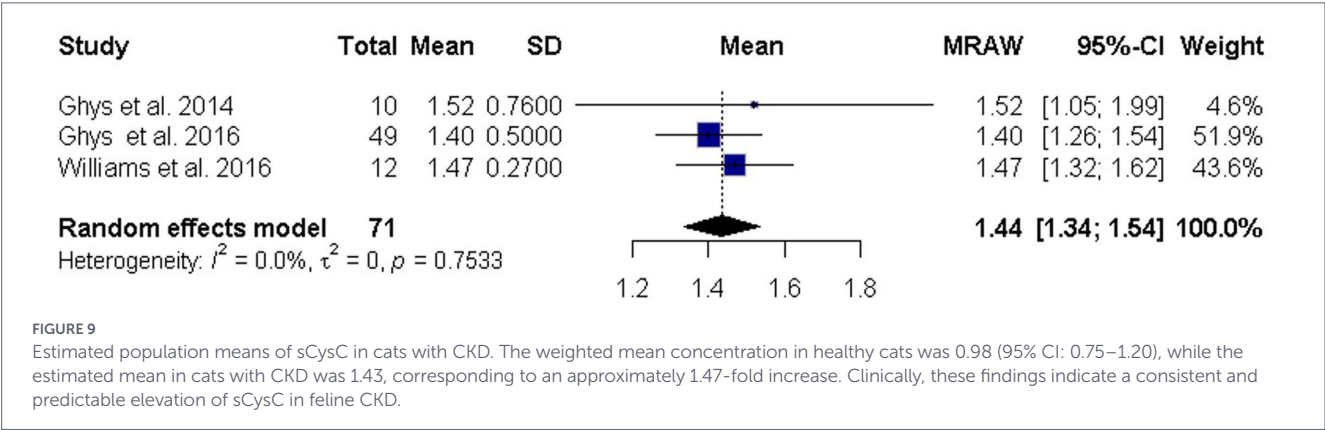
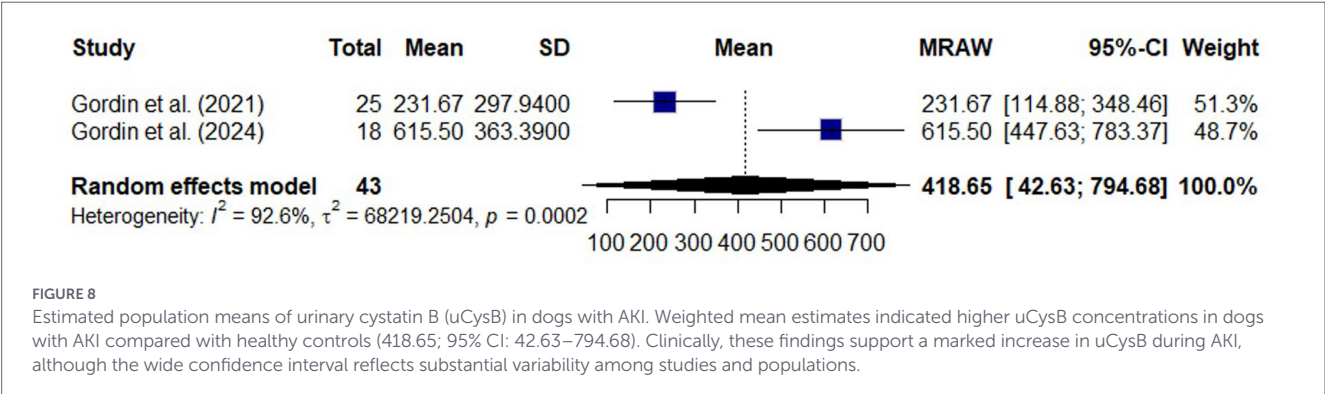
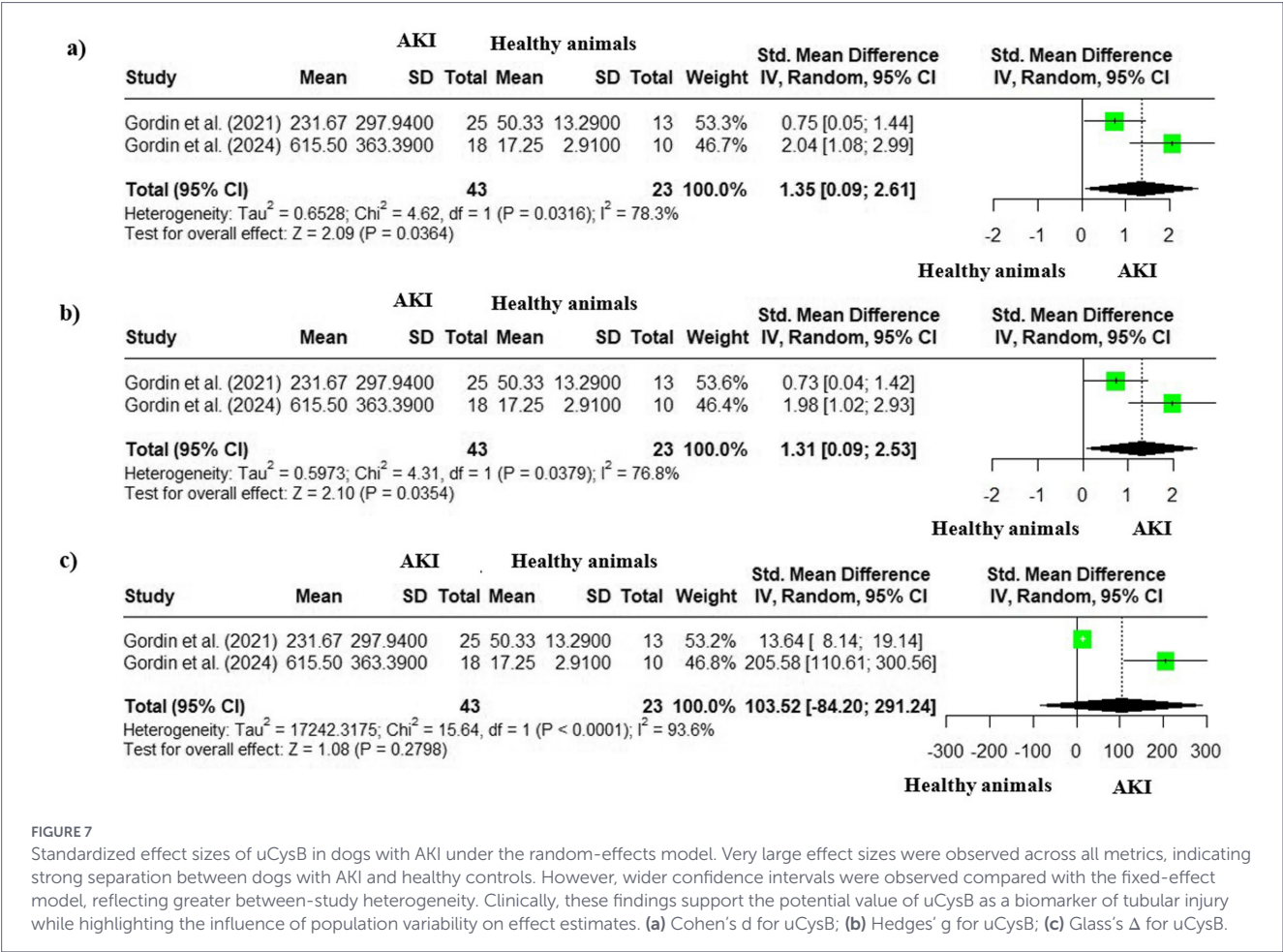
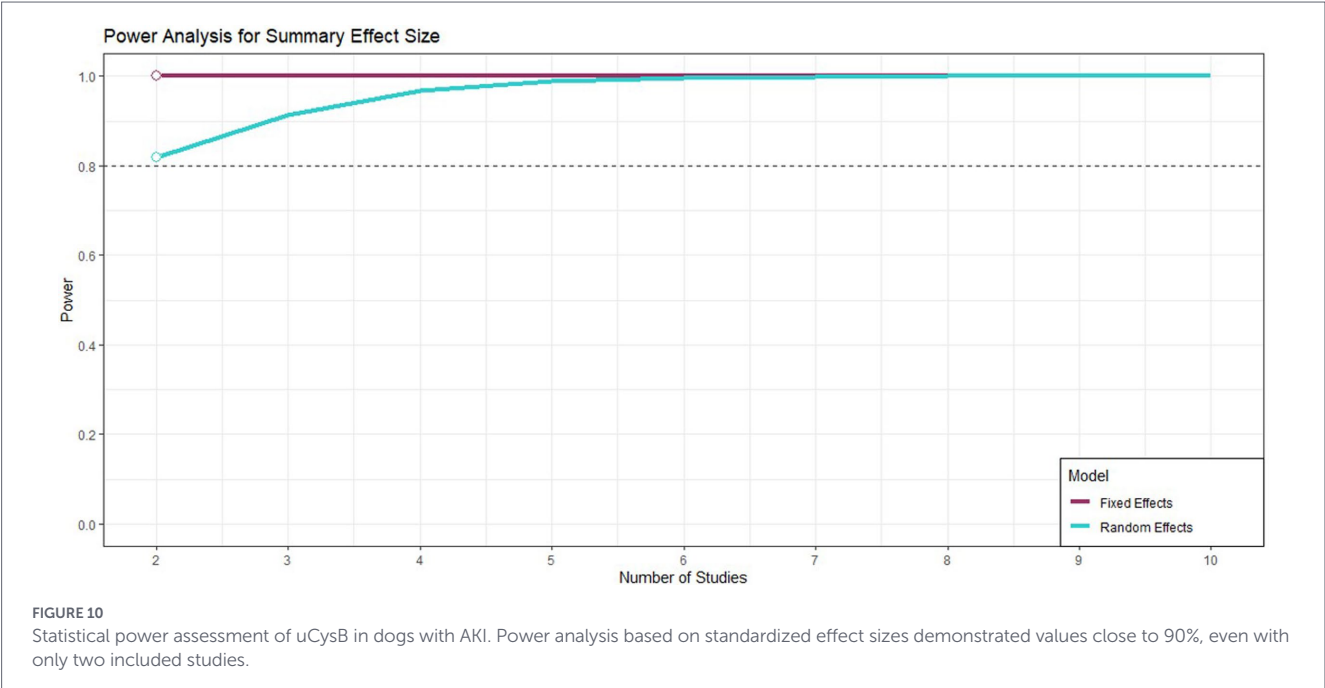


TABLE 3 Comparative overview of renal biomarkers used for the assessment of glomerular filtration and tubular injury.

Biomarkers	Clinical performance	Diagnostic value	Key limitations	Clinical availability
Creatinine	Late marker of GFR decline	Low sensitivity for early renal dysfunction; high specificity for established reduction in GFR	Strongly influenced by muscle mass, diet, age, sex, and hydration status. Increases only after substantial loss of renal function (~ > 60–75% GFR reduction).	Widely available, low cost, routinely automated in clinical laboratories
SDMA	Early marker of reduced GFR	High sensitivity for mild-to-moderate decreases in GFR; moderate specificity depending on population and cutoff used	Not clearly superior to creatinine, and may yield more false positives at certain cutoffs. Biological variability in inflammatory and catabolic states.	Commercially available in veterinary diagnostic platforms; moderate cost acceptable precision, and minimal interference, enabling high-throughput clinical use
sCysC	Endogenous estimator of GFR	Good correlation with measured GFR in human medicine; moderate-to-high diagnostic performance in veterinary studies with variability	Influenced by inflammation, thyroid dysfunction, and corticosteroid therapy. Lack of full standardization across veterinary species.	Limited availability; higher cost; specialized laboratories
uCysB	Experimental biomarker of tubular injury	Insufficient evidence for pooled estimates of sensitivity or specificity. Preliminary studies suggest potential association with early tubular damage	High inter-study heterogeneity and small sample sizes. Lack of established reference intervals and validation in veterinary medicine.	uCysB is currently under clinical evaluation in veterinary medicine, although it is already used in clinical routine in some centers in the USA.

The table summarizes clinical performance, diagnostic value based on current evidence, key limitations, and clinical availability of SDMA, sCysC, and uCysB. While creatinine and SDMA are established markers of GFR, sCysC shows variable performance depending on species and clinical context. uCysB remains an experimental biomarker with limited and heterogeneous evidence, precluding definitive estimates of diagnostic accuracy and routine clinical application.



injury and subclinical renal dysfunction (19, 24). Furthermore, dogs with progressive stage 1 CKD demonstrated significantly higher uCysB concentrations than both stable CKD and control dogs, whereas no significant differences were observed for SDMA, systolic blood pressure, or urine specific gravity (20). These findings suggest that uCysB may identify ongoing renal injury before conventional markers detect functional deterioration (59, 61).

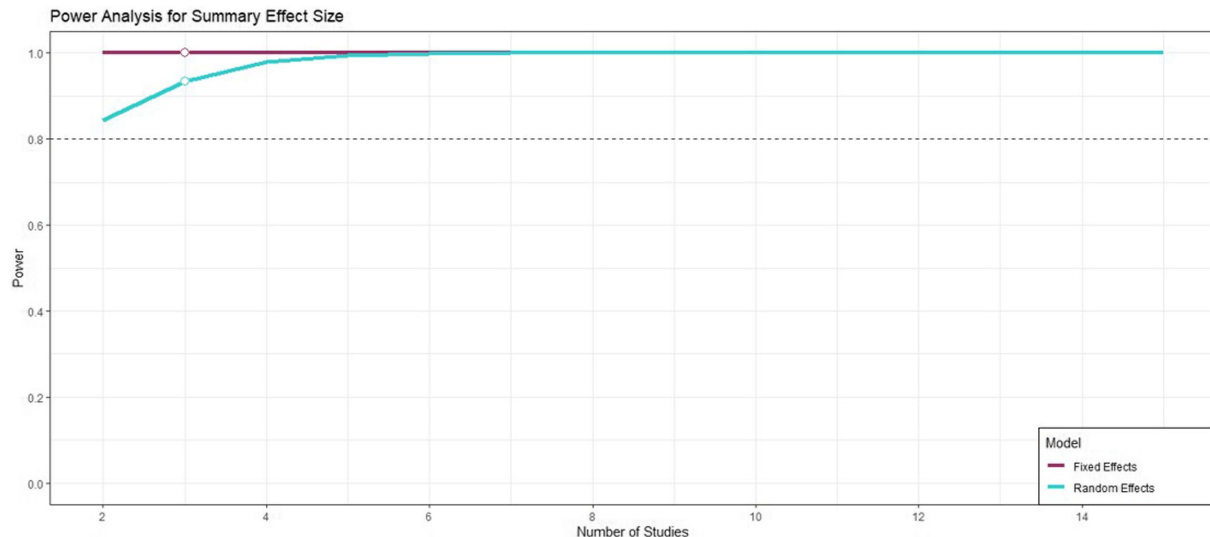


FIGURE 11

Statistical power assessment of sCysC in cats with CKD. Power analysis based on standardized effect sizes demonstrated values close to 90%, despite the limited number of included studies.

Although quantitative evidence for uCysB was predominantly derived from canine studies, the only eligible feline study also suggested potential diagnostic and prognostic relevance of this biomarker in cats with AKI. Higher uCysB concentrations were associated with poorer clinical outcomes; however, most evaluated cats already exhibited elevated creatinine concentrations at the time of diagnosis, indicating that the biomarker was assessed primarily during established stages of renal dysfunction. Therefore, although increased uCysB concentrations may reflect greater tubular injury and disease severity, the currently available feline data do not clearly demonstrate whether uCysB increases earlier than traditional functional biomarkers or provides superior diagnostic performance. Consequently, further prospective studies involving non-azotemic cats and serial monitoring throughout hospitalization are required to determine whether uCysB elevation truly precedes changes in creatinine and other conventional markers of renal function (18, 22, 23, 25).

Despite these promising findings, interpretation of uCysB requires caution. Most studies compared dogs with AKI exclusively to healthy controls, which may overestimate the discriminatory capacity of the biomarker and does not fully reflect real-world clinical decision-making. In clinical practice, differentiating AKI from other causes of azotemia is generally more relevant than distinguishing diseased animals from healthy individuals. This limitation may partially explain the substantial heterogeneity observed across studies ($I^2 = 93.7\%$). Variability in analytical platforms, assay calibration, AKI staging criteria, disease etiology, study populations, breed distribution, and duration of injury likely contributed to the observed dispersion of results (23, 26). Consequently, future studies should prioritize standardized assay methodologies, consensus-based AKI classification systems, and clinically relevant comparator groups to improve reproducibility and facilitate clinical translation.

Additional limitations remain regarding the biological specificity of uCysB. Correlations between urinary concentrations and renal histopathological findings are still scarce, as studies

incorporating renal biopsy remain limited (27). Moreover, increased uCysB concentrations have also been reported in humans with several neoplastic conditions, potentially reflecting release from cancer cells rather than exclusively tubular injury (28–30). Therefore, although the biomarker appears promising as an indicator of active renal injury, further investigation is required to define its specificity and diagnostic positioning within broader clinical contexts.

sCysC demonstrated a distinct but equally relevant pattern. Cats with CKD consistently exhibited higher sCysC concentrations than healthy controls, with population estimates indicating an average increase of approximately 1.4–1.5-fold. The relatively narrow confidence intervals and lower heterogeneity compared with uCysB suggest a more reproducible biomarker response across studies. This finding is biologically plausible, as progressive reductions in glomerular filtration rate are expected to result in gradual accumulation of circulating sCysC (31).

One of the main theoretical advantages of sCysC is its reduced dependence on muscle mass compared with creatinine (32, 33). This characteristic has generated considerable interest in both human and veterinary nephrology, particularly for patients affected by sarcopenia or altered body composition (53). However, much of the evidence supporting this advantage originates from human medicine, where combinations of creatinine and sCysC have demonstrated utility in identifying sarcopenic individuals (34, 56, 57). In veterinary medicine, direct comparisons between sarcopenic and non-sarcopenic populations remain unavailable, and robust validation against reference methods for glomerular filtration rate, such as inulin or iothexol clearance, is still limited (54, 58).

In human medicine, sCysC has shown improved sensitivity for detecting moderate reductions in glomerular filtration rate and stronger correlations with reference GFR measurements than creatinine, especially at values close to 60 mL/min/1.73 m², in addition to showing better correlation with reference methods for measuring glomerular filtration, such as radioisotopic clearance (35–38). However, comparable veterinary evidence remains

scarce. Consequently, although sCysC appears to be a promising complementary marker of renal function, current data do not demonstrate clear superiority over established biomarkers such as creatinine or SDMA (48–52). At present, SDMA remains the best-validated alternative when creatinine interpretation is potentially confounded by changes in muscle mass (39, 41–47). Similar to human medicine, sCysC may currently be more appropriately viewed as an adjunctive marker of risk stratification rather than a replacement for established diagnostic tools (41, 62).

Species-specific interpretation is essential when translating these findings into clinical practice. The evidence supporting uCysB is predominantly derived from canine studies, whereas only a single feline study met the inclusion criteria of this review. Although this feline study was included qualitatively and suggested potential diagnostic and prognostic relevance, quantitative synthesis was not possible, preventing robust species-specific conclusions for cats. Conversely, the evidence regarding sCysC is primarily derived from feline CKD populations. Therefore, direct extrapolation of findings between species is not currently justified and should be approached with caution (55).

This study presents both strengths and limitations. Although the number of eligible studies was limited, statistical power approached 90% for both biomarkers, indicating a low probability of type II error and supporting the biological validity of the observed associations. Nevertheless, substantial methodological heterogeneity, differences in assay methodologies, variations in disease classification criteria, and the predominance of comparisons between diseased and healthy animals remain important limitations.

From a clinical perspective, the findings suggest that uCysB may represent a promising biomarker of active tubular injury in dogs, with potential utility for detecting renal injury before conventional functional markers become abnormal. In contrast, sCysC appears to function primarily as a complementary biomarker of CKD in cats, although current evidence does not support its replacement of creatinine or SDMA in routine clinical practice. Future research should focus on establishing standardized analytical methodologies, clinically actionable cut-off values, prognostic thresholds, and comparisons with other causes of azotemia. Such efforts will be essential to determine the true clinical value of cystatin-based biomarkers and facilitate their integration into routine veterinary nephrology.

5 Conclusion

sCysC and uCysB demonstrated the ability to distinguish cats and dogs with nephropathy from healthy individuals. However, sCysC in cats presented moderate elevations in CKD and, crucially, its clinical validation is limited by the scarcity of comparative studies between nephropathic animals and those with other non-renal pathologies. This methodological gap, combined with the absence of proven superiority over creatinine or SDMA, discourages its routine use in current staging and therapeutic management. On the other hand, uCysB in dogs and cats has proven to be a promising biomarker in AKI, with elevated concentrations associated with unfavorable outcomes, conferring both diagnostic and prognostic potential. Nevertheless, the implementation of this marker for early staging and

personalization of therapeutic protocols in AKI still requires more robust clinical investigations.

6 Limitations

Despite providing a comprehensive synthesis of the available evidence, this study has limitations that should be acknowledged. The small number of eligible studies restricts species-specific conclusions and limits subgroup analyses. Substantial methodological heterogeneity was observed, including differences in assays, disease classification criteria, study populations, and AKI/CKD etiologies, which likely contributed to variability in effect estimates. In addition, most studies compared diseased animals with healthy controls, potentially overestimating diagnostic performance in real clinical settings where differentiation among disease states is required. Nevertheless, despite the limited number of included studies, statistical power analyses consistently demonstrated values close to 90% for both biomarkers, supporting that the observed associations likely represent true biological effects. These findings suggest that the currently available studies may already be sufficient to validate the existence of biomarker alterations in CKD and AKI, although additional prospective and standardized investigations remain necessary to refine prognostic utility, and clinical applicability.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

BS: Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Writing – original draft, Writing – review & editing. BM: Investigation, Methodology, Writing – original draft. SC: Supervision, Validation, Writing – review & editing. CP: Investigation, Methodology, Writing – review & editing. LC: Supervision, Validation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Urinary KIM-1 as a Marker of Renal Tubular Injury Associated with Urethral Obstruction in Non-Azotemic Cats

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Simple Summary

Renal tubular injury associated with urethral obstruction may occur in cats before alterations in conventional renal biomarkers become apparent. In this study, urinary concentrations of kidney injury molecule-1 (KIM-1), a biomarker of tubular epithelial injury, were evaluated in non-azotemic cats with urethral obstruction and compared with those of healthy cats. Cats with urethral obstruction presented significantly higher urinary KIM-1 concentrations despite having serum creatinine and blood urea nitrogen concentrations within reference intervals. These findings suggest that urinary KIM-1 may provide additional information regarding renal tubular injury associated with urethral obstruction and support further investigation of its clinical utility in feline medicine.

Keywords: renal injury, renal biomarkers, KIM-1

Abstract

Cats with urethral obstruction may develop tubular damage before conventional indicators of renal dysfunction become abnormal, making early recognition challenging when relying solely on conventional renal biomarkers. This study evaluated urinary kidney injury molecule-1 (KIM-1) concentrations in non-azotemic cats with urethral obstruction and compared its expression with conventional renal biomarkers. Twenty-four male cats were prospectively enrolled and allocated into a control group ($n = 12$) and a urethral obstruction group ($n = 12$), all presenting serum creatinine concentrations within the reference interval. Urinary KIM-1 concentrations were quantified by enzyme-linked immunosorbent assay (ELISA) and normalized to urinary creatinine concentrations. Serum creatinine and blood urea nitrogen (BUN) did not differ significantly between groups. In contrast, urinary KIM-1 concentrations were significantly higher in cats with urethral obstruction compared with healthy controls ($p < 0.001$). These findings suggest that increased urinary KIM-1 expression may be associated with renal tubular injury in non-azotemic cats with urethral obstruction, even in the absence of alterations in conventional renal biomarkers. Therefore, urinary KIM-1 may represent a useful non-invasive marker of renal tubular injury associated with obstructive urinary disease. Further prospective studies incorporating additional renal biomarkers, histopathological evaluation, and longitudinal follow-up are warranted to better define its clinical applicability.

1. Introduction

Acute kidney injury (AKI) can be defined as a spectrum of diseases associated with a sudden onset of renal parenchyma injuries, more typically characterized by generalized kidney failure to meet the body's excretory, metabolic and endocrine demands [1,2]. The main causes of AKI include the prolonged and indiscriminate use of nephrotoxic drugs, such as non-steroidal anti-inflammatory and chemotherapy drugs, infectious diseases, vasculitis, neoplasms, and obstruction of the urinary tract [1,3,4]. In cats, urethral obstruction is the primary cause of AKI [5].

The clinical signs of AKI can range from acute uremia to anuria that can lead to renal parenchymal injury and the development of chronic kidney disease (CKD) [1,3]. Therefore, identifying this condition in cats at early stages is crucial to prevent kidney disease progression to kidney failure. However, early diagnosis of AKI in this species is challenging due to the current limitations of diagnostic approaches in detecting subclinical kidney lesions [6].

The advent of urinary biomarkers and the classification system for AKI are important milestones in understanding the bidirectional relationship between AKI and CKD in humans [4].

Epidemiological studies have shown that even transient kidney dysfunction, such as prerenal azotemia caused by dehydration, is a risk factor for the development of CKD if the cause of azotemia is not previously corrected [7,8]. Clinical studies in humans and cats critically ill with AKI report only a slight increase in serum creatinine (sCr) concentration (0.1–0.2 mg/dL), while studies evaluating early biomarkers such as kidney injury molecule-1 (KIM-1) and galectin-3 reveal an increase in concentration before increases in sCr [9].

However, despite advances in veterinary nephrology, the early diagnosis of AKI in cats remains challenging, particularly in non-azotemic patients. Conventional renal biomarkers, such as serum creatinine and blood urea nitrogen (BUN), primarily reflect glomerular filtration function and typically increase only after substantial loss of functional renal mass [4,9]. In contrast, structural biomarkers, including kidney injury molecule-1 (KIM-1), provide direct information regarding tubular epithelial injury and may be detected before measurable changes in glomerular filtration occur [10,11,12]. Consequently, structural biomarkers have attracted increasing interest as adjunctive tools for the detection of renal injury in both human and veterinary medicine [4,10,11].

Recently, the feline Kidney Injury Molecule 1 (KIM-1) gene and its protein were described by Bland et al. [10]. KIM-1 is a tubular transmembrane glycoprotein that is not detected in the urine of patients without kidney injury [13]. The occurrence of AKI induces KIM-1 expression on the surface of undifferentiated epithelial cells of the proximal convoluted tubule. This enabled the detection of KIM-1 in the urine and tissue of cats in cases of ischemic injury, which has the potential to be used as a marker of tubular injury associated with AKI, critical disease and suspected AKI due to associated ischemic processes, including urethral obstruction [11,12].

There are few studies on the expression of early biomarkers of AKI in cats. The study by Woerde et al. [14] demonstrated increased serum galectin-3 (sGal-3) concentrations in azotemic cats with ureteral obstruction. Likewise, studies evaluating urinary KIM-1 in cats with AKI reported increased expression in cases of urinary obstruction, sepsis, and other causes of renal injury [11,14]. However, information regarding urinary KIM-1 concentrations in non-azotemic cats with naturally occurring urethral obstruction remains limited.

Most previous studies evaluating renal biomarkers have focused on established kidney injury, chronic kidney disease, or experimental models. Therefore, the present study contributes additional information regarding urinary KIM-1 expression in cats with urethral obstruction before the development of overt azotemia, improving our understanding of renal tubular injury

associated with this common feline emergency. We hypothesized that non-azotemic cats with urethral obstruction would present increased urinary KIM-1 concentrations compared to healthy cats, even in the absence of alterations in conventional renal biomarkers such as serum creatinine and BUN, suggesting the presence of early renal tubular injury.

Therefore, the present study aimed to evaluate urinary KIM-1 concentrations as a marker of renal tubular injury associated with urethral obstruction in non-azotemic cats and to compare its expression with conventional renal biomarkers.

2. Materials and Methods

2.1. Study Population

Twenty-four male cats (*Felis catus*) of different ages attended at the Professor Sylvio Barroso Veterinary Hospital, State University of Ceará, Brazil, were prospectively enrolled between February 2019 and February 2021. Cats were divided into two groups: a control group (CG), composed of clinically healthy cats ($n = 12$), and a urethral obstruction group (UOG), composed of non-azotemic cats with urethral obstruction ($n = 12$).

The inclusion criteria for cats in the UOG were the presence of urethral obstruction associated with clinical signs of lower urinary tract disease, including dysuria, stranguria, pollakiuria, hematuria, and urinary house soiling, in addition to serum creatinine concentrations < 1.6 mg/dL, blood urea nitrogen (BUN) concentrations between 20 and 65 mg/dL, and urine specific gravity (USG) < 1.030 .

Information regarding the duration of clinical signs associated with urethral obstruction was obtained from owner reports at admission. Although the exact duration of obstruction could not be objectively determined in all cases, owners generally reported the onset of clinical signs within approximately 24–48 h before presentation.

The diagnosis of urethral obstruction was established based on compatible clinical signs, absence of urine output, physical examination findings including a large, firm, and painful urinary bladder that could not be manually expressed, and abdominal ultrasonography demonstrating marked bladder distension. Obstruction was further confirmed during urethral catheterization and restoration of urinary flow.

All cats underwent a complete clinical evaluation at admission, including physical examination, complete blood count, serum biochemical analysis, urinalysis, quantitative urine culture, and abdominal ultrasonography. Laboratory and imaging findings were used to support the diagnosis of urethral obstruction, assess the overall clinical status of the patients, and exclude concurrent diseases that could interfere with study outcomes. Abdominal ultrasonography was performed to confirm bladder distension associated with urethral obstruction and to exclude urethrolithiasis, neoplasia, congenital urinary tract abnormalities, and ultrasonographic evidence of chronic kidney disease.

Following sample collection, cats with urethral obstruction received standard clinical management, including bladder decompression, urethral catheterization, intravenous fluid therapy, analgesia, and supportive care according to their clinical condition.

Exclusion criteria included urethrolithiasis, neoplasia, congenital urinary tract abnormalities detected by ultrasonography, urinary tract infection based on quantitative urine culture, previous or current history of lower urinary tract infection, diabetes mellitus, and hyperthyroidism. Information regarding previous lower urinary tract infections was obtained from owner reports during the admission interview and, when available, from medical records.

Cats with a history of antimicrobial or anti-inflammatory drug administration within the six months preceding admission were also excluded from the study. All urine cultures were performed prior to any therapeutic intervention, and only cats with negative urine culture results were eligible for inclusion in the study.

For the CG, clinically healthy cats presented for routine vaccination and/or elective castration were included. All healthy cats underwent a complete physical examination, and hematological, biochemical, and urinary parameters were within reference intervals for the species.

All procedures were conducted according to the ethical principles of animal experimentation approved by the Ethics Committee on the Use of Animals of the State University of Ceará (protocol 02875696/2019). All owners provided written informed consent authorizing biological sample collection, abdominal ultrasonography, and the use of clinical data for research purposes.

2.2. Analytical Methods

Blood samples (5 mL) were punctured from the jugular vein and separated into two aliquots: 1 mL was placed in a tube with EDTA for automatic blood count (ABX Micros ESV 60, Horiba Medical, Kyoto, Japan), and 4 mL were centrifuged for serum biochemical analysis. Serum creatinine (sCre), blood urea nitrogen (BUN), aspartate aminotransferase (AST), alanine aminotransferase (ALT), albumin and alkaline phosphatase (AF) were obtained on a biochemical analyzer (Cobas C111, Roche[®], Basel, Switzerland) based on the enzymatic colorimetric method and spectrophotometry, following the manufacturer's recommendations and using BioSystems[®] (Quezon City, Philippines) reagents. Reference intervals adopted by the Clinical Pathology Laboratory of the Veterinary Hospital were 0.8–1.8 mg/dL for serum creatinine and 20–65 mg/dL for blood urea nitrogen (BUN).

All blood and urine samples from cats with urethral obstruction were collected immediately upon admission and before any therapeutic intervention, including urethral catheterization, bladder decompression, intravenous fluid therapy, or stabilization procedures. Following initial clinical evaluation and abdominal ultrasonography, urine samples were obtained by ultrasound-guided cystocentesis for laboratory analyses.

Urine samples were collected by cystocentesis, quantifying a total volume of 10 mL, and urine specific gravity (USG) was determined by refractometry (analog optical refractometer RSG32, Akso, São Leopoldo, Brazil). A conversion factor for cats was then applied using the equation: Feline USG = (0.846 × USG reading in human refractometry) + 0.154 by Reppas and Foster [15].

In addition, urine samples were analyzed using reagent strips and a semi-automatic urine analyzer (Combilizer 13, Human Diagnostics Worldwide, Wiesbaden, Germany) to obtain measurements of pH, protein, glucose, ketone, bilirubin, leukocytes, and erythrocytes in the urinalysis. Thus, an aliquot of 3 mL was centrifuged (3 min at 365× *g*), and the pellet was analyzed by light microscopy after 30 min [15].

Therefore, the measurement of urinary protein and urinary creatinine was performed with a semi-automatic biochemical analyzer (A15 BioSystems, Barcelona, Spain). The protein present in urine was determined using a commercial BioSystems[®] kit, which is based on the pyrogallol red principle, in an endpoint reaction and reading at 620 nm. Urinary creatinine (uCre) was measured using the Jaffé alkaline picrate principle, with the commercial kit (BioSystems[®] kit) in the endpoint reaction and reading at 505 nm. For this purpose, the urine samples were diluted 1:50 in distilled water to reach the linearity range of the test, with the final value being duly corrected. The urinary protein/creatinine ratio was the relationship between protein and creatinine values expressed in the same unit (mg/dL). Test validation was performed by determining repeatability.

All urine samples were analyzed qualitatively and quantitatively for aerobic bacteria culture, and samples with bacterial growth ≥ 105 colony-forming units/mL were considered positive [16].

2.3. Determination of Urinary Levels of KIM-1 by Enzyme-Linked Immunosorbent Assay

Urinary KIM-1 levels were quantified using the sandwich enzyme-linked immunosorbent assay (ELISA) (Human Kit from Boster Biological Technology, Fremont, CA, USA) for antigen quantification. Samples were measured by spectrophotometry at 450 nm, in triplicate and their values were expressed as means. A standard curve was plotted with values ranging from 12.5 to 500 pg/mL.

The detection range for KIM-1 was 12.5–500 pg/mL, and the intra-assay coefficient of variation was 3.2% expressed as a ratio to the uCre concentration. Urinary KIM-1 levels were normalized by dividing their values by the uCre values of the same sample and expressed as pg/mg of uCre (KIM-1/uCre). All measurements were performed in triplicate, and the mean value was used for statistical analysis to ensure analytical reliability.

2.4. Statistical Analysis

The software R (4.2.1) was used to perform all statistical tests. The means of serum creatinine, BUN, urinary density, urinary protein creatinine ratio, and KIM-1 of the CG and UOGs were compared using Student's *t*-test. Generalized linear models (GLM) were used to test the effects of clinical status, age, weight, and castration on biochemical parameters. Firstly, the possibility of multicollinearity was excluded by calculating the Variance Inflation Factor (VIF) with the “car” package (3.1-5). All factors presented $VIF < 2$, therefore, none were considered problematic in the model. To evaluate the equality of variances of fixed categorical factors (clinical condition and castration), Levene's test was used [17]. When residual diagnostics indicated non-normality of residuals, the response variable was transformed using the Boxcox function from the MASS package (7.3-65) prior to model refitting [18].

Five models were built for the following variables (response factors): serum creatinine, BUN, urinary density, urinary protein creatinine ratio, and KIM-1. In all variables, clinical status (CG $\times$ UOG), age, weight, castration, and interactions were included as fixed factors. According to Burnham and Anderson [19], these factors were removed sequentially to select the model with the lowest Akaike information criterion and correction for small sample sizes (AICs). All values were

expressed as mean $\pm$ standard deviation (SD) and differences with p -value < 0.05 were considered significant. The individual data used for all statistical analyses are provided in the [Supplementary Materials \(Table S1\)](#).

3. Results

3.1. Study Population

The mean $\pm$ SD ages of cats in the CG and UOG were 2.54 ± 1.70 and 2.08 ± 1.31 years, respectively. Mean body weight was 4.45 ± 0.90 kg in the CG and 4.50 ± 0.50 kg in the UOG. No significant differences were observed between groups regarding age ($t = 0.74$; $p = 0.47$) or body weight ($t = 0.15$; $p = 0.88$).

No specific breed predisposition was identified among the evaluated cats. In the UOG, nine cats were neutered (75%) and three were intact (25%), whereas in the CG, seven cats were neutered (58.33%) and five were intact (41.67%).

Feline idiopathic cystitis associated with inflammatory urethral plugs was considered the presumptive cause of urethral obstruction in all cats included in the UOG. Obstruction was attributed to inflammatory processes associated with epithelial desquamation and urethral plug formation. Urethrolithiasis, neoplasia, congenital abnormalities, and urinary tract infection were excluded through abdominal ultrasonography, urine culture, and clinical evaluation.

All UOG cats were promptly treated, stabilized, and managed through the unblocking process. They were hospitalized and received full support to stabilize the obstructive and AKI condition, including intravenous fluid therapy, urethral catheterization, and post-procedural analgesia [methadone 0.2 mg/kg intravenously (IV) three times daily]. Urethral catheterization was performed after stabilization, asepsis, and anesthesia of the patient, with the administration of methadone 0.2 mg/kg intramuscularly (IM) with acepromazine 0.02 mg/kg (IM) initially, followed by propofol 4 mg/kg (IV), induction and maintenance with isoflurane via tracheal tube. After hospital discharge, the animals were submitted to hydrotherapy, analgesic and adjuvant therapy, according to their specific laboratory and clinical needs.

3.2. Measurement of Renal Markers

The data of serum and urinary renal biomarkers for CG and UOG are shown in [Table 1](#).

Table 1.

Evaluation of biochemical parameters in clinically healthy cats (CG) and cats with urethral obstruction (UOG).

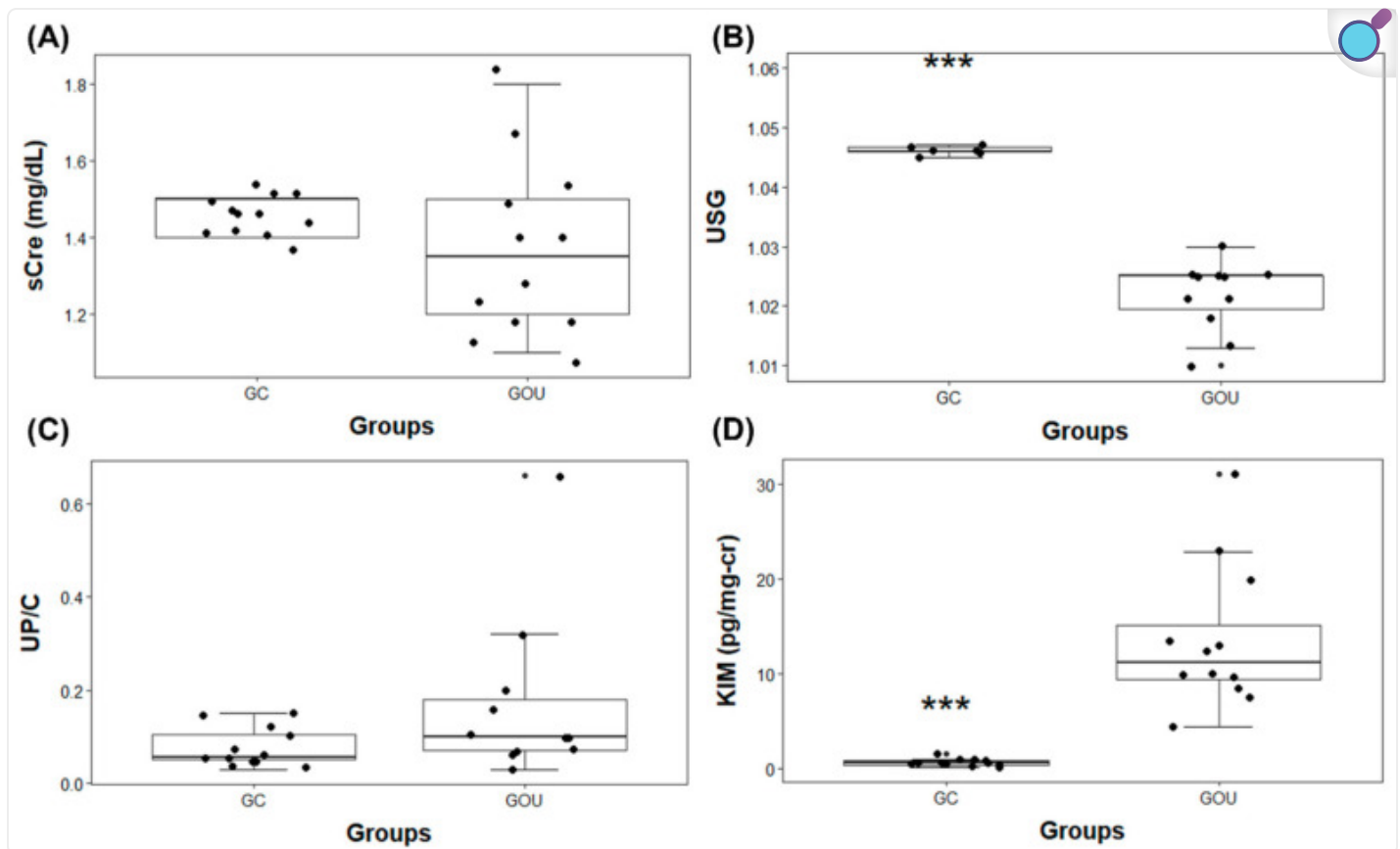
Parameters	CG (<i>n</i> = 12)	CI 95%	OUG (<i>n</i> = 12)	CI 95%
sCre (mg/dL)	1.46 ± 0.05	1.43–1.49	1.37 ± 0.23	1.22–1.51
BUN (mg/dL)	46.33 ± 8.86	40.71–51.96	50.92 ± 9.56	44.84–56.99
USG	1.054 ± 0.010	1.048–1.061	1.022 ± 0.006	1.018–1.026
UP/C	0.077 ± 0.042	0.050–0.104	0.170 ± 0.182	0.048–0.292
KIM-1 (pg/mg-Cr)	0.61 ± 0.38	0.37–0.85	13.54 ± 7.53	8.75–18.32

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Serum creatinine (sCre), blood urea nitrogen (BUN), urine specific gravity (USG), urine protein to creatinine ratio (UP/C) and kidney injury molecule-1 (KIM-1) in domestic cats. Mean ± SD; 95% confidence interval (CI).

The sCre models did not differ from the null model, indicating that there was no effect of clinical status, age, weight, and castration on sCre levels. Furthermore, animals from the CG did not show statistical differences in relation to UOG (*p* = 0.12; [Figure 1A](#)). Similarly, the model using UP/C as predictor did not differ from the null model, with no effect of clinical status, age, weight, and castration on UP/C levels ([Figure 1C](#)).

Figure 1.



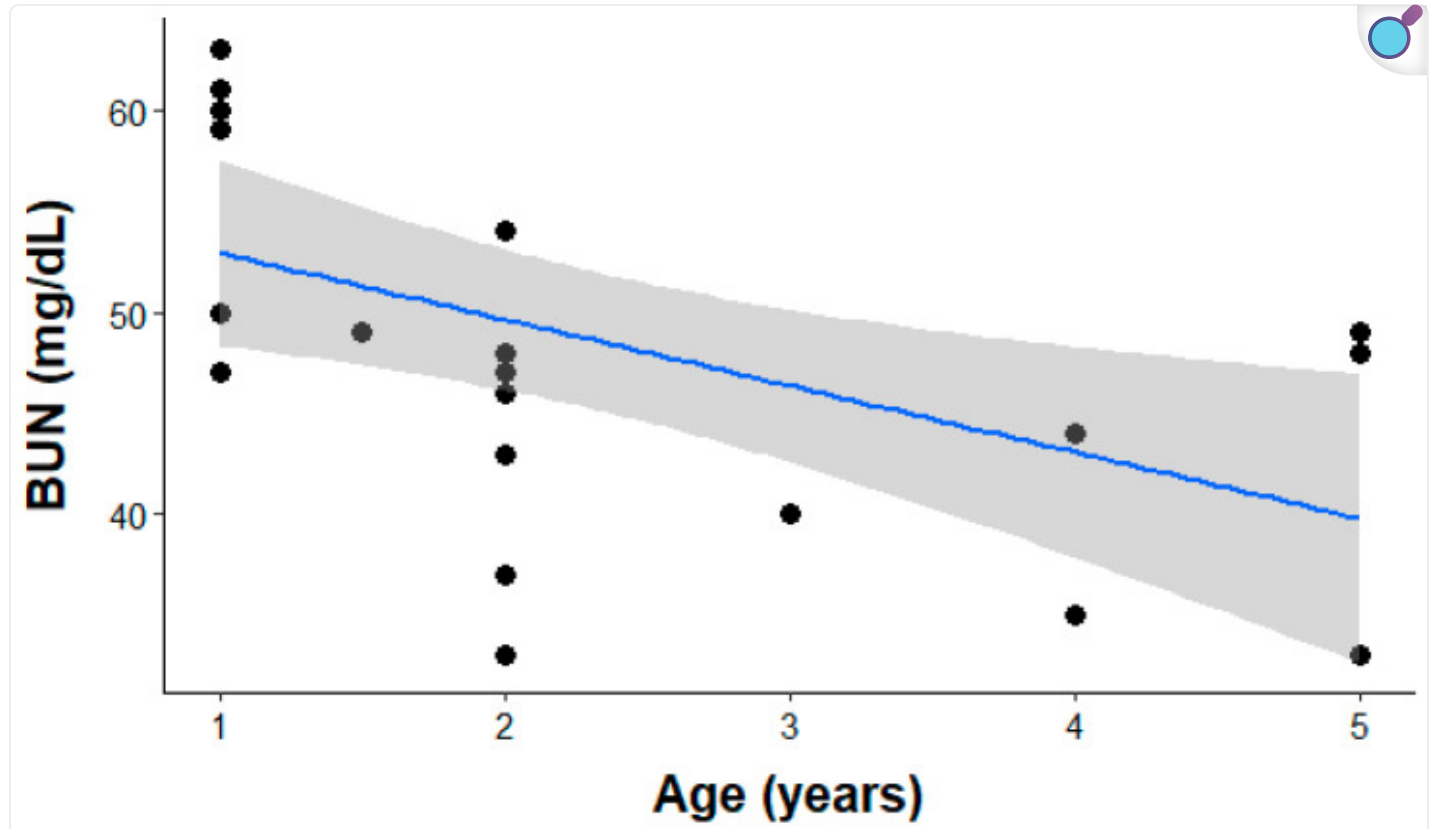
[Open in a new tab](#)

Effect of clinical status on (A) Serum creatinine concentration (sCre); (B) urine specific gravity (USG); (C) urine protein to creatinine ratio (UP/C) and (D) kidney injury molecule-1 (KIM-1) in domestic cats. Control group (CG); Urethral obstruction group (UOG) ($n = 12$, each). *** $p < 0.05$.

For the model including USG as a response factor, the best model showed a significant effect of clinical status, with higher values in the CG (GLM: -0.05 ± 0.005 ; $t = -10.09$; $p < 0.01$) than in the UOG (Figure 1B). In addition, the model using KIM-1 as a response factor revealed a significant effect of clinical status, with higher expression of KIM-1 in UOG animals (GLM: 0.20 ± 0.016 , $t = 12.55$, $p < 0.001$) than in GC animals (Figure 1D). There was no effect of weight, age, and castration in any of the models, therefore these factors were excluded from the final model.

A last model using BUN as a response factor demonstrated a significant negative effect with age (GLM: -3.28 ± 1.12 ; $t = -2.92$; $p = 0.008$) (Figure 2). Weight, clinical status and castration had no effect in any of the models and were excluded from the final model.

Figure 2.



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Effect of age on blood urea nitrogen (BUN) in domestic cats. (Generalized linear models—GLM: -3.28 ± 1.12 ; $t = -2.92$; $p = 0.008$). Trend line (blue line) and confidence intervals (gray band).

4. Discussion

The present findings indicate that cats with urethral obstruction exhibited markedly increased urinary KIM-1 concentrations despite normal conventional renal biomarkers, suggesting an association between urinary KIM-1 excretion and tubular injury. These findings are consistent

with previous studies in humans and experimental animal models of ischemia–reperfusion injury, in which KIM-1 expression increases during the early stages of renal injury [20,21].

An important aspect of the present study is that urine samples were collected before urethral catheterization, bladder decompression, fluid therapy, and other therapeutic interventions. Therefore, urinary KIM-1 concentrations reflected the clinical status of the cats at presentation, minimizing the potential influence of treatment-related factors on biomarker expression and strengthening the interpretation that increased KIM-1 levels were primarily associated with the obstructive condition itself.

Creatinine and blood urea nitrogen (BUN) are widely used markers for renal function evaluation; however, they primarily reflect glomerular filtration rather than structural renal injury. Although BUN is a sensitive marker of nitrogenous waste accumulation, its concentration may be influenced by several non-renal factors. Likewise, in cases of reduced glomerular filtration, tubular secretion of creatinine may increase, potentially masking early renal injury through an overestimation of renal function. Therefore, the need for clinically available biomarkers capable of assessing tubular health and injury is justified [4].

In the present study, neither serum creatinine nor BUN concentrations were reliable in diagnosing renal dysfunction in non-azotemic cats with urinary obstruction (UOG), with both values remaining within the reference values for the species (Table 1). We also found that creatinine was not affected by clinical status, even when controlling for age, weight, and castration. In contrast, we found that BUN was affected by age, with higher values in younger animals. This could be related to the higher protein content used in the diets of young animals [22]. Although these biomarkers are currently used to assess kidney function, sCre and BUN are considered late markers of kidney injury [9].

The lower USG observed in UOG cats may reflect impaired urinary concentrating ability associated with tubular dysfunction [15]. Furthermore, persistent hyposthenuria is suggestive of a loss of urinary concentrating capacity in the tubular segments of the nephrons [15,23]. In cats, USG $\leq$ 1.025 has been considered an indicator of decreased kidney function [12], and our results support this.

The inclusion criteria used in the present study were designed to select cats with urethral obstruction that remained non-azotemic according to serum creatinine concentrations, while presenting reduced urine concentrating ability. Importantly, cats were not evaluated solely on the basis of clinical history and ultrasonography. All animals underwent complete clinical and

laboratory screening, including physical examination, complete blood count, serum biochemical analysis of renal and hepatic parameters, urinalysis, quantitative urine culture, and abdominal ultrasonography. Apart from findings related to the obstructive urinary condition, the remaining clinical, hematological, biochemical, urinary, and imaging parameters did not suggest systemic disease or chronic kidney disease.

Reduced urine specific gravity in cats with urethral obstruction may be physiologically explained by transient tubular dysfunction associated with increased intravesical and intrarenal pressure, altered renal perfusion, medullary washout, and impairment of tubular concentrating mechanisms [5,12,15,23,24]. Previous clinical data in cats with urethral obstruction have shown reduced urine specific gravity in obstructed cats and have also indicated that not all affected cats present overt azotemia at the time of diagnosis [12]. Therefore, the combination of serum creatinine concentrations within the reference interval and reduced USG does not necessarily indicate pre-existing chronic kidney disease. However, because SDMA, cystatin-C, histopathological evaluation, and longitudinal renal follow-up were not available, early or subclinical chronic kidney disease cannot be completely excluded. Consequently, a degree of misclassification bias remains possible and should be considered when interpreting urinary KIM-1 concentrations in cats with urethral obstruction.

Xavier Júnior et al. [12], in a clinical study evaluating dosages of KIM-1 and urinary GGT as biomarkers of acute kidney injury in cats, observed that the concentration of KIM-1 in urine is not equivalent to the concentration of conventional markers used in clinical routine; probably, this biomarker is influenced and expressed by the degree of tubular cell injury.

Previous investigations in human and experimental models have consistently reported increased KIM-1 expression following tubular injury. Some studies have also shown that KIM-1 can help differentiate ischemic AKI from prerenal azotemia and CKD and be useful in differentiating the various subtypes of AKI [25,26,27].

In the present study, we observed an increase in KIM-1 values in cats with post-renal AKI in non-azotemic patients. This increase can be explained due to the urethral obstruction, obliteration, and retention of urinary flow, which cause an increase in intravesical, ureteral, and renal pressure. This increase sustained by intrarenal pressure results in the loss of renal parenchyma and the influx of leukocytes into the renal parenchyma, releasing cytokines. As a result, there is a development of electrolyte imbalances, fibrosis in parenchyma due to the lesion, and renal ischemia secondary to dehydration caused by urethral obstruction [5]. This ischemia will lead to renal injury due to the

absence of ATP, and the Na⁺/K⁺ ATPase pump will have its functioning altered, causing the entrance of calcium in tubular cells, inducing apoptosis and acute tubular necrosis [5,9,26].

Current criteria for diagnosis and classification of AKI heavily depend on sCr changes [2]. However, the timing of changes in sCr level and concomitant changes in glomerular filtration rate does not allow for an accurate estimate of time to injury or severity of dysfunction [1,8,24]. Thus, the early phase of AKI may not be associated with a significant increase in sCr levels due to several factors, including water overload, high catabolism, sepsis, and ischemic insults, resulting in delayed diagnosis and intervention. Therefore, the search for new biomarkers of AKI, especially in its initial phase, has intensified in recent years [28].

Urethral obstruction can trigger a sequence of pathophysiological events that may culminate in renal tubular injury. Urine retention increases intravesical pressure, which may be transmitted to the ureters and renal pelvis, resulting in increased intrarenal pressure and reduced renal perfusion [4,24]. This process can lead to renal ischemia, tubular epithelial cell injury, oxidative stress, inflammatory responses, and, in more severe or prolonged cases, apoptosis and tubular necrosis [4,24]. These mechanisms provide a biological rationale for the increased urinary excretion of tubular injury biomarkers, such as KIM-1, in cats with urethral obstruction [20,21,24].

The increased urinary KIM-1 concentrations observed in the present study may reflect tubular epithelial injury secondary to ischemia and increased intrarenal pressure associated with urethral obstruction. Ischemia plays a central role in the pathophysiology of obstructive renal injury, and although reperfusion is essential to restore tissue oxygenation and prevent irreversible cellular damage, it may also contribute to additional oxidative and inflammatory injury [4,24]. The reintroduction of oxygenated blood into previously ischemic tissues can intensify cellular damage through the generation of reactive oxygen species and inflammatory mediators [4,24].

Potential confounding factors should also be considered when interpreting urinary KIM-1 concentrations. Dehydration and hemodynamic alterations commonly observed in cats with urethral obstruction may contribute to reduced renal perfusion and tubular stress. In addition, all cats in the UOG presented feline idiopathic cystitis associated with inflammatory urethral plugs, and the potential contribution of lower urinary tract inflammation to urinary biomarker excretion cannot be completely excluded. Likewise, the presence of subclinical kidney disease cannot be definitively ruled out, despite the inclusion of young non-azotemic cats with no clinical or ultrasonographic evidence of chronic kidney disease. Therefore, increased urinary KIM-1 concentrations should be interpreted within the clinical context and not as an isolated indicator of renal tubular injury.

Another consideration is that urinary KIM-1 concentrations were measured while cats were still obstructed. Although urinary retention may theoretically influence urinary biomarker concentrations, KIM-1 is primarily recognized as a marker of proximal tubular epithelial injury [20,21]. Previous studies from our group demonstrated persistently increased urinary KIM-1 concentrations after relief of urethral obstruction, suggesting that elevated KIM-1 concentrations are not solely attributable to urine retention [12]. Nevertheless, the contribution of urinary retention itself cannot be completely excluded, and future longitudinal studies evaluating urinary KIM-1 before and after relief of obstruction are warranted.

This study has some limitations, including the relatively small sample size and the lack of longitudinal follow-up to assess the progression of kidney injury over time. According to IRIS guidelines, AKI diagnosis and staging are largely based on changes in serum creatinine concentration. However, non-azotemic patients with early tubular injury may not fulfill classical staging criteria despite the presence of renal damage. Therefore, cats included in the present study were interpreted as having suspected early AKI associated with urethral obstruction and evidence of early tubular injury before the onset of overt azotemia.

Another limitation of the present study is that the exact duration of urethral obstruction could not be objectively determined in all cats and was based primarily on owner-reported clinical history. Therefore, variation in the duration and severity of obstruction may have contributed to differences in the degree of tubular injury and urinary KIM-1 expression among affected cats.

Additional limitations include the absence of complementary renal biomarkers, such as SDMA and cystatin-C, as well as the lack of histopathological confirmation of tubular injury. Therefore, increased urinary KIM-1 concentrations should be interpreted cautiously as evidence of early renal tubular injury associated with urethral obstruction rather than as definitive confirmation of AKI. Further prospective studies including additional renal biomarkers, histopathological evaluation, and longitudinal follow-up are warranted to better characterize the role of KIM-1 in early feline AKI.

5. Conclusions

The results obtained in this study indicate that urinary KIM-1 may provide clinically relevant information regarding renal tubular injury in non-azotemic cats with urethral obstruction. Increased urinary KIM-1 concentrations were observed despite normal serum creatinine and BUN concentrations, supporting its potential application in the assessment of renal tubular injury associated with urethral obstruction. However, given the pilot nature of this study and the absence

of longitudinal outcome assessment, further prospective studies incorporating additional renal biomarkers and long-term follow-up are warranted to better define the clinical role of KIM-1 in feline acute kidney injury.

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Abbreviations

The following abbreviations are used in this manuscript:

AKI	Acute Kidney Injury
CKD	Chronic Kidney Disease
sCre	Serum Creatinine
KIM-1	kidney injury molecule-1
CG	Control Group
UOG	Obstruction Group
BUN	Blood Urea Nitrogen
AST	Aspartate Aminotransferase
ALT	Alanine Aminotransferase
AF	Albumin and Alkaline Phosphatase
USG	Urine Specific Gravity
uCre	Urinary Creatinine
GLM	Generalized Linear Models
VIF	Variance Inflation Factor
AICs	Criterion and Correction for Small Sample Sizes
SD	Mean $\pm$ Standard Deviation
IV	Intravenously
IM	Intramuscularly
CI	Confidence Interval
UP/C	Urine Protein to Creatinine Ratio

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Supplementary Materials

The following supporting information can be downloaded at:

<https://www.mdpi.com/article/10.3390/ani16121907/s1> [↗](#), Table S1. Individual clinical and

laboratory data of all cats included in the study, including age, body weight, castration status, serum creatinine (sCre), blood urea nitrogen (BUN), urine specific gravity (USG), urine protein-to-creatinine ratio (UP/C), and urinary kidney injury molecule-1 (KIM-1) concentrations used for statistical analyses.

[animals-16-01907-s001.zip](#) ^(12.1KB, zip)

Author Contributions

Conceptualization, F.A.FX.J. and J.S.A.M.E.; methodology, F.A.FX.J., T.L.S., A.M.C.M. and I.N.G.d.S.; software, T.H.d.S.P. and T.L.S.; validation, F.A.FX.J., T.H.d.S.P., T.L.S. and A.M.C.M.; formal analysis, F.A.FX.J., S.L.A., N.B.d.M., A.R.A.P., I.O.d.C. and H.N.d.A.J.; investigation, F.A.FX.J., S.L.A., N.B.d.M., A.R.A.P., I.O.d.C. and H.N.d.A.J.; resources, F.A.FX.J., T.L.S., A.M.C.M. and I.N.G.d.S.; data curation, T.L.S. and T.H.d.S.P.; writing—original draft F.A.FX.J., S.L.A., T.H.d.S.P., A.R.A.P. and H.N.d.A.J.; preparation, F.A.FX.J., S.L.A., T.H.d.S.P., A.R.A.P. and H.N.d.A.J.; writing—review and editing, F.A.FX.J., S.L.A., T.H.d.S.P., A.R.A.P. and H.N.d.A.J.; visualization, F.A.FX.J., S.L.A., T.H.d.S.P., A.R.A.P. and H.N.d.A.J.; supervision, F.A.FX.J., T.L.S. and N.B.d.M.; project administration, F.A.FX.J. and J.S.A.M.E.; funding acquisition, J.S.A.M.E., I.N.G.d.S. and A.M.C.M. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement

All experiments were carried out according to the ethical principles of animal experimentation adopted by the Ethics Committee on the Use of Animals of the State University of Ceará (02875696/2019, approved on 2 April 2019).

Informed Consent Statement

All cat owners were informed about the study and signed a consent form to authorize the collection of biological samples as well as abdominal ultrasonography, and the use of the data for publications.

Data Availability Statement

The data presented in this study are available on request from the corresponding author.

Conflicts of Interest

The authors declare no conflicts of interest.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

[animals-16-01907-s001.zip](#) (12.1KB, zip)

Data Availability Statement

The data presented in this study are available on request from the corresponding author.

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Evaluation of urine glutathione peroxidase 4 in cats with chronic kidney disease

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Introduction: Ferroptosis is a distinct form of regulated cell death characterized by iron-dependent lipid peroxidation that damages cellular membranes and leads to the end of a cell's life. Glutathione peroxidase 4 (GPX4), the only enzyme capable of the reduction of lipid peroxidation products within cells, is a key regulator of this process.

Aims: The role of GPX4 in feline chronic kidney disease (CKD) has not been previously investigated. This study aims to determine whether urine GPX4 levels are associated with CKD severity in cats and to assess their potential as a progression biomarker.

Methods: Urine GPX4 levels were measured using a commercial feline ELISA kit. The urine-GPX4-to-creatinine ratio (UGCR) was calculated. Fifteen healthy cats, 61 cats with CKD, and six cats with acute-on-chronic kidney disease (ACKD) were included in the study.

Results: Compared with the control group (urine GPX4, median [IQR]: 25.21 [18.99–26.91]; UGCR: 0.072 [0.057–0.101] × 10⁻⁴) and the early-stage CKD group (urine GPX4: 24.31 [22.00–24.07]; UGCR: 0.134 [0.070–0.260] × 10⁻⁴), cats with late-stage CKD showed significantly higher levels of urine GPX4 (26.89 [25.11–31.66]; *p* = 0.011) and UGCR values (0.271 [0.197–0.457] × 10⁻⁴; *p* < 0.001). Within the CKD subgroups, UGCR was significantly higher in cats with proteinuria, hypertension, anemia, and those receiving iron supplementation (all *p* < 0.003). Serum creatinine levels and WBC counts were identified as independent variables that were correlated with UGCR. Cats in the CKD progression group had higher UGCR than non-progressors, and an elevated UGCR was associated with an increased risk of CKD progression (hazard ratio [HR], 1.75; 95% CI, 1.20–2.54; *p* = 0.003).

Conclusion and clinical importance: UGCR increased with the severity of CKD and was significantly associated with serum creatinine concentration and disease progression. Urine GPX4 may thus serve as a novel biomarker for monitoring renal deterioration and progression in cats with CKD.

KEYWORDS

feline, ferroptosis, progression, renal disease, urine biomarkers

1 Introduction

Several cell death pathways contribute to the development and progression of both acute kidney injury and chronic kidney disease (1). Among these pathways, ferroptosis can be distinguished by its dependence on iron-driven lipid peroxidation that ultimately leads to oxidative damage to cell membranes, which then drives the pathogenesis of multiple disease states (2). Ferroptosis is a distinct form of regulated cell death and is different from apoptosis, autophagy, and necrosis. Morphologically, it is characterized by shrunken mitochondria, increased membrane density, and a reduced or absent of mitochondrial cristae; this is without any of the nuclear changes that are typically associated with apoptosis (3, 4). Accumulating evidence indicates that ferroptosis is implicated in various forms of pathological cell death, including ischemia–reperfusion injury (5, 6), neurodegenerative disease (7, 8) and cancer cell death (9, 10).

Ferroptosis is primarily linked to an overloading of intracellular iron that facilitates lipid peroxidation (1). In addition to abnormalities in iron metabolism, the inhibition of intracellular antioxidant defenses can also trigger ferroptosis. Under physiological conditions, a transmembrane protein complex known as system xc[−], mediates the exchange of extracellular cystine and intracellular glutamate across the plasma membrane, thereby supporting the biosynthesis of glutathione peroxidase 4 (GPX4) (3). This process is critical to glutathione production and oxidative protection within cells, as lipid peroxides are able to be metabolized by GPX4-catalyzed reduction reaction (4, 11–13). Consequently, GPX4 serves as a central regulator of ferroptosis.

Proper kidney integrity depends on the presence of functional GPX4 protein (12). In an inducible GPX4-depleted mouse model, massive renal tubule cell death and acute kidney injury were observed within 2 weeks (12). In addition, kidney tubular cells are perhaps the most ferroptosis-sensitive cells in mammals (12). Ferroptosis has been shown to be a key driver of both cisplatin-induced acute kidney injury (AKI) (14), and ischemia–reperfusion–associated renal necrosis (5). Emerging evidence suggests that this pathway is involved in the progression of chronic kidney disease (CKD) in humans (15). A decreased abundance of GPX4 has been observed in renal tubular cells of CKD patients (16), as well as in murine models of renal fibrosis that were induced by unilateral ureteral obstruction (UUO) (16, 17) and ischemia–reperfusion injury (18, 19). Inhibition of ferroptosis has been shown to attenuate kidney injury and fibrosis in various CKD animal models (16–18, 20). Furthermore, GPX4 expression has been proposed as a potential predictor of disease progression in diabetic kidney disease (21).

Despite these advances involving human and experimental models, the investigation of the role of ferroptosis in veterinary CKD is still largely unexplored. Accordingly, the present study evaluated urine GPX4 levels in cats across different stages of CKD and acute-on-chronic kidney disease (ACKD) to determine whether there is any association with disease severity and to explore the potential involvement of ferroptosis in renal deterioration.

2 Materials and methods

2.1 Animals

Retrospectively, we enrolled feline patients presenting with an elevated renal index or that were undergoing routine health

examinations at the National Taiwan University Veterinary Hospital between September 2019 and March 2022. The patients were classified into the control group, the CKD group, and the ACKD group, based on clinical assessment.

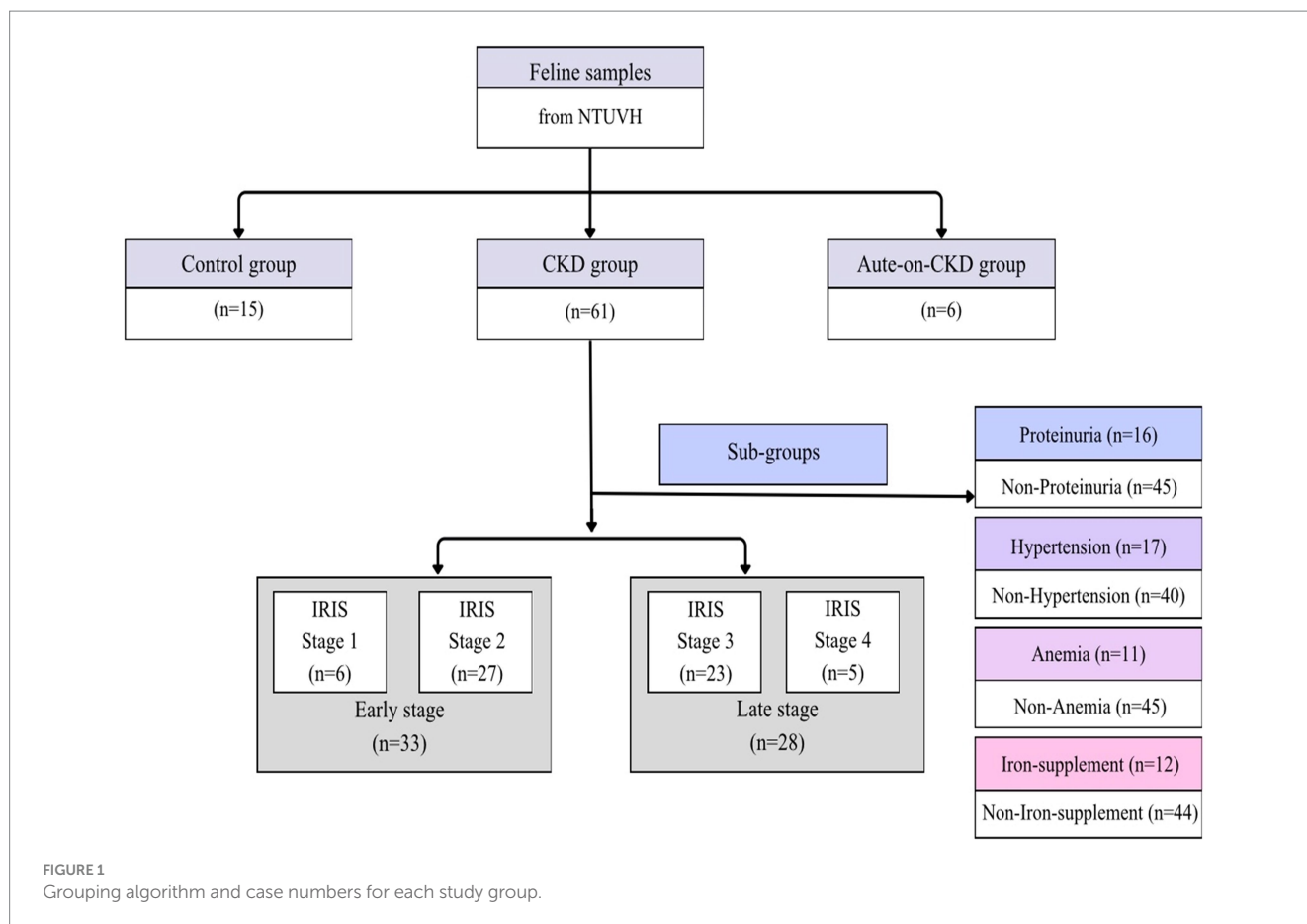
The control group contains cats without any clinical signs or abnormal physical examination results. These cats also had no medical history of renal-related, neuro-related, cardio-related, oncology-related, or endocrine-related symptoms. No medication was administered to the animals at the sample collection time point, except for the monthly heartworm and ectoparasite prevention. In addition to normal blood urea nitrogen (BUN, <30 mg/dL) and creatinine concentration (<1.6 mg/dL) being measured as part of their routine blood examination, other parameters were assessed, namely SDMA, hematocrit, red blood cell counts (RBC), white blood cells counts (WBC), plasma alanine aminotransferase level (ALT), plasma aspartate transaminase level (AST), plasma alkaline phosphatase level (ALKP), plasma albumin level (Alb), total protein (TP), plasma glucose level, plasma sodium level, plasma potassium level, and plasma chloride level, and these needed to be within the appropriate reference ranges for a cat to be included in the control group. Additionally, the results of an abdominal X-ray, urinalysis, and urinary tract ultrasound revealed no significant findings for the control group cats.

Cats classified into the chronic kidney disease (CKD) group met at least one of the following criteria. Firstly, persistent azotemia, defined as a plasma creatinine concentration ≥ 1.6 mg/dL for a duration of at least 3 months. Secondly, persistent proteinuria, defined as a urine protein-to-creatinine ratio (UPC) > 0.4 on at least three occasions, each separated by more than 2 weeks. Finally, radiological abnormalities consistent with CKD, such as small, irregular kidneys, reduced corticomedullary distinction, and/or polycystic kidney disease.

All CKD cases were staged according to the International Renal Interest Society (IRIS) staging system (Figure 1) into stages 1–4. For each case, initial symptoms, history, physical examination findings, diagnostic imaging results, and clinicopathological data were recorded. Cats diagnosed with neoplasia, endocrine disorders, or systemic infectious diseases, such as, blood parasites or viral infections, including feline leukemia virus (FeLV), Feline Immunodeficiency Virus (FIV) or feline infectious peritonitis (FIP), urinary tract infection (UTI), or post-renal azotemia (e.g., ureteral or urethral obstruction), when confirmed by ultrasonographic renal pelvic dilatation (>0.3 cm), were excluded.

Cats were categorized into subgroups based on the presence of CKD-related complications, including proteinuria, hypertension, anemia, and iron supplementation. Proteinuria was defined as a urine protein-to-creatinine ratio (UPC) > 0.4 on three separate occasions at least 2 weeks apart. Hypertension was defined as systolic blood pressure > 180 mmHg on two separate measurements or by the documented administration of amlodipine. Anemia was diagnosed when at least two of the following three hematologic parameters were below the reference range: hematocrit, hemoglobin, or red blood cell count. Cats that had received oral or intramuscular iron supplementation for a minimum of 1 month were classified into the iron supplementation group.

To assess factors associated with CKD progression, cats were further divided into two groups: progression and non-progression. Progression was defined as a 15% or greater increase in plasma



creatinine concentration from baseline (22) over a 180-day follow-up period.

2.2 Sample collection and storage

Blood was collected through venipuncture of the jugular or cephalic vein. Serum samples were obtained from tubes containing a clot activator. The samples were next centrifuged at 1,500×g for 3 min, and then the supernatants were stored at −80 °C until further analysis. Hematological values were analyzed using the Exigo hematology analyzer and the IDEXX Procyte Dx* hematology analyzer. Plasma chemistry profiles were measured using the Ortho Clinical Diagnostics™ VITROS™ 350 system. Plasma electrolytes were determined using the Roche Cobas b 121 system. Urinalysis was performed using the Roche Cobas b 411 system.

2.3 Measurement of urine GPX4 by feline commercial GPX4 ELISA kit

Urine GPX4 levels in the cats were measured using a species-specific commercial ELISA kit (MBS093863, MyBioSource, USA) by following the manufacturer's instructions. Standards and urine samples were tested in duplicate. After adding the HRP-conjugate reagent, plates were incubated at 37 °C for 1 h. The wells were then washed four times with 1× wash buffer. Finally chromogenic substrates

(50 µL each of Substrate/Chroomogen A and B) were added sequentially, followed by incubation at 37 °C for 15–20 min while protected from light. The reactions were terminated with 50 µL of Stop Solution, and absorbance was read at 450 nm within 15 min using a microplate reader. GPX4 concentrations were derived from the relevant standard curves. According to the manufacturer's datasheets, the feline GPX4 ELISA kit has a sensitivity of 1.0 ng/mL and a detection range of 3–100 ng/mL or 12–100 ng/mL, with intra-assay or inter-assay coefficients of variation (CVs) of less than 15% (Supplementary Table S1). The urine-GPX4-to-creatinine ratio (UGCR) was also measured.

2.4 Statistical analysis

We used R (version 4.1.2, R Foundation for Statistical Computing, Vienna, Austria) for data analysis and figure plotting. Numeric continuous variables were evaluated for normality through visual inspection of histograms and by the Shapiro–Wilk test. Most data were not normally distributed; therefore, for consistency, all numerical data are presented as medians with interquartile ranges (25th–75th percentiles). The Wilcoxon rank-sum test was used to compare parameters between two groups. Kruskal–Wallis tests, followed by Dunn's multiple comparison tests, were employed for comparisons among three or more groups. The Chi-square test or Fisher's exact test was performed to assess associations between categorical variables. Simple linear regression and multivariate linear regression analyses

were conducted to evaluate the associations between UGCR and clinical parameters. Multivariate linear regression analyses were performed using a backward stepwise approach that including variables with a p -value less than 0.05. Strength of association was interpreted using R^2 values with 0.1 approximately, indicating a weak association, 0.2–0.3, indicating a moderate association, and ≥ 0.4 , indicating a strong association.

To identify predictors of CKD progression, Cox proportional hazards regression analysis was conducted to estimate the hazard ratios of clinical parameters linked to disease progression. Receiver operating characteristic (ROC) analysis was conducted to assess the ability of the urine GPX4-to-creatinine ratio (UGCR) to discriminate cats that experienced CKD progression within a 180-day follow-up period, and area under the ROC curve (AUROC) was compared with the null hypothesis of an AUROC of 0.5. Additionally, the optimal cut-off value was determined using maximally selected rank statistics (*surv_cutpoint* function, R package *survminer*), which identifies the threshold that maximizes the standardized log-rank statistic while accounting for multiple testing. Based on this cut-off value, CKD cats were stratified into groups and compared using Kaplan–Meier survival analysis with the log-rank test. A p -value < 0.05 was considered statistically significant.

3 Results

3.1 The performance characteristics of the ELISA kits

Standard calibration curves were successfully established. The calibration curves were generated using five concentrations (3.12, 6.25, 12.5, 50, and 100 ng/mL), and demonstrated excellent linearity within the manufacturer's specified detection range ($R^2 = 0.9998$). A representative standard curve is shown in [Supplementary Figure S1](#). All duplicate measurements showed intra-assay coefficients of variation below 15% (12.7%), confirming acceptable repeatability.

3.2 Study population

A total of 82 cats were enrolled in this study, including 15 healthy cats, 61 CKD cats and 6 ACKD cats. Among CKD subgroups, there were 6 cats in the IRIS stage 1 subgroup, 27 cats in the IRIS stage 2 subgroup, 23 cats in the IRIS stage 3 subgroup, 5 cats in the IRIS stage 4 subgroup, and 6 cats in the ACKD subgroup. Because of the small size of IRIS stage groupings, the cats were also separated into an early-stage group (IRIS stages 1 and 2) and a late stage group (IRIS stages 3 and 4), which contained 33 and 28 cats, respectively ([Figure 1](#)).

In the control group, the most common breed was domestic shorthair ($n = 12$), followed by American shorthair ($n = 1$), British shorthair ($n = 1$), and Ragdoll ($n = 1$). In the CKD group, domestic shorthair was also predominant ($n = 45$), followed by Scottish Fold ($n = 3$), American shorthair ($n = 2$), domestic longhair ($n = 1$), Exotic Shorthair ($n = 4$), mixed-breed ($n = 3$), Persian ($n = 2$), Ragdoll ($n = 4$), and breed not recorded ($n = 3$).

In order to assess the influence of the complications associated with CKD in this study, cats with CKD were identified as having proteinuria, hypertension, and anemia. Additionally, the cats taking

or not taking iron supplements were also considered. There were 16 cats with proteinuria and 45 without proteinuria; 17 with hypertension and 40 without hypertension; 11 with anemia and 45 without anemia; and 12 cats receiving iron supplementation compared with 44 not receiving iron supplementation ([Figure 1](#)).

3.3 Urine GPX4 levels, UGCR and clinical parameters

The urine GPX4 and UGCR measurements for the control group, the various IRIS CKD stages, and the ACKD group are shown in [Figure 2](#). Urine GPX4 showed an increasing trend in the later stages of CKD, although the difference did not reach statistical significance. Of note, after adjustment for urine creatinine, UGCR rose significantly in parallel with a decline in renal function thus achieving a markedly elevated level during the stages 3 and 4 as well as when ACKD was present.

For further analysis, stages 1 and 2 of CKD were grouped together as early stage CKD, while stages 3 and 4 were grouped together as late stage CKD. The results for GPX4, UGCR, and various relevant clinical parameters across the control, early, and late CKD groups are summarized in [Table 1](#). Compared with the control group (urine GPX4, median [IQR]: 25.21 [18.99–26.91]; UGCR: 0.072 [0.057–0.101] $\times 10^{-4}$) and the early-stage group (urine GPX4: 24.31 [22.00–24.07]; UGCR: 0.134 [0.070–0.260] $\times 10^{-4}$), both urine GPX4 (26.89 [25.11–31.66]) and UGCR (0.271 [0.197–0.457] $\times 10^{-4}$) were significantly increased in cats with late-stage CKD ($p = 0.011$) and ACKD group ($p < 0.001$), respectively.

Compared with the controls, cats in both early and late CKD stages were older ($p < 0.001$) and had higher BUN ($p < 0.001$) and plasma creatinine ($p < 0.001$) concentrations, along with a lower urine specific gravity (USG; $p < 0.001$) than the control group. Additionally, BUN ($p < 0.001$), plasma creatinine ($p < 0.001$), and phosphate ($p = 0.034$) and WBC ($p = 0.033$) were significantly higher in the late-stage group compared with the early-stage group.

3.4 Linear regression of UGCR and other clinical parameters

Simple regression analysis showed that UGCR was positively correlated with age (weak-to-moderate association), serum creatinine (strong association), WBC (weak association), phosphate (moderate association), and potassium (weak association), and was negatively correlated with hematocrit (moderate association). Among these, serum creatinine exhibited the strongest association ($R^2 = 0.46$). A stepwise multiple regression analysis identified serum creatinine and WBC as independent variables related to UGCR, while the effects of age and other parameters were not significant ([Table 2](#)).

3.5 Subgroup analysis

UGCR was significantly higher in the proteinuric group compared to the non-proteinuric group ($p < 0.001$). Similarly, cats with hypertension exhibited markedly elevated UGCR compared with the non-hypertension group ($p < 0.001$). UGCR was also higher in anemic

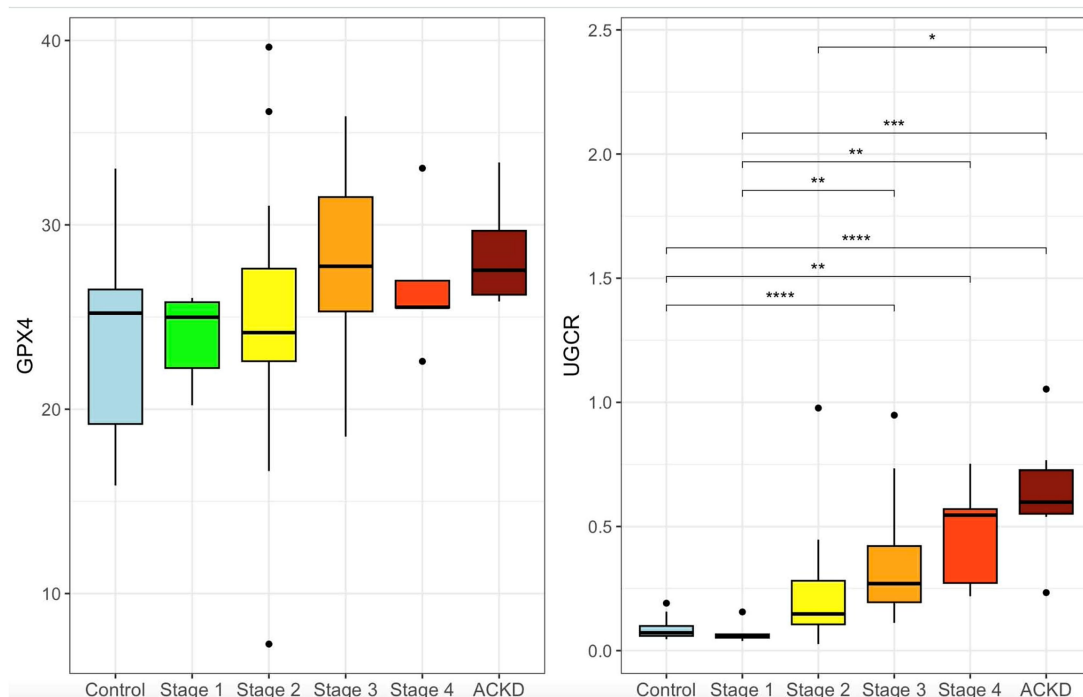


FIGURE 2

Urine GPX4 levels (ng/mL) and GPX4-to-creatinine ratio (UGCR) ($\times 10^{-4}$) in cats with different stages of chronic kidney disease (CKD) and acute-on-chronic kidney disease (ACKD). (A) Urine GPX4 levels showed no significant differences among groups, although a mild decrease was observed in advanced CKD and ACKD. (B) UGCR increased progressively with CKD severity and was highest in ACKD ($p < 0.05$, $p < 0.01$, $p < 0.001$, $p < 0.0001$).

cats than in non-anemic cats ($p = 0.001$). Additionally, UGCR was significantly increased in cats with CKD who had been receiving iron supplementation for at least 1 month ($p = 0.003$) (Figure 3). To evaluate whether CKD stage influenced these findings, cats in both the anemia and non-anemia groups were further classified into the early and late stages of CKD. In the late-stage group, UGCR remained significantly higher in cats with anemia ($p = 0.038$), while in the early-stage group, UGCR showed a non-significant upward trend ($p = 0.067$) (Supplementary Figure S2).

3.6 Assessment of progression

In the cats with CKD that progressed within a 180-day follow-up period, UGCR was significantly higher in the progression group ($p = 0.023$) compared to the non-progression group. The clinical parameters of the progression and non-progression groups are summarized in Table 3. BUN ($p = 0.001$), plasma creatinine ($p = 0.001$), phosphate ($p = 0.004$), potassium ($p = 0.012$), and UPC ($p = 0.004$) were significantly higher in the progression group. Additionally, in the progression group, albumin ($p = 0.02$), hematocrit ($p = 0.007$), hemoglobin ($p = 0.01$), and RBC count ($p = 0.009$) were significantly lower than in the non-progression group. However, in most cases, the hematocrit, hemoglobin, and RBC count were within the reference ranges for both groups.

Elevated UGCR was significantly associated with an increased risk of CKD progression (Hazard ratio [HR], 1.75; 95% CI, 1.20–2.54; $p = 0.003$). Higher BUN (HR, 1.80; 95% CI, 1.43–2.27; $p < 0.001$), serum creatinine (HR, 4.17; 95% CI, 2.29–7.59; $p < 0.001$), and

phosphate (HR, 3.04; 95% CI, 1.69–5.46; $p < 0.001$) were also positively associated with CKD progression. In contrast, lower albumin (HR, 0.33; 95% CI, 0.16–0.66; $p = 0.002$) and hematocrit (HR, 0.19; 95% CI, 0.08–0.43; $p < 0.001$) were negatively associated with progression risk (Figure 4).

ROC analysis demonstrated a modest but statistically significant discriminative performance of UGCR for CKD progression, with an AUROC of 0.707 (standard error 0.075, 95% CI 0.560–0.854, $p = 0.023$) (Supplementary Figure S3). Given the time-dependent nature of CKD progression, further prognostic evaluation focused on survival-based analyses.

A UGCR threshold of 0.2058 was applied to stratify cats into low and high UGCR groups. Kaplan–Meier survival analysis revealed a significant difference in the progression-free interval between the two groups (log-rank $p = 0.0019$), with cats in the high UGCR group showing a shorter time to progression (Figure 5).

4 Discussion

To the best of our knowledge, this is the first study to investigate the relationship between urine GPX4 and the clinical parameters of CKD in cats. We observed that there was a significant increase in UGCR as renal function deteriorated at the late stages of CKD in cats, and that this was correlated with serum creatinine concentration. It is important to note that—this trend differs from human studies, which have reported reduced GPX4 expression in renal tissue biopsies of individuals with CKD (16). A similar reduction in renal GPX4 has also been documented in mouse models of fibrotic kidney disease

TABLE 1 Parameters of interest and clinical characteristics of the control group, early CKD group, and late CKD group.

Parameters	Control (N = 15)	CKD groups (N = 61)				p value
		Early-stage CKD ^a (IRIS stages 1–2; N = 33)		Late-stage CKD ^b (IRIS stages 3–4; N = 28)		
GPX4 (ng/mL)	25.21 (18.99–26.91)	24.31 (22.00–24.07)	N = 33	26.89 ^{c,d} (25.11–31.66)	N = 28	0.011
UGCR (×10 ^{−4})	0.072 (0.057–0.101)	0.134 (0.070–0.260)	N = 33	0.271 ^{c,d} (0.197–0.457)	N = 28	<0.001
Age (years)	1.7 (1.0–6.0)	8.0 (4.0–12.0) c	N = 33	13.5 ^c (7.5–15.5)	N = 28	<0.001
Gender (% Female)	73.3	48.6	N = 33	42.9	N = 28	0.148
BUN (mg/dL)	21 (20–28)	35 (25–48)	N = 33	42 ^{c,d} (35.5–60.5)	N = 28	<0.001
Creatinine (mg/dL)	1.5 (1.3–1.6)	2.1 ^c (1.7–2.4)	N = 33	3.5 ^{c,d} (3.3–4.3)	N = 28	<0.001
Hematocrit (%)	37.2 (33.6–45.5)	39.3 (36.2–43.2)	N = 33	36.6 (28.6–42.0)	N = 28	0.092
Hemoglobin (g/L)	13.5 (11.5–15.7)	13.4 (12.5–14.5)	N = 33	12.5 (9.85–14.2)	N = 28	0.073
RBC (10 ⁶ /μL)	8.59 (6.92–9.51)	8.32 (7.65–9.22)	N = 33	7.7 (6.27–8.81)	N = 28	0.036
MCV (fl)	46.7 (42.2–51.4)	34.2 (33.2–34.8)	N = 33	34.3 (32.8–35.2)	N = 28	0.987
MCHC (g/dL)	34.1 (33.7–35.7)	34.2 (33.2–34.8)	N = 33	34.3 (32.8–35.2)	N = 28	0.890
PLT (K/μL)	305 (235–377)	271 (220–356)	N = 33	273 (189–390)	N = 28	0.868
WBC (/μL)	7,500 (4,800–9,000)	7,000 (5,600–9,200)	N = 33	10,000 ^{c,d} (6,950–11,250)	N = 28	0.046
Albumin (g/dL)	3.6 (3.3–3.8)	3.2 (3.1–3.6)	N = 30	3.3 (2.9–3.6)	N = 23	0.058
Phosphate (mg/dL)	-	4.2 (3.6–4.6)	N = 24	4.5 (4.0–7.1)	N = 28	0.034
Sodium (mEq/L)	155.7 (154.7–157.6)	154.4 (153.4–156.3)	N = 33	155.5 (153.6–156.9)	N = 28	0.101
Potassium (mEq/L)	3.80 (3.63–3.87)	3.75 (3.51–3.90)	N = 33	3.90 (3.60–4.28)	N = 28	0.306
Chloride (mEq/L)	115.7 (114.4–117.7)	116.9 (114.9–118.6)	N = 33	116.8 (115.7–119.1)	N = 28	0.607
ALT (U/L)	43.0 (34.0–55.0)	45.0 (34.0–64.3)	N = 20	42.5 (30.3–150.3)	N = 10	0.912
USG	1.046 (1.041–1.053)	1.021 ^c (1.013–1.035)	N = 33	1.012 ^{c,d} (1.009–1.017)	N = 28	<0.001
UPC	0.05 (0.02–0.06)	0.07 (0.04–0.23)	N = 33	0.30 ^c (0.08–0.63)	N = 26	0.001
Urine pH	6.22 (5.93–6.91)	6.03 (5.86–6.36)	N = 33	5.93 (5.51–6.18)	N = 28	0.103

Numerous variables were tested with Kruskal Wallis Test and represented as the median (IQR). The post hoc analysis was Dunn's test. Category variables were tested with χ^2 test.

^aIRIS stage 1 and stage 2.

^bIRIS stage 3 and stage 4.

^cAchieved significance with the control group.

^dAchieved significance with the early stage group.

GPX4, Glutathione peroxidase 4; UGCR, urine-GPX4-to-creatinine ratio; BUN, blood urea nitrogen; RBC, red blood cell; MCV, mean corpuscular volume; MCHC, mean corpuscular hemoglobin concentration; PLT, platelet count; WBC, white blood cell; ALT, alanine aminotransferase; USG, urinary specific gravity; UPC, urine protein-to-creatinine ratio.

Phosphate concentrations were not available for the control group because phosphate testing was not routinely performed in clinically healthy cats with normal renal indices and SDMA values.

TABLE 2 Simple and multivariable linear regression analysis of clinical parameters against UGCR.

Parameters	Simple regression analysis			Stepwise multiple regression analysis	
	Adjusted β	p value	R ²	Adjusted β	p value
Age (years)	0.432	<0.001	0.19	0.199	0.065
Creatinine (mg/dL)	0.675	<0.001	0.46	0.557	<0.001
Hematocrit (%)	−0.505	<0.001	0.26		
WBC (/μL)	0.416	<0.001	0.17	0.259	0.019
Phosphate (mg/dL)	0.472	<0.001	0.22		
Potassium (mEq/L)	0.312	0.006	0.097		

GPX4, Glutathione peroxidase 4; UGCR, urine-GPX4-to-creatinine ratio; WBC, white blood cell.

(16–18). These studies suggest that ferroptosis occurs in CKD, as GPX4 is a primary regulator of ferroptosis, and this can lead to the release of large amounts of lipid peroxides, potentially causing changes in the phenotype of tubular epithelial cells (16, 23, 24). Importantly, prior studies have assessed GPX4 primarily in kidney tissue, whereas neither urine GPX4 detection nor GPX4's relationship to tubular GPX4 has been previously investigated. Two hypotheses may be applied to explain the apparent discrepancy observed in relation to our study. First, as renal function declined, tubular epithelial injury and/or ferroptotic stress may lead to the leakage or

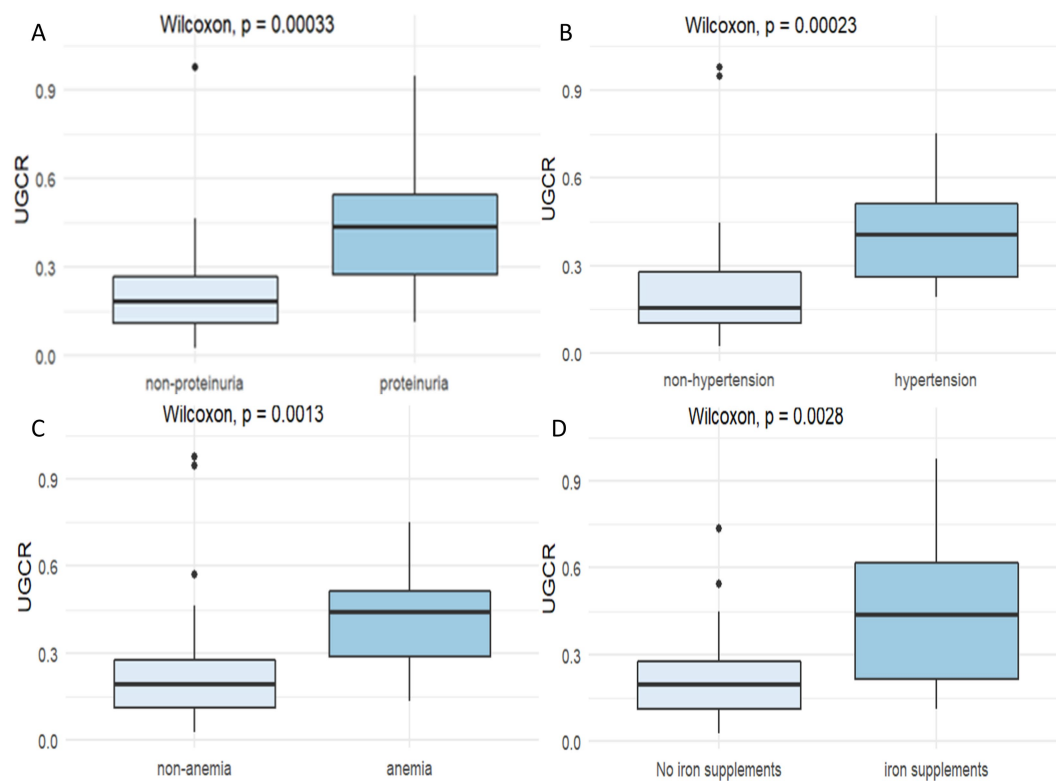


FIGURE 3

Box-and-whisker plots of the urine GPX4-to-creatinine ratio (UGCR) ($\times 10^{-4}$) in the subgroups of cats with CKD. (A) Proteinuric and non-proteinuric groups. (B) Hypertensive and nonhypertensive. (C) Anemic and non-anemic groups. (D) Cats receiving and not receiving iron supplementation. Boxes represent the interquartile range (IQR), the horizontal line within each box indicates the median, and whiskers denote the range. Statistical comparisons between two groups were performed using the Mann–Whitney U test. GPX4, glutathione peroxidase 4; UGCR, urine GPX4-to-creatinine ratio.

TABLE 3 Parameters of interest and clinical characteristics of progression and non-progression groups.

Parameters	Non-progression (N = 33)		Progression (N = 15)		p value
GPX4 (ng/mL)	25.71 [22.85–29.9]	N = 33	26.82 [22.6–30.7]	N = 15	0.764
UGCR ($\times 10^{-4}$)	0.176 [0.105–0.363]	N = 33	0.27 [0.219–0.546]	N = 15	0.023
Age (years)	9 [5–14]	N = 33	13 [8–15]	N = 15	0.378
BUN (mg/dL)	22.5 [17.4–30]	N = 33	60 [41–103]	N = 15	0.001
Creatinine (mg/dL)	2.4 [1.8–3.3]	N = 33	3.7 [2.9–7.6]	N = 15	0.001
Hematocrit (%)	40.2 [36.7–43.9]	N = 31	30.1 [24–39.1]	N = 14	0.005
Hemoglobin (g/L)	13.7 [12.2–14.9]	N = 31	10.3 [8.3–14]	N = 14	0.008
RBC ($10^6/\mu\text{L}$)	8.39 [7.45–9.4]	N = 31	6.64 [5.57–8.36]	N = 14	0.006
WBC ($/\mu\text{L}$)	8,100 [5,850–10,425]	N = 31	8,600 [7,100–11,000]	N = 14	0.608
Albumin (g/dL)	3.2 [3.1–3.7]	N = 22	3.1 [2.8–3.5]	N = 12	0.026
Phosphate (mg/dL)	4.25 [3.7–4.63]	N = 22	6.7 [5.1–9.3]	N = 12	0.005
Sodium (mEq/L)	154.7 (153.4–156.4)	N = 22	155.3 (153.2–156.6)	N = 12	0.818
Potassium (mEq/L)	3.71 [3.47–3.9]	N = 22	4.39 [3.88–4.7]	N = 12	0.009
Chloride (mEq/L)	117.1 [114.9–118.6]	N = 22	117.05 [113.5–118.5]	N = 12	0.0765
UPC	0.13 (0.05–0.35)	N = 31	0.51 (0.2–1.96)	N = 14	0.005

GPX4, Glutathione peroxidase 4; UGCR, urine-GPX4-to-creatinine ratio; BUN, blood urea nitrogen; RBC, red blood cell; MCV, mean corpuscular volume; MCHC, mean corpuscular hemoglobin concentration; PLT, platelet count; WBC, white blood cell; ALT, alanine aminotransferase; USG, urinary specific gravity; UPC, urine protein-to-creatinine ratio.

shedding of intracellular GPX4 into urine, thus resulting in lower tissue expression but higher urinary levels. Therefore, elevated UGCR may reflect the combined effects of renal tubular injury and oxidative

stress, and thus cannot be considered a specific indicator of ferroptosis in isolation. Second, species-specific differences in renal GPX4 biology cannot be excluded, as GPX4 expression has not been

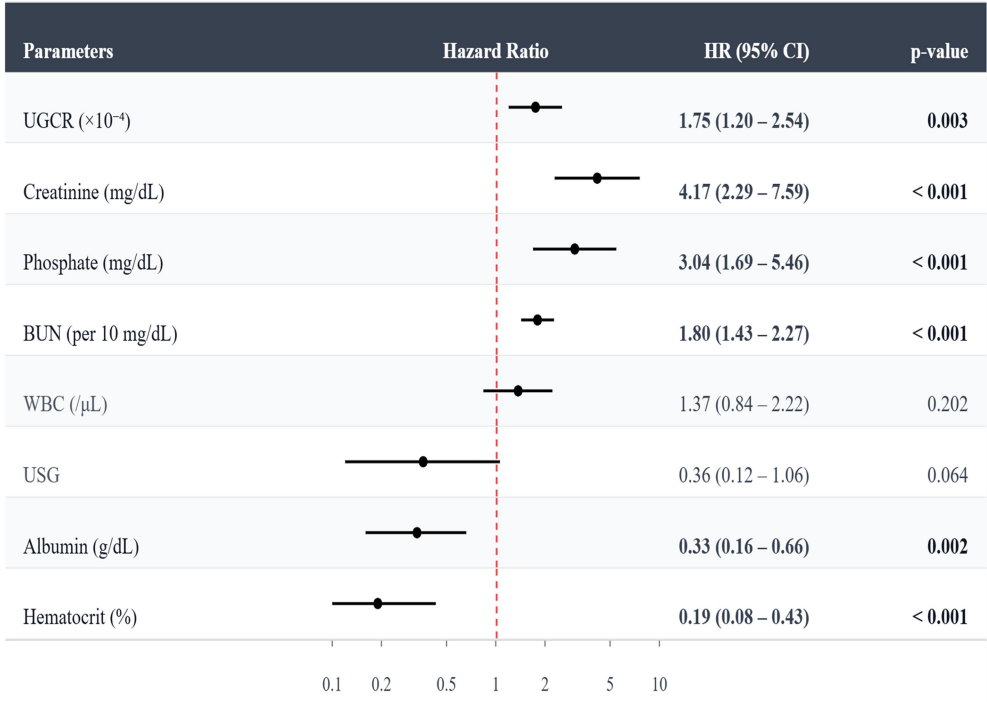


FIGURE 4
Univariable Cox proportional hazards models for predictors of CKD progression. 95% CI, 95% confidence interval; CKD, chronic kidney disease; UGCR, urine-GPX4-to-creatinine ratio; USG, urinary specific gravity; BUN, blood urea nitrogen; WBC, white blood cell count.

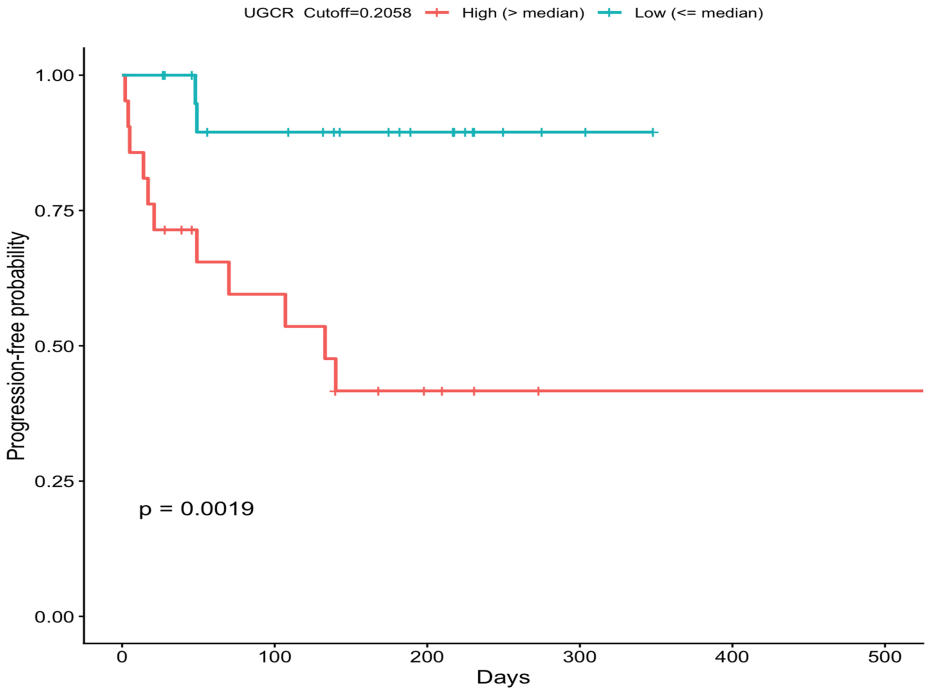


FIGURE 5
Kaplan–Meier curves for the time period to the progression of CKD in cats with UGCR $\leq 0.2058 \times 10^{-4}$ (green line) and $>0.2058 \times 10^{-4}$ (red line); UGCR, urine-GPX4-to-creatinine ratio.

characterized in feline kidneys. Further studies are needed to elucidate the mechanisms underlying these important potential differences.

In the present study, UGCR was significantly higher in the proteinuric group compared to the non-proteinuric group. Previous research has shown that GPX4 plays a reno-protective role in tubular

cells, with GPX4-null mice exhibiting marked increases in tubular cell death and proteinuria (12). Mice with neutrophil-specific GPX4 haploinsufficiency also developed a range of clinical features such as autoantibodies, neutropenia, and proteinuria, and treatment with a specific ferroptosis inhibitor significantly ameliorates disease severity (25). In humans, the gene expression pattern of GPX4 was also found to be significantly correlated with proteinuria in human focal segmental glomerulosclerosis samples (26). Assuming that urine GPX4 level rises as renal function declines, our findings are consistent with these previous studies.

Growing evidence suggests that an increase in reactive oxygen species (ROS) production and oxidative stress may be involved in the pathogenesis of hypertension (27, 28). Oxidative stress has been demonstrated to induce hypertension by activating the renin-angiotensin system (RAAS), promoting renal vasoconstriction, and impairing endothelial function (29). A genome-wide association study in a Spanish population identified GPX4 as a gene linked to CKD-related hypertension. GPX4 has been suggested to be involved in protecting cells against oxidative damage (30), which is consistent with its established protective role. However, this also needs further detailed investigation.

In the current study, UGCR was higher in CKD cats with hypertension compared to their normotensive counterparts. This finding again contradicts previous studies (16–18). However, these earlier studies measured GPX4 directly in kidney tissue, whereas our findings reflect GPX4 excreted into urine. This distinction raises the possibility that declining renal GPX4 content and increasing urine GPX4 levels may represent different facets of the same pathological process, namely that GPX4 depletion from tubular cells is accompanied by its leakage or shedding into urine during oxidative or ferroptotic injury. Nonetheless, species-specific differences between humans, rodents, and cats may also contribute to these divergent patterns, underscoring the need for further comparative studies.

In the present study, UGCR was significantly higher in CKD cats with anemia compared to those without anemia. Experimental evidence supports a link between GPX4 and erythropoiesis. This is because mice with hematopoietic cell-specific GPX4 deletion develop anemia that is characterized by an increased proportion of erythroid precursor cells and reticulocytes (31). Subsequent studies have demonstrated that GPX4 is essential for preventing anemia by inhibiting receptor-interacting protein kinase-3 (RIP3)-dependent necroptosis in erythroid precursor cells (32). These findings highlight the essential role of GPX4 in regulating the survival and maturation of erythroid cells. To date, however, direct evidence relating GPX4 to CKD-related anemia remains limited. Our results are the first to demonstrate an association between urine GPX4 levels and anemia in cats with CKD, although the underlying mechanism remains unclear. Furthermore, UGCR was significantly higher in cats receiving iron supplementation than in those without supplementation. When we separated the cohort into cats receiving iron supplementation and those not, the iron-supplemented group showed a less pronounced increase in UGCR across CKD stages, whereas the group without iron supplementation retained a clearer stage-dependent trend (Supplementary Figure S4). These findings suggest that iron supplementation may act as a potential confounder; however, the observed association between increasing UGCR and CKD severity is not solely attributable to iron supplementation. In a CKD mouse model, iron dextran treatment increased GPX4 transcription in

kidney tissue (33), which suggests that iron therapy may influence GPX4 expression. Because anemic cats are more likely to receive iron supplements, both the underlying anemia and the effects of iron supplementation may contribute to the observed increase in urine GPX4 levels.

WBC count was identified as an independent variable associated with UGCR in our analysis. Previous studies have reported that GPX4 regulates neutrophil survival and inflammation by limiting ferroptosis (25). In addition, GPX4, together with vitamin E, protects hematopoietic stem and progenitor cells from lipid peroxidation, thereby maintaining white blood cell homeostasis (34). The positive association between WBC count and UGCR observed in this study suggests that leukocyte-related processes may influence urine GPX4 levels. However, the underlying mechanism remains unclear and warrants further investigation.

UGCR was highest in cats with ACKD, exceeding the levels observed in both the control group and all CKD subgroups. Previous studies have implicated ferroptosis in the pathogenesis of AKI (35, 36). Experimental studies have shown that mice with GPX4 deletion spontaneously develop AKI (12), whereas GPX4 upregulation prevents renal injury (37). Similarly, overexpression of GPX4 in diabetic rat kidneys was shown to alleviate ferroptosis-induced renal damage (38). These findings indicate that GPX4 plays a protective role in AKI. The markedly increased urine GPX4 levels observed in cats with ACKD may therefore reflect an active renal response involving GPX4 regulation. Nevertheless, the specific relationship between GPX4 in feline urine and AKI in feline CKD remains to be elucidated.

In cats with CKD that have progressed over a 180-day follow-up period, UGCR was significantly higher than in the non-progression group. ROC analysis of UGCR showed statistically significant results in relation to progression, even though the overall discriminative performance was modest. However, the univariate Cox proportional hazard model also showed that high UGCR was linked to a higher hazard ratio of progression; additionally, the survival curve also showed that the cats with high UGCR had a shorter progression time. Increased oxidative stress has been demonstrated to drive the progression of CKD (39). In human studies, GPX4, which is well known as a key antioxidant gene, has also been linked to CKD progression (30, 40). In an animal model of diabetic kidney disease, transcriptional profiling using a PCR array revealed that antioxidant therapy upregulated GPX4 expression, which in turn attenuated disease progression in diabetic rats (41).

Our study has several limitations. First, it was a single-center, retrospective analysis, which may limit the generalizability of the findings to other clinical settings. Misclassification bias and missing data are also possible. Second, the number of cases, especially in the ACKD and advanced CKD groups, was relatively small, which may have reduced the statistical power and increased the possibility of a type II error. Third, due to the retrospective study design, breed information was missing for a subset of cases. While this precludes the evaluation of breed-specific effects, all included cats met rigorous clinical and diagnostic criteria, ensuring the reliability of the renal data. Nonetheless, the influence of breed as a potential confounding factor cannot be entirely ruled out. Additionally, a more detailed analysis of iron supplementation, including the route, dose, and duration of administration, may also be informative but was not feasible with the available sample size. Furthermore, the observational nature of the study precludes conclusions regarding causality between

UGCR and disease progression. Additionally, the possibility of an age-related effect cannot be ruled out without age-matched controls. Lastly, the lack of renal biopsy prevented the pathological confirmation of kidney disease and of an assessment of GPX4 expression in kidney tissue.

In conclusion, UGCR was found to be closely linked to renal function and inflammation status in cats, and showed significant increases in late-stage CKD and even higher levels in those with acute-on-CKD. UGCR was also related to important clinical issues like anemia, proteinuria, and hypertension. These findings suggest that UGCR may serve as a valuable biomarker for assessing CKD severity and may have potential utility for the monitoring disease progression, thus helping clinical risk stratification and ongoing management of feline patients.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Ethics statement

The animal studies were approved by Institutional Animal Care and Use Committee (IACUC) National Taiwan University. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent was obtained from the owners for the participation of their animals in this study.

Author contributions

W-LH: Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. S-HH: Conceptualization, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft. Z-YH: Formal analysis, Methodology, Visualization, Writing – review & editing. V-CW: Data curation, Validation, Visualization, Writing – review & editing. Y-HC: Data curation, Writing – original draft. Y-JL: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fvets.2025.1756038/full#supplementary-material>

SUPPLEMENTARY FIGURE S1

Standard calibration curve for measuring feline urinary GPX4 concentrations by ELISA.

SUPPLEMENTARY FIGURE S2

Box-and-whisker plot of the urine GPX4-to-creatinine ratio (UGCR) in anemia and non-anemia groups, further stratified into early- and late-stage CKD groups.

SUPPLEMENTARY FIGURE S3

ROC curve of UGCR for predicting progression within the 180-day follow-up period. Sen: Sensitivity; Spe: specificity.

SUPPLEMENTARY FIGURE S4

Distribution of cats by CKD stage and iron supplementation status. Cats were categorized according to IRIS CKD stages 1–4 and further stratified by iron supplementation. The number of cats without versus with iron supplementation was as follows: stage 1 (5 vs 0), stage 2 (22 vs 3), stage 3 (15 vs 6), and stage 4 (2 vs 3).

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Evaluation of equation as a classifier for kidney function in domestic cats: a proof-of-concept study

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Abstract

In human medicine, estimated glomerular filtration rate (eGFR) is routinely used to assess renal function. In cats, reliable eGFR formulas are not yet available. To test the hypothesis that a human eGFR equation with four metabolites can effectively distinguish cats with different renal functions, two cohorts of cats were retrospectively enrolled. Cohort 1 included 34 colony cats and cohort 2 had 29 client-owned cats. The metabolite markers were quantified in cat serum and renal function was estimated using the adapted equation (eRF_{fel}). Receiver Operating Characteristic Area Under the Curve for eRF_{fel} as a binary classifier were 0.9714 and 0.9958, and 0.9731 and 0.9643 as a ternary classifier for cohort 1 and cohort 2, respectively. In cohort 1, eRF_{fel} was 1.30 ± 0.04 (Mean $\pm$ SEM, mL/min/kg), 0.78 ± 0.06 , and 0.36 ± 0.05 for healthy control, IRIS Chronic Kidney Disease (CKD) stages 1 and 2 (CKD1/2), and stages 3 and 4 (CKD3/4), respectively, and 1.30 ± 0.06 , 0.72 ± 0.05 , and 0.47 ± 0.04 for healthy control, CKD stage 2, and CKD stage 3 in cohort 2, respectively (both $P < 0.0001$). Although the original equation was trained on human data, eRF_{fel} was associated positively with urine specific gravity, but negatively with serum symmetric dimethylarginine and

creatinine in cats (all $P < 0.0001$). The preliminary results demonstrated the effectiveness of eRF_{fel} as a classifier for feline renal function.

Subject terms: Diseases, Medical research, Nephrology

Introduction

Chronic kidney disease (CKD) is a common disease of cats, with a marked increase in prevalence in older cats aged 12 years or older, and is reported to be the most common cause of death in cats over five years of age^{1,2}. Typically, an initial diagnosis of CKD is made using blood and urine markers (serum creatinine, serum symmetric dimethylarginine [SDMA], urine specific gravity [USG]), followed by staging in accordance with International Renal Interest Society (IRIS) guidelines^{3,4}. In IRIS stage 1 CKD, cats are non-azotemic, with many cats showing no noticeable clinical signs despite already experiencing a significant loss of nephron function^{4,5}. The transition from IRIS stage 2 to stage 4 CKD in cats is characterized by a progressive decline in kidney function, with increasing levels of azotemia and associated clinical signs. While stage 2 typically presents mild or asymptomatic azotemia, stage 4 refers to severe azotemia and is considered end-stage renal failure^{4,5}. Indeed, approximately 75% of functioning nephrons need to be lost before serum creatinine levels rise above the reference interval⁶. Serum creatinine levels are also influenced by non-renal factors, such as lean body mass and body weight⁷. Hall et al. reported that serum SDMA increased earlier than creatinine⁶. Brans et al. were unable to confirm the superiority of SDMA to creatinine as a reliable marker for glomerular filtration rate (GFR).

Although GFR measurement is regarded as the gold standard for assessing renal function, it currently involves the use of an intravenous clearance marker, followed by multiple samplings, a process that is stressful for both pets and their owners⁸⁻¹⁰. Thus, additional diagnostic markers and tools that can aid in the early detection and monitoring of kidney dysfunction are crucial for effective management of feline CKD.

In human medicine, estimated GFR (eGFR) measurements are routinely made to assess renal function, as an alternative to using exogenous markers to measure GFR directly. These eGFR assessments are often made using equations based on concentrations of serum creatinine, cystatin C or a combination of both¹¹. Unfortunately, exploration of similar creatinine-based eGFR formulas for cats, have not been shown to be reliable¹². Over recent years, many human studies have evaluated metabolomic approaches to the investigation of CKD both to understand its pathogenesis better and also to find alternative assays that will improve the existing eGFR assays, which still show bias^{11,13-21}. Among these, Freed and colleagues²¹ developed a human machine

learning model for eGFR using measurements of pseudouridine (Ψ), phenylacetylglutamine (PAGln), N-acetylthreonine (AcThr), and tryptophan (Trp), and a large dataset of CKD patients. The authors demonstrated their assay was more accurate than the eGFR assays based on either creatinine or cystatin C alone. Pseudouridine (Ψ) is synthesized through isomerization of RNA uridine residues^{22,23}, and has been reported to be a biomarker for kidney function and CKD progression in humans¹³⁻¹⁸. PAGln and AcThr are derivatives of the amino acid phenylalanine and threonine, respectively, with PAGln recognized as a gut microbiota-produced uremic toxin associated with the risk of cardiovascular diseases, cerebrovascular disorders, and CKD in humans²⁴⁻²⁷. Studies have also associated AcThr, Ψ and C-glycosylated Trp with renal function decline and diabetes^{19,20}.

Standard methods for quantification of circulating Ψ , PAGln, and AcThr have been developed in humans^{21,26}, but were not so far been applied to cats. As a proof-of-concept study, we wanted to apply these assays, along with a modification of the eGFR equation derived for their use by Freed and colleagues²¹, to see if their measurement in feline serum would be able to distinguish between healthy cats, cats with early and late stages CKD. We hypothesized that the adapted model would effectively discriminate between cats with different levels of renal function and that based on this, further investigations would be warranted.

Results

Physical and clinical characteristics of cats

Cats in cohort 1 were all domestic shorthair from the same colony. The mean age of the healthy control group was younger than that of IRIS stage 1 and stage 2 CKD (CKD1/2, $P = 0.002$), while the mean body weight of the healthy control group was greater than those of both CKD1/2 and CKD3/4 (IRIS stage 3 and 4 CKD) group. ($P = 0.001$) (Table [1](#)). No significant association was observed between sex and CKD group ($P = 0.42$).

Table 1.

Physical and clinical descriptions of cats in cohort 1

	Healthy	CKD1/2	CKD3/4	P value
Sample size	14	14	6	n/a
Age (years)	10.9 ± 0.7 ^a	15.4 ± 0.7 ^b	11.7 ± 0.6 ^{ab}	0.002
Sex (F/M)	9/5	7/7	5/1	0.42
Weight (kilograms)	4.7 ± 0.3 ^a	3.6 ± 0.2 ^b	2.9 ± 0.3 ^b	0.001
Breed				
DSH	14	14	6	n/a
Clinical measurements				
Serum creatinine (mg/dL)	1.36 ± 0.06 ^a	1.69 ± 0.11 ^a	3.80 ± 0.50 ^b	<0.001
Serum SDMA (µg/dL)	11.0 ± 0.6 ^a	17.0 ± 1.0 ^b	31.0 ± 2.4 ^c	<0.001
USG*	1.050 ± 0.002 ^a	1.015 ± 0.002 ^b	1.011 ± 0.001 ^b	<0.001
UPC*	0.33 ± 0.19 ^a	0.35 ± 0.06 ^a	2.35 ± 1.14 ^b	0.002

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CKD1/2, IRIS stages 1 ($n = 2$) and 2 ($n = 12$) CKD; CKD3/4, IRIS stages 3 ($n = 4$) and 4 ($n = 2$) CKD. DSH, domestic short-haired; SDMA, symmetric dimethylarginine; UPC, urine protein to creatinine ratio; USG, urine specific gravity. P values were obtained from ANOVA for continuous variables or Fisher's Exact Test for categorical variables. Different superscripted letters denote statistical difference between groups. Age, weight, and clinical variables are reported as mean ± SEM. All cats were neutered.

*Urine sample from one CKD3/4 cat was not collected.

Cats in cohort 2 were client-owned. The mean age of the healthy control group was younger than those of IRIS stage 2 CKD (CKD2) and IRIS stage 3 CKD (CKD3) ($P = 0.003$) (Table 2). No difference in body weight or sex was observed between groups ($P = 0.15$ and $P = 0.47$, respectively).

Table 2.

Physical and clinical descriptions of cats in cohort 2

	Healthy	CKD2	CKD3	P value
Sample size	9	10	10	n/a
Age (years)	13.0 ± 0.8 ^a	17.7 ± 0.6 ^b	16.2 ± 1.1 ^b	0.003
Sex (F/M)	6/3	4/6	4/6	0.47
Weight (kilograms)	3.93 ± 0.23	3.38 ± 0.15	3.84 ± 0.23	0.15
Breed				
DSH	9	6	7	n/a
DLH	0	2	1	n/a
Birman	0	1	0	n/a
Devon Rex	0	1	0	n/a
BSH	0	0	1	n/a
Tiffany	0	0	1	n/a
Clinical measurements				
Serum creatinine (mg/dL)	1.27 ± 0.06 ^a	1.93 ± 0.08 ^b	3.56 ± 0.23 ^c	<0.001
Serum SDMA (µg/dL)	10.8 ± 0.8 ^a	17.9 ± 1.5 ^b	21.3 ± 0.9 ^b	<0.001
USG	1.045 ± 0 ^a	1.018 ± 0 ^b	1.015 ± 0 ^b	<0.001
UPC	0.15 ± 0.02	0.36 ± 0.10	0.48 ± 0.33	0.91

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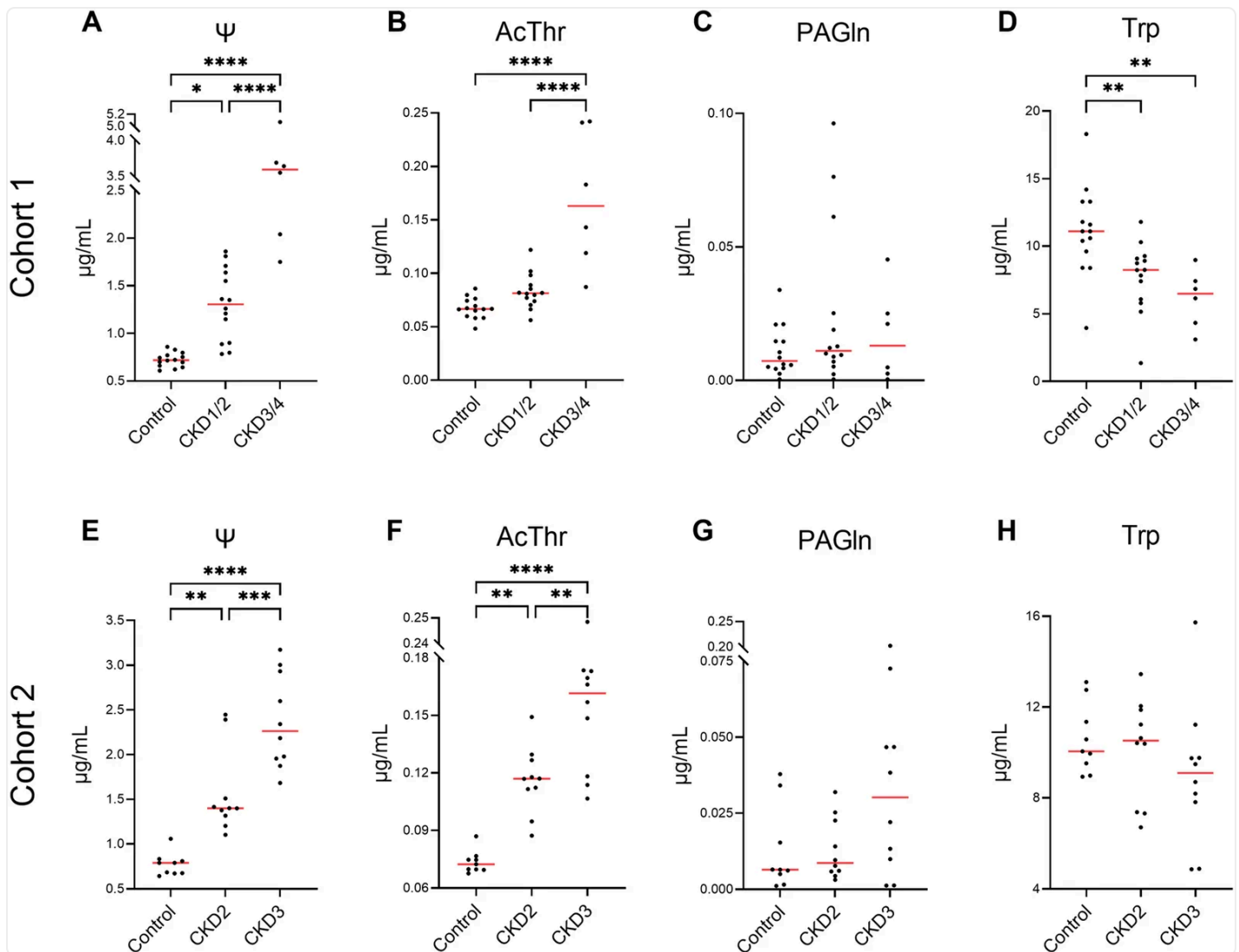
CKD2, IRIS stage 2; CKD3, IRIS stage 3. DSH, domestic short-haired; DLH, domestic long-haired; BSH, British short-haired. SDMA, symmetric dimethylarginine; UPC, urine protein to creatinine ratio; USG, urine specific gravity. *P* values were obtained from ANOVA for continuous variables or Fisher's Exact Test for categorical variables. Different superscripted

letters denote statistical difference between groups. Age, weight, and clinical variables are reported as mean $\pm$ SEM. All cats were neutered.

Quantification of metabolite markers

In cohort 1, the median (IQR) concentrations of Ψ ($\mu\text{g/mL}$) for control, CKD1/2, and CKD3/4 groups were 0.72 (0.66, 0.78), 1.31 (0.90, 1.66), and 3.60 (1.97, 4.04), respectively ($P < 0.0001$, Fig. [1A](#)). Similarly, the median concentrations of AcThr ($\mu\text{g/mL}$) for control, CKD1/2, and CKD3/4 groups were 0.07 (0.06, 0.07), 0.08 (0.07, 0.09), and 0.16 (0.11, 0.24), respectively ($P < 0.0001$, Fig. [1B](#)). No difference in PAGln between groups was observed ($P = 0.25$, Fig. [1C](#)). To the contrary, the concentration of Trp was decreased in CKD1/2 and CKD3/4 compared to the control ($P = 0.0011$, Fig. [1D](#)).

Fig. 1. Quantifications of metabolites.



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The concentrations of pseudouridine (Ψ), acetylthreonine (AcThr), phenylacetylglutamine (PAGln), and tryptophan (Trp) in cohort 1 **A-D**, and cohort 2 **E-H**, respectively. The horizontal red lines represent the medians. The means among groups were compared using ANOVA test followed by Tukey's HSD multiple comparisons. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$. CKD1/2, IRIS stage 1 and stage 2 CKD; CKD3/4, IRIS stage 3 and stage 4 CKD.

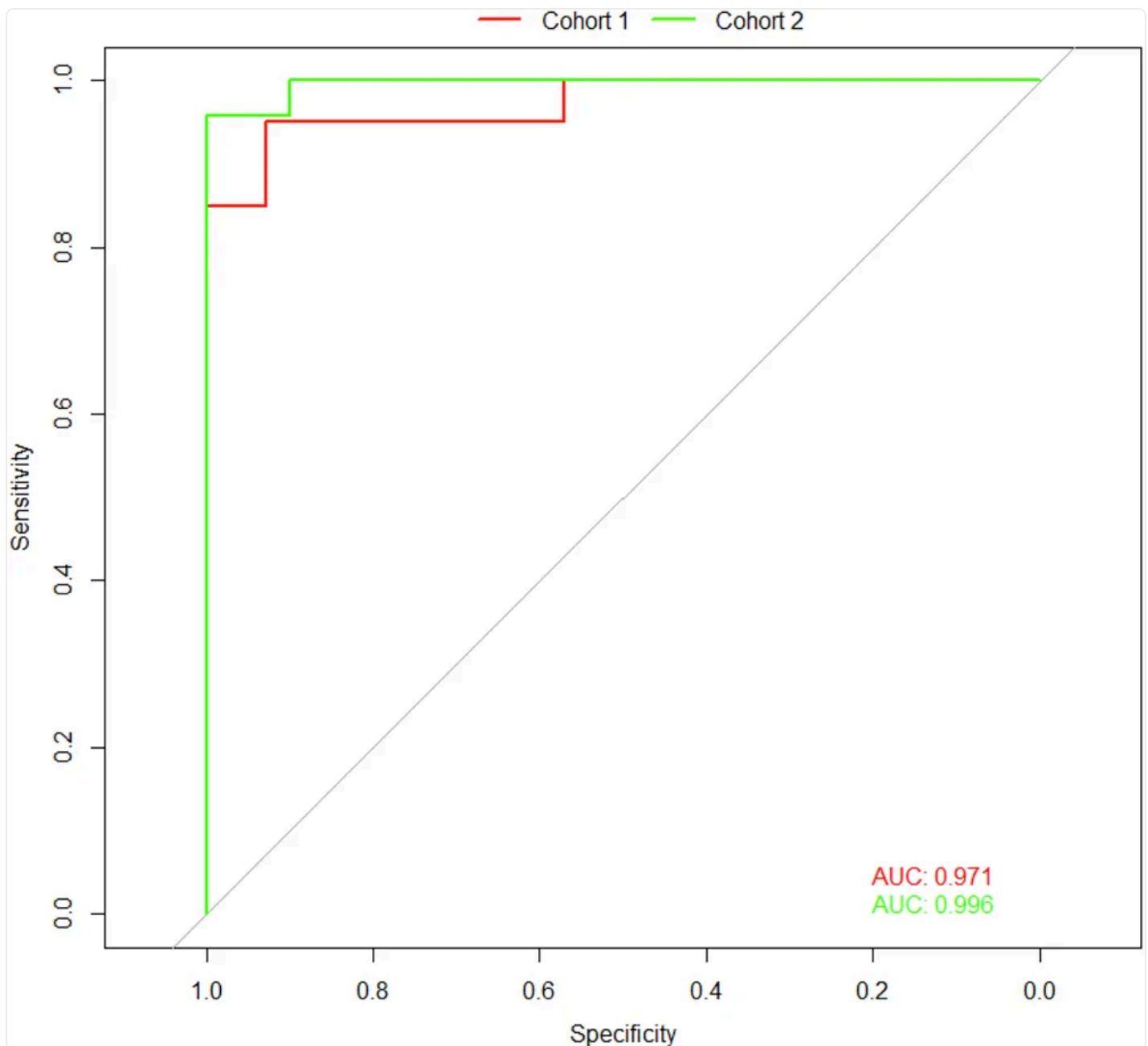
Similar observations were made in cohort 2. The median concentrations of Ψ ($\mu\text{g/mL}$) for control, CKD2 and CKD3 were 0.79 (1.29, 1.73), 1.40 (1.29, 1.73), and 2.26 (1.93, 2.95), respectively ($P < 0.0001$, Fig. [1E](#)). The levels of AcThr were also higher in both CKD groups compared to the control group ($P < 0.0001$, Fig. [1E](#)). No difference in PAGln or Trp between groups was observed ($P = 0.09$, $P = 0.37$ respectively, Fig. [1G-H](#)).

The concentrations of Ψ , AcThr, PAGln, Trp, and the estimated renal function (eRF_{fel} as defined in the Method) of the healthy non-CKD cats in cohort 1 were not different from those in cohort 2 (Supplemental Table [S1](#)).

Performance of eRF_{fel} as a classifier

The overall effectiveness of eRF_{fel} as a binary classifier to correctly separate cats as either being healthy or CKD was assessed using the Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) analysis. The ROC AUC values were 0.9714 (95% CI: 0.9235-1) for cohort 1 and 0.9958 (0.9843-1) for cohort 2 (Fig. [2](#)).

Fig. 2. Receiver Operating Characteristic plot and Area Under the Curve.



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Receiver Operating Characteristic (ROC) curves were generated for cohort 1 (red) and cohort 2 (green). A ROC curve is a graphical representation of true positive rate (sensitivity) against the false positive rate (1-specificity) at various thresholds. The Area Under the Curve (AUC) of a ROC curve is a metric used to evaluate the performance of a machine

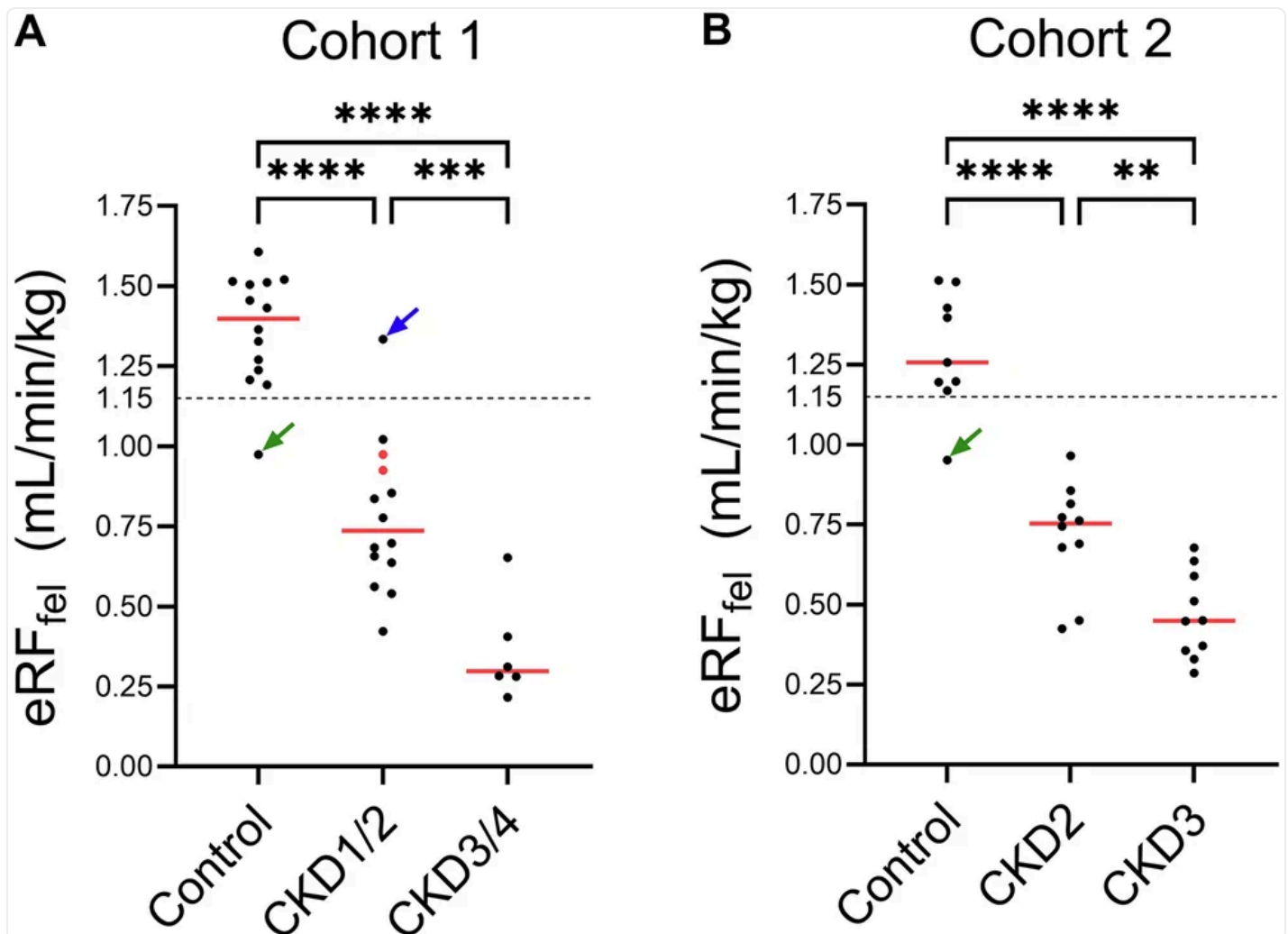
learning model. An AUC of 1.0 indicates perfect classifications, while an AUC of 0.5 (grey diagonal line) indicates no discrimination. A value above 0.9 is considered excellent.

Further, the overall performances of eRF_{fel} as a ternary classifier to correctly classify cats in cohort 1 (healthy, CKD1/2, and CKD3/4) and in cohort 2 (healthy, CKD2, and CKD3) were 0.9731 and 0.9643, respectively.

Values obtained by eRF_{fel}

In cohort 1, the median eRF_{fel} values (mL/min/kg) of the control, CKD1/2, and CKD3/4 groups were 1.34 (1.23, 1.51), 0.74 (0.62, 0.94), and 0.30 (0.26, 0.47), respectively ($P < 0.0001$, Fig. [3A](#)). The eRF_{fel} values of all CKD cats were < 1.15 (below the reference interval for feline GFR) except for one, whose eRF_{fel} was 1.33 (Fig. [3A](#), blue arrow). The eRF_{fel} values of all healthy cats were > 1.15 except one, whose eRF_{fel} was 0.97 (Fig. [3A](#), green arrow). The median eRF_{fel} value of the CKD1/2 group was greater than that of CKD3/4 group ($P < 0.001$). Notably, the eRF_{fel} values of the two cats with IRIS stage 1 CKD were 0.97 and 0.93 (Fig. [3A](#), red circles).

Fig. 3. Values obtained by eRF_{fel} .



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The eRF_{fel} values of **A** healthy control, CKD1/2, and CKD3/4 groups in cohort 1 and **B** healthy control, CKD2, and CKD3 groups in cohort 2. The horizontal red lines in the boxplots represent the medians. The two IRIS stage 1 CKD cats the CKD1/2 group of cohort 1 are labeled in solid red circles. The green arrows in **A**, **B** indicate healthy cats whose eRF_{fel} values were outside the reference range (**A**, creatinine=1.34 mg/dL and SDMA = 16 μ g/dL; **B**, creatinine=1.43 mg/dL and SDMA = 12 μ g/dL). The blue arrow in **A** indicates an IRIS stage 2 CKD cat whose eRF_{fel} value was within the reference range (creatinine=1.61 mg/dL and SDMA = 21 μ g/dL). The greyish dotted line, $y = 1.15$ mL/min/kg represents the lower limit of the reference range proposed by the Veterinary Diagnostic Laboratory at

Michigan State University. Group means were compared using the ANOVA test followed by Tukey's HSD multiple comparisons. * $P < 0.05$, ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$.

For client-owned cats in cohort 2, the median values of eRF_{fel} (mL/min/kg) for the control, CKD2, and CKD3 groups were 1.3 (1.2, 1.5), 0.75 (0.62, 0.83), and 0.45 (0.35, 0.60), respectively ($P < 0.0001$) (Fig. [3B](#)). The eRF_{fel} of healthy cats were within the reference ranges for feline GFR (> 1.15 mL/min/kg) except for one, whose eRF_{fel} was 0.95 (Fig. [3B](#), green arrow).

Association between eRF_{fel} and physical and clinical features

In cohort 1, eRF_{fel} showed negative correlations with SDMA and creatinine (Spearman correlation coefficient $\rho = -0.85$ and -0.68 respectively, both $P < 0.0001$), but a positive one with USG ($\rho = 0.82$, $P < 0.0001$) (Supplemental Fig. [S1A](#), Supplemental Fig. [S2A-C](#)). Similarly, significant associations between eRF_{fel} and SDMA, creatinine, and USG were observed in cohort 2 ($\rho = -0.81$, -0.89 , and 0.75 , respectively, all $P < 0.0001$) (Supplemental Fig. [S1B](#), Supplemental Fig. [S2D-F](#)).

In cohort 1, eRF_{fel} values were negatively associated with age (Spearman $\rho = -0.38$, $P = 0.028$) and positively associated with weight ($P < 0.0001$, $\rho = 0.67$) (Supplemental Fig. [S1A](#)). In cohort 2, a negative correlation was observed between eRF_{fel} and age ($\rho = -0.56$, $P = 0.002$), but there was no association with body weight (Supplemental Fig. [S1B](#)).

Discussion

We applied a validated machine learning model developed to estimate GFR in humans^{[21](#)} to two cohorts of cats, by introducing a “feline factor” to account for species-specific differences in kidney functions, the differences in measurement units and reference ranges between humans and domestic cats. The adapted equation (eRF_{fel}), which uses four metabolite markers as predictor variables, was used to assess renal function in cats. But from our current data, it cannot be assumed to provide an estimate of feline GFR.

The two cohorts of cats, one from a research colony and the other consisting of client-owned cats, were retrospectively enrolled using stored serum samples to assess eRF_{fel} equation's performance as a classification model to differentiate between healthy cats and cats with CKD and between cats with different severities of CKD. Our results confirmed our hypothesis and demonstrated the potential for using these metabolites in a classification model in cats. While the results of our

study provide evidence to support the potential value of assaying these metabolites in the diagnosis and staging of feline CKD, larger prospective studies incorporating directly measured GFR for comparison will be required to fully assess their potential and to establish whether they can be used to estimate GFR in cats. Longitudinal data on healthy senior cats and those suspected of early-stage (i.e. IRIS stage 1 and early stage 2) CKD, will also enhance our understanding of these metabolite assays. Wider metabolomic studies in such cats may also yield other metabolite markers that may be of interest in feline CKD, as has been shown in humans¹⁴⁻¹⁹.

One of the key differences between the model used by Freed and colleagues and previous ones is that it does not depend on creatinine levels²¹. Although creatinine is widely recognized as a standard surrogate marker for renal function, its levels remain within the normal interval until approximately 75% of nephrons are nonfunctioning⁶. In addition, creatinine production can be influenced by non-renal factors such as lean muscle mass. Pseudouridine (Ψ), a nucleoside whose levels are not dependent on muscle mass, has been implicated as a biomarker for kidney function and CKD progression¹³⁻¹⁸. In cats, elevated blood Ψ levels have also been reported in CKD, and more broadly, in kidney diseases²⁸. Pseudouridine is the most abundant posttranscriptional modifications of cellular RNAs, and is produced via Ψ synthases through isomerization of uridine residues that are already part of an RNA chain^{22,23}. Although Ψ was discovered more than a century ago, its enzymatic mechanism and function of this modification are still not fully elucidated. Phenylacetylglutamine and AcThr are both amino acid derivatives. PAGln is gut microbiota-derived uremic toxin that has been associated with the risk of cardiovascular diseases, cerebrovascular disorders, and CKD in humans²⁴⁻²⁷. Studies have associated AcThr, Ψ and C-glycosylated tryptophan (Trp) with renal function decline in human patients with type 1 diabetes with impaired renal function¹⁹ and in patients with diabetic kidney disease²⁰. To date, the absolute concentrations of Ψ , AcThr, or PAGln in the circulation of cats have not been reported. Trp, an essential amino acid for cats, is primarily metabolized through three pathways: the indole pathway, the serotonin pathway, and the predominant kynurenine pathway. Our study suggests that post-transcriptional and post-translational alterations of these metabolites may potentially contribute to the pathophysiology of CKD in cats. One of the exciting potential benefits of the metabolomic approach to exploring markers of CKD is that they may yield both more accurate data for eGFR, but also may provide better diagnostics for the advancement of CKD, as some metabolites may infer a greater risk of progressive disease^{13-15,18,25,26,29}.

The reference interval for serum iothelol clearance for cats provided by the Veterinary Diagnostic Laboratory at Michigan State University (MSU) is 1.15-2.73 mL/min/kg. Heiene et al. showed a slightly wider reference range of 1.0-3.5 mL/min/kg for iothelol in 51 clinically normal, non-azotemic cats⁸. The eRF_{fel} values of 93% (13/14) of the healthy cats in cohort 1 and 89% (8/9)

healthy cats in cohort 2 fell within the reference interval for eGFR by MSU, while eRF_{fel} of 95% (20/21) of the CKD cats in cohort 1 and all CKD cats in cohort 2 were below the interval. Significantly, the eRF_{fel} values of the two IRIS stage 1 CKD cats in cohort 1 were close to the lower threshold of the normal ranges proposed by MSU and Heiene and colleagues. Because cats in the study were retrospectively enrolled, measured GFRs were not available. An important next step would be to understand the accuracy of this eRF_{fel} equation by comparing it with measured GFR. Additionally, evidence of the ability to predict renal function decline in comparison to current markers will strengthen its potential for future clinical uses.

There were some important differences between the two cohorts of cats used in this study. The cats in cohort 1 lived in the same environment in North America, whilst cats in cohort 2 lived in different homes in Europe with different environmental variables. In cohort 1, cats with CKD had significant increases in UPC, while no difference in UPC was observed in cohort 2. Similar to human studies, we were unable to rule out cats with subclinical conditions, which might confound our classification of the cats. Fasting blood samples were collected in cohort 1, but not cohort 2. Diet composition is considered as one of the key factors that influence the circulating metabolome in humans³⁰ and pets³¹. These differences in environment and fasting conditions could potentially contribute to the variations found in the measured metabolites. Diet and medications are other potential confounding factors. Cats with CKD are usually on special renal diets designed to support the kidney function. Many CKD cats experience inappetence and body weight loss. So dry or wet or both diet types are offered to encourage food consumptions. Moreover, some CKD cats were on medications to regulate high BP or suppress hyperthyroidism.

We further explored the relationships between eRF_{fel} and some key physical and clinical features associated with feline CKD. Although the equation was trained on human data and the algorithm did not have any feline-related knowledge, the eRF_{fel} showed a strong correlation with creatinine, SDMA, and USG. Additionally, the equation was associated to a lesser extent with age and body weight. We believe that the association with these physical, biochemical and clinical features is a reflection of the health status of these cats. For example, weight was not significantly different between the groups in cohort 2, and similarly its correlation with eRF_{fel} became insignificant. Hence, we considered the observation as an encouraging sign, rather than as a source of concern for confounding effects. Nevertheless, the performance of this equation as an estimator for feline renal function and against the current renal markers such as creatinine have yet to be determined.

In conclusion, this study has demonstrated that evaluating the combination of four metabolites previously shown to have collectively provided a good assessment of eGFR in humans, was able to provide good distinction between healthy cats and cats with different stages of CKD. This strongly

suggests that further studies should be undertaken to evaluate these metabolites as potential markers for eGFR in cats and potential markers of progressive disease in this species.

Methods

Prediction model for feline renal function

A multivariate machine learning model developed and trained for human eGFR²¹ was modified and adapted for cats. The model depends on four metabolite markers, Ψ , AcThr, PAGln, and Trp, as predictor variables.

The reference ranges for serum iohexol clearance for cats provided by Michigan State University Veterinary Diagnostic Laboratory (<https://cvm.msu.edu/vdl> ²¹) are 1.15-2.73 mL/min/kg. In humans, a GFR of 90 mL/min/1.73 m² or higher is considered normal according to the guidelines from the National Kidney Foundation. A “feline factor” (f) was therefore introduced, based on these GFR values, to modify the original equation for use in cats.

Original eGFR equation reported by Freed and colleagues²¹:

$$\text{eGFR}_{\text{met}} = 28.106 \times \Psi^{-0.805} \times \text{PAGln}^{-0.047} \times \text{AcThr}^{-0.075} \times \text{Trp}^{0.266}$$

New estimated feline renal function (eRF_{fel}) formula used here:

$\text{eRF}_{\text{fel}} = f \times (28.106 \times \Psi^{-0.805} \times \text{PAGln}^{-0.047} \times \text{AcThr}^{-0.075} \times \text{Trp}^{0.266})$ Where the “feline factor” f is a constant, $f = \frac{1.15}{90} = 0.0128 \text{ m}^2/\text{kg}$, and Ψ , PAGln, AcThr, and Trp represent β -pseudouridine, phenylacetylglutamine, N-acetylthreonine, and tryptophan, respectively. All metabolites are expressed in $\mu\text{g/mL}$.

Animals

In this study, cohort 1 were colony cats, and cohort 2 were client-owned cats, both being enrolled retrospectively^{32,33}. In cohort 1, there were 34 colony cats, all domestic short-hair, including 14 healthy non-CKD cats (control group), 14 with early-stage CKD (CKD1/2 group, including 2 IRIS stage 1 and 12 IRIS stage 2 cats), and 6 with late-stage CKD (CKD3/4 group, including 4 IRIS stage 3 and 2 IRIS stage 4) (Table [1](#)). Cohort 2 consisted of 29 client-owned cats, including 9 healthy

control cats, 10 IRIS stage 2 (CKD2) and 10 IRIS stage 3 cats (CKD3) (Table [2](#)). All cats were neutered.

All cats received a complete physical examination with a body condition score^{[34](#)}, muscle condition score^{[35](#)}, medical history evaluation, complete blood count, serum biochemistry profile including SDMA, indirect systolic blood pressure (sBP), and urinalysis including urine protein to creatinine ratio (UPC). A cat was diagnosed with CKD if USG < 1.035, and had elevated serum creatinine or SDMA levels on at least two different assessments, where the medical history, and physical examination findings supported the diagnosis. Cats diagnosed with CKD with regulated hyperthyroidism and hypertension were allowed to enroll. Cats diagnosed with CKD were then staged using the IRIS guidelines. A cat was assigned IRIS stage 1 if its serum creatinine concentrations was <1.6 mg/dL, otherwise IRIS stages 2-4 as appropriate. A cat was considered healthy if the following criteria were met: (1) USG $\geq$ 1.035, (2) serum SDMA < 18 μ g/dL, (3) serum creatinine <1.6 mg/dL, (4) sBP <160 mmHg, and (5) no clinical evidence of kidney disease, urinary tract infection or any other major systemic diseases such as cancer, heart disease, or diabetes mellitus at the time of examination. Cats within each cohort were evaluated by the same board-certified veterinary internists. The study protocol was reviewed and approved by the Institutional Animal Care and Use Committee of the Nestlé Purina PetCare Company with the approval number 7427. An owner consent was obtained for all client-owned cats.

Sample collection

Overnight fasting blood samples were collected from the colony cats that lived in North America (cohort 1). Client-owned cats that lived in Europe (cohort 2) were not fasting at the time of blood draw. Serum samples were immediately stored at -80°C until use.

LC/MS quantification of metabolites

Cat serum samples were analyzed by liquid chromatography-tandem mass spectrometry (LC-MS/MS) method. Serum metabolites of cohort 1 were quantified by a commercial laboratory (Metabolon Inc., Morrisville, NC). Serum samples from cohort 2 were quantified at the Analytical Resources Core at Colorado State University (Fort Collins, CO), and the detailed protocol was described below.

Reference standards N-acetyl-L-threonine was purchased from Santa Cruz Biotechnology (Dallas TX), β -pseudouridine and phenylacetyl L-glutamine were from Cayman Chemical (Ann Arbor, MI),

tryptophan was from Sigma-Aldrich (St.Louis, MO), phenylacetyl-d₅-L-glutamine and β-pseudouridine-¹³C,¹⁵N₂ were from LGC (Manchester, NH), N-acetyl-d₃-L-threonine-2,3-d₂ was from CDN Isotopes (Quebec, Canada), tryptophan-¹³C₁₁ was from Cambridge Isotopes (Tewksbury, MA). All solvents mentioned elsewhere in section 4.4. were purchased from Fisher Scientific (Hampton, NH). To extract the metabolites, serum samples (50 μL) were mixed with ice-cold methanol (200 μL) and internal standard mix (10 μL containing 600 ng/mL of N-acetyl-d₃-L-threonine-2,3-d₂, 80 ng/mL of phenylacetyl-d₅-L-glutamine, 3000 ng/mL of β-pseudouridine-¹³C,¹⁵N₂, 400 ng/mL of tryptophan-¹³C₁₁ in 50% methanol/water). The mixture was vortexed for 3 s, incubated at -20°C overnight, followed by centrifugation at 15,000 *g* x 10 min and 4°C. The supernatant was analyzed on a Waters ACQUITY Classic UPLC coupled to a Waters Xevo TQ-S triple quadrupole mass spectrometer (Milford, MA). A quality control (QC) sample that was pooled from study samples was injected after every six study samples. Chromatographic separations were carried out on a Waters Atlantis Premier BEH Z-HILIC VanGuard FIT column (2.1 ×100 mm, 1.7 μM). Mobile phases were (A) 10 mM ammonium formate in water with 0.1% formic acid and (B) acetonitrile with 0.1% formic acid. The LC gradient was as follows: time=0 min, 80% B; time=0.5 min, 80% B; time=1.6 min, 35% B; time=1.61 min, 5% B; time=3.61 min, 5% B; time=3.62 min, 80% B; time=6 min, 80% B. Flow rate was 0.4 mL/min and injection volume was 2 μL. Samples were held at 6°C in the autosampler, and the column was operated at 40°C. Mass detector was operated in ESI positive mode with the capillary voltage set to 0.7 kV. Inter-channel delay was set to 3 msec. Source temperature was 150°C and desolvation gas (nitrogen) temperature 500°C. Desolvation gas flow was 1000 L/h, cone gas flow was 150 L/h, and collision gas (argon) flow was 0.25 mL/min. Nebulizer pressure (nitrogen) was set to 7 Bar. The MS acquisition functions were scheduled by retention times. The retention time, multiple reaction monitoring (MRM) transitions, dwell time, cone energy and collision energy of each compound were described in Supplemental Table [S2](#). Data were processed using Skyline open source software^{[36](#)}. Peak areas of target compounds detected in biological samples were normalized to the peak areas of the corresponding internal standard. The linear fit of calibration curve was used to extrapolate the concentrations of target compounds.

Receiver Operating Characteristic (ROC) Area Under the Curve (AUC)

The overall performance of the eRF_{fel} as a binary classifier to discriminate between healthy cats and cats with CKD and a ternary classifier to distinguish among three classes (e.g. healthy non-CKD, CKD1/2, and CKD3/4) were assessed using ROC curves. The ternary AUC is defined as the mean of several pairwise AUC scores using the Hand and Till method^{[37](#)}. The ROC AUC was calculated as a performance matrix using the statistical package *pROC* in R^{[38](#)}.

Statistical analysis

Within each cohort, group means were compared using ANOVA, followed by Tukey's HSD post-hoc multiple comparisons to control the type I error. To determine whether a statistically significant association existed between two categorical variables, Fisher's exact test was performed. Metabolite concentrations were presented as medians with interquartile ranges (IQR). Between cohorts, the mean concentrations of the four metabolites and eRF_{fel} in healthy non-CKD cats were compared using the Welch's t-test. Spearman's correlation analysis was performed to examine the associations between eRF_{fel} , age, body weight, and key renal markers SDMA, creatinine, and USG. Data analysis and visualization were performed in statistical software R (version 4.4.1) and GraphPad Prism (version 10.4.0).

Supplementary information

[Supplemental Files](#) (357.8KB, pdf)

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Author contributions

QL developed the hypothesis, designed the experiments, performed data acquisition, analysis and visualization, and wrote the manuscript. LY developed the targeted assays and performed LC/MS quantification of the metabolites. Both authors reviewed the manuscript and consented for publication.

Data Availability

The datasets generated and analyzed during the current study are not publicly available due to pet privacy concerns but are available from the corresponding author on reasonable request.

Competing interests

QL is an employee of Nestlé Research. QL has a patent pending for the equation eRF_{fel} (Application No. 19/299108), but has no financial interest in the outcome of this study or financial gain from the pending application. LY declared no competing interests.

Footnotes

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Supplementary information

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

[Supplemental Files](#) (357.8KB, pdf)

Data Availability Statement

The datasets generated and analyzed during the current study are not publicly available due to pet privacy concerns but are available from the corresponding author on reasonable request.

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Increased serum amyloid A and urinary kidney injury marker - 1 concentrations in cats with pyelonephritis: a diagnostic accuracy study

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Abstract

Objectives

To investigate the discriminatory potential of serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and assess renal tubular injury in cats with pyelonephritis.

Materials and Methods

Diagnostic accuracy study using samples from prospectively enrolled cats and stored material. Cats were classified according to diagnosis as pyelonephritis, ureteral obstruction, chronic kidney disease, subclinical bacteriuria, cystitis, or healthy. Serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and urinary kidney injury marker-1 were compared across groups. Receiver operating characteristic curve analysis assessed the performance of inflammatory

biomarkers. Serum amyloid A and urinary kidney injury marker-1 were measured consecutively in a subset of cats with pyelonephritis.

Results

Samples from 71 cats (pyelonephritis $n = 22$, other urological conditions $n = 35$, healthy $n = 14$) were included. Median serum amyloid A concentration was significantly higher in cats with pyelonephritis, overlapping minimally with other groups. At a threshold of 49.1 mg/L, serum amyloid A showed a positive likelihood ratio of 32, sensitivity of 0.95 (95% CI: 0.74 to 0.99) and specificity of 0.97 (95% CI: 0.85 to 0.99), outperforming neutrophil:lymphocyte ratio. Serum amyloid A normalised in 12/14 cats post-treatment. Urinary kidney injury marker-1 was elevated in more than 50% of cats with pyelonephritis and ureteral obstruction, significantly higher than in healthy cats and those with subclinical bacteriuria or cystitis. Variation in urinary kidney injury marker-1 was observed during follow-up.

Conclusions

At a threshold of 49 mg/L, serum amyloid A is a highly accurate biomarker for differentiating pyelonephritis from other urological conditions. Elevated urinary kidney injury marker-1 suggests frequent proximal tubular injury.

Clinical Significance

Pyelonephritis is challenging to diagnose. Serum amyloid A can improve diagnostic precision and support timely treatment decisions. Urinary kidney injury marker-1 helps identify tubular injury; however, the clinical implications remain to be investigated.

INTRODUCTION

Pyelonephritis (PN) is an infection of the renal pelvis and parenchyma, most commonly resulting from ascending bacteria originating in the lower urinary tract. Like other urinary tract infections in cats, PN is mostly seen in elderly female cats, and underlying kidney disease appears to be common (Whitehouse et al., [2024](#)). Despite its potential severity, the clinical presentation of PN can be non-specific, and the diagnosis is rarely straightforward (Weese et al., [2019](#)). To obtain a diagnosis with certainty, a positive culture on urine obtained by pyelocentesis is required, a technically challenging procedure beyond routine reach. Thus, in most clinical settings, the

diagnosis relies on a positive bladder urine culture coupled with clinical, laboratory and/or ultrasonographic signs suggestive of upper urinary tract infection.

Unfortunately, PN shares numerous characteristics with other urological disorders, especially chronic kidney disease (CKD) and ureteral obstruction (UO), resulting in considerable diagnostic overlap. Additionally, cases involving coexisting urological conditions, such as subclinical bacteriuria (SBU) or cystitis in cats with CKD, are not uncommon and are particularly complex to differentiate from PN.

Serum amyloid A (SAA) is a major acute-phase protein which increases rapidly in response to an inflammatory stimulus. Serum amyloid A is produced in the liver, and increases are detectable in blood within 8 hours (Rossi, [2023](#)). High magnitude increases of SAA have been demonstrated in cats with inflammatory conditions such as upper respiratory disease, pneumonia, pyometra, feline infectious peritonitis and sepsis (Troia et al., [2017](#); Yuki et al., [2020](#)). Recently, significant increases in SAA concentrations were also demonstrated in 17 cats with pyelonephritis when compared to cats with other urological conditions (Kurtz et al., [2024](#)), suggesting that SAA might be a promising discriminatory biomarker in this population of cats.

Neutrophilia, monocytosis and lymphopaenia are common haematological findings in cases of systemic inflammation. Neutrophil:lymphocyte ratio (NLR) and monocyte:lymphocyte ratio (MLR) can be considered as indicators of the immune-inflammatory response. They are easily calculated ratios, combining cellular parts of the innate and adaptive immune system, with the components available from a routine complete blood count (CBC) as performed by most automated laboratory haematological analysers. Alterations of NLR and MLR have been demonstrated in cats with various inflammatory and infectious conditions (Donato et al., [2023](#); Gori et al., [2021](#)), and leukocyte ratios could potentially serve as discriminatory markers in cats with PN.

Kidney injury marker-1 (KIM-1) is a scavenger receptor of the proximal tubular epithelial cells and a highly specific marker of ischaemic or toxic acute kidney injury in people, cats and rodents. KIM-1 is well conserved across species, and the KIM-1 molecule has been identified in urine and renal tissue of cats with confirmed and suspected acute kidney injury (Bland et al., [2014](#); Bland et al., [2017](#)). The extent of tubular damage in cats with PN has not been investigated.

The primary aim of this study was to investigate and assess the discriminatory potential of selected non-invasive inflammatory biomarkers, SAA, NLR and MLR. We hypothesised that cats with PN would have systemic inflammation and thus increased concentration of SAA as well as

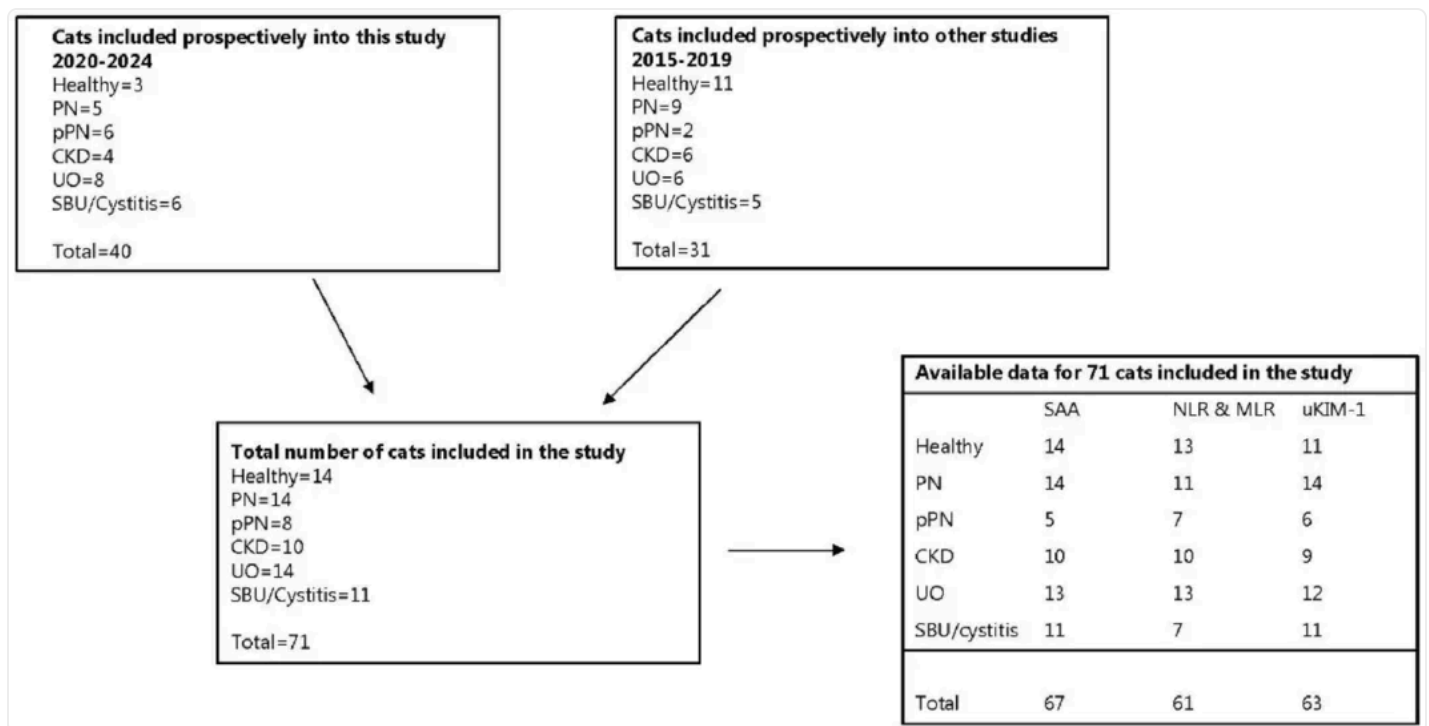
higher NLR and MLR than cats with other urological conditions, such as UO, CKD, cystitis or SBU or a combination of these. We also hypothesised that SAA would decrease in response to therapy in cats with PN. The secondary aim was to investigate renal tubular injury in cats with PN and to assess whether urinary KIM-1 (uKIM-1) can serve as a biomarker to identify cats with PN, discriminating these from cats with other urological conditions. We hypothesised that cats with PN and cats with UO are at risk for proximal tubular injury and thus would have higher uKIM-1 concentrations compared to healthy cats, cats with bacterial infection (cystitis) or colonisation (SBU) of the lower urinary tract, and cats with CKD.

MATERIALS AND METHODS

Study design and enrolment of cats

The study was designed as a diagnostic accuracy study, with the reporting following the STARD criteria (Table [S1](#)). Cats were either prospectively enrolled into this study from 2020 to 2024 or enrolled into previous studies at the University Hospital for Companion Animals (UHCA) at the University of Copenhagen, Denmark from 2015 to 2019 (Godia et al., [2022](#); Jessen et al., [2020](#); Langhorn et al., [2019](#)) with stored urine and blood samples and medical record data included in the present study (Fig [1](#)). Cats with urological conditions were a convenience sample consisting of cats presenting to the UHCA or Blue Star Animal Hospital in Gothenburg (BSAH), Sweden, and fulfilling the criteria described below, whereas healthy control cats were recruited for the purpose of research. Recruitment took place at the UHCA, except for three cats that were recruited from the BSAH. Ethical permission was obtained from the Ethical Committee at the UHCA. Informed consent was obtained from owners of all cats participating in this study.

FIG 1.



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Number of cats included and data available for analysis of serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and urinary kidney injury marker-1. CKD Chronic kidney disease; PN Pyelonephritis; pPN Presumptive pyelonephritis; SBU Subclinical bacteriuria; UO Ureteral obstruction.

Classification of cats

Cats were classified as PN (confirmed PN) based on a positive urine culture obtained by pyelocentesis. Cats were classified as pPN (presumptive PN) based on clinical AND/OR clinicopathological findings suggestive of PN in combination with a positive urine culture obtained by cystocentesis. Clinical and clinicopathological findings suggestive of PN were defined as (i) signs of systemic inflammation such as pyrexia AND/OR unspecific signs of systemic illness such as lethargy/inappetence/vomiting AND/OR haematological signs of systemic inflammation such as neutrophilia, neutropenia, presence of band neutrophils AND (ii) signs of renal involvement (renal pain AND/OR azotaemia). In previous studies aimed at evaluating the microRNAome in cats

with pyelonephritis (Godia et al., [2022](#); Jessen et al., [2020](#)), increased acute phase proteins were part of the inclusion criteria for cats with pPN. Cats with pPN included before 2020 were, therefore, excluded from the evaluation of SAA as a predictive marker in this study.

Cats were classified as having other urological conditions if diagnosed with either CKD, UO, cystitis, SBU, or a combination of these. A diagnosis of CKD required stable azotaemic renal disease (defined as serum creatinine above 174 $\mu\text{mol/L}$) with a urine specific gravity (USG) below 1.035, or degenerative changes on ultrasonography, and a negative cystocentesis urine bacterial culture. For cats that were retrospectively included from previous prospective urological research projects, a diagnosis of CKD made by the researchers of the previous study was considered acceptable if the medical record was not available for evaluation. UO (complete or partial) was diagnosed if cats had uni- or bilateral ureteral obstructive disease on ultrasonographic evaluation with a concurrently negative pyelocentesis urine bacterial culture. Cats were diagnosed with cystitis when presented with clinical signs of lower urinary tract infection (pollakiuria, dysuria, stranguria, periuria and/or haematuria) in the presence of a positive urine culture obtained by cystocentesis, whereas SBU was recorded if a positive cystocentesis urine culture was obtained in the absence of these clinical signs. Finally, clinically healthy cats ≥ 1 year of age were classified as healthy based on an unremarkable clinical examination and clinicopathological findings (complete blood count, biochemistry and urinalysis). For all cats, it was an exclusion criterion if the cats had been receiving antibiotics ≤ 72 hours before evaluation for study inclusion.

Diagnostic work - up

Diagnostic work-up of animals with PN, pPN and cats with other urological conditions was performed according to the discretion of the attending clinician, except for post-treatment SAA and uKIM-1 analysis, which was performed for the purpose of the study.

For each cat, a CBC, serum biochemistry, urinalysis and urine culture were performed at the Veterinary Diagnostic laboratory, University of Copenhagen, Denmark. Quantitative aerobic urine culture was performed at SUND Vet Diagnostik, University of Copenhagen, Denmark and at the BSAH diagnostic laboratory, Sweden. Bacterial species identification was performed by matrix-assisted laser desorption/ionisation-time of flight (MALDI-TOF) mass spectrometry (Vitek MS RUO; BioMérieux). For cats with PN and UO, abdominal ultrasonography was also performed, including the collection of urine by pyelocentesis in case of pyelectasia (>2 mm dilation) for aerobic culture.

NLR and MLR

A CBC was analysed at the time of inclusion. Depending on location and time of admission, analysis was performed with either the Advia 2120i (Siemens Healthineers, Erlangen, Germany) or the Procyte R (IDEXX Laboratories, Inc., Westbrook, Maine). Neutrophilia was defined as $>12.6 \times 10^9/L$ using either instrument, and left shift was defined as an increased band neutrophil count $>0.3 \times 10^9/L$ (confirmed microscopically). Neutrophil:lymphocyte and monocyte:lymphocyte ratios were calculated for each cat.

Serum amyloid A

Serum amyloid A was measured at the time of inclusion as part of the biochemical profile. The analysis was run within 48 hours of sample collection, except for the three cats from BSAH, for which SAA was measured in serum stored at -80°C for 9 years after collection. In cats with PN, SAA was measured again within 1 week of treatment cessation. During treatment, SAA measurement was not part of the protocol but could be performed at the discretion of the attending clinician. The concentration of SAA was analysed at the Veterinary Diagnostic Laboratory at the UCPH (LZ-SAA (Eiken chemicals A/S, Tokyo, Japan) on Advia 1800, SIEMENS, before June 26, 2022, then VET-SAA (Eiken chemicals A/S, Tokyo, Japan) on Atellica analyser, SIEMENS). Increased SAA was defined as a concentration $> 5 \text{ mg/L}$ using either instrument. The microbiologists performing cultures were blinded to the result of the SAA analysis, whereas attending clinicians and investigators were not blinded.

Urinary KIM - 1

Excess urine was aliquoted in 1 mL plastic vials and stored at -80°C . Samples were batched and shipped on ice with courier to the University of Guelph, Canada. Sample identity was disguised before shipping. For analysis, urine samples were thawed and briefly centrifuged. A $20 \mu\text{L}$ aliquot of urine was mixed with $400 \mu\text{L}$ of buffer, and $150 \mu\text{L}$ of this mixture was applied to a feline-specific lateral flow assay. After 15 minutes of incubation at room temperature, the KIM-1 of the test sample and of the positive control impregnated into the membrane was measured with a handheld radiofrequency identification reader (Bland et al., [2019](#)). The ratio of the test to control sample optical density was calculated and normalised to a USG of 1.035.

Statistical analysis

Statistical analyses were performed using MedCalc® Statistical Software version 22.016 (MedCalc Software Ltd, Ostend, Belgium; <https://www.medcalc.org>; 2023) and GraphPad Prism 10.0.3.

Statistical significance was set at $P < .05$. Tests for normality and log normality were performed using D'Agostino and Pearson tests.

Continuous variables (SAA, NLR, MLR) are presented as medians with 25th and 75th percentiles. A Kruskal-Wallis test was applied to detect statistically significant differences between groups of cats, followed by Dunn's *post hoc* analysis. In one cat with pPN and in three cats with other urological disorders, SAA concentrations were below the lower reporting limit (<5 mg/L) of the VET-SAA assay, and the SAA concentrations were assigned a value of 0.

Receiver operating characteristic analysis was employed to calculate the area under the curve (AUC) of SAA, NLR and MLR to differentiate between cats with PNs (confirmed and presumptive) and cats with other urological conditions. The 95% confidence interval (95% CI) was calculated. To obtain estimates of the optimal cut-off value for SAA, LNR and MLR, the cut-off value associated with the maximum value of the Youden index ($\text{sensitivity}_c + \text{specificity}_c - 1$; with c representing all possible cut-off values) was assigned as the optimal cut-off value. Because the optimal cut-off value associated with the maximum Youden index relies on a disease prevalence of 50%, giving equal weight to sensitivity and specificity and ignoring costs of various decisions, the optimal cut-off value was also estimated using the cost approach in MedCalc. Briefly, sensitivity (SE), specificity (SP) and positive likelihood ratio (LR+) were calculated for disease prevalences of 10%, 33% and 50% and assuming costs of false positive and false negative of 1 compared to costs of true positive and true negative of 0.

RESULTS

A total of 71 cats were included in the study (Fig [1](#)). Demographic data, key clinical features and routine laboratory data indicative of systemic inflammation and renal involvement are listed in Tables [1](#) and [2](#), respectively. Comorbidities and underlying abnormalities in cats participating in the study are listed in Table [S2](#). Stratified data for cats with other urological conditions are shown in Table [S3](#).

Table 1.

Demographic data for the 71 included cats

	PN	pPN	Other	Healthy cats
<i>n</i>	14	8	35	14
Breeds	Abyssinian (1)	Birman (3)	Abyssinian	DSH (10)
	Bengal (1)	Burmilla (1)	(1)	Norwegian forest cat (2)
	Birman (1)	DSH (4)	Birman (2)	Ragdoll (2)
	DSH (5)		British shorthair (1)	
	Maine Coon (2)		DLH (2)	
	Oriental (2)		DSH (20)	
	Ragdoll (1)		Maine Coon mix (2)	
	Siberian forest cat (1)		Maine Coon (2)	
			Norwegian forest cat (1)	
			Ocicat (1)	
			Oriental (1)	
			Ragdoll (1)	
			Unknown (1)	

	PN	pPN	Other	Healthy cats
Age (years)	8 (2 to 14)	8.5 (4 to 17)	8 (1 to 17)	6 (2 to 16)
Sex	FE (2)	FE (2)	FE (4)	FE (2)
	FN (12)	FN (4)	FN (14)	FN (7)
	ME (0)	ME (0)	ME (0)	ME (1)
	MN (0)	MN (2)	MN (17)	MN (4)

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Ages are presented as median (range)

FE Female entire, FN Female neutered, ME Male entire, MN Male neutered, PN

Pyelonephritis, pPN Presumptive pyelonephritis

Table 2.

Key features of 14 cats with confirmed PN, 8 cats with pPN, 35 cats with other urological diseases and 14 healthy cats

	PN	pPN	Other	Healthy cats
Abdominal pain	7/13, 53.8%	4/8, 50%	5/34, 14.7%	0/14, 0%
Rectal temperature	39.3 (37.4 to 40.4)	39.25 (37.4 to 40.5)	38.1 (37.1 to 39.5)	38.5 (37.6 to 39.2)
Pyrexia (<i>n/n</i> , %)	8/13, 61.5%	3/8, 37.5%	2/19, 10.5%	0/5, 0%
Creatinine ($\mu\text{mol/L}$) (range)	204 (102 to 735)	276.5 (125 to 419)	197 (62 to 1470)	120 (86 to 133)
Creatinine > 140 ($\mu\text{mol/L}$) (<i>n/n</i> , %)	12/13, 92.3%	7/8, 87.5%	25/35, 71.4%	0/14, 0%
Azotaemia (<i>n/n</i> , %)	8/13, 62%	6/8, 75%	20/35, 57%	0/14, 0%
USG	1024.5 (1006 to 1051)	1018.5 (1013 to 1050)	1020.5 (1010 to 1052)	1050 (1042 to 1050)
Neutrophils	14.9 (11.8 to 66.27)	18.22 (4.99 to 33.66)	5.67 (1.56 to 11.57)	4.19 (1.63 to 6.76)
Neutrophilia (<i>n/n</i> , %)	10/11, 91%	5/8, 62.5%	0/32, 0%	0/15, 0%
Left shift (<i>n/n</i> , %)	6/11, 54%	4/7, 57.1%	0/29, 0%	0/15, 0%
Uropathogens	<i>E. coli</i> (11) <i>Enterococcus faecalis</i> (1) (this cat also had <i>E. coli</i> in the bladder)	- <i>E. coli</i> (5)	<i>E. coli</i> (5) <i>Staphylococcus felis</i> (1)	

	PN	pPN	Other	Healthy cats
	<i>Staphylococcus felis</i> (1) <i>Staphylococcus</i> spp. (1)	Haemolytic and non-haemolytic <i>E. coli</i> (1) <i>Enterococcus faecalis</i> (1) <i>Enterococcus</i> spp. (1)	<i>Staphylococcus aureus</i> subspecies <i>aureus</i> (1) <i>Staphylococcus</i> spp. (1) <i>Streptococcus</i> spp. (1) <i>Micrococcus luteus</i> (1) <i>Bacillus</i> (1)	

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Azotaemia is defined as creatinine >17 µmol/L. Numerical variables are presented as median values (range). Pyrexia is defined as $T \geq 39.2^{\circ}\text{C}$. USG > 1050 was recorded as 1050 and USG < 1020 was recorded as 1019 in the calculation of the median USG Urine specific gravity

SAA, NLR and MLR

The median SAA concentration was significantly higher in cats with PN (180.3 mg/L) and pPN (607.4 mg/L) when compared to healthy cats (0.0 mg/L) and cats with other urological conditions (0.4 mg/L) (Table 3). There was minimal overlap of SAA values between groups of cats (Fig 2A).

Table 3.

Serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio measurements in healthy cats, cats with other urological conditions, cats with pyelonephritis and cats with presumptive pyelonephritis

	PN	pPN	Other	Healthy cats
SAA (mg/L)	<i>N</i> = 14 Median: 180.3 ₋ [*] (167.2 to 205.8)	<i>N</i> = 5 Median: 607.4 ₋ [*] (134.9 to 690.5)	<i>N</i> = 34 Median: 0.4 (0.0 to 3.1)	<i>N</i> = 14 Median: 0.0 (0.0 to 0.2)
NLR	<i>N</i> = 11 Median: 15.20 ₋ [*] (6.36 to 22.75)	<i>N</i> = 7 Median: 11.85 (3.63 to 25.75)	<i>N</i> = 30 Median: 2.33 (1.85 to 5.72)	<i>N</i> = 13 Median: 1.66 (1.52 to 2.54)
MLR	<i>N</i> = 11 Median: 0.31 ₋ [*] (0.12 to 0.65)	<i>N</i> = 7 Median: 0.16 ₋ [*] (0.15 to 0.63)	<i>N</i> = 30 Median: 0.08 (0.06 to 0.23)	<i>N</i> = 13 Median: 0.07 (0.04 to 0.10)

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SAA, NLR and MLR are presented as median (25th and 75th percentile). Median values between groups were statistically assessed by the Kruskal–Wallis test with Dunn's *post hoc* test

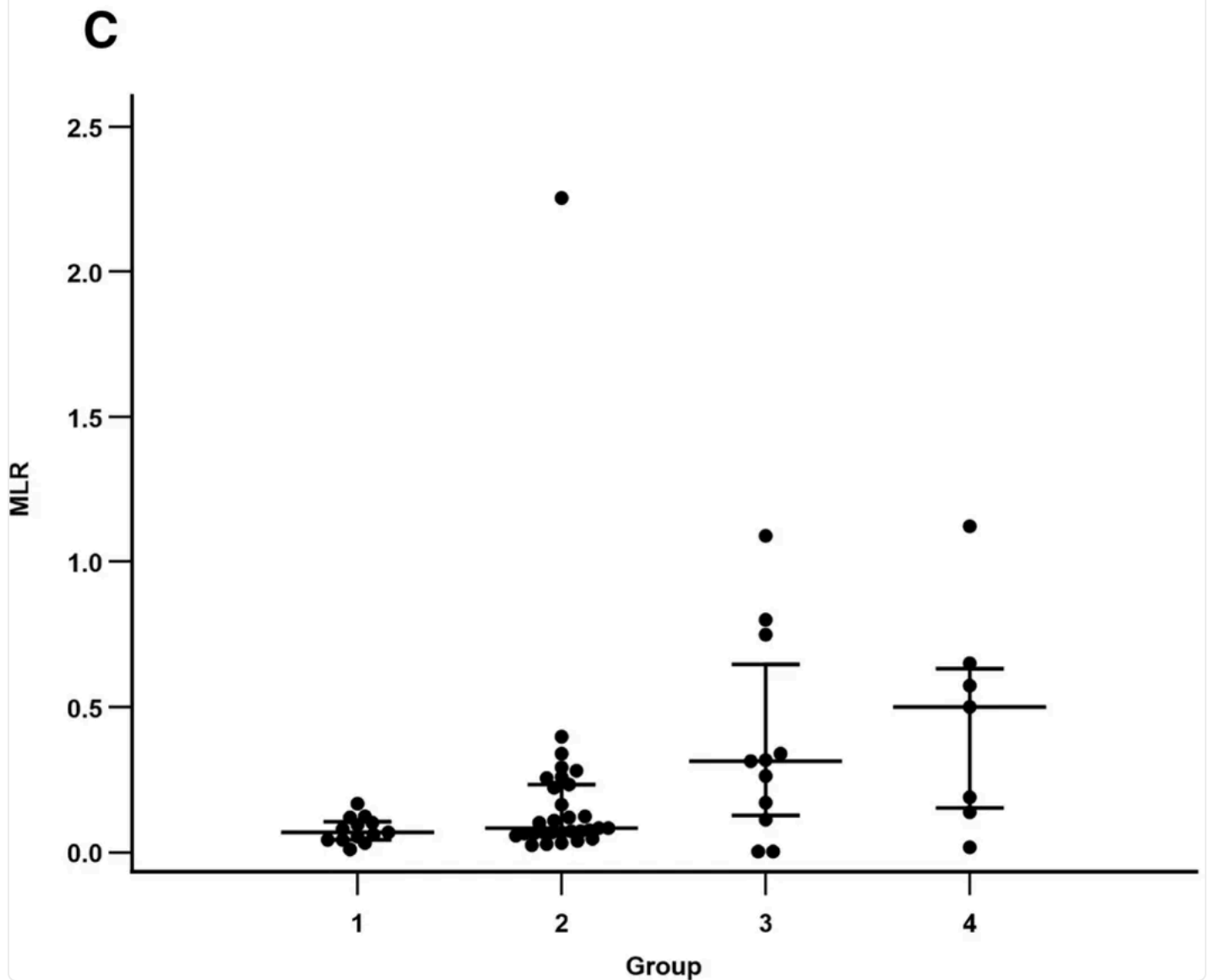
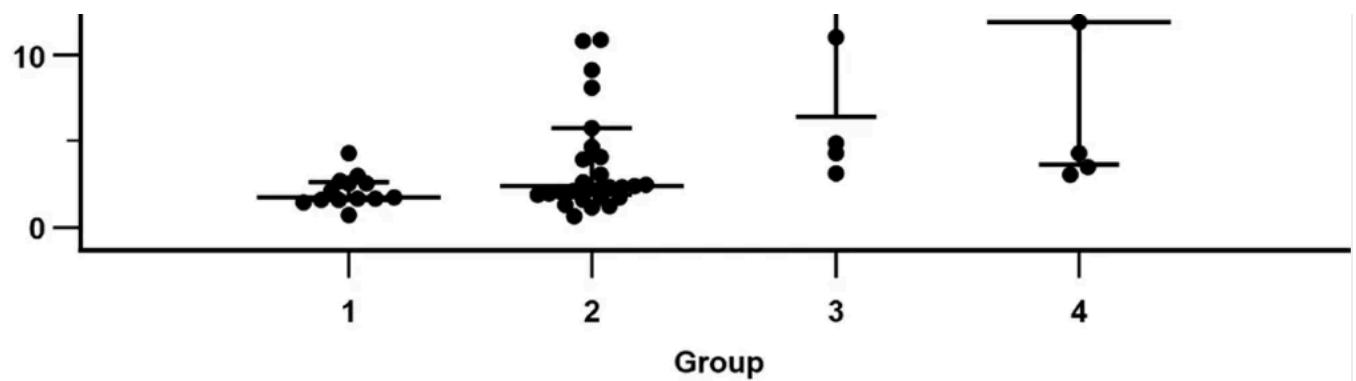
MLR Monocyte:lymphocyte ratio, *N* Number of cats, NLR Neutrophil:lymphocyte ratio, PN Pyelonephritis (confirmed), pPN Presumptive pyelonephritis, SAA Serum amyloid A

*Median concentrations of SAA in cats with confirmed pyelonephritis were significantly higher than those of healthy cats and cats with other urological conditions ($P < .001$).

Median concentrations of SAA in cats with presumptive pyelonephritis were significantly

higher than those of healthy cats ($P = .004$) and cats with other urological conditions ($P = .021$). For NLR, cats with confirmed pyelonephritis had significantly higher ratios than healthy cats ($P = .006$). For MLR, cats with confirmed pyelonephritis and cats with presumptive pyelonephritis had significantly higher ratios than healthy cats ($P = .039$)

FIG 2.



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(A) Serum amyloid A (SAA) in 14 clinically healthy cats (Group 1), 34 cats with other urological conditions (Group 2), 14 cats with confirmed pyelonephritis (Group 3) and 5 cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Median concentrations of SAA in cats with confirmed pyelonephritis (Group 3) were significantly higher than those in healthy cats (Group 1) and cats with other

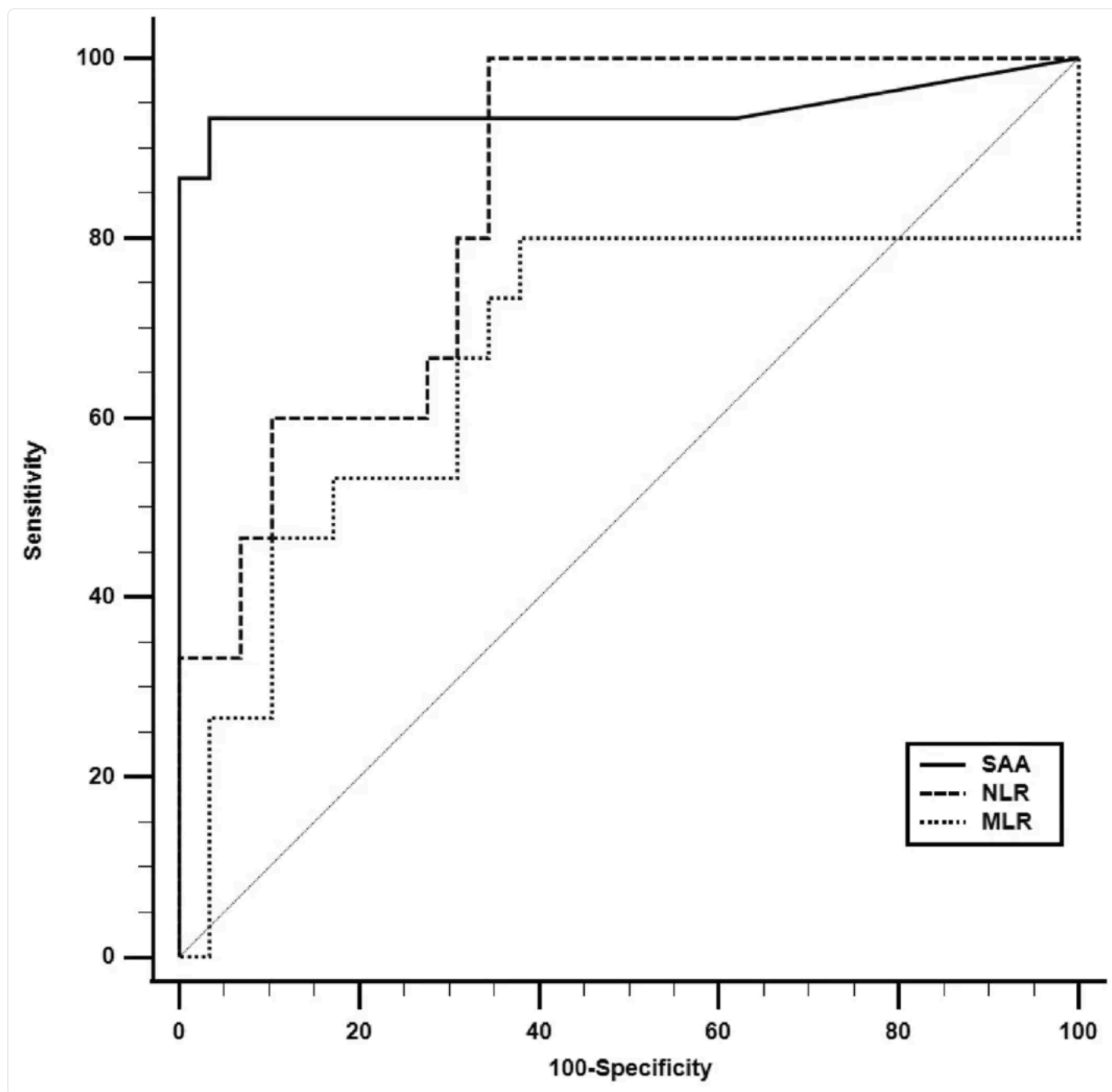
urological conditions (Group 2) ($P < .001$). Median concentrations of SAA in cats with presumptive pyelonephritis (Group 4) were significantly higher than those in healthy cats (Group 1) ($P = .004$) and cats with other urological conditions ($P < .021$). (B) Neutrophil:lymphocyte ratio (NLR) in 13 clinically healthy cats (Group 1), 30 cats with other urological conditions (Group 2), 11 cats with confirmed pyelonephritis (Group 3) and seven cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Cats with confirmed pyelonephritis (Group 3) had significantly higher NLR ratios than healthy cats (Group 1) ($P = .006$). (C) Monocyte:lymphocyte ratio (MLR) in 13 clinically healthy cats (Group 1), 30 cats with other urological conditions (Group 2), 11 cats with confirmed pyelonephritis (Group 3) and seven cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Cats with confirmed pyelonephritis (Group 3) and cats with presumptive pyelonephritis (Group 4) had significantly higher MLR ratios than healthy cats (Group 1) ($P = .039$).

The median NLR was significantly higher in cats with confirmed PN (15.2) when compared to healthy cats (1.66), but not compared to cats with other urological conditions (2.33) (Table [3](#)). The median MLR was significantly higher in cats with PN (0.31) and pPN (0.16) when compared to healthy cats (0.07), but not compared to cats with other urological conditions (0.08) (Table [3](#)). There was substantial overlap of NLR and MLR between groups of cats (Fig [2B,C](#)).

Stratified data for cats with other urological conditions are shown in Table [S4](#).

Receiver operating characteristic curve analysis (Fig [3](#)) revealed that SAA and NLR discriminated between cats with PN (confirmed and presumptive) and cats with other urological conditions, with SAA having superior accuracy to NLR. Area under curves, optimal cut-off, sensitivity, specificity and positive likelihood ratios for SAA, NLR and MLR are listed in Table [4](#), and disease prevalence-directed cut-offs are shown in Table [5](#).

FIG 3.



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Receiver operating characteristic (ROC) curve for serum amyloid A (SAA), neutrophil:lymphocyte ratio (NLR) and monocyte:lymphocyte ratio (MLR) in cats with pyelonephritis and presumptive pyelonephritis *versus* cats with other urological conditions.

Table 4.

Comparison of the area under receiver operating characteristic curves for serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio in distinguishing between cats with confirmed and presumptive pyelonephritis and cats with other urological conditions, likely to be clinically relevant differential diagnoses to pyelonephritis

	AUC (95% CI)	Cut-off	Sens (95% CI)	Spec (95% CI)	LR+
SAA	0.957 (0.862 to 0.994)	49.1	0.95 (0.74 to 0.99)	0.97 (0.85 to 0.99)	32.2
<i>N</i> = 19 (PN and pPN) versus 32 Other					
NLR	0.867 (0.737 to 0.947)	3.01	1.00 (0.82 to 1.00)	0.63 (0.44 to 0.80)	2.7
<i>N</i> = 18 (PN and pPN) versus 30 Other					
MLR	0.702 (0.572 to 0.840)	0.29	0.56 (0.31 to 0.79)	0.90 (0.74 to 0.98)	5.6
<i>N</i> = 18 (PN and pPN) versus 30 Other					

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The group of cats with pyelonephritis consisted of cats with a confirmed diagnosis of pyelonephritis (*n* = 11) and cats with presumptive pyelonephritis (*n* = 8)

AUC Area under the curve, LR+ Positive likelihood ratio, MLR Monocyte:lymphocyte ratio, NLR Neutrophil:lymphocyte ratio, pPN Presumptive pyelonephritis, PN Confirmed pyelonephritis, SAA Serum amyloid A, Sens Sensitivity, Spec Specificity

Table 5.

Optimal cut-off value for serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio in distinguishing between cats with pyelonephritis and cats with other urological conditions likely to be clinically relevant differential diagnoses to pyelonephritis

Prevalence (%)	Optimal cut-off value based on disease prevalence and costs [*]	Optimal cut-off value based on Youden index and assuming that the ratio of cats with pyelonephritis to cats with relevant differential diagnoses reflects the disease prevalence (29%)
SAA		
10	Cut-off value: 143.9 mg/L SE: 0.90 [0.67 to 0.99] SP: 1.00 [0.90 to 1.00]	Cut-off value: 49.1 mg/L SE: 0.95 [0.74 to 0.99] SP: 0.97 [0.85 to 0.99]
33	Cut-off value: 143.9 mg/L SE: 0.90 [0.67 to 0.99] SP: 1.00 [0.90 to 1.00]	
50	Cut-off value: 49.1 mg/L SE: 0.95 [0.74 to 0.99] SP: 0.97 [0.85 to 0.99]	
NLR		
10	Cut-off value: 18.49 SE: 0.39 [0.17 to 0.64]	Cut-off value: 3.01 SE: 1.00 [0.82 to 1.00]

Prevalence (%)	Optimal cut-off value based on disease prevalence and costs [*]	Optimal cut-off value based on Youden index and assuming that the ratio of cats with pyelonephritis to cats with relevant differential diagnoses reflects the disease prevalence (29%)
MLR	33	SP: 1.00 [0.88 to 1.00] Cut-off value: 10.83
		SE: 0.67 [0.41 to 0.87]
	50	SP: 0.90 [0.74 to 0.98] Cut-off value: 3.01
		SE: 1.00 [0.82 to 1.00]
		SP: 0.63 [0.44 to 0.80]
MLR	10	Cut-off value: 0.39
		SE: 0.39 [0.17 to 0.64]
		SP: 0.97 [0.83 to 0.99]
	33	Cut-off value: 0.29
		SE: 0.56 [0.31 to 0.79]
		SP: 0.90 [0.74 to 0.98]
MLR	50	Cut-off value: 0.29
		SE: 0.56 [0.31 to 0.79]
		SP: 0.90 [0.74 to 0.98]

Values in brackets are the associated 95% confidence intervals

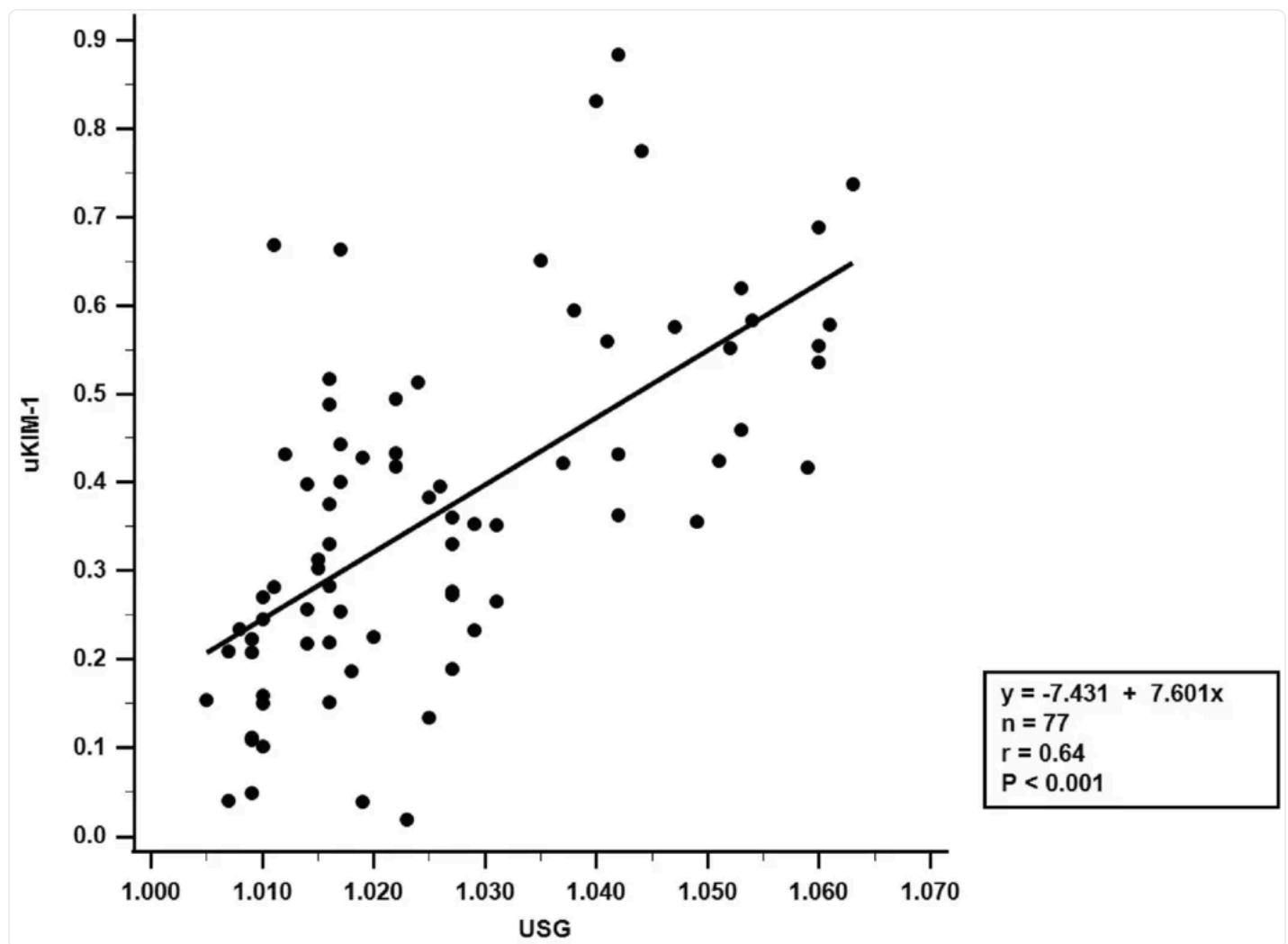
MLR monocyte:lymphocyte ratio, NLR neutrophil:lymphocyte ratio, SAA Serum amyloid A, SE Sensitivity, SP Specificity

*Based on the disease prevalence and estimated cost of false positive: 1; cost of false negative: 1; cost of true positive: 0; and cost of true negative: 0

Urinary KIM - 1

KIM-1 analyses were performed on stored urine samples from 63 cats. The correlation between USG and uKIM-1 is shown in Fig 4. When compared to the 90th percentile of healthy cats, mild to marked uKIM-1 increases were detected in 12/20 and 7/12 cats with pyelonephritis and UO, respectively. Urinary KIM-1 was significantly higher in cats with PN and pPN and in cats with UO compared to healthy cats and cats with SBU/cystitis. There was no statistically significant difference between uKIM-1 in cats with CKD and any other group (Table 6, Fig 5).

FIG 4.



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Correlation between urinary specific gravity and urinary kidney injury marker-1 in 63 cats.

Table 6.

Urinary kidney injury marker-1 in healthy cats, cats with pyelonephritis (confirmed and presumptive) and cats with chronic kidney disease, ureteral obstruction and cystitis/subclinical bacteriuria

	Healthy cats (<i>n</i> = 11)	PN/pPN (<i>n</i> = 20)	CKD (<i>n</i> = 9)	UO (<i>n</i> = 12)	Cystitis/SBU (<i>n</i> = 11)
uKIM-1 (T/C)	0.36 [0.31 to 0.42]	0.73 [*] ₋ [0.49 to 0.86]	0.62 [0.54 to 0.67]	0.76 [*] ₋ [0.49 to 0.93]	0.33 [0.29 to 0.42]

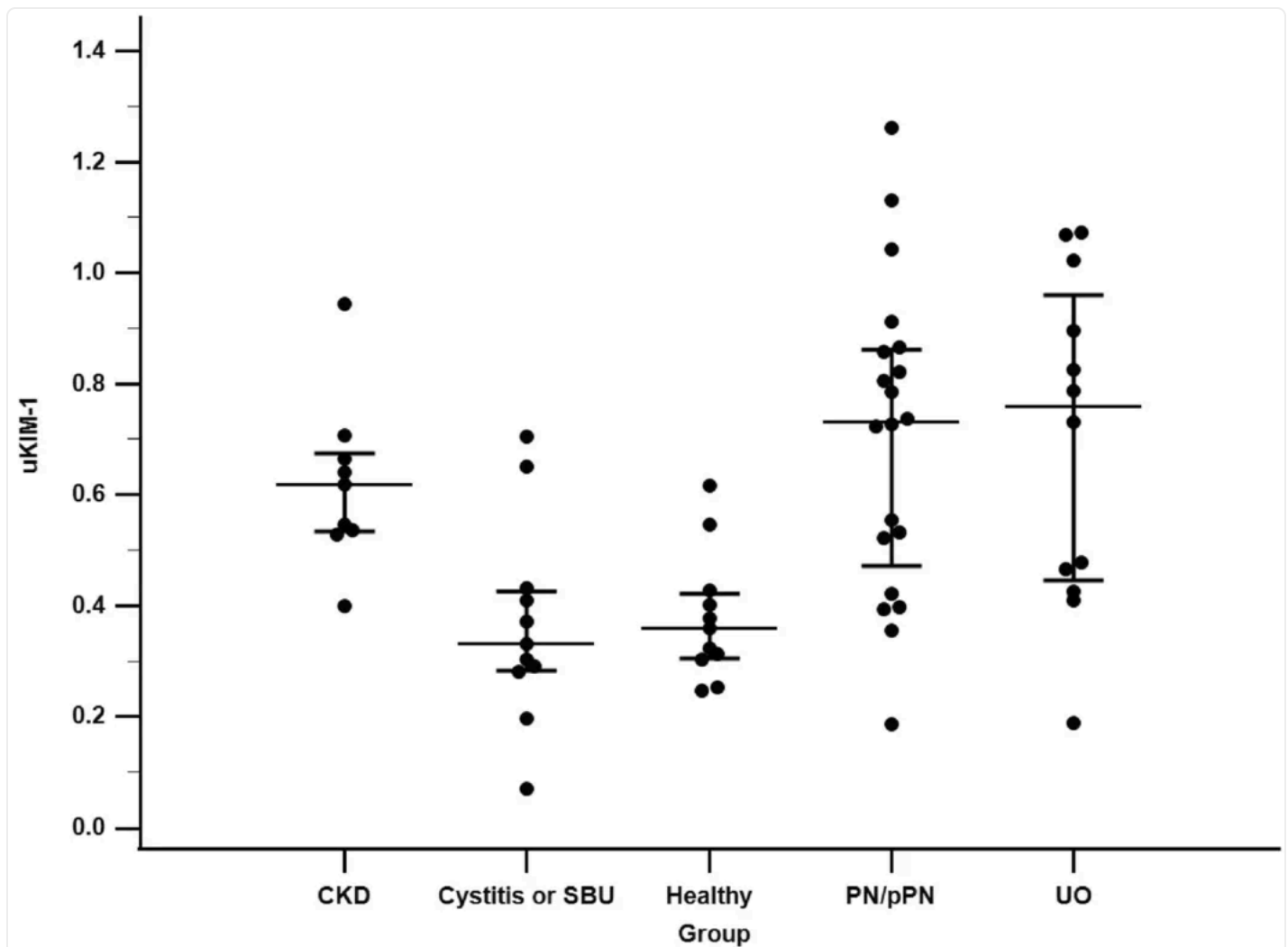
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Numbers are presented as median (25th to 75th percentile). Urinary KIM-1 values are normalised to a urinary specific gravity of 1035 and expressed as test/control ratios; therefore, being unitless

CKD Chronic kidney disease, PN Confirmed pyelonephritis, pPN Presumptive pyelonephritis, SBU Subclinical bacteriuria, T/C Test/control ratio, UO Ureteral obstruction

*Median values between groups were assessed by the Kruskal-Wallis test with Dunn's *post hoc* test indicating statistically significant differences between PN/pPN and healthy cats ($P = .009$), PN/pPN and cats with cystitis/SBU ($P = .008$), UO and healthy cats ($P = .019$), UO and cats with cystitis/SBU ($P = .018$)

FIG 5.



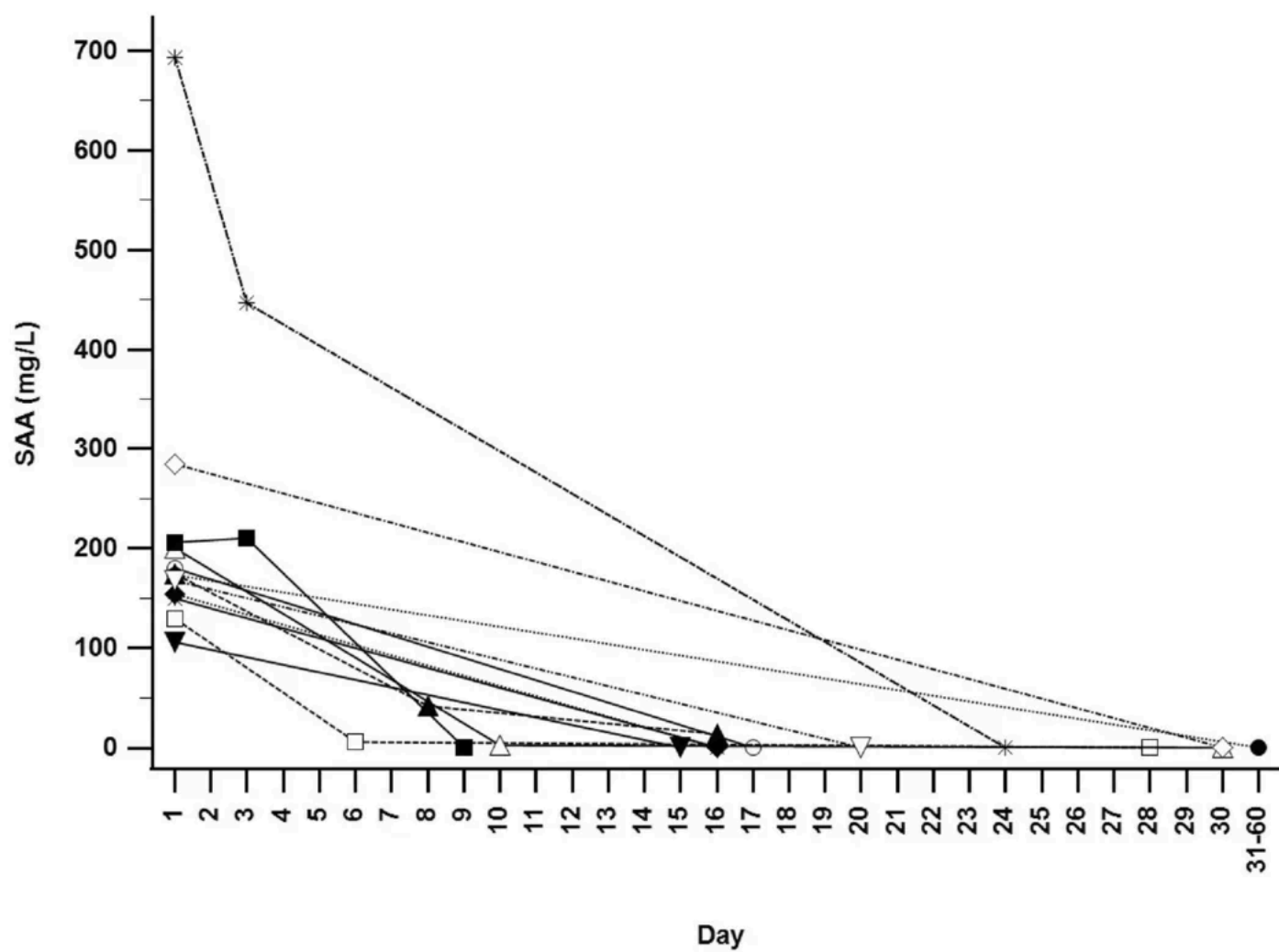
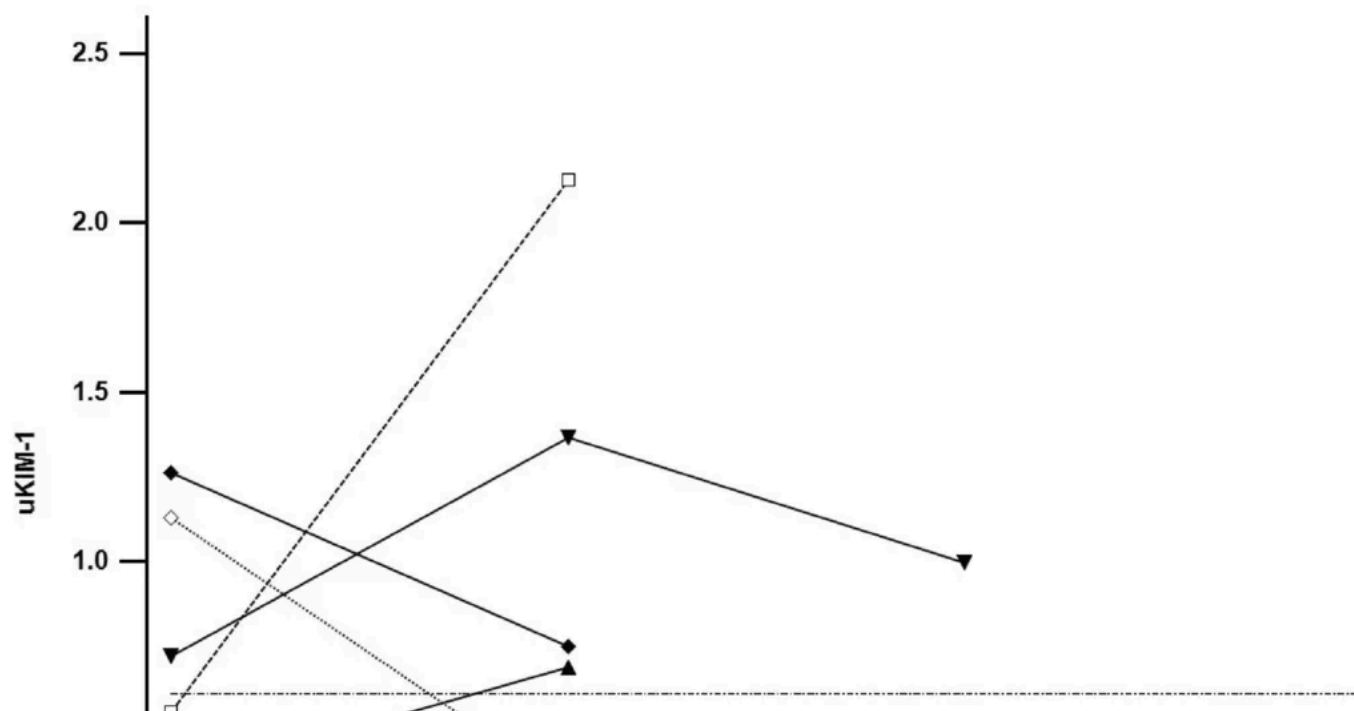
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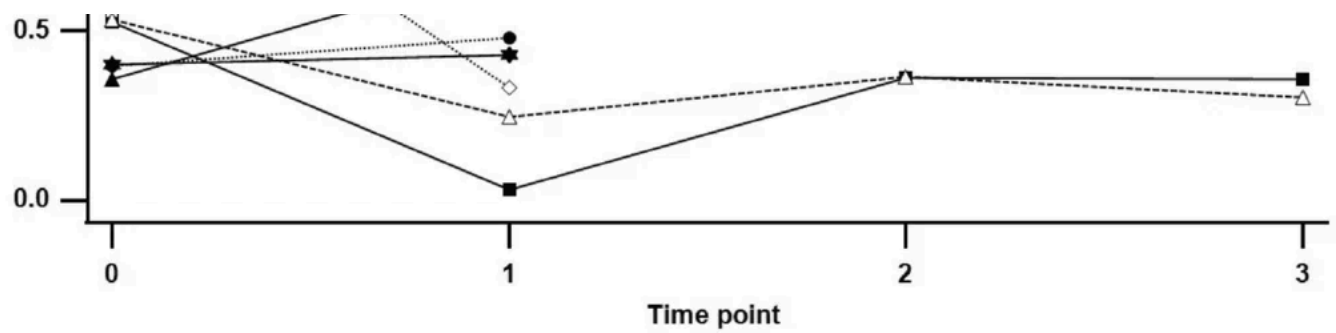
Urinary kidney injury marker-1 (uKIM-1) in 11 clinically healthy cats, 20 cats with confirmed (PN) or presumptive pyelonephritis (pPN), 12 cats with ureteral obstruction (UO) and 11 cats with cystitis or subclinical bacteriuria (SBU). Horizontal lines represent medians and 25th to 75th percentiles. uKIM-1 concentrations are normalised to urine specific gravity of 1035 and presented as control/test ratios. Median uKIM-1 was significantly different between PN/pPN and healthy cats ($P = .009$), PN/pPN and cats with cystitis/SBU ($P = .008$), UO and healthy cats ($P = .019$), UO and cats with cystitis/SBU ($P = .018$).

Follow - up SAA and urinary KIM - 1

Follow-up SAA concentrations were available for 14 cats with PN or pPN after treatment, some of which had a follow-up SAA concentration measured more than once. In 12/14 cats, SAA was completely normalised. In the remaining two cats, SAA was reduced to 5.8 and 13 mg/L at the last follow-up at Day 6 and Day 16, respectively (Fig [6A](#)). Follow-up uKIM-1 values were available for nine cats with PN or pPN (Fig [6B](#)).

FIG 6.

A**B**



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(A) Serum amyloid A (SAA) in consecutive samples from nine cats with confirmed pyelonephritis (solid lines and solid symbols) and five cats with presumptive pyelonephritis (dotted lines and open symbols). Day 1 is the day of inclusion. (B) Urinary kidney injury marker-1 (uKIM-1) in consecutive samples from six cats with confirmed pyelonephritis (solid lines and solid symbols) and three cats with presumptive pyelonephritis (dotted lines and open symbols). The horizontal dotted line represents the upper end of the interval for healthy cats. The time interval between time points was: T0–T1: median 18 days (12 to 137 days), T1–T2: median 22 days (5 to 37 days), T2–T3: 90 to 630 days.

DISCUSSION

The main aim of our study was to assess the ability of SAA concentration and leukocyte ratios to discriminate cats with PN from cats with other urological conditions constituting relevant differential diagnoses. We found a substantial (21- to 169-fold) increase in SAA in all cats with PN, and in all but one cat with pPN. Furthermore, we found minimal overlap in SAA concentrations between cats with PN and cats with other urological conditions. Serum amyloid A proved to be a more sensitive marker of inflammation than the clinical signs, as only 48% of the pyelonephritis cats (PN + pPN) were pyretic (cut-off 39.2°C). During our data processing, another study supporting the use of SAA as a biomarker in cats with PN was published from France (Kurtz et al., [2024](#)). The study population was overall fairly similar to ours, though the group of cats with other urological diseases was wider, and the classification methods varied (classified by panel). When comparing cats with PN/pPN to cats with other urological diseases, Kurtz and coworkers also found a significantly higher SAA concentration in cats with PN, and interestingly, they found an optimal cut-off value of 51.3 mg/L. These results are consistent with ours, indicating reproducibility and robustness of the SAA assay despite different laboratory set-ups, and thus strengthening our confidence in the conclusions. At first glance, our findings and those of Kurtz

et al. appear to conflict with previous studies on SAA in cats with CKD and lower urinary tract disease. However, this discrepancy mainly reflects differences in cut-off values applied. Previous studies (Javard et al., [2017](#); Sasaki et al., [2003](#); Yuki et al., [2020](#)) reported systemic inflammation and increased SAA in cats with CKD and/or lower urinary tract disease, using a cut-off of ≤ 5 mg/L, the upper limit for healthy cats. When applying a higher threshold of 50 mg/L – an approximation of the cut-offs calculated in our study and the study by Kurtz and coworkers – most CKD and lower urinary tract cases fall below this cut-off. In Sasaki et al.'s short communication, an SAA concentration > 50 mg/L was found in only 2/27 cats with lower urinary tract disease and in 12/75 with renal failure (Sasaki et al., [2003](#)). No details were provided on disease duration, concurrent upper urinary tract involvement for the cats with lower urinary tract disease, or comorbidities. Yuki et al. reported significantly higher SAA concentrations in cats with CKD and lower urinary tract disease than in healthy cats, but individual values were not shown. Median SAA concentration in lower urinary tract disease was 0 mg/L (IQR: 0 to 53.8), and in CKD, the IQR was 0 to 5.9 mg/L, suggesting most concentrations fell below 50 mg/L (Yuki et al., [2020](#)). Likewise, Jarvard et al. reported a mean SAA concentration of 8.1 mg/L (SD 6.5) in 38 cats with CKD, supporting the notion that inflammation in chronic renal disease does not elicit a high magnitude acute phase response (Javard et al., [2017](#)). Degenhardt et al. studied SAA in cats with renal azotaemia (acute kidney injury (AKI), CKD and acute-on-chronic kidney disease (AOC)). Of 44 cats, 15 had SAA concentrations > 50 mg/L; four had PN and the remaining 11 cats were diagnosed with AOC, AKI or CKD plus a variety of comorbid conditions, some of which are well known to be associated with systemic inflammation (pyometra, perforated intestinal foreign body, diabetic ketoacidosis) (Degenhardt et al., [2023](#)). In cats with CKD, an SAA increase was associated with comorbid conditions in all cases, whereas the cause of a high SAA concentration was less obvious in cats with AKI or AOC.

To conclude this argument, causes of increased SAA concentration in cats with azotaemia are obviously not restricted to PN, and this non-specific acute phase protein should be interpreted in the context of other clinical and paraclinical findings. However, in cats with azotaemia and bacteriuria, an increased SAA concentration > 50 mg/dL can be considered indicative of PN, unless another apparent cause of inflammation is identified. Likewise, PN can be considered unlikely in cats with azotaemia and bacteriuria if SAA lies below the 50 mg/L threshold. This is particularly helpful in a clinical setting, as SBU or cystitis in cats with underlying CKD is considered difficult to discriminate from PN due to the unspecific clinical presentation of the latter.

This study further identified that the SAA concentration in cats with PN decreases with antimicrobial treatment. The clinical implications include a potential role for SAA to guide the duration of treatment and thus target antimicrobial therapy to the individual cat. However,

tailoring therapy based on SAA response was not examined in our study, and the SAA concentration as a response biomarker during treatment was not systematically assessed.

Neutrophil:lymphocyte ratio was considered a potentially interesting biomarker as it combined parts of both the innate and adaptive immune system. The NLR was found to be significantly higher in cats with PN compared to other cats. However, values overlapped between groups, and a positive test result only modestly increased the likelihood of PN. It is doubtful that NLR provides any added discriminatory value when compared to other components of the blood count, such as neutrophilia and an increased number of band neutrophils. Neutrophil:lymphocyte ratio may potentially play a part in monitoring of treatment in cats with increased ratios; this, however, was not tested in our study. Of interest, NLR was also assessed in the study by Kurtz and coworkers and was not found to be significantly different between groups (Kurtz et al., [2024](#)).

The second aim of our study was to assess tubular damage in cats with PN, and we detected mild to marked uKIM-1 elevations in more than 50% of cats with PN and UO, confirming that proximal tubular injury is common in these populations of cats. We also detected large variations in uKIM-1 concentrations in cats with PN and UO, which can be attributed to several factors. Production of KIM-1 by proximal renal tubular cells and its shedding into urine depend on the severity, extent and duration of injury. Severe AKI results in markedly increased KIM-1 in the distal segment of the proximal tubular cells after 24 hours and occurs before decreases in USG (Bland et al., [2017](#)). But severe injury can also cause cell swelling, cell death, and obstruction of urine flow. If this occurs unilaterally, low uKIM-1 from a non-affected kidney may mix with smaller volumes of high uKIM-1 from an injured kidney and result in overall mildly to markedly increased, or even normal, uKIM-1 values, which may change as urine flow is re-instituted and/or the epithelium regenerates. In our study, four out of nine cats, from which consecutive samples were available, displayed marked changes in uKIM-1 between initial presentation and the first follow-up visit (Fig [6](#)), stressing that sequential uKIM-1 measurements are likely more useful than single samples to measure the extent of injury and repair.

Another factor of importance in our study concerns the anatomic localisation of the inflammation detected. Ascending infection may be confined to the renal pelvis (pyelitis) or extend into the renal parenchyma (pyelonephritis), conditions that cannot be reliably differentiated in clinical practice. Anatomically, KIM-1 is expressed predominantly by proximal tubular epithelial cells (Bland et al., [2017](#)). Therefore, infections confined to the renal pelvis may not provoke significant KIM-1 expression. The importance of anatomic localisation was highlighted in a preclinical rodent model of AKI, in which uKIM-1 was increased in rats with PN-associated AKI, but not in rats with acute

PN alone (Skowron et al., [2018](#)). Based on histopathology, the authors of that study concluded that localisation of renal inflammation differed between rats with and without uKIM-1 increases.

Lastly, disease chronicity may involve fibrotic or atrophic tubules, leading to reduced or absent KIM-1 production (Zhang et al., [2008](#)). This may be of relevance in our population of cats with PN or UO suffering from multiple coexisting conditions of unknown duration, including CKD. Declining capacity of KIM-1 production with advanced CKD was also proposed in a recent study of cats, in which uKIM-1 concentrations of cats with CKD did not differ from those of healthy cats (Kornya et al., [2025](#)).

There are some limitations to the present study. The sample size was not derived from a sample size calculation but was based on feasibility (PN cases confirmed by pyelocentesis are rare) and available funding. We used a case–control design for enrolment of cats rather than consecutive enrolment of all cats presenting with urological disease. Though case–control enrolment is commonly used in diagnostic studies, it carries an inherent risk of selection bias. However, in our study, the control group of cats with urological disease resembled cats with PN or pPN in terms of complexity (azotaemia and presence of comorbidities), thus increasing the generalizability of the results.

Two different SAA assays were used in the present study, the LZ-SAA and the VET-SAA, the latter used also in the study by Kurtz and coworkers (Kurtz et al., [2024](#)). Published comparisons of the LZ-SAA and VET-SAA assays report high concordance in classifying samples as normal or increased, despite modest differences in absolute values (Waugh et al., [2022](#)). The transition to the VET-SAA assay in June 2022 resulted in SAA quantification for ten animals, six of which had very high SAA values (199 to 848 mg/L) and the remaining 4 had SAA values <5 mg/L. Therefore, while the assay transition may have introduced minor analytical variation, it is unlikely to have impacted the interpretation of results in this study.

Serum samples from three cats were stored for an extensive time period before analysing SAA. In cats, SAA has been shown to remain stable for prolonged periods at –20°C (Ceron et al., [2005](#)). In the present study, samples were stored continuously at –80°C with minimal freeze–thaw cycles. Under these conditions, significant degradation of feline SAA is considered unlikely. Finally, the SAA concentrations in the three samples stored for 9 years were 199, 640 and 848 mg/L, respectively. Even if hypothetically some SAA protein degradation took place during storage it would not have altered the conclusions of this study.

In conclusion, we found that SAA increases in cats with PN and decreases in response to therapy. When applying a threshold of 50 mg/L, SAA increases the likelihood of pyelonephritis substantially in cats with urological diseases. In addition, our results indicate that proximal tubular injury is common in cats with PN and cats with UO. The course of tubular injury is best assessed by consecutive testing; however, the clinical implications of uKIM-1 alterations remain to be investigated.

Author contributions

L. R. Jessen: Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; visualization; writing – review and editing; project administration; data curation; supervision; software; formal analysis. **R. Langhorn:** Investigation; writing – review and editing; writing – original draft. **N. Dupont:** Investigation; writing – review and editing. **A. S. Gravgaard:** Investigation; writing – review and editing. **N. Bak-Jacobsen:** Investigation; writing – review and editing. **B. J. Reezigt:** Investigation; writing – review and editing. **L. N. Nielsen:** Writing – original draft; methodology; validation; formal analysis; writing – review and editing; software. **D. Bienzle:** Investigation; writing – original draft; writing – review and editing; supervision; formal analysis; software; validation; methodology. **F. S. Moberg:** Investigation; writing – review and editing. **A. L. Jensen:** Writing – original draft; methodology; formal analysis; data curation; writing – review and editing; software.

Conflict of interest

D. Bienzle is a co-inventor of the feline KIM-1 assay. None of the authors of this article has a financial or personal relationship with other people or organisations that could inappropriately influence or bias the content of the paper.

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Supporting information

Table S1. STARD 2015 reporting guidelines.

[JSAP-67-810-s004.docx](#) (22.1KB, docx)

Table S2. Comorbidities in 14 cats with confirmed PN, 8 cats with presumptive PN, 35 cats with other urological diseases (10 with CKD, 14 with UO, 11 with SBU/cystitis) and 14 healthy cats.

[JSAP-67-810-s003.docx](#) (16.9KB, docx)

Table S3. Key features of 35 cats with other urological diseases.

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Table S4. Serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio measurements in cats with chronic kidney disease, cats with ureteral obstructions and cats with bacterial cystitis or subclinical bacteriuria.

[JSAP-67-810-s002.docx](#) (16.2KB, docx)

The last two authors have shared last authorship.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Table S1. STARD 2015 reporting guidelines.

[JSAP-67-810-s004.docx](#) ^(22.1KB, docx)

Table S2. Comorbidities in 14 cats with confirmed PN, 8 cats with presumptive PN, 35 cats with other urological diseases (10 with CKD, 14 with UO, 11 with SBU/cystitis) and 14 healthy cats.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Untargeted metabolomic profiling of serum from client-owned cats with early and late-stage chronic kidney disease

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Abstract

Evaluation of the metabolome could discover novel biomarkers of disease. To date, characterization of the serum metabolome of client-owned cats with chronic kidney disease (CKD), which shares numerous pathophysiological similarities to human CKD, has not been reported. CKD is a leading cause of feline morbidity and mortality, which can be lessened with early detection and appropriate treatment. Consequently, there is an urgent need for early-CKD biomarkers. The goal of this cross-sectional, prospective study was to characterize the global, non-targeted serum metabolome of cats with early versus late-stage CKD compared to healthy cats. Analysis revealed distinct separation of the serum metabolome between healthy cats, early-stage and late-stage CKD. Differentially abundant lipid and amino acid metabolites were the primary contributors to these differences and included metabolites central to the metabolism of fatty acids, essential amino acids and uremic toxins. Correlation of multiple lipid and amino acid metabolites with clinical metadata important to CKD monitoring and patient treatment (e.g. creatinine, muscle condition score) further illustrates the relevance of exploring these metabolite classes further for their capacity to serve as biomarkers of early CKD detection in both feline and human populations.

Introduction

Chronic kidney disease (CKD) is a common medical condition of cats associated with significant physiological and metabolic disturbances, including cachexia, malnutrition, deranged amino acid metabolism, and oxidative stress [1, 2]. The International Renal Interest Society (IRIS) recommends staging based on disease severity (Stage 1–4) using surrogate markers of glomerular filtration rate (i.e. creatinine and symmetric dimethylarginine [SDMA]) [3]. Unfortunately, caregivers of early-stage CKD cats may not detect subtle clinical signs, such as weight loss and polyuria, and cats can have only slight changes on routine laboratory tests [4, 5]. Therefore, recognizing CKD in early disease stages (Stage 1 and 2), before overt clinical signs of advanced disease develop, is a challenge for veterinarians. Timely recognition would permit more frequent monitoring and treatment earlier in the course of disease, which may slow disease progression and allow detection of disease sequelae prior to developing complications [5, 6].

Metabolites are made and used during normal cellular functions, and disease causes disruption of biochemical pathways creating specific metabolomic profiles. These profiles can be discerned by characterizing the metabolome in a biological matrix using untargeted metabolomics and comparing healthy versus disease samples. This method has been extensively performed in people with CKD and was successfully used for biomarker discovery for early diagnosis and etiology identification [7]. Various feline untargeted metabolomics studies have been performed [8–12]. The studies used plasma [9, 10], serum [11], feces [8, 9], and urine [12] collected from cats with early-stage CKD. Most studies used purpose-bred research cats [8–11] and examined dietary impacts on the metabolome [8, 9, 11].

To date, no published veterinary studies compare metabolite profiles between client-owned early-stage and late-stage CKD in cats. The ability to identify metabolic drivers of early versus late-stage CKD, despite variable environmental conditions (e.g. diet, medical treatments, household) and pathophysiological conditions (naturally-occurring CKD with and without comorbidities including chronic enteropathy, cardiomyopathy, etc.), is imperative for biomarkers to be applied successfully to diverse patient populations. The objective of this study was to identify disease- and stage-specific metabolites to improve understanding of CKD pathophysiology, particularly in early-stage disease, and thereby suggest potential therapeutic targets and biomarkers of early-stage CKD. To better define metabolic disturbances in CKD cats, we applied untargeted serum metabolomics to compare client-owned cats with early- and late-stage CKD and healthy cats.

Results

Clinical evaluation of healthy, early-stage and late-stage CKD cats

Of the 56 cats enrolled in this study, 25 healthy cats (median, nine years; range, 1-14 years) and 30 CKD cats (median, 14 years; range, 2.5-19 years) were included in analyses. Among CKD cats, 17 had early-stage CKD (three cats IRIS CKD stage 1; 14 cats IRIS CKD stage 2), and 13 cats had late-stage CKD (nine IRIS CKD stage 3; four IRIS CKD stage 4). One cat with Stage 4 CKD and severe azotemia (creatinine 13.1 mg/dL) was classified as an outlier upon analysis of clinical and metabolomic datasets, and it was removed from all analyses. All 55 cats included in analyses were neutered. Healthy cats (14 male, 11 female) were domestic short-, medium-, or long-haired cats (22/25), Siamese (2/25), or Himalayan (1/25). All CKD cats (17 male, 13 female) were short-, medium-, or long-haired cats. All cats except for two with CKD (one cat with early-stage and one with late-stage CKD), were fasted for at least 10 hours prior to sample collection.

Physical examination and laboratory parameters for healthy, CKD stage 1 and 2, and CKD stage 3 and 4 cats are presented in **Table 1** and demographic information for individual cats is provided in **Supplementary File 1**. The majority of healthy cats (68%; 17/25) had normal muscle mass; 24% (6/25 cats) had mild muscle loss and 8% (2/25 cats) had moderate muscle loss. All Stage 1 and 2 CKD cats, except one had either mild (53%; 9/17), or moderate (35%; 6/17) muscle loss. Similarly, based on recorded MCS, all cats with Stage 1 and 2 except one had either mild (33%; 4/12), moderate (42%; 5/12), or severe (16%; 2/12) muscle loss. Most cats, except two healthy and two CKD cats, had serum thyroxine concentrations measured, and all were below or within the laboratory reference range. No cats had a history of current or past hyperthyroidism. Most cats (23/25 healthy; 29/31 CKD) had a blood pressure between 120-160 mmHg with a normal direct fundic examination. One healthy cat and three CKD cats had a blood pressure between 160-180 mmHg with a normal fundic examination. Fecal sugar centrifugation was performed in 41/55 cats and all were negative for parasite ova. A urine protein to creatinine (UPC) ratio was performed in most healthy cats (22/25) and was normal (range, 0.06-0.29). Most CKD cats had a UPC ratio performed (25/30 cats) and only two cats had proteinuria (UPC ratio, 0.7 and 1.4).

Most healthy cats were not receiving medications, except for topical flea and heartworm preventative (selamectin) in three cats and oral glucosamine in one cat. Eight CKD cats were on one or more medications or supplements, including aluminum hydroxide (two cats), potassium gluconate (two cats), probiotic (five cats), polyethylene glycol 3350 (two cats), topical selamectin and oral glucosamine (one cat each). All healthy cats were fed a commercial diet formulated to meet the Association of American Feed Control Officials Nutritional Profile for adult feline maintenance [13]. For CKD cats, 16 were fed one or more commercial renal therapeutic diets, ten were fed an over-the-counter diet marketed for adult or senior cats, and two were fed a combination of a renal therapeutic diet and an over-the-counter adult maintenance diet. The diet was unknown in two CKD cats.

1. Patient demographics, physical examination, and laboratory variables

Variable	Healthy (n=25)	Early-Stage CKD (Stages 1 and 2) (n=17)	Late-Stage CKD (Stages 3 and 4) (n=13)
Sex	14 MC, 11 FS	9 MC; 8 FS	9 MC; 5 FS
Age (years)	9 (1-14)	14 (4-17)	13 (3-19)
Body weight (kg)	4.6 (2.8-8.1)	3.9 (3.2-6.4)	4.3 (2.4-5.8)
BCS (1-9)	5 (4-9)	5 (4-7)	5 (2-6)
MCS (0-3)*	0 (0-2) ^a	1 (0-3) ^b	1.5 (0-3) ^b
Hematocrit (%)	38 (30-47) ^a	35 (31-43)	36 (23-42) ^b
Creatinine (mg/dL)	1.4 (1.1-2.2) ^a	2 (1.3-2.7) ^b	3.7 (3.1-7.4) ^c
BUN (mg/dL)	22 (18-31) ^a	39 (22-56) ^b	55 (42-90) ^c
Calcium (mg/dL)	9.7 (8.8-10.4) ^a	10.1 (9.2-14.4)	10.8 (9.8-13.3) ^b
Phosphorus (mg/dL)	3.7 (2.9-4.6) ^a	3.9 (2.3-6.2)	4.1 (3.1-8.9) ^b
Potassium (mEq/L)	4.2 (3.5-5.2)	4.7 (3.7-5.3)	4.6 (2.4-5.1)
Albumin (g/dL)	3.7 (2.9-4.4)	3.5 (3.2-4.0)	3.6 (3.2-3.9)
USG	1.049 (1.038-1.073) ^a	1.018 (1.010-1.039) ^b	1.016 (1.009-1.025) ^b

Values outside of parentheses represent the median value for each patient group, and values inside of parentheses show the range. For each variable, columns within each row with a different superscript letter were statistically different from each other ($p < 0.05$). Abbreviations: BCS = body condition score; BUN = blood urea nitrogen; CKD = chronic kidney disease; FS = female spayed; MC = male castrated; MCS = muscle condition score; USG = urine specific gravity. Muscle condition score: 0 = normal muscle mass; 1 = mild muscle loss; 2 = moderate muscle loss; 3 = severe muscle loss

Healthy, early-stage CKD and late-stage CKD cats have distinct serum metabolomes

The global, non-targeted serum metabolome was evaluated in 55 cats. **Table 2** shows the distribution of chemical classes, including numbers of differentially abundant metabolites when comparing the three groups. **Supplementary File 2** provides fold differences and pairwise p-values for all differentially abundant metabolites between each group. Across all samples, 918 metabolites were detected and

included 830 named and 88 unknown metabolites. Lipids represented ~40.1% of the total metabolome and accounted for ~43.3% of differentially abundant metabolites when comparing healthy versus early-stage CKD cats, ~ 44.6% of differences between healthy and late-stage CKD cats, and ~32.2% of differences between early- and late-stage CKD cats. Amino acids were the second most abundant class, comprising ~22.4% of the metabolome, and they accounted for ~25.8% of differentially abundant metabolites when comparing healthy and early-stage CKD cats, ~22.8% of differentially abundant metabolites when comparing healthy and late-stage CKD cats, and ~30.5% of differentially abundant metabolites when comparing early- and late-stage CKD cats.

Partial least squares discriminant analysis (PLS-DA) showed clear separation when comparing metabolite profiles of healthy cats to those with CKD Stages 1-4, with 11.1% and 7.0% of metabolome differences between CKD and healthy explained by components 1 and 2 respectively (**Figure 1A**). Further separation of early-stage versus late-stage CKD cats was additionally observed, with 13.9% and 8.9% of metabolome differences explained by components 1 and 2 respectively (**Figure 1B**). A full list of differentially abundant metabolites, their normalized abundance, and p-values following statistical testing is included in **Supplementary File 2**. To assess major metabolite contributors to differences between healthy, early-stage CKD, and late-stage CKD cats, PLS-DA with hierarchical clustering analysis (HCA) (**Figure 1C**) was used to identify metabolites that most readily discriminated patient groups. Clear separation between all three groups was observed when the top 50 most discriminating PLS-DA ranked metabolites were included in the HCA projection (**Figure 1C**). These metabolites included 17 unknown metabolites, 12 lipids, eight amino acids, five xenobiotics, three nucleotides, three vitamins/cofactors, one carbohydrate, and one energy metabolite.

Differentially Abundant Metabolites In Healthy Cats and Cats with Early versus Late Chronic Kidney Disease

Chemical Class	Healthy versus Early-Stage CKD (Stages 1 and 2)	Healthy versus Late-Stage CKD (Stages 3 and 4)	Early-Stage (Stages 1 and 2) versus Late-Stage CKD (Stages 3 and 4)
Amino Acids (206)	62 (↑ 34, ↓ 28)	100 (↑ 59, ↓ 41)	55 (↑ 35, ↓ 20)
Peptides (36)	2 (↑ 1, ↓ 1)	9 (↑ 5, ↓ 4)	4 (↑ 1, ↓ 3)
Carbohydrates (22)	5 (↑ 2, ↓ 3)	10 (↑ 6, ↓ 4)	5 (↑ 4, ↓ 1)
Enzymes and Cofactors (30)	13 (↑ 4, ↓ 9)	16 (↑ 6, ↓ 10)	5 (↑ 2, ↓ 3)
Energy Metabolism (9)	6 (↑ 5, ↓ 1)	8 (↑ 5, ↓ 3)	2 (↑ 0, ↓ 2)
Lipids (369)	104 (↑ 66, ↓ 38)	195 (↑ 115, ↓ 80)	58 (↑ 30, ↓ 28)
Nucleotides (51)	13 (↑ 7, ↓ 6)	26 (↑ 14, ↓ 12)	12 (↑ 8, ↓ 4)
Xenobiotics (100)	12 (↑ 8, ↓ 4)	29 (↑ 13, ↓ 16)	17 (↑ 7, ↓ 10)
Unknown and Partially Characterized Metabolites (95)	23 (↑ 11, ↓ 12)	44 (↑ 25, ↓ 19)	22 (↑ 13, ↓ 9)
TOTAL (918)	240 (↑ 138, ↓ 102)	437 (↑ 248, ↓ 189)	180 (↑ 100, ↓ 80)

Numbers next to chemical class indicates total number of identified metabolites. For each comparison, numbers refer to the total number of differentially abundant metabolites (p < 0.05) when comparing each pair of treatments. Numbers in parentheses specify how many differentially abundant metabolites were increased (↑) in the first treatment relative to the second treatment and decreased (↓) in the first treatment relative to the second treatment. Statistical significances are based on Kruskal-Wallis testing of median-scaled log-transformed metabolite abundances, and significance was defined as p < 0.05 following Bonferroni-Hochberg posthoc analysis.

Abbreviations: CKD = chronic kidney disease.

Lipid metabolism is a key driver of metabolome differences between healthy, early-stage and late-stage CKD cats.

Lipids, which comprised most of the serum metabolome in healthy, early-stage, and late-stage CKD cats, were examined further to identify the metabolic pathways and metabolites contributing to the largest differences between disease states (**Figure 2**). Given the diversity of lipids and metabolic pathways within this chemical class, pathway enrichment scores (PES) were used to identify key lipid pathways and metabolites contributing to differences between patient groups. When comparing healthy and early-stage CKD cats, 26 pathways were identified as significant contributors to metabolite differences (**Figure 2A**).

Among these metabolic pathways, fatty acid (FA) (amino and short chain fatty acid/SCFA) and lactosylceramide metabolism each had a PES of 3.76, which was the highest PES observed between these two patient groups. When ranking lipid metabolites by their magnitude of fold difference between healthy and early-stage CKD cats, the FA metabolites 2-aminooctanoate (1.73-fold decrease in early-stage CKD versus healthy, $p = 0.0053$), butyrate/isobutyrate (2.05-fold decrease in early-stage CKD versus healthy, $p = 4.80 \text{ E-}5$) and valerate (2.90-fold increase in healthy versus early-stage CKD, $p = 7.89 \text{ E-}4$) were among the top 20 most differentially abundant lipid metabolites between healthy and early-stage CKD cats (**Figure 2B**). Of note, the phosphatidylinositol metabolism (PES 2.68) metabolite 1-palmitoyl-2-linoleoyl-GPI was 144.70-fold increased in healthy versus early-stage CKD cats ($p = 0.035$).

In healthy versus late-stage CKD cats, 31 lipid metabolic pathways were significant contributors to metabolome differences. Similar to healthy versus early-stage CKD cats, FA metabolic pathways (acyl carnitine, amino, short-chain) were also significant contributors to differences among healthy versus late-stage CKD cats (all PES 2.44). Plasmalogen and glycerolipid metabolism (both PES 2.44) and branched chain FA metabolism (PES 1.22) further contributed several of the highest-magnitude differentially abundant metabolites for healthy versus late-stage CKD cats (**Figure 2A**). These metabolites included the glycerolipid glycerol-3-phosphate (57.55 fold-decrease in late-stage CKD versus healthy cats, $p = 0.033$), the branched fatty acid (16 or 17)-methylstearate (123.40-fold decreased in late-stage CKD versus healthy, $p = 0.015$), the amino fatty acid 2-aminoheptanoate (3.74-fold increase in healthy versus late-stage CKD, $p = 8.0 \text{ E-}4$), and the plasmalogen 1-(1-enyl-palmitoyl)-2-oleoyl-GPE (40.19-fold increase in healthy versus late-stage CKD, $p = 2.0 \text{ E-}4$) (**Figure 2C**).

When comparing early versus late-stage CKD cats, 14 lipid metabolic pathways were significant contributors to metabolome differences. FA metabolic pathways accounted for several of these pathways and included acyl carnitine and dicarboxylate FA (PES 3.53), acyl choline (PES 2.94), and SCFA metabolism (PES 2.94) (**Figure 2A**). Among the most differentially abundant lipid metabolites within these pathways included the acyl carnitine and dicarboxylate metabolite pimeloylcarnitine/3-methyladipoylcarnitine (2.77-fold increase in early-stage versus late-stage CKD, $p = 0.011$), the acyl choline metabolite oleoylcholine (2.27-fold decrease in late-stage CKD versus early-stage CKD, $p = 0.042$), and the short-chain FA metabolite valerate (2.49-fold decrease in late-stage CKD versus early-stage CKD, $p = 0.011$) (**Figure 2D**). The glycerolipid (PES 2.94) metabolite glycerol-3-phosphate was 656.88-fold decreased in late-stage CKD cats relative to early-stage CKD cats ($p = 0.012$) and it was the largest lipid fold difference among all lipids and group comparisons in the dataset.

Compared to healthy cats, derangements in serum amino acids are observed in early-stage and late-stage CKD cats.

Amino acids were the second largest chemical class contributing to differences between healthy and CKD cats, including between early versus late-stage CKD cats. Considering the prevalence of cachexia and lean muscle loss in cats with CKD [14], differences in the 11 essential feline amino acids (EAA) were compared between groups (**Figure 3, Table 3**). Five of the 11 EAAs were significantly decreased in both

early and late-stage CKD versus healthy cats. This included arginine, histidine, phenylalanine, threonine, and tryptophan. Ten of the 11 EAAs were significantly decreased in late-stage versus early-stage CKD. This included arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. None of the examined amino acids were significantly increased in late-stage CKD relative to early-stage CKD. One of the EAA, taurine, exhibited no differences between groups.

Essential Amino Acid Differences Across Healthy Cats and Cats with Early versus Late-Stage Chronic Kidney Disease			
Acid	Healthy versus Early-Stage CKD (Stages 1 and 2)	Healthy versus Late-Stage CKD (Stages 3 and 4)	Early-Stage (Stages 1 and 2) versus Late-Stage CKD (Stages 3 and 4)
Arginine	↑ 1.03 (0.0062)	↑ 1.08 (< 1.00 E-6)	↑ 1.04 (0.018)
Asparagine	↑ 1.05 (0.010)	↑ 1.17 (< 1.00 E-6)	↑ 1.11 (0.010)
Cysteine	1.01 (0.38)	↑ 1.08 (0.00015)	↑ 1.06 (0.0037)
Glutamine	1.01 (0.25)	↑ 1.09 (0.000021)	↑ 1.07 (0.0021)
Glutamate	1.01 (0.47)	↑ 1.12 (0.000053)	↑ 1.10 (0.011)
Proline	1.02 (0.36)	↑ 1.09 (0.00033)	↑ 1.07 (0.0071)
Phenylalanine	↑ 1.02 (0.0091)	↑ 1.07 (< 1.00 E-6)	↑ 1.04 (0.011)
Serine	0.97 (0.36)	0.98 (0.50)	1.01 (0.50)
Tryptophan	↑ 1.05 (0.0034)	↑ 1.14 (< 1.00 E-6)	↑ 1.08 (0.015)
Tyrosine	↑ 1.03 (0.011)	↑ 1.11 (< 1.00 E-6)	↑ 1.07 (0.010)
Valine	1.01 (0.24)	↑ 1.09 (0.000057)	↑ 1.08 (0.0044)
For this comparison, numbers indicate the fold difference when comparing each group of metabolites. These fold differences were calculated by dividing the first group by the second group. Differences and statistical significances are based on Kruskal-Wallis testing of median-log-transformed metabolite abundances, and significance was defined as p < 0.05 using Benjamini-Hochberg posthoc analysis. Italic numbers in parentheses represent the p-value for this test. Bold with ↑ indicates that the amino acid metabolite was significantly increased in the first group relative to the second group. Abbreviations: CKD = chronic kidney disease.			

Uremic toxin metabolism differs between healthy versus CKD cats and when comparing early versus late-stage CKD cats

Given the link between the gut microbiome and uremic toxins, differences in ten metabolites involved in the metabolism of major gut-derived uremic toxins were evaluated in **Figure 4**. When comparing early-stage

CKD and healthy cats, significant decreases were observed for tryptophan (0.96-fold decrease, $p = 0.011$), tyrosine (0.94-fold decrease, $p = 0.0020$), and phenylalanine (0.97-fold decrease, $p = 0.0092$). Tryptophan is a metabolite precursor to the uremic toxins indoxyl-3-sulfate and indoleacetate, tyrosine is the precursor for the uremic toxins phenol sulfate and p-cresol sulfate, and phenylalanine is the precursor for the uremic toxin phenylacetate. No significant differences in metabolite abundance were observed for these uremic toxins when comparing healthy versus early-stage CKD cats. When comparing late-stage CKD and healthy cats, significant decreases in tryptophan (0.89-fold decrease in late-stage CKD versus healthy cats, $p < 1.00 \text{ E-}6$), phenylalanine (0.92-fold decrease, $p < 1.00 \text{ E-}6$), and tyrosine (0.90-fold decrease, $p = 0.00034$) were similarly observed. Additionally, methyl indole-3-acetate was increased in late-stage CKD cats compared to healthy cats (1.81-fold increase, $p = 0.0069$). While there were no differences in choline abundance between groups, significant increases in the abundance of its downstream metabolite, the uremic toxin trimethylamine-N-oxide (TMAO), were observed when comparing early-stage CKD to healthy cats (1.17-fold increase, $p = 0.0014$) and in late-stage CKD versus healthy cats (1.21-fold increase, $p = 0.00010$).

Disease severity further impacted uremic toxin metabolism. Compared to early-stage CKD, late-stage CKD cats exhibited significant decreases in tryptophan (0.93-fold decrease, $p = 0.011$) and phenylalanine (0.95-fold decrease, $p = 0.012$) as well as significant increases in methyl indole-3-acetate (1.72-fold increase, $p = 0.0069$). Interestingly, the uremic toxin 3-indoxyl sulfate (also known as indoxyl-sulfate), which when increased in serum, has previously been reported as a marker of CKD progression [15, 16], did not achieve statistical significance when comparing between early-stage versus late-stage CKD ($p = 0.74$).

Correlations between selected metabolites with creatinine and muscle condition score

Associations between the 110 metabolites assessed in Figures 1-4 and selected clinical variables were further evaluated using Spearman and Pearson correlations (**Table 4, Supplementary File 3**). Given their roles in GFR estimation and muscle metabolism respectively [17, 18], metabolites strongly and moderately correlated with creatinine and MCS were examined further (**Table 4**). Seven metabolites were strongly correlated with creatinine including the vitamin C metabolite gulonate ($r = 0.77$, $p < 1.00 \text{ E-}6$), the nucleotides 4-ureidobutyrate ($r = 0.72$, $p = 0.0010$) and orotidine ($r = 0.72$, $p = 0.0010$), the unknown metabolite X-21283 ($r = 0.71$, $p < 1.00 \text{ E-}6$), the dicarboxylate FA suberoylcarnitine ($r = 0.71$, $p < 1.00 \text{ E-}6$), and the EAAs threonine ($r = -0.71$, $p < 1.00 \text{ E-}6$) and phenylalanine ($r = -0.74$, $p < 1.00 \text{ E-}6$). Eleven metabolites were significantly moderately correlated with MCS. These included the positively-correlated carbohydrate 1,5-anhydroglucitol ($r = 0.63$, $p < 1.00 \text{ E-}6$), the dicarboxylate FA 3,4-dihydroxybutyrate ($r = 0.58$, $p = 0.000006$), the phospholipid trimethylamine N-oxide ($r = 0.52$, $p = 0.000064$), the unknown metabolites X-25387 ($r = 0.51$, $p = 0.000079$) and X-12730 ($r = 0.51$, $p = 0.000087$), and the vitamin C metabolite gulonate ($r = 0.51$, $p = 0.000098$). Metabolites moderately negatively correlated with MCS included the dicarboxylate FA octadecenedioate ($r = -0.53$, $p = 0.000036$), the amino FA 2-aminooctanoate ($r = -0.53$, $p = 0.000044$), the branched FA (9-or-10)-methylundecanoate ($r = -0.52$, $p = 0.000056$), the EAA

threonine ($r = -0.51$, $p = 0.000079$), and the diacylglycerol linoleoyl-linolenoyl-glycerol ($r = -0.50$, $p = 0.00011$).

Table 4. Correlations of Selected Metabolites with Creatinine and Muscle Condition Score

	Metabolic Pathway	Metabolite	Creatinine	MCS
d	Alanine, Aspartate	N-carbamoyl-alanine	<i>0.54 (p = 0.000016)</i>	<i>0.47 (0.00033)</i>
	Glycine, Serine, Threonine	Threonine	<i>-0.71 (p <1.00 E-6)</i>	<i>-0.51 (p = 0.000079)</i>
	Guanidino, Acetamido	1-methylguanidine	<i>0.66 (p < 1.00 E-6)</i>	<i>0.35 (p = 0.010)</i>
		Guanidinosuccinate	<i>0.58 (p = 0.000003)</i>	<i>0.47 (p = 0.00037)</i>
	Histidine	Histidine	<i>-0.68 (p <1.00 E-6)</i>	<i>-0.34 (p = 0.013)</i>
	Leucine, Isoleucine, Valine	Isoleucine	<i>-0.55 (p = 0.000011)</i>	<i>-0.25 (p = 0.067)</i>
		Leucine	<i>-0.59 (p = 0.000002)</i>	<i>-0.28 (p = 0.043)</i>
		Valine	<i>-0.54 (p = 0.000015)</i>	<i>-0.33 (p = 0.013)</i>
	Lysine	Lysine	<i>-0.53 (p = 0.000025)</i>	<i>-0.10 (p = 0.44)</i>
	Methionine, Cysteine, S-Adenosyl-methionine, Taurine	Methionine	<i>-0.54 (p = 0.000016)</i>	<i>-0.21 (p = 0.12)</i>
		S-methylcysteine	<i>-0.57 (p = 0.00004)</i>	<i>-0.27 (p = 0.049)</i>
		Tyrosine	<i>-0.63 (p <1.00 E-6)</i>	<i>-0.48 (p = 0.00025)</i>
	Phenylalanine	Phenylalanine	<i>-0.73 (p < 1.00E-6)</i>	<i>-0.44 (p = 0.00096)</i>
	Tryptophan	Indoleacetylglutamine	<i>0.57 (p = 0.000004)</i>	0.14 (p = 0.29)
		Tryptophan	<i>-0.69 (p <1.00 E-6)</i>	<i>-0.47 (p = 0.00035)</i>
ite	Urea Cycle, Arginine, Proline	Arginine	<i>-0.67 (p <1.00 E-6)</i>	<i>-0.40 (p = 0.0027)</i>
		Dimethylguanidino valeric acid	<i>0.62 (p <1.00 E-6)</i>	<i>0.37 (p = 0.0069)</i>
	Glycolysis, Gluconeogenesis, Pyruvate	1,5-anhydroglucitol	<i>0.35 (p = 0.0078)</i>	<i>0.62 (p <1.00 E-6)</i>
	Ascorbate, Aldarate	Gulonate	<i>0.76 (p <1.00 E-6)</i>	<i>0.76 (p <1.00 E-6)</i>
		Retinal	<i>0.61 (p <1.00 E-6)</i>	<i>0.34 (p = 0.012)</i>
	Pantothenate, Co-enzyme A	Pantoate	<i>0.51 (p <1.00 E-6)</i>	<i>0.32 (p = 0.017)</i>
	Diacylglycerol	linoleoyl-linolenoyl-glycerol [1]	<i>-0.45 (p = 0.00056)</i>	<i>-0.50 (p = 0.00011)</i>
	Endocannabinoid	Azeloyltaurine	<i>0.58 (p = 0.000003)</i>	0.21 (p = 0.12)
		Oleoyl ethanolamide	<i>-0.51 (p = 0.000053)</i>	<i>-0.28 (p = 0.043)</i>
	Fatty Acid, Acyl Choline	Arachidonoylcholine	<i>-0.53 (p =</i>	<i>-0.10 (p =</i>

		<i>0.000023)</i>	0.46)
	Dihomo-linolenoyl-choline	<i>-0.53 (p = 0.000026)</i>	-0.23 (p = 0.098)
	Linoleoylcholine	<i>-0.60 (p < 1.00 E-6)</i>	<i>-0.31 (p = 0.022)</i>
	Oleoylcholine	<i>-0.53 (p = 0.000026)</i>	-0.24 (p = 0.082)
Fatty Acid, Amino	2-aminooctanoate	<i>-0.51 (p = 0.000067)</i>	<i>-0.53 (p = 0.000044)</i>
Fatty Acid, Branched	(9 or 10)-methylundecanoate	<i>-0.51 (p = 0.000064)</i>	<i>-0.52 (p = 0.000056)</i>
Fatty Acid/BCAA	2-methylmalonylcarnitine	<i>0.61 (p < 1.00E-6)</i>	0.20 (p = 0.14)
Fatty Acid (Acyl Carnitine, Dicarboxylate)	Pimeloylcarnitine/3-methyladipoylcarnitine	<i>0.59 (p = 0.000002)</i>	0.34 (p = 0.011)
	Suberoylcarnitine	<i>0.71 (p < 1.00 E-6)</i>	0.26 (p = 0.055)
Fatty Acid, Dicarboxylate	Octadecenedioate	<i>-0.68 (p < 1.00 E-6)</i>	<i>-0.53 (p = 0.000036)</i>
Fatty Acid, Dihydroxy	3,4-dihydroxybutyrate	<i>0.44 (p = 0.00077)</i>	<i>0.58 (p = 0.000006)</i>
Fatty Acid, Long Chain Polyunsaturated (n3 and n6)	Docosapentaenoate	<i>-0.51 (p = 0.000049)</i>	-0.063 (p = 0.65)
Fatty Acid, Long Chain Saturated	Arachidate	<i>-0.60 (p < 1.00 E-6)</i>	-0.16 (p = 0.23)
Fatty Acid, Monohydroxy	13-HODE + 9-HODE	<i>-0.62 (p < 1.00 E-6)</i>	<i>-0.37 (p = 0.0056)</i>
Phosphatidylcholine	Stearoylcholine	<i>-0.51 (p = 0.000053)</i>	-0.23 (p = 0.095)
Phosphatidylethanolamine	1-palmitoyl-2-arachidonoyl-GPE	<i>-0.51 (p < 1.00 E-6)</i>	-0.14 (p = 0.29)
Phosphatidylinositol	1-palmitoyl-2-arachidonoyl-GPI	<i>-0.58 (p = 0.000003)</i>	<i>-0.28 (p = 0.042)</i>
Phospholipid	Trimethylamine-N-oxide	<i>0.53 (p = 0.000026)</i>	<i>0.52 (p = 0.000064)</i>
Plasmalogen	1-(1-enyl-palmitoyl)-2-oleoyl-GPE	<i>-0.50 (p = 0.000089)</i>	<i>-0.36 (p = 0.0086)</i>
e Purine, Adenine	N6-succinyladenosine	<i>0.57 (p = 0.000004)</i>	<i>0.38 (p = 0.0048)</i>
	Pyrimidine, Orotate	<i>0.72 (p < 1.00 E-6)</i>	<i>0.44 (p = 0.0010)</i>
	Pyrimidine, Uracil	<i>0.72 (p < 1.00 E-6)</i>	<i>0.44 (p = 0.0010)</i>
l Unknown	BDP, C17H18N2O4 (2)	<i>-0.56 (p = 0.000006)</i>	-0.25 (p = 0.065)
	BDP, C17H18N2O4 (3)	<i>-0.53 (p = 0.000028)</i>	-0.25 (p = 0.073)
	Branched/straight/cyclopropyl 10:1 FA (1)	<i>-0.64 (p < 1.00 E-6)</i>	<i>-0.47 (p = 0.00033)</i>
	X-12117	<i>0.50 (p = 0.000095)</i>	<i>0.37 (p = 0.0057)</i>
	X-12730	<i>0.42 (p =</i>	<i>0.51 (p =</i>

		<i>0.0013)</i>	<i>0.000087)</i>
	X-17351	<i>0.55 (p = 0.000009)</i>	0.18 (p = 0.18)
	X-21283	<i>0.71 (p < 1.00 E-6)</i>	0.25 (p = 0.069)
	X-24935	<i>-0.55 (p = 0.000011)</i>	<i>-0.27 (p = 0.046)</i>
	X-24937	<i>-0.59 (p = 0.000002)</i>	<i>-0.30 (p = 0.029)</i>
	X-25387	<i>0.43 (p = 0.00081)</i>	<i>0.51 (p = 0.000079)</i>

selected for correlation analysis included the 110 metabolites identified as key patient differentiators from HCA, essential amino acid, and uremic toxin analyses. For each comparison, values show the correlation coefficient (parentheses) from Spearman's correlations tests. Significance was defined as $p < 0.05$. Strongly correlated were defined as those that had a correlation coefficient between 0.70-0.89, and moderately-correlated metabolites those with a correlation coefficient between 0.50-0.69. Box colors indicate the strength and direction of correlated where green shades indicate positive correlations (dark green = strongly correlated metabolites, light green = moderately-correlated metabolites), red shades indicate negative correlations (red = strongly correlated metabolites, pink = moderately-correlated metabolites), and beige boxes indicate metabolites that were weakly to poorly correlated with these metabolites. Bold and italics indicate metabolites that achieved statistical significance for the comparison of interest. Abbreviations: BCAA = branched chain amino acid, BDP = bilirubin degradation product, FA = fatty acid, GPC = glycerophosphorylcholine, GPE = glycerophosphorylethanolamine, GPI = glycerophosphorylinositol, HODE = hydroxyoctadecadienoic acid, MCS = muscle condition score.

Discussion

The purpose of this study was to compare the serum metabolome of client-owned cats that were healthy to those with naturally-occurring CKD to identify metabolites that readily distinguished between health and early versus late-stage disease. This is the first published study that characterizes the serum metabolome in client-owned cats. While some conventional serum biomarkers, such as creatinine and SDMA, are well correlated with GFR and routinely used for CKD diagnosis and staging, there are limitations to their clinical use. Creatinine is affected by extra-renal disorders, including lean muscle mass, endocrinopathies, diet, and it can fall within normal reference intervals during early-stage CKD [19–21]. SDMA is a more sensitive marker of GFR than creatinine and is less affected by extra-renal disorders, however mild increases can occur in the absence of kidney disease [22–24]. There is growing interest in identifying biomarkers that are highly discriminative between CKD and other diseases and that distinguish early-stage CKD from both healthy and late-stage CKD cats. Early detection could reduce disease morbidity and improve survival by improving monitoring and allowing for earlier interventions prior to the development of overt clinical signs and complications. Additionally, little is known about metabolic disturbances that occur in cats with early-stage CKD, a time when clinical signs and laboratory abnormalities can be subtle. A better understanding of these disturbances could improve understanding of disease pathophysiology and support treatments to implement early in the disease to improve outcome. Serum is a routinely obtained sample used for feline health evaluations, as well as in CKD diagnosis and staging. With the expansion of high-throughput metabolomics in veterinary medicine and the capacity of this tool to elucidate molecular mechanisms of disease, analysis of the serum metabolome represents a feasible next step to advance feline early-stage CKD biomarker discovery.

This study examined 55 client-owned cats with a confirmed diagnosis of CKD at enrollment. The significant increases observed in BUN and creatinine ($p < 0.05$ when comparing healthy versus early-stage CKD, healthy versus late-stage CKD, and early-stage versus late-stage CKD) as well as shifts in phosphorus, total calcium, hematocrit, and MCS ($p < 0.05$ when comparing healthy versus either early-stage or late-stage CKD) (Table 1) are similar to changes observed in feline CKD populations examined elsewhere [25], supporting that this population is appropriately representative of client-owned cats with CKD. Within this population, distinct differences in the serum metabolome were observed when comparing healthy cats to those with early and late-stage CKD (Fig. 1a, 1b). Similarly, Ruberti *et al.* showed distinct serum metabolomes between healthy and Stage 1 and 2 CKD cats, further supporting detectable metabolic disturbances in early-stage disease [11]. When comparing global metabolite differences between patient groups, lipid and amino acid metabolites contributed to the majority ($> 60\%$) of differentially abundant metabolites between healthy and early-stage CKD, healthy and late-stage CKD, as well as between early and late-stage CKD (Table 2). The importance of both lipids and amino acids in discriminating between patient groups was additionally revealed via unsupervised HCA, which showed that lipids and amino acids contributed to the majority of the metabolites that most-readily separated healthy versus CKD cats as well as between early and late-stage CKD cats (Fig. 1c). Collectively, these serum metabolome changes in lipids and amino acids could be potential therapeutic targets in the management of CKD, and these chemical classes merit further investigation for biomarker discovery.

Although the roles of lipid dysmetabolism in CKD onset and progression is an active area of investigation in human medicine, little is known in dogs and cats [26]. Dogs with CKD have dyslipidemia based on lipoprotein electrophoresis [27]. Data presented herein represents the first publication to highlight systemic lipid disturbances in cats with CKD. In this study, 36 lipid metabolic pathways were found to be altered when comparing between healthy and early or late-stage CKD cats and between early and late-stage CKD cats (Fig. 2a-2c), where the majority of these pathways are either directly or indirectly linked to FA metabolism. Despite differences in the major etiologies of CKD between humans and cats, accumulation of lipid in the renal cortex, particularly tubular epithelium, has been documented in both people and cats with CKD [26, 28]. In people, FA dysmetabolism is associated with increased FA deposition into the renal parenchyma, leading to increased inflammation and progressive tubular destruction [29]. In the current study, significant differences in the SCFAs butyrate, isobutyrate, and valerate distinguished early-stage CKD cats from healthy cats (Figs. 2a, 2c). In animals and people with CKD, increased butyrate, isobutyrate and valerate were associated with reduced systemic and renal pro-inflammatory cytokine production [29], suggesting that decreased levels of these SCFAs may be early indicators of renal damage. The significant fold differences of these SCFAs when comparing healthy and early-stage CKD cats supports that these metabolites could be further examined for their capacities to identify cats with early-stage CKD and as a potential therapeutic target. Furthermore, the FA-derived metabolite glycerol-3-phosphate, which has been explored for its roles in dysregulated phosphorus metabolism and vitamin-D synthesis during renal damage [30], was markedly decreased in late-stage CKD versus early-stage CKD, indicating that some serum FA biomarkers may decrease in abundance in the serum with advancing CKD stage, despite decreases to GFR. Although these lipid metabolites were

only weakly-correlated with clinical and laboratory parameters (**Supplementary File 3**), the roles that lipid metabolism plays in CKD, especially in its progression from early-stage to late-stage disease, are largely unknown [26, 28], such that their levels in serum may increase or decrease via mechanisms that are not proportionate to conventional monitoring parameters. In turn, both increases and decreases in serum lipid metabolites should be considered as biologically meaningful changes in cats with CKD.

EAAAs are potential diagnostic and therapeutic biomarkers in CKD as they are easily measured and uniquely modified in CKD cats. Understanding the interrelationships between CKD and amino acid dysmetabolism is a second area of active investigation within veterinary medicine, where alterations to amino acid metabolism are linked to cachexia and uremic toxin production in feline CKD [14, 31, 32]. These processes are strongly tied to morbidity and mortality among CKD cats, and adjunctive therapies are routinely prescribed for these derangements during CKD management [17, 33, 34], highlighting the need to better understand the molecular mechanisms driving these pathologies. CKD-induced cachexia is believed to occur due to a combination of increased nausea and appetite dysregulation, decreased total caloric and protein intake, and protein malassimilation in the gut [33]. Significant decreases in MCS were appreciated when comparing healthy cats to early-stage CKD and late-stage CKD cats (Table 1), supporting that this population of cats was ideally suited for examining serum amino acid derangements occurring secondary CKD-induced cachexia. The BCAA leucine has been widely characterized in people for its roles in protein turnover and muscle synthesis, and it decreases with cachexia and sarcopenia in elderly adults [35], making it a promising biomarker to evaluate for early-onset cachexia that may be occurring with feline CKD. In this study, leucine was not significantly correlated with muscle condition, and thus additional exploration would be necessary to determine if it has value as a biomarker of feline cachexia. However, there were no differences in leucine abundance when comparing early-stage and late-stage CKD cats, suggesting that it also may not be suitable as an early-stage feline CKD serum biomarker.

Decreases in five amino acids were identified in the serum of cats with early-stage CKD versus healthy cats (arginine, histidine, phenylalanine, threonine, tryptophan) (Fig. 3, Table 3). These amino acids were further decreased when comparing late-stage CKD to early-stage CKD cats, suggesting that there may be progressive changes occurring with disease advancement. In a previous investigation, serum phenylalanine, tryptophan, and threonine levels decreased proportionately with CKD stage [14]. In addition to decreasing proportionately with CKD stage, threonine was moderately negatively correlated with MCS (Table 4). As an EAA, threonine in cats is an important source of cellular glucose, as an energy source for gut microbes, and as a major contributor to proteins critical for muscle development, immunity, and gastrointestinal health [36, 37]. In states of decreased threonine intake, lean muscle catabolism is one method that increases threonine bioavailability to carry out these vital processes [36], thereby explaining one mechanism for why it may decrease proportionately to MCS during feline CKD. In a study assessing dietary amino acid supplementation in Stage 1 and 2 feline CKD, threonine intake was positively-correlated with lean body mass changes [32].

The European Uremic Toxin Work Group identified > 100 uremic toxins that are classified as either free water-soluble low molecular weight (< 500 Da), middle molecules ($\geq$ 500 Da), or protein-bound [38]. Some major protein-bound uremic toxins originate from gut microbial metabolism of aromatic amino acids including indoxyl sulfate, p-cresol sulfate, phenol sulfate, indolacetate, and phenylacetate. Trimethylamine-n-oxide is a free water-soluble low molecular weight uremic toxin that is formed in the gut via microbial fermentation of choline, phosphatidylcholine, L-carnitine, and betaine [39]. These uremic toxins are linked to systemic inflammation and oxidative stress, cardiovascular disease, immunodeficiency, cachexia, intestinal barrier damage and endotoxemia, and renal injury [40]. In this study, significantly increased abundances of indolacetate and TMAO differentiated early-stage CKD from both late-stage CKD and healthy cats (Fig. 4). TMAO was moderately positively correlated with creatinine and MCS (higher MCS indicates more muscle wasting). The relative abundance of indoxyl sulfate did not change between the three groups (Fig. 4), which differs from previous publications showing higher serum concentrations in CKD versus healthy cats [31, 41]. It is important to note that these previous studies utilized targeted metabolomics to compare absolute concentrations of indoxyl sulfate to a purified standard [31], versus the global-non-targeted approach herein using relative abundances normalized across a matrix of several hundred metabolites. Consequently, these different methods should be considered when comparing studies. Collectively, these results support that monitoring serum uremic toxin levels across disease stages can be linked to clinically-meaningful parameters that can be monitored for earlier intervention to abate disease progression.

The strengths of this study include applying a global metabolomics approach to healthy client-owned cats and those with both early and late-stage CKD using serum, which is a routinely-used and non-invasive diagnostic sample. While this study was not sufficiently powered to compare cats between separate IRIS stages or sub-stages due to low enrollment of Stage 1 and 4 CKD cats, separation of cats into early-stage CKD (Stages 1 and 2) and late-stage CKD (Stages 3 and 4) groups still allows for broader comparisons; this grouping has been applied previously to compare CKD cats [14, 42]. The diversity of diets, medications, and supplements consumed likely contributes to metabolome variation, due to differential effects on host and gut microbial metabolism [43]. The impact of these factors on metabolite abundances cannot be readily discerned in the current study, as specific formulations, frequency and duration of use are not known for all study cats. Two cats were not fasted before serum collection, which may contribute to some metabolome variation (**Supplementary File 1**). However, despite this, the serum metabolome robustly distinguished between healthy, early-stage, and late-stage CKD cats, highlighting the clinical relevance of applying this matrix to diverse, client-owned populations of cats and/or human populations. Furthermore, the stage of disease in CKD cats was based on serum creatinine concentrations, which are negatively correlated with muscle mass [44]. This potentially impacted the disease stage cats were given.

The serum metabolome readily differentiates healthy cats from those with early and late-stage CKD and supports that substantial metabolic derangements occur in early-stage CKD. Lipid and amino acid metabolites are key determinants of disease severity and are generally well-correlated with clinical variables used to guide CKD clinical decision-making CKD. Therefore, there is immense promise in

investigating if these metabolites can serve as discriminative and feasibly measured biomarkers that distinguish healthy versus early-stage CKD. Given the similarities between feline and human CKD, biomarkers identified for feline early-stage CKD detection may advance human diagnostics, resulting in improved life quality for animals and people with CKD.

Methods

Study population

The study included serum samples from 55 cats sourced from previous studies. All studies were approved by either the Clinical Review Board (VCS 2018 – 168; VCS 2019 – 198) at Colorado State University or the Institutional Animal Care and Use Committee (IACUC) at Oregon State University (IACUC 2020-0069; IACUC 2020-0065) and were performed in accordance with relevant guidelines and regulations. The study findings are reported adhering to ARRIVE guidelines [45]. Healthy cats and cats diagnosed with CKD were recruited from clients of the Colorado State University Veterinary Teaching Hospital from 2018 to 2019 and Oregon State University Veterinary Teaching Hospital from 2020 to 2021. To be eligible for inclusion, cats underwent a thorough evaluation that included a client history, review of past medical record, and physical examination performed by a board-certified internal medicine specialist. Cats had a complete blood count, serum biochemistry panel, urinalysis and in most cases, urine protein-to-creatinine (UPC) ratio, fecal sugar centrifugation parasite screen, serum total thyroxine concentration, and blood pressure measurement by Doppler sphygmomanometry was performed. A nine-point body condition score (BCS; Nestle Purina, St. Louis, MO, USA) and MCS was obtained (0 = normal; 1 = mild muscle loss; 2 = moderate muscle loss; 3 = severe muscle loss) [46, 47]. Cats were considered healthy based on unremarkable medical history including receiving no medications (besides flea and tick preventatives), physical examination, and normal laboratory testing. Cats with CKD were staged (CKD Stage 1–4) based on serum creatinine concentration using International Renal Interest Society (IRIS) staging guidelines [42]. Exclusion criteria included systemic disease including diabetes mellitus, hyperthyroidism, liver disease, known or suspect gastrointestinal disease and antibiotic therapy within the past two weeks.

Sample collection

Sera was collected as part of unrelated studies approved by Colorado State University Veterinary Teaching Hospital Clinical Review Board (2018 – 168) and the Institutional Animal Care and Use Committee at Oregon State University (2020-0065). Sera was frozen within 4–6 hours of collection and stored in -80°C for up to four years. All participating cat owners provided written consent prior to sample collection. Serum samples were collected via jugular or medial saphenous venipuncture, frozen on site within one hour of collection, and stored at -80°C freezer until analysis. These samples were stored for up to three years prior to utilization in the present study.

Metabolome sample preparation

Analysis of global serum metabolic profiles was performed by a commercial laboratory (Metabolon Inc ©, Morrisville, NC) as previously described. Briefly, serum was submitted in triplicate for each patient on dry ice. Upon arrival, samples were stored at -80°C until processing. Ice cold methanol (80% methanol v/v solution maintained at -80°C) was added to each sample to precipitate proteins and release small molecules prior to chromatographic injection. The resultant samples were vortexed for two minutes (Glen Mills GenoGrinder 2000), centrifuged, and placed on a TurboVap © (Zymark) to remove any remaining ice-cold methanol solvent. Each sample was next subdivided into five separate aliquots for ultra-high performance liquid chromatography tandem mass spectrometry (UPLC-MS/MS) analysis. These aliquots included: two aliquots for reverse-phase UPLC-MS/MS with positive electrospray ionization (ESI), one aliquot for reverse-phase UPLC-MS/MS with negative ESI, a fourth aliquot for hydrophilic interaction liquid chromatography tandem mass spectrometry (HILIC/UPLC-MS/MS) with negative ESI, and a fifth aliquot saved as a spare. All aliquots were stored under nitrogen prior to downstream analysis.

UPLC-MS/MS analysis

Each aliquot was dried and reconstituted in solvents optimized for one of four chromatographic extraction methods. This setup included one aliquot optimized for elution of hydrophilic metabolites that was gradient-eluted through a C18 column (Waters UPLC BEH C18-2.1x100 mm, 1.7 µm) with a mobile phase of water and methanol with 0.05% perfluoropentanoic acid and 0.1% formic acid. The second aliquot was optimized for hydrophobic metabolite extraction through the same C18 column with a gradient elution in methanol, acetonitrile, and water with 0.05% perfluoropentanoic acid and 0.01% formic acid. The third aliquot was optimized for basic metabolite extraction, through a separate C18 column, via gradient elution through a mobile phase of water and methanol titrated to a pH of 8 with 6.5 mM ammonium bicarbonate. The fourth aliquot was optimized for elution through a HILIC column (Waters UPLC BEH Amide 2.1x150 mm, 1.7 µm) with a mobile phase gradient of water and acetonitrile titrated to a pH of 10.8 with 10 mM of ammonium formate. Following chromatographic extraction, eluted metabolites from each sample underwent fragmentation using a heated electrospray ionization (HESI-II) source ran in either positive or negative ion mode. The first two aliquots were analyzed using positive ion mode conditions and the last two aliquots were analyzed using negative ion mode conditions. Metabolite detection was performed using a Thermo Scientific Q-Exactive high resolution/accurate mass spectrometer integrated with an Orbitrap mass analyzer set to a 35,000-mass resolution, which alternated between MS and data-dependent MSⁿ scans using dynamic exclusion. The scan range for this machine covered mass to charge ratios (m/z) of ~ 70-1000 m/z.

For all analyses, a blank was included in each sample run, where initial extraction of ultra-pure water was used to establish the baseline charge signal running through the mass spectrometer. Positive controls included external standards comprised of well-characterized human plasma samples previously run on the chromatography and mass spectrometry equipment as well as a pooled sample consisting of equal

amounts of each experimental sample. A third control consisted of an internal standard cocktail that consisted of metabolites at known concentrations that would not interfere with endogenous metabolite analysis (i.e., metabolites not found in serum). Negative controls included aliquots of the solvents used in the various sample extractions. To control for chromatographic drift, sample elution through the chromatographer was randomized and control samples were proportionately spaced between experimental sample injections.

Data extraction, compound identification and quantitation

Mass spectral data was extracted and processed using proprietary technologies owned by Metabolon Inc. Briefly, mass spectral peaks are identified as compounds through peak comparison to an internal library of over 4,500 purified standards and recurrent unknowns (i.e., metabolites identified only to the m/z level). Compound identification was made based on retention index matches, a mass match +/- 10 ppm, and evaluation of forward and reverse MS/MS scores. To determine metabolite raw abundances, area under the curve was quantified for each peak after controlling for mass spectrometer current differences within runs and between runs. Pathway enrichment scores (**PES**) for selected metabolic pathways were calculated using the following equation:

$$PathwayEnrichmentScore = \frac{\binom{k}{m}}{\binom{n}{N}}$$

Where “k” is the number of significant metabolites in the metabolic pathway, “m” is the number of identified metabolites in the pathway, “n” is the number of significant metabolites in the comparison of interest (e.g. healthy versus early-stage CKD, healthy versus late-stage CKD, or early-stage versus late-stage CKD), and “N” is the total number of metabolites in the dataset.

Statistical Analysis

For both clinical metadata and the serum metabolome, CKD cats were grouped as early-stage CKD (Stages 1 and 2) and late-stage CKD (Stages 3 and 4) for the purpose of statistical analysis. Clinical variables (bloodwork parameters, weight, BCS, MCS, age) were compared between healthy (n = 25), early-stage CKD (n = 17), and late-stage CKD (n = 13) cats using either a one-way ANOVA or Kruskal-Wallis test with either Tukey post-hoc correction or Dunn’s multiple comparison test, respectively. Normality was confirmed with Shapiro-Wilk test and evaluation of QQ plots was performed. Statistical analysis was performed in GraphPad Prism (Version 9.5.1, GraphPad Software LLC, Boston, MA).

Statistical analysis and visualization for metabolomics data was performed using Metaboanalyst 5.0 and GraphPad Prism [48]. All metabolites were normalized within Metaboanalyst by dividing each raw abundance by the median raw abundance for that metabolite across the dataset, followed by log base 10 transformation. Differential abundance analysis comparing normalized abundances between healthy cats (n = 25), early-stage CKD cats (n = 17), and late-stage (n = 13) CKD cats was performed in GraphPad

Prism using a Kruskal-Wallis test with a Benjamini-Hochberg adjustment for multiple comparisons. Random forest analysis was performed in Metaboanalyst, where feature ranks were assigned based on mean decrease accuracy values generated from a model using 500 trees and seven predictors with the default randomness parameter turned “on”. Unsupervised hierarchical clustering analysis (HCA) was performed in Metaboanalyst using the top 50 most discriminating metabolites (highest variable importance scores) from partial least-squares discriminant analysis (PLS-DA). Heatmaps were generated in Metaboanalyst using normalized data with all features autoscaled (i.e., mean-centered and then divided by the standard deviation of each feature). Sample distances were calculated using Euclidean distances and clustered using Ward algorithms. R code for Metaboanalyst analysis and visualization is provided in **Supplementary File 4**. Spearman’s correlations were used to evaluate the associations between clinical metadata parameters that were continuous or ranked ordinal variables. Pearson’s correlations were used to evaluate the associations between binary (yes/no) variables. Correlation strength was defined based on the following criteria [49]: Very weak correlation ($r = 0.00-0.19$), weak correlation ($r = 0.20-0.49$), moderate correlation ($r = 0.50-0.69$), strong correlation ($r = 0.70-0.89$), and very strong correlation ($0.90-1.00$). For all analyses, significance was defined as $p < 0.05$ following any post-hoc adjustments for multiple comparisons where indicated. For all statistical analyses, p-values smaller than 1.00×10^{-6} were recorded as $p < 1.00 \times 10^{-6}$.

Abbreviations

Bilirubin degradation product: BDP

Body condition score: BCS

Branched chain amino acid: BCAA

Blood urea nitrogen: BUN

Chronic kidney disease: CKD

Fatty acid: FA

Female, spayed: FS

Glomerular filtration rate: GFR

Glycerophosphorylcholine: GPC

Glycerophosphorylethanolamine: GPE

Glycerophosphorylinositol: GPI

Hierarchical clustering analysis: HCA

Hydroxyoctadecadienoic acid: HODE

International Renal Interest Society: IRIS

Male, castrated: MC

Mass-to-charge ratio: m/z

Medium chain fatty acid: MCFA

Mono-unsaturated fatty acid: MUFA

Muscle condition score: MCS

Partial least squares discriminant analysis: PLS-DA

Pathway enrichment score: PES

Short chain fatty acid: SCFA

Symmetric dimethyl arginine: SDMA

Trimethylamine N-oxide: TMAO

Urine specific gravity: USG

Ultra-high performance liquid chromatography tandem mass-spectrometry: UPLC-MS/MS

Declarations

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Author Contributions:

Funding for this study was secured by S.S. and J.Q. This study was conceptualized by J.Q. and S.S. Collection of clinical samples and patient metadata was performed by JQ and SS. Curation and statistical analysis of clinical metadata was performed by S.S., and by N.J.N. for metabolomics data. N.J.N, S.S., J.A.W. and J.Q. contributed to data interpretation. N.J.N., S.S., and J.A.W. were responsible for figure and table creation. Manuscript writing was completed by N.J.N and S.S. Manuscript reviewing and

editing was performed by N.J.N, S.S., J.A.W., and J.Q. All authors reviewed and endorsed the final manuscript for publication.

Data Availability Statement:

Metabolomics and/or clinical metadata analyzed in this study is available by request and under the discretion of the corresponding author.

Additional Information:

Competing Interests

N.J.N. has no competing interests to declare. S.S. is a research consultant for IDEXX Laboratories, Inc. and previous work funded by Nestle Purina and IDEXX Laboratories, Inc. She has received a speaker honorarium from Royal Canin, IDEXX Laboratories, Inc., and Boehringer-Ingelheim. Preliminary results from this analysis were presented in abstract form at the 2022 Annual Forum of American College of Veterinary Internal Medicine, held in Austin, Texas (ePoster NU30: Untargeted Metabolomic Profiling of Serum from Cats with Chronic Kidney Disease). J.Q.'s work has been funded by EveryCat Health Foundation, Morris Animal Foundation, Nestle Purina, Trivium Vet, Zoetis. She has received compensation as a member of the scientific advisory board of Nestle Purina, Elanco, Zoetis. She also has consulted or served as a key opinion leader for Boehringer Ingelheim, Dechra, Elanco, Gallant, Heska, Hill's, IDEXX, Nestle Purina, Royal Canin, SN Biomedical, Vetoquinol, Zoetis and received compensation. J.A.W.'s laboratory has been funded by EveryCat Health Foundation, Morris Animal Foundation, American Kennel Club's Canine Health Foundation, Nestle Purina, FDA, and National Institutes of Health. She has received speaker honorariums from Royal Canin, Nestle Purina, and DVM360.

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Figures

Figure 1

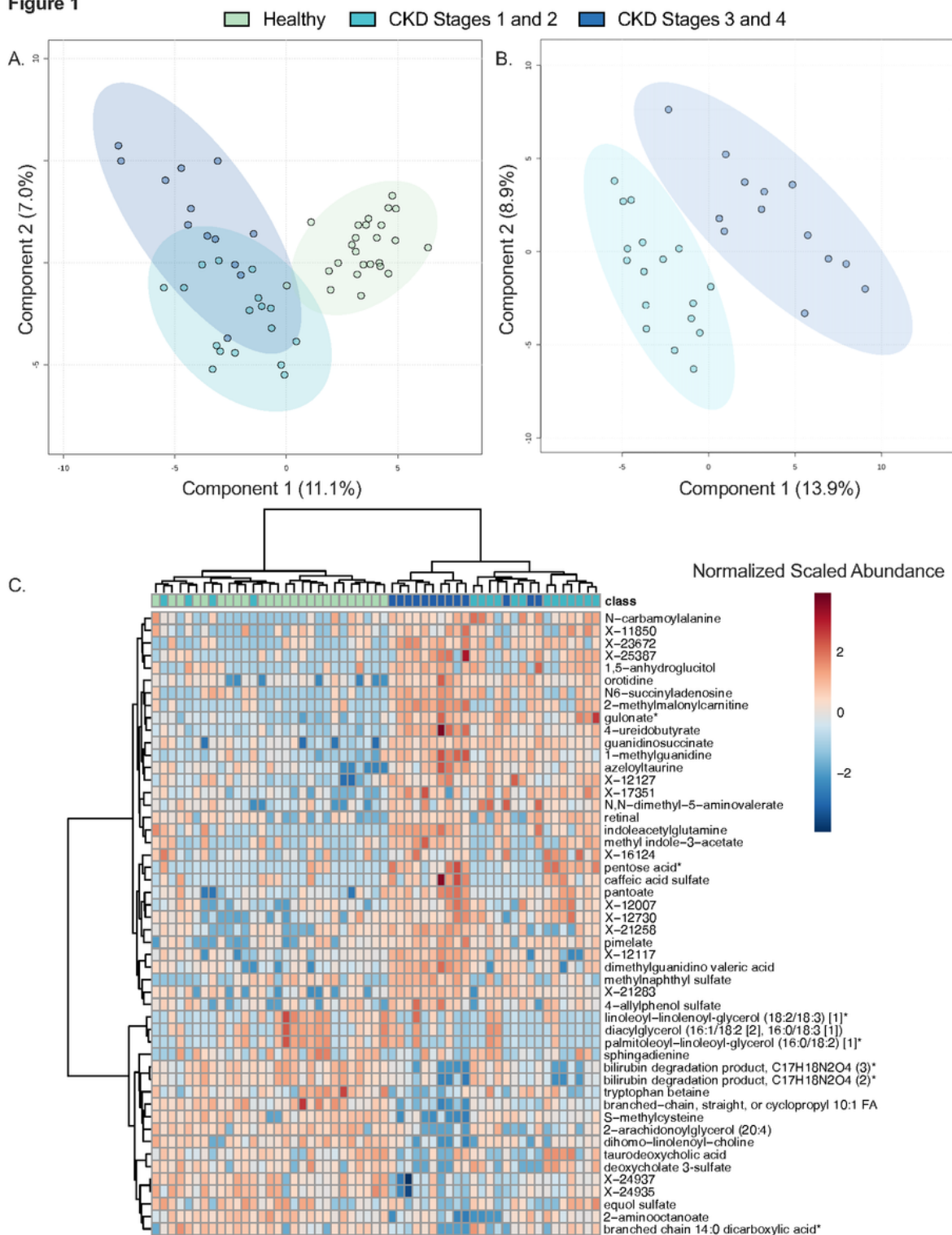


Figure 1

Disease stage distinctly differentiates the serum metabolome of healthy, early-stage CKD, and late-stage CKD cats. Partial least squares discriminant analysis (PLS-DA) projections of healthy cats, cats with early-stage CKD (Stages 1 and 2), and cats with late-stage CKD (Stages 3 and 4) (a) and cats with early-stage CKD versus late-stage CKD (b). Each circle represents the serum metabolome of one cat. Shaded ellipses surrounding each patient group represent 95% confidence intervals. Unsupervised hierarchical

clustering analysis and heatmap of the 50 serum metabolites with the largest PLS-DA mean decrease accuracy scores (c). Each column represents one cat and each box represents one metabolite. Class boxes refer to the disease state of each cat, where green boxes indicate healthy cats, teal boxes represent early-stage CKD cats, and navy boxes represent late-stage CKD cats. Metabolite box colors reflect the normalized, scaled relative abundance of each metabolite when scaled across the dataset, where red boxes reflect an increased normalized abundance relative to the dataset median and blue boxes show metabolites with decreased normalized abundance relative to the dataset median. Branch points were calculated using Euclidean distances where longer branches indicate larger differences between cats. Abbreviations: CKD = Chronic kidney disease; FA = Fatty acid. [1] and [2] in metabolite names are used to indicate isomers and * in names indicates metabolite identities were made using *in-silico* annotations.

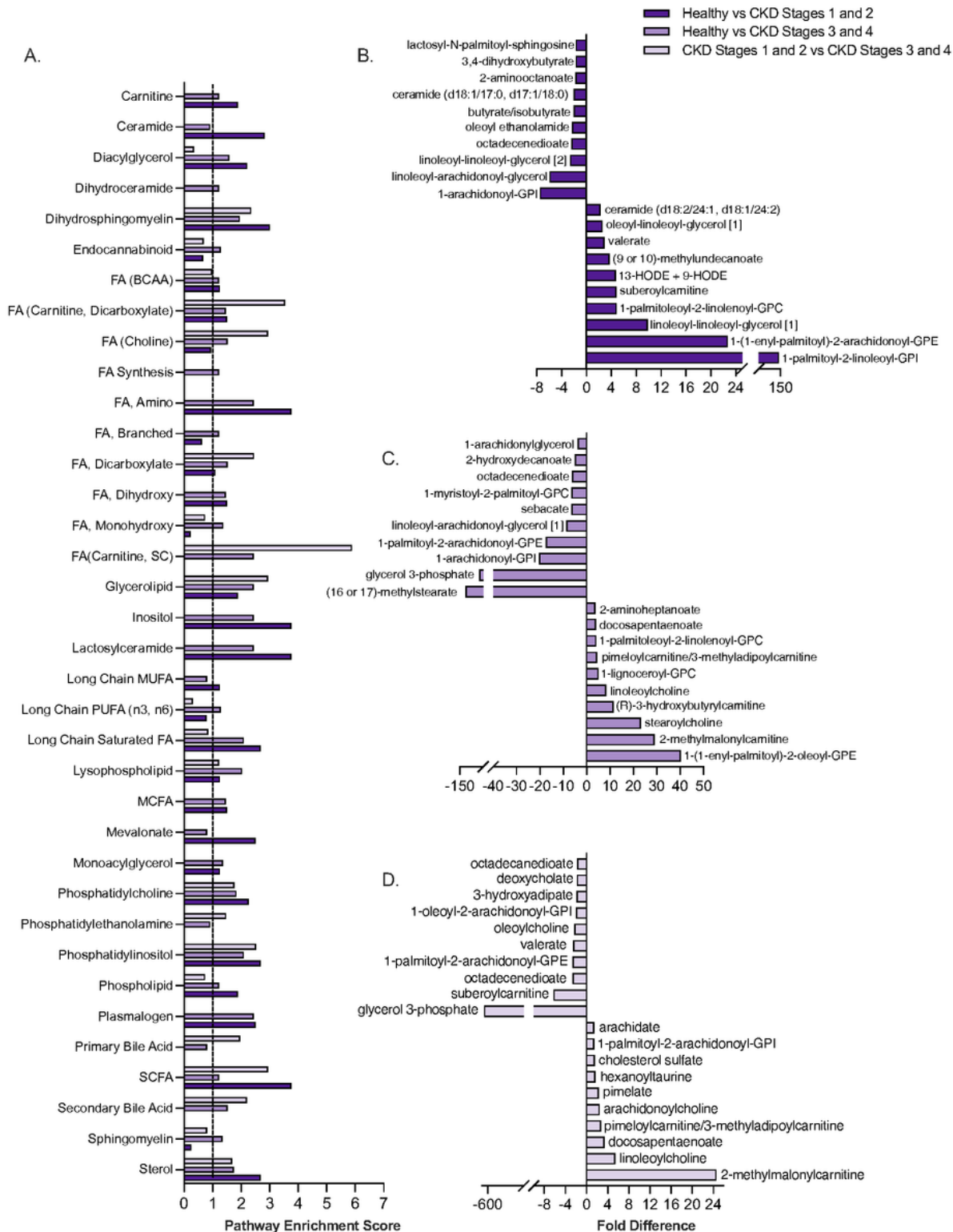


Figure 2

Lipid metabolism is a key driver of serum metabolome differences between healthy, early-stage CKD and late-stage CKD cats. Pathway enrichment scores of lipid metabolic pathways comparing healthy cats, early-stage (Stages 1 and 2) and late-stage (Stages 3 and 4) cats (**a**). Dotted line at 1.0 shows metabolic pathways that were defined as meaningful contributors to patient group differences (pathway enrichment score of ≥ 1.0 in at least one patient group). Differentially abundant serum lipids with the 20 largest fold

differences when comparing healthy cats versus early-stage CKD cats (**b**), healthy cats versus late-stage CKD cats (**c**) and early-stage versus late-stage CKD cats (**d**). Significance was defined as $p \leq 0.05$ following Benjamini-Hochberg adjustments to a Kruskal-Wallis test comparing normalized, scaled abundances of each metabolite across the three patient groups. Abbreviations: BCAA = Branched chain amino acid; CKD = Chronic kidney disease; FA = Fatty acid; GPC = glycerophosphorylcholine; GPE = glycerophosphorylethanolamine; GPI = glycerophosphorylinositol; HODE = Hydroxyoctadecadienoic acid; MCFA = Medium chain fatty acid; MUFA = Mono-unsaturated fatty acid; SC = Short chain; SCFA = Short chain fatty acid. [1] and [2] in metabolite names are used to indicate isomers.

Healthy CKD Stages 1 and 2 CKD Stages 3 and 4

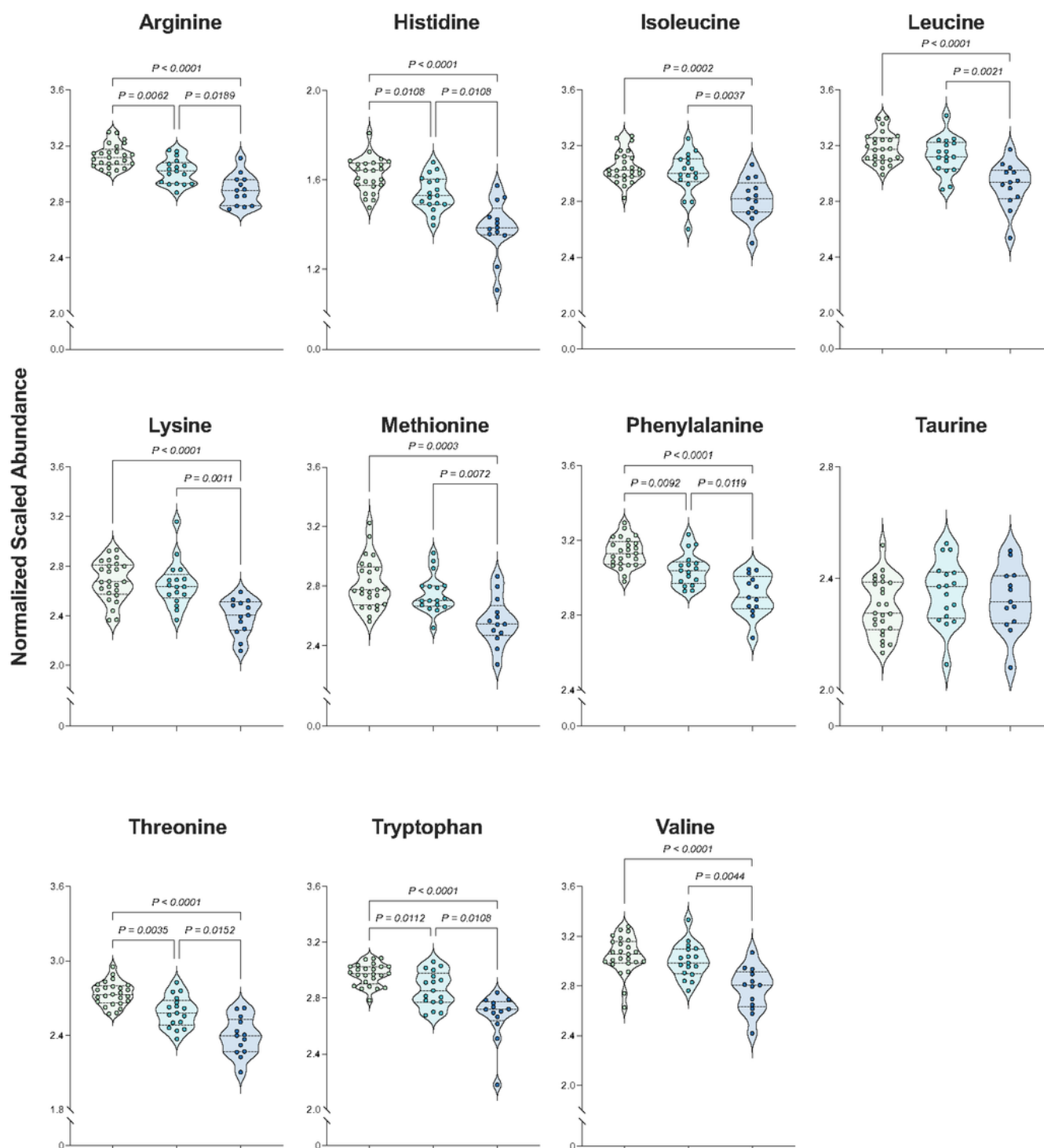


Figure 3

Cats with late-stage CKD exhibit decreased serum abundances of essential amino acids compared to healthy cats and those with early-stage CKD. Normalized, scaled abundances of 11 feline essential amino acids. Each circle represents one cat, where colors refer to patient groups: Green = Healthy cat; Teal = Early-stage CKD cat (Stages 1 and 2); Navy = Late-stage CKD cat (Stages 3 and 4). Dotted lines on each violin plot show the 25th, 50th (median) and 75th percentiles of normalized, scaled metabolite abundance

distributions for each amino acid. Significance was defined as $p \leq 0.05$ following Benjamini-Hochberg adjustments to a Kruskal-Wallis test comparing normalized, scaled abundances of each metabolite across the three patient groups. Abbreviations: CKD = chronic kidney disease.

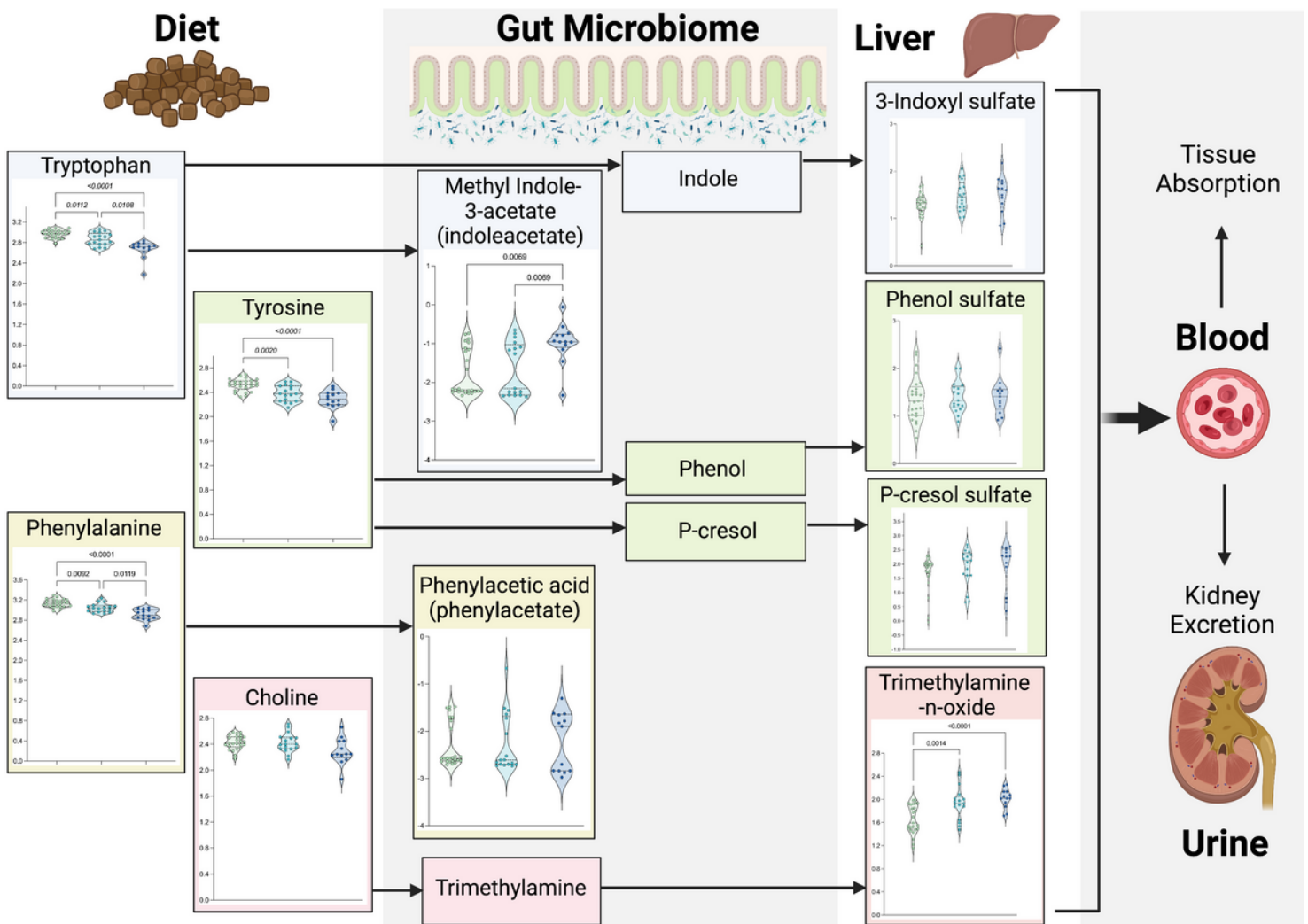


Figure 4

Increased abundances of uremic toxins are present in the serum of cats with late-stage versus early-stage CKD and healthy cats. Normalized, scaled abundances of ten uremic toxins. Each circle represents one cat, where colors refer to patient groups: Green = Healthy cat; Teal = Early-stage CKD cat (Stages 1 and 2); Navy = Late-stage CKD cat (Stages 3 and 4). Dotted lines on each violin plot show the 25th, 50th (median) and 75th percentiles of normalized, scaled metabolite abundance distributions for each amino acid. Arrows between metabolites indicate their relationships to each other in uremic toxin metabolic pathways, where metabolites to the left of an arrow are upstream metabolites (precursors) to the metabolites on the right side of arrows. Significance was defined as $p \leq 0.05$ following Benjamini-Hochberg adjustments to a Kruskal-Wallis test comparing normalized, scaled abundances of each metabolite across the three patient groups. Figure created with BioRender.com. Abbreviations: CKD = chronic kidney disease.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [FileS1CKDSupplementaryDemographicMetadata10.27.23FINAL.xlsx](#)
- [FileS2DifferentiallyAbundantMetabolitesStats10.27.23FINAL.xlsx](#)
- [FileS3FelineCKDSpearmanCorrelations10.27.23FINAL.xlsx](#)
- [FileS4RScriptMetaboanalyst10.27.23FINAL.r](#)

STANDARD ARTICLE

Small Animal Internal Medicine
Nephrology/Urology

Alterations in gut-derived uremic toxins before the onset of azotemic chronic kidney disease in cats

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Abstract

Background: Although gut-derived uremic toxins are increased in azotemic chronic kidney disease (CKD) in cats and implicated in disease progression, it remains unclear if augmented formation or retention of these toxins is associated with the development of renal azotemia.**Objectives:** Assess the association between gut-derived toxins (ie, indoxyl-sulfate, p-cresyl-sulfate, and trimethylamine-N-oxide [TMAO]) and the onset of azotemic CKD in cats.**Animals:** Forty-eight client-owned cats.**Methods:** Nested case-control study, comparing serum and urine gut-derived uremic toxin abundance at 6-month intervals between initially healthy cats that developed azotemic CKD (n = 22) and a control group (n = 26) that remained healthy, using a targeted metabolomic approach.**Results:** Cats in the CKD group had significantly higher serum indoxyl-sulfate (mean [SD], 1.44 [1.06] vs 0.83 [0.46]; $P = .02$) and TMAO (mean [SD], 1.82 [1.80] vs 1.60 [0.62]; $P = .01$) abundance 6 months before the detection of azotemic CKD. Furthermore, logistic regression analysis indicated that indoxyl-sulfate (odds ratio [OR]: 3.2; 95% confidence interval [CI]: 1.2–9.0; $P = .04$) and TMAO (OR: 3.9; 95% CI: 1.4–11; $P = .03$) were predictors for the onset of azotemia 6 months before diagnosis. However, renal function biomarkers creatinine, symmetric dimethylarginine, and urinary specific gravity were significantly correlated with indoxyl-sulfate and TMAO abundance, causing a loss in predictive significance after correction for these factors.**Conclusions:** Impaired gut-derived uremic toxin handling is apparent at least 6 months before the diagnosis of azotemia, likely reflecting an already ongoing decrease in GFR, tubular function, or both. A direct causal relationship between gut-derived uremic toxicity and the initiation of CKD in cats is still lacking.**Abbreviations:** CKD, chronic kidney disease; IQR, inter quartile range; IRIS, International Renal Interest Society; FDR, false discovery rate; GFR, glomerular filtration rate; QCS, quality Control samples; SDMA, symmetric dimethylarginine; sCr, serum creatinine; TMA, trimethylamine; TMAO, trimethylamine-N-oxide; UHPLC-HRMS, ultrahigh-performance liquid chromatography-high-resolution mass spectrometry; UPC, urine protein to creatinine ratio; USG, urine-specific gravity.This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.© 2024 The Author(s). *Journal of Veterinary Internal Medicine* published by Wiley Periodicals LLC on behalf of American College of Veterinary Internal Medicine.

KEYWORDS

CKD development, early intervention, feline chronic kidney disease, pathophysiology, uremic toxicity

1 | INTRODUCTION

In CKD of cats, it is well-established that circulating concentrations of specific gut-derived uremic toxins (indoxyl-sulfate, p-cresyl-sulfate, and trimethylamine-N-oxide [TMAO]) start to accumulate with declining renal function.^{1,2} The formation of such toxins involves intestinal bacterial modification from dietary precursors.¹⁻⁵ Because these toxins (especially indoxyl-sulfate) are well-recognized mediators of tubular inflammation and have been associated with progression to advanced renal dysfunction,⁶⁻⁸ therapeutic reduction of retained uremic toxins has been an important topic in recent research.^{9,10} Interestingly, different approaches for decreasing uremic toxin concentrations in cats with CKD have been introduced including: decreasing dietary protein content,¹¹ altering the gut microbiome,^{12,13} and binding intestinal toxin precursors.¹⁴

Despite the observation that uremic toxins accumulate in azotemic CKD in cats and are implicated in disease progression, it remains unclear if increased formation or retention of these toxins is associated with the development of renal azotemia. In the progression from initial functional decline to azotemic CKD, a high dietary protein content typical of diets for cats may play a considerable role.^{11,15,16} Therefore, it can be hypothesized that formation or retention of specific gut-derived uremic toxins derived from dietary precursors highly abundant in the diet may underlie this observation in the early disease process.^{17,18} Addressing this knowledge gap may provide supporting argumentation for the potential involvement of these toxins in the initial pathological mechanisms underlying CKD in cats and may support the use of toxin lowering strategies in early disease stages.

Therefore, our main aim was to investigate the relationship between gut-derived uremic toxins and the development of renal azotemia in cats. We focused on measuring serum and urine abundance of indoxyl-sulfate, p-cresyl-sulfate, and TMAO, as well as their dietary precursors, in cats before the onset of azotemic CKD.¹⁹⁻²¹ By conducting longitudinal follow-up of healthy senior cats, we hoped to identify differences in gut-derived uremic toxin pathways between cats that developed azotemic CKD and those that remained healthy. Using a targeted metabolomic approach we specifically intended to determine if increased abundance of these toxins could predict the onset of azotemic CKD. Ultimately, our goal was to establish whether impaired handling of gut-derived uremic toxins is evident before the diagnosis of azotemic CKD in cats.

2 | MATERIALS AND METHODS

2.1 | Ethics statement

The study protocol was approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering

(EC2020-081). All specimens from cats in the study were obtained prospectively from privately owned domestic cats using a proactive recruitment process. Written informed consent was obtained before inclusion in the study.

2.2 | Study design

Our nested case-control study comprised a prospective cohort investigation involving a target population of healthy domestic senior cats (>10 years of age) at baseline. The evaluated outcome for the target population was development of azotemic CKD. The key criteria for CKD diagnosis were serum creatinine concentration (sCr) >1.83 mg/dL (>161.8 µmol/L), urinary specific gravity (USG) <1.035, and compatible history. The study population consisted of initially healthy senior cats that were recruited via the research facility of the Small Animal Clinic at the Faculty of Veterinary Medicine, between September 2021 and March 2023. Health at baseline was determined by a physical examination, systolic blood pressure (SBP) measurement (Doppler method), CBC, serum biochemistry (including sCr, symmetric dimethylarginine [SDMA], electrolytes), feline leukemia virus antigen testing (FeLV), feline immunodeficiency virus serology (FIV), and complete urinalysis including urinary protein-to-creatinine ratio (UPC) determination and bacterial culture, following established protocols.²²⁻²⁷ The cats were screened every 6 months using the same tests (except FIV and FeLV screening). The study population was further defined based on health reevaluation every 6 months, with development of azotemic CKD as the outcome of interest. Cats showing evidence of any other clinically relevant health abnormalities (except situational hypertension) on the listed examinations at baseline or during follow-up were excluded.

The study group consisted of:

1. A CKD development cohort including all cats from the study population that developed azotemic CKD during the study period based on the defined criteria (sCr >1.83 mg/dL, USG <1.035).²⁸ Data on cats that developed CKD was tracked back to 6 and 12 months before their diagnosis (ie, timepoints –12 months, –6 months, and time of CKD onset). Cats suspected of acute kidney injury were not included. The sCr cut-off in the study was based on a previously established laboratory-specific reference interval consisting of 130 healthy cats.²⁹
2. A control cohort of randomly selected cats from the study population that remained healthy throughout 1 year of follow-up. Data on cats that remained healthy was tracked back to 6 and 12 months before health confirmation (ie, timepoints –12 months, –6 months, and maintained health). Additionally,

cats with renal function abnormalities (sCr >1.83 mg/dL, SDMA >14 µg/dL, USG <1.035, UPC >0.3 or a clinically relevant increase over time in sCr, SDMA, or both [$>25\%$] within the reference interval) were not eligible for inclusion in the control cohort.²⁷ Controls were matched to cases in an approximately 1 : 1 ratio by simple random sampling. The general study design is presented in Figure 1.

All cats, regardless of their cohort, were fed commercially available diets. Specific renal formula diets were introduced only after the onset of azotemic CKD.

The primary objective of our nested case-control study was to assess the association between circulating gut-derived uremic toxins and the development of azotemic CKD before diagnosis. Furthermore, the longitudinal evolution in gut-derived uremic toxin metabolism and

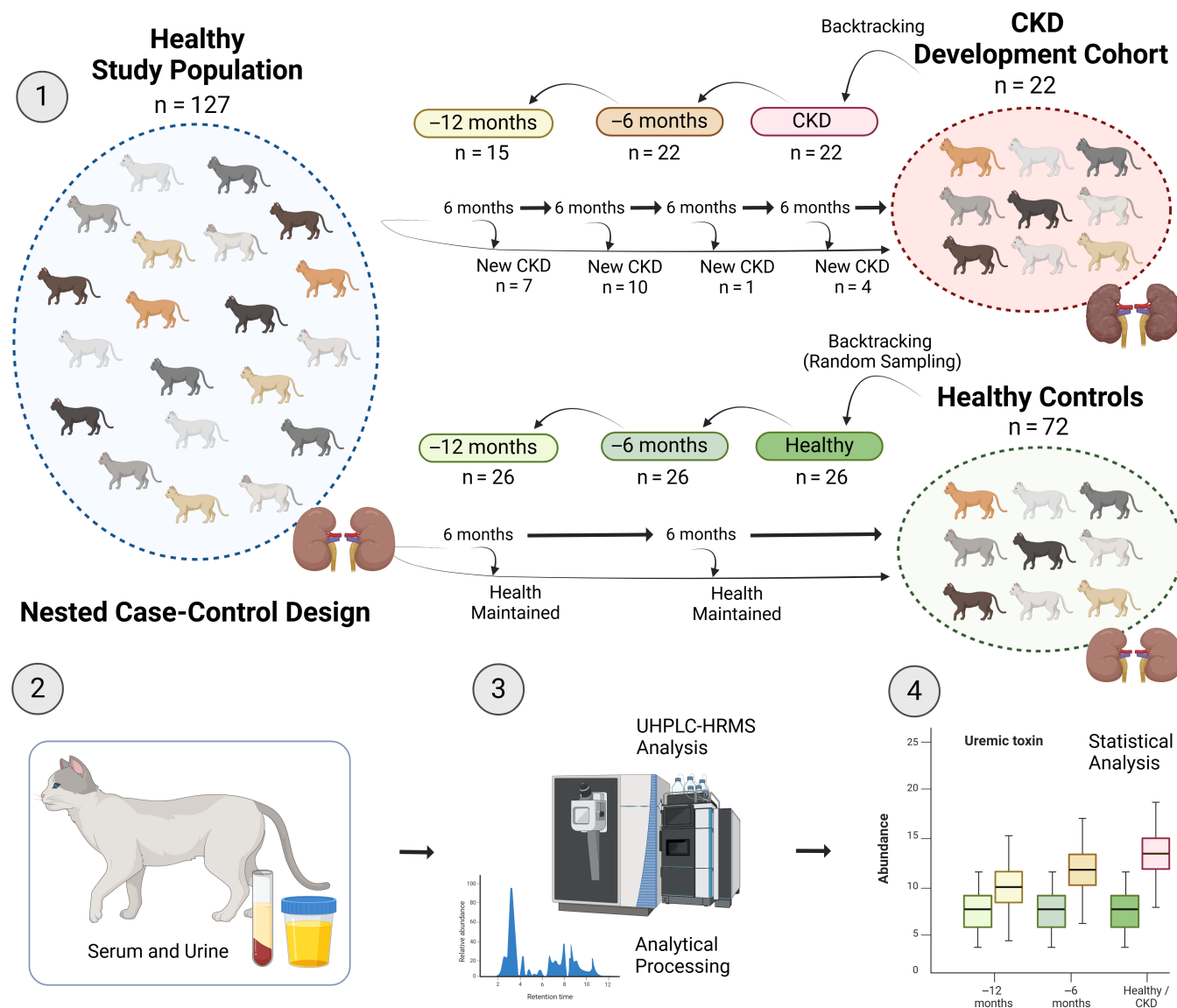


FIGURE 1 (1) This prospective nested case-control study consisted of a healthy population of cats that was followed longitudinally, with assessments conducted at 6-month intervals. Group 1 consisted of all cats that developed azotemic CKD during the study (further referred to as CKD development cohort). Data on cats that developed CKD was tracked back to 6 and 12 months before their diagnosis (ie, timepoints -12 months, -6 months, and CKD onset). Group 2 consisted of a control cohort of cats that remained healthy throughout 1 year follow-up. Likewise, data on cats that remained healthy was tracked back to 6 and 12 months before health confirmation (ie, timepoints -12 months, -6 months, and maintained health). (2) Serum and urine samples were collected, followed by chemical extraction according to feline specific optimized and validated methods. (3) Samples from both matrices were analyzed using ultrahigh-performance liquid chromatography coupled to high-resolution mass spectrometry (UHPLC-HRMS). (4) The primary objective of our nested case-control study was to assess the association between circulating gut-derived uremic toxins and the development of azotemic CKD, 6 and 12 months before diagnosis. Statistical analyses also aimed to compare metabolite abundance between the CKD development cohort and the healthy cohort (at timepoints -12 months, $n = 15$ vs $n = 26$; -6 months, $n = 22$ vs $n = 26$ and CKD onset or maintained health, $n = 22$ vs $n = 26$). Furthermore, the trend of altering metabolites was evaluated over time in the CKD development cohort ($n = 15$).

excretion during the development of azotemic CKD (ie, 12 and 6 months before and at time of azotemic CKD diagnosis) was of specific interest and compared to the healthy cohort at each timepoint (ie, at timepoints –12 months, –6 months, and maintained health). To this end, we employed a targeted metabolomic approach aimed at analyzing (I) specific gut-derived uremic toxins, (II) their intermediates, and (III) dietary precursors. Such an approach allows for the elucidation of metabolic changes in uremic toxin-related pathways that are associated with development of overt CKD in cats.^{1,2} In this pathophysiological context, respectively (I) indoxyl-sulfate, TMAO, and p-cresyl-sulfate; (II) indole, trimethylamine (TMA), and p-cresol; and (III) tryptophan, carnitine, betaine, and tyrosine emerged as important metabolites.^{19–21} Therefore, they were selected to assess their evolution in preazotemic CKD stages.

2.3 | Sample collection

Blood samples were acquired by venipuncture, specifically from the cephalic or jugular veins (23-gauge needle), and collected into plain serum tubes (VACUETTE Serum Tubes with Gel). The samples were allowed to clot for maximum 15 minutes at room temperature, followed by centrifugation (5 minutes; 1931×g) at 4°C. Within a narrow time window (15 minutes), sterile urine samples were obtained by ultrasound-guided cystocentesis, utilizing a 22-gauge needle and transferred to plain tubes. The samples then were centrifuged (3 minutes; 355×g) at room temperature. After centrifugation, both serum and urine supernatants were aliquoted into Eppendorf tubes and preserved at –80°C until further analysis.

2.4 | Metabolomics analyses

All 10 analytical standards (Table S1) and internal standard (ISTD) were purchased from reputable sources (Sigma-Aldrich [St Louis, Missouri, USA], ICN Biomedicals Inc [Ohio, USA]).

All solvents utilized in this experiment (ie, acetonitrile, acetone, and methanol) were of liquid chromatography-mass spectrometry grade for extraction purposes and sourced from Thermo Fisher Scientific (Loughborough, UK) and VWR International (Merck, Darmstadt, Germany). Ultrapure water was obtained by the utilization of a purified water system, Arium 611UV (Sartorius, Göttingen, Germany).

We designed, optimized, and validated specific metabolomics extraction protocols for feline serum and urine samples targeting the selected metabolites.³⁰ For both biomatrices (urine, serum), the same ultrahigh-performance liquid chromatography (UHPLC) and high-resolution mass spectrometry (HRMS) settings were applied.^{1,30} For chromatographic separation, a Vanquish Horizon UHPLC system (Thermo Fisher Scientific, San José, CA, USA) was used with an Acquity UHPLC HSS C18 column T3 1.8 µm (150 mm × 2.1 mm; Waters, Manchester, UK) at a column temperature of 45°C. An injection volume of 10 µL at a maximum injection time of 150 ms was applied. Ultrapure water and acetonitrile, both acidified with 0.1%

formic acid, were used as analytical solvents. A constant flow rate of 0.4 mL/min was maintained throughout the analysis. An Exploris 120 stand-alone bench top orbitrap HRMS (Thermo Fisher Scientific, San José, CA, USA), equipped with a heated electrospray ionization source (HESI II), operating in polarity switching full scan mode at a scan range of 53–800 Da and with a mass resolution of 120 000 full width at half maximum (1 Hz), was used. Before mass spectrometric detection, initial instrument calibration was achieved by infusing ready-to-use calibration mixtures (Thermo Fisher Scientific, San José, CA, USA).

For each biomatrix, quality control samples (QCs) were generated by pooling sample extracts specific to the respective matrix. To adjust for instrumental fluctuations, QCs were run in duplicate after every 10 samples. Furthermore, at the start of the analyses, samples from the QC pool were injected to facilitate column conditioning, tailored to the specific biomatrix under investigation. An analytical standard mixture containing all a priori selected analytical standards (n = 10); (Table S1) was injected before, in the middle, and at the end of the analyses. This procedure enabled the assessment of the operational parameters of the device and ensured accurate identification of the designated metabolites.

2.5 | Data analysis

Targeted data processing was carried out using Xcalibur 3.0 (Thermo Fisher Scientific, San José, CA, USA). In this process, the identification of compounds relied on their *m/z*-value, carbon-isotope (C-isotope) profile, MS/MS fragmentation spectra, and retention time in relation to that of the internal standard.^{30,31} Compound abundance was determined based on the total area under the curve. Subsequently, the data underwent QC normalization. In the case of urine data, an additional normalization step was applied, where the data was adjusted by urinary creatinine concentration.^{32,33} We employed R (version 4.2.1) and SPSS Statistics 28 for statistical analyses. Group comparisons were conducted using QC-normalized abundance, without absolute quantification of metabolites.

Serum and urine abundance of gut-derived uremic toxins and their precursors were compared between the healthy cohort and the CKD development cohort at 3 timepoints: 12 months prior, 6 months prior, and at the time of azotemic CKD onset or maintained health. To do so, a nonparametric independent Mann-Whitney *U*-test (for non-normally distributed data) or students *t*-test was employed (for normally distributed data). The Friedman test was applied to evaluate the progression of specific metabolites over time in the CKD development cohort, using only data from those cats with available measurements at all timepoints (ie, 12 months prior, 6 months prior, and at the onset of azotemia). Normality was evaluated using the Shapiro-Wilk test, and the data distribution was visualized using histograms and QQ-plots. To address the issue of multiple comparisons, a post hoc correction procedure based on false discovery rates (FDR), specifically the Benjamini-Hochberg method, was applied. To assess the predictive ability and associations of circulating uremic toxins (indoxyl-sulfate,

p-cresyl-sulfate, and TMAO) with the development of azotemic CKD, univariate binary logistic regression analysis was conducted. Each model was constructed to forecast the binary outcome for CKD development vs remaining healthy (control group) based on the abundance of the specific uremic toxins: indoxyl-sulfate, TMAO, and p-cresyl-sulfate, assessed 6 months before the onset of azotemic CKD. Age, sCr, SDMA, USG, and UPC were included in our logistic regression analysis because they were reasoned to confound the association between the selected uremic toxins and CKD. Especially, the influence of decreasing glomerular filtration rate (GFR), estimated by sCr and SDMA and the effect of renal tubular function, estimated by USG and UPC on the circulating abundance of the selected gut-derived uremic toxins were of specific interest before development of azotemic CKD. The effect of age per se was added to exclude potential effects other than renal function on circulating gut-derived uremic toxins. Spearman correlation analysis was performed to assess the underlying relationship between the studied gut-derived uremic toxins and the selected covariates in the prediction of azotemic CKD onset.

3 | RESULTS

3.1 | Inclusion

The eligible population included 171 cats that appeared healthy and initially were presented by their owners. Of these, 127 cats were included as the study population for longitudinal follow-up. The main reasons for exclusion at baseline were hyperthyroidism ($n = 25$), FIV-positive status ($n = 9$), preexisting CKD diagnosis ($n = 6$), and other morbidities ($n = 4$). After the first follow-up at 6 months, 7 cats developed azotemic CKD; after 12 months, an additional 10 cats; at 18 months, 1 cat; and after 24 months, another 4 cats. Consequently, a total of 22 cats developed CKD, with backtracked samples and data available for 6 months before diagnosis, and for 15 cats, data was available 12 months before diagnosis. After 12 months, 72 cats from the initial cohort of healthy cats remained in the study. The main reasons for exclusion from the healthy group were CKD diagnosis ($n = 17$), development of other morbidities ($n = 13$), owner noncompliance ($n = 11$), death or euthanasia ($n = 8$), and hyperthyroidism ($n = 6$). Of these 72 eligible cats, 26 cats were randomly selected and backtracked as the control cohort, meeting the specified criteria regarding renal function (Figure 1).

3.2 | Routine renal parameters

The CKD development cohort was on average 2 years older than the healthy control cohort at baseline (Table 1). Body weight did not differ between groups at baseline, 6 months before and at the time of CKD diagnosis, and no clear trend was observed regarding weight loss in the CKD development cohort before onset of azotemia (Table S2). Furthermore, no significant differences were found between groups regarding SBP at all timepoints. Symmetric dimethylarginine but not

sCr was significantly higher at baseline in the CKD development cohort. However, 6 months before azotemia developed, both SDMA and sCr were higher in the CKD group compared with the healthy cohort. Importantly, even at this timepoint, SDMA remained within the reference range ($8\text{--}14\text{ }\mu\text{g/dL}$) for 82% (18/22) of the cats (Table 1). Serum creatinine concentrations showed a clear continuous increase from baseline to the development of azotemia (Table 1), particularly during the 6 months leading up to the onset of azotemia (Table S2). The SDMA concentrations began to increase only between 6 months prior and at the time of CKD diagnosis (Table S2), without showing a significant overall increasing trend during the 1-year period (Table 1). Urinary specific gravity was markedly decreased at baseline in the CKD development cohort, falling around the threshold of optimal urine concentration, and showing a significant downward trend over time within this group (Table 1). Conversely, UPC showed no differences between the 2 groups at any timepoint and remained stable over time within the CKD development cohort (Tables 1 and S2).

3.3 | Uremic toxins and associated metabolites

In the indole pathway, both the bacterial tryptophan catabolite indole and its uremic toxin derivative, indoxyl-sulfate, showed significantly increased serum, but not urine, abundance 6 months before CKD diagnosis compared with the healthy controls (Tables 2, 3, and Figure 2). These increases in serum abundance were even more pronounced at the time of diagnosis. Over the different timepoints, a significantly increasing pattern in serum indole abundance was observed within the CKD development cohort, especially between 12 and 6 months before diagnosis (Tables 2 and S3). Despite indoxyl-sulfate having a relatively high abundance already observed at baseline, a significant increase was observed between 12 months prior and the time of CKD diagnosis (Table S3). However, changes in indole and indoxyl-sulfate abundance 12 months before the onset of azotemia remained statistically insignificant in comparison with the healthy cohort. Tryptophan showed a significant decrease in urinary abundance at the time of azotemic CKD diagnosis, but no significant changes in serum were observed.

A significant increase in serum TMAO, but not TMA was observed 6 months before the onset of azotemia (Table 2 and Figure 2). Trimethylamine-N-oxide also showed augmented urinary excretion at that time (Table 3). At the time of diagnosis, the serum abundance and urinary excretion of TMAO further increased and serum trimethylamine (TMA) also became significantly increased. Both serum and urine TMAO showed significant upward patterns during azotemic CKD development, especially between 12 and 6 months before diagnosis for serum abundance (Tables 2, 3, S3, and S4). Serum carnitine and betaine showed no clear changes in abundance between the CKD and control group, but, a significant upward trend was observed for carnitine over time within the CKD development cohort attributed to the increase between 6 months prior and the time of diagnosis (Tables 2 and S3). Also, serum betaine abundance decreased in the CKD cohort over time, especially between 6 months prior and the time of diagnosis (Table S3).

TABLE 1 Comparison of baseline routine parameters related to renal health between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
Age (years)					
–12 months	11.7 (1.3), n = 26	11.5 (11–13)	13.3 (2.8), n = 15	13.5 (12–15)	.08 ^b
Body weight (kg)					
–12 months	4.79 (1.14), n = 26	4.50 (3.95–5.16)	4.33 (1.03), n = 15	3.97 (3.55–4.95)	.14 ^c
–6 months	4.71 (1.11), n = 26	4.55 (3.89–5.10)	4.45 (0.96), n = 22	4.27 (3.80–4.80)	.71 ^c
–Healthy/CKD	4.61 (1.08), n = 26	4.55 (3.79–5.00)	4.30 (1.03), n = 22	4.12 (3.45–4.88)	.71 ^c
P value (within CKD group n = 15) ^d				.247	
Systolic blood pressure (mm Hg)					
–12 months	147 (26), n = 23	150 (125–160)	168 (19), n = 13	160 (150–185)	.06 ^b
–6 months	152 (25), n = 23	150 (130–170)	155 (19), n = 17	150 (143–165)	.68 ^b
Healthy/CKD	146 (20), n = 21	150 (126–160)	156 (22), n = 19	160 (140–170)	.17 ^b
P value (within CKD group n = 15) ^d				.231	
Creatinine (μmol/L)					
–12 months	119.6 (21.3), n = 26	121.6 (104.3–137.2)	131.9 (17.9), n = 15	130.8 (125.2–145.9)	.09 ^b
–6 months	115.1 (14.8), n = 26	112.8 (105.2–125.7)	138.6 (15), n = 22	137.9 (130–152.5)	<.01 ^b
Healthy/CKD	122.7 (16.4), n = 26	125.5 (114–131.3)	197.3 (39.2), n = 22	183.9 (174.6–199.4)	<.01 ^c
P value (within CKD group n = 15) ^d				<.001	
Symmetric dimethylarginine [SDMA] (μg/dL)					
–12 months	9.5 (2.3), n = 26	9.4 (7.4–11.3)	11.3 (2.4), n = 15	11.5 (9.2–13.1)	.04 ^b
–6 months	8.8 (2.2), n = 26	8.35 (7.0–10.9)	11.6 (2.9), n = 22	11.2 (9.6–13.6)	<.01 ^b
Healthy/CKD	8.0 (2.0), n = 26	7.3 (6.7–9.8)	15.5 (7.4), n = 22	13.4 (10.5–18.4)	<.01 ^c
P value (within CKD group n = 15) ^d				.165	
Urinary specific gravity [USG]					
–12 months	1.047 (1.006), n = 26	1.046 (1.042–1.052)	1.038 (1.011), n = 15	1.036 (1.032–1.045)	.01 ^b
–6 months	1.048 (1.007), n = 26	1.047 (1.043–1.051)	1.031 (1.009), n = 22	1.030 (1.025–1.037)	<.01 ^b
Healthy/CKD	1.047 (1.007), n = 26	1.045 (1.042–1.051)	1.020 (1.007), n = 22	1.018 (1.015–1.025)	<.01 ^c
P value (within CKD group n = 15) ^d				<.001	
Urinary protein : creatinine [UPC]					
–12 months	0.16 (0.07), n = 24	0.15 (0.12–0.20)	0.17 (0.06), n = 14	0.17 (0.12–0.19)	.72 ^c
–6 months	0.16 (0.05), n = 25	0.16 (0.12–0.19)	0.18 (0.08), n = 21	0.16 (0.11–0.24)	.76 ^c
Healthy/CKD	0.16 (0.05), n = 26	0.14 (0.13–0.18)	0.15 (0.07), n = 19	0.14 (0.09–0.19)	.74 ^c
P value (within CKD group n = 15) ^d				.320	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health, respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of the Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of normally distributed data, presented as mean/SD.

^cP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

^dRepresents P value for comparison of different timepoints in the CKD development cohort.

Tyrosine-derived uremic toxins p-cresol and p-cresyl-sulfate only increased in serum at the time of azotemic CKD diagnosis, without a significant upward pattern during CKD development (Table 2, S3, and Figure 2). No significant changes in urinary abundance were observed (Tables 3 and S4). Serum and urine tyrosine

also underwent no significant differences before the onset of azotemic CKD.

Univariate binary logistic regression analyses (Table 4) identified significant Chi-square statistics (indoxyl-sulfate: $\chi^2 = 7.369$, $P = .01$; TMAO: $\chi^2 = 9.069$, $P < .01$; p-cresyl-sulfate: $\chi^2 = 4.281$, $P = .04$),

TABLE 2 Comparison of serum abundance of specific gut-derived uremic toxins, their dietary precursors, and intermediates between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
L-Tryptophan					
–12 months	0.91 (0.19), n = 26	0.92 (0.79–1.03)	0.82 (0.17), n = 15	0.83 (0.72–0.96)	.15 ^b
–6 months	0.87 (0.17), n = 26	0.87 (0.72–.99)	0.79 (0.18), n = 22	0.78 (0.66–0.95)	.18 ^b
Healthy/CKD	0.89 (0.16), n = 26	0.86 (0.78–1.00)	0.76 (0.18), n = 22	0.81 (0.66–0.90)	.08 ^b
P value (within group n = 15) ^c				.420	
Indole					
–12 months	0.62 (0.50), n = 26	0.53 (0.32–0.70)	1.60 (1.98), n = 15	0.67 (0.42–2.09)	.1 ^d
–6 months	0.81 (1.06), n = 26	0.32 (0.23–1.14)	1.73 (1.30), n = 22	1.26 (0.84–2.78)	.02 ^d
Healthy/CKD	0.61 (0.38), n = 26	0.46 (0.29–1.03)	2.09 (1.60), n = 22	1.39 (1.05–3.38)	<.01 ^d
P value (within group n = 15) ^c				.027	
Indoxyl-sulfate					
–12 months	0.83 (0.70), n = 26	0.69 (0.42–1.07)	1.18 (0.74), n = 15	1.00 (0.63–1.62)	.08 ^d
–6 months	0.83 (0.46), n = 26	0.83 (0.50–1.07)	1.44 (1.06), n = 22	1.01 (0.72–2.26)	.02 ^b
Healthy/CKD	0.76 (0.44), n = 26	0.72 (0.51–1.02)	2.02 (1.19), n = 22	1.76 (1.19–2.61)	<.01 ^d
P value (within group n = 15) ^c				.334	
Carnitine					
–12 months	1.10 (0.68), n = 26	0.94 (0.55–1.50)	1.09 (0.61), n = 15	0.92 (0.58–1.45)	1 ^d
–6 months	1.09 (0.77), n = 26	0.96 (0.46–1.54)	1.08 (0.63), n = 22	1.00 (0.51–1.50)	1 ^d
Healthy/CKD	1.08 (0.57), n = 26	1.04 (0.63–1.39)	1.48 (0.90), n = 22	1.23 (0.64–2.19)	.26 ^d
P value (within group n = 15) ^c				.011	
Betaine					
–12 months	0.89 (0.51), n = 26	0.62 (0.51–1.16)	0.94 (0.63), n = 15	0.62 (0.50–1.32)	1 ^d
–6 months	0.99 (0.51), n = 26	0.75 (0.59–1.24)	0.91 (0.66), n = 22	0.71 (0.48–1.08)	1 ^d
Healthy/CKD	1.08 (0.63), n = 26	0.93 (0.57–1.47)	0.90 (0.72), n = 22	0.60 (0.41–1.15)	1 ^d
P value (within group n = 15) ^c				.057	
Trimethylamine (TMA)					
–12 months	1.19 (0.77), n = 26	1.20 (0.98–1.29)	1.22 (0.37), n = 15	1.13 (1.05–1.31)	1 ^d
–6 months	1.13 (0.21), n = 26	1.09 (0.99–1.34)	1.17 (0.26), n = 22	1.17 (1.00–1.26)	.76 ^b
Healthy/CKD	1.10 (0.30), n = 26	1.08 (0.88–1.22)	1.37 (0.34), n = 22	1.29 (1.17–1.54)	.02 ^d
P value (within group n = 15) ^c				.449	
Trimethylamine-N-oxide (TMAO)					
–12 months	1.49 (0.77), n = 26	1.32 (0.87–1.95)	1.43 (0.63), n = 15	1.43 (1.01–1.65)	.79 ^b
–6 months	1.60 (0.62), n = 26	1.31 (0.70–1.64)	1.82 (1.80), n = 22	1.80 (1.33–2.31)	.01 ^b
Healthy/CKD	1.20 (0.61), n = 26	1.06 (0.78–1.68)	2.02 (0.87), n = 22	1.83 (1.58–2.14)	<.01 ^d
P value (within group n = 15) ^c				.031	
Tyrosine					
–12 months	0.90 (0.21), n = 26	0.89 (0.77–1.07)	0.86 (0.15), n = 15	0.84 (0.73–0.97)	.69 ^b
–6 months	0.85 (0.20), n = 26	0.85 (0.70–0.93)	0.89 (0.22), n = 22	0.86 (0.71–1.03)	.89 ^d
Healthy/CKD	0.91 (0.21), n = 26	0.88 (0.77–1.08)	0.83 (0.18), n = 22	0.83 (0.70–0.97)	.62 ^b
P value (within group n = 15) ^c				.127	
p-Cresol					
–12 months	0.99 (1.32), n = 26	0.60 (0.26–1.07)	1.24 (1.23), n = 15	0.96 (0.14–1.85)	.91 ^d
–6 months	0.63 (0.55), n = 26	0.53 (0.15–1.03)	1.33 (1.74), n = 22	0.83 (0.14–1.50)	.34 ^d
Healthy/CKD	0.71 (0.64), n = 26	0.55 (0.16–1.24)	1.55 (1.10), n = 22	1.47 (0.62–2.45)	.01 ^d
P value (within group n = 15) ^c				.449	

(Continues)

TABLE 2 (Continued)

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
p-Cresyl-sulfate					
–12 months	0.98 (1.26), n = 26	0.62 (0.27–1.09)	1.22 (1.18), n = 15	0.95 (0.16–1.85)	.93 ^d
–6 months	0.63 (0.54), n = 26	0.54 (0.15–1.01)	1.30 (1.67), n = 22	0.84 (0.15–1.50)	.32 ^d
Healthy/CKD	0.71 (0.62), n = 26	0.56 (0.17–1.26)	1.52 (1.07), n = 22	1.43 (0.63–2.42)	.01 ^d
P value (within group n = 15) ^c				.449	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of normally distributed data, presented as mean/SD.

^cRepresents P value for Friedman's comparison of different timepoints in the CKD development cohort.

^dP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

indicating that circulating gut-derived uremic toxins individually serve as predictors for CKD risk 6 months before diagnosis. Furthermore, the pseudo- R^2 values of the models suggest that variations in indoxyl-sulfate, TMAO, and p-cresyl-sulfate account for approximately 19%, 23%, and 11.4% of the variance, respectively, in the development of azotemic CKD in cats. The odds ratios (ORs) indicated that for each unit increase in the abundance of indoxyl-sulfate (95% CI: 1.2–9.0, $P = .04$), TMAO (95% CI: 1.4–11, $P = .03$), and p-cresyl-sulfate (95% CI: 0.9–4.1, $P = .1$), the odds of developing CKD increase by 3.2, 3.9, and 1.9 times, respectively. However, the outcome of p-cresyl-sulfate was not significant.

The predictive significance of indoxyl-sulfate and TMAO changed upon adjusting for specific covariates related to renal health (Table 5). After accounting for sCr and SDMA, the predictive value of both indoxyl-sulfate and TMAO lost significance. More specifically, adjustment for USG strongly attenuated the relationship between indoxyl-sulfate and azotemic CKD development, but only mildly for TMAO. Age and UPC only slightly decreased the association between indoxyl-sulfate, TMAO, and the onset of azotemia. However, the predictive value of indoxyl-sulfate narrowly lost its significance when adjusted for age.

To elucidate the confounding effects on predictive capacity, the relationship between indoxyl-sulfate, TMAO, and selected clinico-pathological variables related to renal health was assessed (Table 6). Serum creatinine and SDMA concentrations showed a positive linear relationship with both indoxyl-sulfate and TMAO. Moreover, a negative correlation was found between USG and both indoxyl-sulfate and TMAO.

4 | DISCUSSION

Our main objective was to examine metabolic alterations in gut-derived uremic toxins (ie, indoxyl-sulfate, p-cresyl-sulfate, and TMAO) and related pathways associated with the development of azotemic

CKD in cats. A nested case-control design was applied, employing targeted metabolomics.

We showed that the serum abundance of indole, indoxyl-sulfate, and TMAO begins to increase at least 6 months before diagnosis of CKD in cats. In contrast, no significant changes were observed in serum or urine abundance of p-cresyl-sulfate at the onset of azotemic CKD. Indoxyl-sulfate, p-cresyl-sulfate, and TMAO rank among the most studied uremic toxins originating from the dietary precursors tryptophan, tyrosine, and carnitine, betaine, and choline, respectively. These dietary precursors undergo partial bacterial degradation in the gut, resulting in byproducts such as indole, p-cresol, and TMAO.^{3,4,34} After passing the intestinal tight junction barrier and entering circulation, these intermediates undergo metabolism in the liver, resulting in the formation of the aforementioned toxins.^{34,35} Excretion of indoxyl-sulfate, p-cresyl-sulfate, and TMAO involves renal tubular transporters, but a certain fraction is eliminated by glomerular filtration.^{34–36} Systemic accumulation of these toxins has been shown to occur in azotemic CKD in cats, worsening in more advanced disease stages.^{1,2}

Considering the dynamics of the indoxyl-sulfate pathway, several mechanisms may theoretically underlie this observation. In the context of impaired renal function, decreased tubular transport capacity, decreasing GFR or both may be the main underlying factors. Our results suggest that urinary excretion corresponds with higher circulating concentrations during the development of CKD. Despite this, excretion of indoxyl-sulfate may be unable to match further increasing abundance in the bloodstream at the time of azotemic CKD diagnosis. Moreover, in previous work, we demonstrated that the urinary abundance of indoxyl-sulfate, p-cresyl-sulfate, and TMAO remains unchanged in more advanced CKD stages, whereas serum concentrations were significantly correlated with CKD International Renal Interest Society (IRIS) stage.¹ These observations may be consistent with findings in human patients in whom lower fractional excretion of indoxyl-sulfate occurred at more advanced CKD stages, indicative of a

TABLE 3 Comparison of urine abundance of specific gut-derived uremic toxins, their dietary precursors, and intermediates between a CKD development cohort and a cohort of healthy controls.

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
L-Tryptophan					
–12 months	1.01 (.67), n = 26	0.93 (0.73–1.16)	0.80 (0.42), n = 15	0.70 (0.46–1.25)	.25 ^b
–6 months	1.01 (.67), n = 26	0.78 (0.60–1.11)	0.76 (0.44), n = 22	0.75 (0.40–0.96)	.29 ^b
Healthy/CKD	1.86 (4.53), n = 26	0.83 (0.53–1.39)	0.52 (0.32), n = 22	0.44 (0.29–0.72)	<.01 ^b
P value (within group n = 15) ^c				.091	
Indole					
–12 months	0.86 (0.61), n = 26	0.80 (0.45–1.12)	1.21 (0.85), n = 15	1.12 (0.46–1.90)	.71 ^b
–6 months	1.05 (0.72), n = 26	0.76 (0.52–1.35)	1.30 (0.97), n = 22	1.10 (0.60–2.13)	.75 ^b
Healthy/CKD	1.23 (1.04), n = 26	0.84 (0.53–1.56)	1.56 (1.48), n = 22	1.05 (0.52–2.05)	.57 ^b
P value (within group n = 15) ^c				.807	
Indoxyl-sulfate					
–12 months	0.99 (0.77), n = 26	0.91 (0.48–1.28)	1.22 (0.55), n = 15	1.11 (0.82–1.77)	.09 ^b
–6 months	0.96 (0.53), n = 26	1.03 (0.52–1.32)	1.56 (0.95), n = 22	1.44 (0.78–2.08)	.06 ^b
Healthy/CKD	1.08 (0.69), n = 26	0.92 (0.57–1.62)	1.51 (0.70), n = 22	1.43 (0.93–2.15)	.06 ^b
P value (within group n = 15) ^c				.420	
Carnitine					
–12 months	0.68 (0.72), n = 26	0.44 (0.27–.78)	0.80 (0.95), n = 15	0.40 (0.18–0.91)	.78 ^b
–6 months	0.97 (0.92), n = 26	0.56 (0.36–1.53)	0.63 (0.54), n = 22	0.51 (0.22–0.81)	.39 ^b
Healthy/CKD	0.95 (0.75), n = 26	0.70 (0.42–1.28)	0.92 (1.20), n = 22	0.48 (0.32–0.70)	.39 ^b
P value (within group n = 15) ^c				.936	
Betaine					
–12 months	0.47 (0.32), n = 26	0.37 (0.28–0.55)	0.87 (1.11), n = 15	0.47 (0.36–0.81)	.13 ^b
–6 months	0.73 (0.78), n = 26	0.43 (0.39–0.70)	0.89 (1.04), n = 22	0.60 (0.42–0.80)	.13 ^b
Healthy/CKD	0.65 (0.83), n = 26	0.47 (0.37–0.55)	0.95 (0.94), n = 22	0.58 (0.48–1.13)	.03 ^b
P value (within group n = 15) ^c				.155	
Trimethylamine (TMA)					
–12 months	1.02 (0.45), n = 26	0.93 (0.72–1.27)	1.38 (1.72), n = 15	0.98 (0.63–1.47)	.9 ^b
–6 months	0.85 (0.36), n = 26	0.78 (0.51–1.11)	1.62 (2.08), n = 22	1.08 (0.80–1.67)	.1 ^b
Healthy/CKD	1.24 (1.27), n = 26	0.98 (0.54–1.38)	1.29 (0.56), n = 22	1.13 (0.92–1.54)	.17 ^b
P value (within group n = 15) ^c				.165	
Trimethylamine-N-oxide (TMAO)					
–12 months	0.95 (0.28), n = 26	0.93 (0.73–1.10)	0.97 (0.44), n = 15	1.00 (0.71–1.32)	.92 ^d
–6 months	0.90 (0.32), n = 26	0.93 (0.65–1.08)	1.32 (0.71), n = 22	1.13 (0.91–1.88)	.05 ^b
Healthy/CKD	1.02 (0.46), n = 26	0.96 (0.71–1.33)	1.77 (0.81), n = 22	1.50 (1.26–2.15)	<.01 ^b
P value (within group n = 15) ^c				.002	
Tyrosine					
–12 months	0.91 (0.40), n = 26	0.85 (0.73–1.06)	0.76 (0.40), n = 15	0.63 (0.53–0.96)	.24 ^b
–6 months	1.00 (0.58), n = 26	0.90 (0.62–1.11)	0.76 (0.38), n = 22	0.71 (0.50–0.90)	.25 ^b
Healthy/CKD	1.58 (3.60), n = 26	0.80 (0.53–1.21)	0.70 (0.43), n = 22	0.58 (0.38–0.88)	.18 ^b
P value (within group n = 15) ^c				.127	
p-Cresol					
–12 months	1.01 (.83), n = 26	0.82 (0.42–1.28)	1.07 (0.84), n = 15	1.03 (0.25–1.84)	.86 ^b
–6 months	0.72 (0.46), n = 26	0.77 (0.23–1.11)	1.13 (1.00), n = 22	0.87 (0.21–1.77)	.43 ^b
Healthy/CKD	1.08 (1.32), n = 26	0.68 (0.29–1.53)	1.13 (0.70), n = 22	1.26 (0.55–1.61)	.83 ^b
P value (within group n = 15) ^c				.819	

(Continues)

TABLE 3 (Continued)

	Healthy cohort		CKD development cohort		P value (between groups) ^a
	Mean (SD)	Median (IQR)	Mean (SD)	Median (IQR)	
p-Cresyl-sulfate					
–12 months	1.01 (0.83), n = 26	0.82 (0.42–1.28)	1.07 (0.85), n = 15	1.02 (0.20–1.84)	.86 ^b
–6 months	0.72 (0.46), n = 26	0.76 (0.22–1.10)	1.13 (1.01), n = 22	0.88 (0.19–1.80)	.43 ^b
Healthy/CKD	1.08 (1.32), n = 26	0.69 (0.29–1.55)	1.13 (0.70), n = 22	1.25 (0.55–1.61)	.8 ^b
P value (within group n = 15) ^c				.819	

Note: Comparisons were made between the CKD development cohort and the healthy cohort at timepoints –12 months, –6 months, and azotemic CKD onset or maintained health respectively. Furthermore, comparisons were made in the CKD development cohort between the different timepoints to assess the behavior of the metabolites over time.

^aRepresents P value for comparison of Healthy and CKD development cohort at timepoints –12 months, –6 months, and CKD onset or maintained health respectively, after FDR correction.

^bP value for comparison of not normally distributed data, presented as median/interquartile range (IQR).

^cRepresents P value for Friedman's comparison of different timepoints in the CKD development cohort.

^dP value for comparison of normally distributed data, presented as mean/SD.

disproportionate decrease in tubular clearance of indoxyl-sulfate compared with GFR.^{35,37} These authors, therefore, suggested that tubular secretory mechanisms become saturated at higher CKD stages.³⁵ It can be hypothesized that decreasing GFR causes an initial retention of indoxyl-sulfate in the bloodstream, which is counteracted by increased tubular secretion, initially leading to increased urinary concentrations of indoxyl-sulfate. However, when CKD progresses compensatory tubular secretion becomes saturated and may further decrease, leading to worsening retention of indoxyl-sulfate in the circulation.^{35,38,39}

Other mechanisms also may be responsible for the early increase in indoxyl-sulfate. Interestingly, together with indoxyl-sulfate, its precursor indole also increases significantly during CKD development. This observation suggests that augmented bacterial formation or increased gut permeability or both may be considerable underlying factors along with decreased renal excretion.^{5,6,9} During azotemic stages of CKD, increasing urea concentrations lead to alterations in intestinal microbiota and disruption of the intestinal epithelial barrier.^{9,40} As a consequence, intestinal dysbiosis and shifts toward proteolytic indole- and p-cresol-forming microbiota are commonly observed in human patients with impaired renal function.^{39,40} More recently, decreased fecal microbiome diversity and richness also was confirmed in cats with CKD,⁵ whereby fecal isovaleric acid, a surrogate marker of excessive intestinal protein fermentation, was increased compared with healthy controls.⁴¹ However, to date intestinal dysbiosis has not been established in preazotemic CKD.

Furthermore, our results indicated that indoxyl-sulfate and TMAO were individually predictive for onset of azotemia 6 months before diagnosis. However, renal function biomarkers creatinine, SDMA, and USG were significantly correlated with indoxyl-sulfate and TMAO abundance, resulting in attenuated predictability after correction for these factors. Therefore, increasing indoxyl-sulfate and TMAO abundance likely reflects an already decreasing GFR rather than being causative factors initiating renal functional decline. On the contrary, in

2 studies of cats with established CKD, retention of indoxyl-sulfate was shown to be associated with impending CKD progression even after correction for renal function markers.^{8,42} At higher concentrations, indoxyl-sulfate starts to accumulate in tubular cells exerting direct toxicity via induction of cellular death and indirectly via promotion of oxidative stress and inflammation.^{43–45} Similarly, TMAO is associated with renal tubulointerstitial fibrosis in a continuous dose-dependent relationship.⁴⁶ As indoxyl-sulfate starts to increase early during the development of azotemia in cats, this toxin may be implicated in early pathophysiological mechanisms involved in the progression to overt CKD. In an animal model progressing from acute kidney injury to CKD, administration of indole exacerbated tubular inflammation and fibrosis, and the authors concluded that early indoxyl-sulfate inhibition is a useful strategy to prevent CKD development.⁴⁷ These results could not consistently be translated to human patients with CKD.⁴⁸ Nevertheless, in several human cohorts, decreases in circulating indoxyl-sulfate induced by PO sorbent treatment delayed deterioration of renal function and subsequent initiation of dialysis.⁴⁹

In human medicine, gut-derived uremic toxins also are recognized for their contribution to CKD-related morbidity, because they increase the risk of cardiovascular disease, contribute to renal anemia, negatively affect bone turnover, intensify muscle atrophy, and accelerate neuroinflammation-associated cognitive decline.^{50–52} Given their well-established toxic properties and association with negative disease outcomes, decrease of circulating indoxyl-sulfate and other gut-derived uremic toxins recently have received more attention in the therapeutic management of CKD in cats.¹⁰ In this context, different strategies have been investigated. Recent evidence suggests that dietary macronutrient composition directly impacts circulating concentrations of gut-derived uremic toxins in healthy adult cats.^{11,17,53} For example, cats fed high-protein diets had significantly increased circulating indoxyl-sulfate concentrations whereas tryptophan itself was not increased.¹⁷ Furthermore, because gut bacteria produce indole, a potential treatment strategy is the use of PO probiotics and

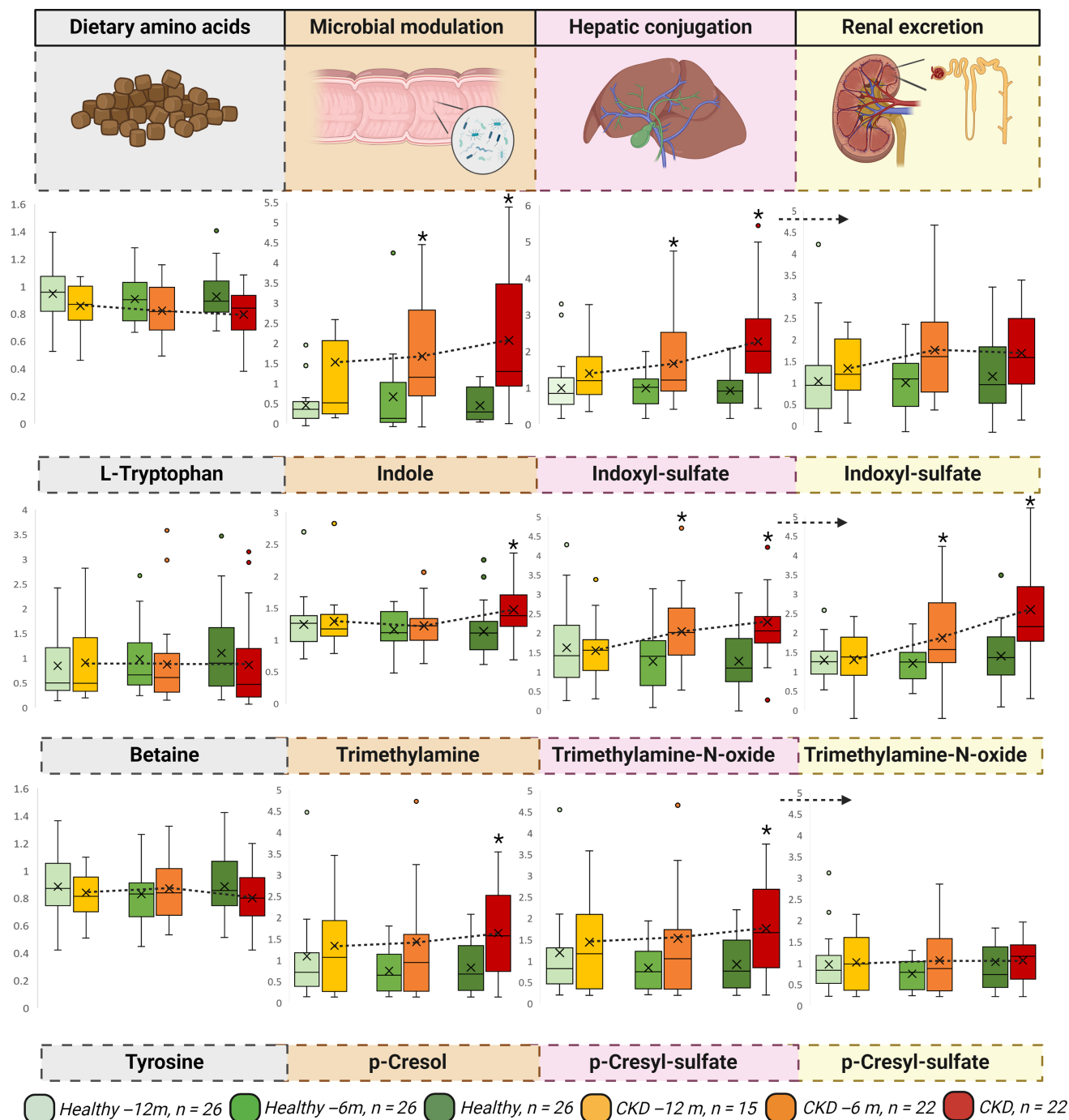


FIGURE 2 Comparisons of (I) serum dietary amino acids tryptophan, tyrosine, carnitine; (II) gut bacterial byproducts indole, p-cresol, trimethylamine (TMA); (III) related uremic toxins indoxyl-sulfate, p-cresyl-sulfate, trimethylamine-N-oxide (TMAO); and (IV) urinary abundance of these compounds between a healthy feline cohort and a cohort of cats developing azotemic CKD over time. Specific comparisons are shown between the healthy cohort and the CKD development cohort at 12 months, 6 months prior, and at the time of azotemic CKD diagnosis or maintained health. Significant P values ($<.05$) are represented by the symbol * after FDR correction.

prebiotics (eg, fermentable fiber) to alter the composition of the microbiome. However, so far, no consistent impact of pro- or prebiotics on indoxyl-sulfate concentrations in the bloodstream has been observed in cats in contrast to studies in humans.^{10,12,13} This difference may be a consequence of insufficient knowledge about potential modulation of the feline-specific microbiome. Especially

in cats with CKD, the microbiome appeared to be more resistant to fermentable fiber-induced change compared with healthy controls.¹³ Lastly, carbon-based adsorbents formulated to bind indole in the gastrointestinal tract showed promising potential in lowering circulating indoxyl-sulfate in cats consistent with findings in humans.^{14,54}

TABLE 4 Results of the binary logistic regression analysis exploring the associations between specific gut-derived uremic toxins and the onset of azotemic feline CKD 6 months before diagnosis.

	Model X^2 (P value)	Model's pseudo- R^2	Odds ratio	95% confidence interval	P value ^a
Indoxyl-sulfate	7.369 (.007)	0.190	3.2	1.2-9.0	.04
Trimethylamine-N-oxide (TMAO)	9.069 (.003)	0.230	3.9	1.4-11	.03
p-Cresyl-sulfate	4.281 (.039)	0.114	1.9	0.9-4.1	.1

^aRepresents P value after FDR correction.

Models	Indoxyl-sulfate			Trimethylamine-N-oxide (TMAO)		
	Odds ratio	95% CI	P	Odds ratio	95% CI	P
Model 1 ^a	3.003	0.992-9.009	.05	3.300	1.119-9.804	.03
Model 2 ^b	1.780	0.556-5.682	.33	1.538	0.461-5.128	.48
Model 3 ^c	2.250	0.791-6.410	.13	2.294	0.737-7.143	.15
Model 4 ^d	1.517	0.239-9.615	.66	6.544	0.847-50.000	.07
Model 5 ^e	2.920	1.050-8.130	.04	4.545	1.414-14.493	.01

^aModel 1 was adjusted for age.^bModel 2 was adjusted for serum creatinine.^cModel 3 was adjusted for serum SDMA.^dModel 4 was adjusted for urine-specific gravity (USG).^eModel 5 was adjusted for urine protein : creatinine ratio (UPC).**TABLE 5** Multivariable logistic regression analysis of indoxyl-sulfate and trimethylamine-N-oxide as predictors of azotemic CKD 6 months before diagnosis, adjusted for various renal health-related factors.

Variables	Indoxyl-sulfate		Trimethylamine-N-oxide (TMAO)	
	Spearman coefficient	P	Spearman coefficient	P
Age	0.074, n = 48	.618	0.298, n = 48	.04
Creatinine	0.395, n = 48	.01	0.534, n = 48	<.01
SDMA	0.389, n = 48	.01	0.498, n = 48	<.01
USG	−0.363, n = 48	.01	−0.414, n = 48	<.01
UPC	0.288, n = 46	.05	0.195, n = 46	.19

TABLE 6 Correlations between indoxyl-sulfate, trimethylamine-N-oxide, and selected clinicopathological variables related to renal health.

Abbreviations: UPC, urine protein : creatinine ratio; USG, urine-specific gravity.

A notable factor regarding interpretation of our study's results is the observation that sCr and especially USG were already significantly altered in the CKD development cohort compared with healthy controls 6 months before the diagnosis of renal azotemia. Additionally, this data underscores the potential usefulness of USG as a clinically applicable tool for serial monitoring of renal function in nonazotemic cats. Such changes may indicate that a substantial decrease in GFR was already present at that time in the CKD development cohort, but was not yet resulting in renal azotemia.²⁹ Notably, SDMA, proposed as an earlier indicator of decreasing GFR compared with sCr, stayed within the reference interval in 18 of 22 cats and remained below the IRIS threshold for stage 2 renal disease (<18 µg/dL) in all cats 6 months before azotemic CKD diagnosis.^{55,56} This observation provides further evidence that the demonstrated alterations in gut-derived uremic toxins appear to occur early in the disease continuum.

The primary limitation of our study was its small sample size, particularly for cats monitored 12 months before the onset of CKD. Given

the occurrence of many nearly significant P values, limited sample size may have hindered the achievement of statistical significance, especially when adjustments were made for multiple comparisons. Another limiting factor is that urine creatinine normalization can be dependent on factors (eg, age, muscle mass, hydration status) other than urinary dilution, and therefore these results should be interpreted with caution.^{32,33} Alternatives such as preanalytical dilution of urine specimens to a fixed creatinine concentration could have provided a more accurate representation of urinary compound abundance. Nonetheless, measuring compound abundance in a 24-hour urinary collection sample remains the gold standard, but is not feasible in a clinical setting.⁵⁷

In conclusion, we determined that the circulating gut-derived uremic toxins indoxyl-sulfate and TMAO increase at least 6 months before the diagnosis of azotemic CKD. However, a direct causal relationship between these toxins and the initiation of azotemic CKD in cats is still lacking. Nevertheless, collectively, our data supports that impaired gut-derived uremic toxin handling occurs early in the CKD continuum in cats, warranting early therapeutic intervention.

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CONFLICT OF INTEREST DECLARATION

Authors declare no conflict of interest.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

Approved by the ethical committee of the faculties of Veterinary Medicine and Bioscience Engineering, Ghent University (EC2020-081).

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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Relationship between Plasma Fibroblast Growth Factor-23 Concentration and Survival Time in Cats with Chronic Kidney Disease

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Abstract

Background

Fibroblast growth factor-23 (FGF-23) and parathyroid hormone (PTH) are commonly increased in cats with azotemic chronic kidney disease (CKD). Both are predictors of survival time in human patients, but these relationships have not previously been examined in the cat.

Objectives

To investigate the relationship between plasma FGF-23 and PTH concentrations at diagnosis of CKD in cats with survival time and with disease progression over 12 months.

Animals

214 azotemic, client-owned cats (≥ 9 years).

Methods

Retrospective study: Biochemical and urinary variables at diagnosis of azotemic CKD, including plasma FGF-23 and PTH concentrations were assessed as predictors of survival time (all-cause mortality) using Cox regression, and as predictors of CKD progression over 12 months using logistic regression.

Results

In the final multivariable Cox regression model, survival was negatively associated with plasma creatinine ($P = .002$) and FGF-23 concentrations ($P = .014$), urine protein-to-creatinine ratio ($P < .001$) and age ($P < .001$). Survival was positively associated with PCV ($P = .004$). In the final multivariable logistic regression model, independent predictors of CKD progression included logFGF-23 and age. Neither plasma phosphate nor PTH was found to be an independent predictor of survival time or of CKD progression.

Conclusions and Clinical Importance

Plasma FGF-23 concentration is a novel prognostic indicator in cats with CKD, independent of other factors including plasma creatinine and phosphate concentrations. Further work is required to assess if FGF-23 contributes directly to CKD progression, but regardless these findings may make FGF-23 a useful biomarker for predicting poorer outcomes in cats with CKD.

Abbreviations

British short-hair
calcium-phosphate
confidence interval
chronic kidney disease
chronic kidney disease-mineral and bone disorders
fibroblast growth factor 23
glomerular filtration rate
hazard ratio
International Renal Interest Society
odds ratio
parathyroid hormone
systolic blood pressure
total calcium
total thyroxine
urine protein-to-creatinine ratio
urine specific gravity
urinary tract infection

Chronic kidney disease (CKD) in cats is highly prevalent; 31% of cats over 15 years of age have azotemic CKD¹ and there is evidence of CKD in 60.4% of cats, with a prevalence of 80.9% in cats over 15 years of age.² CKD causes severe morbidity and mortality, with renal disorders being the most common cause of mortality in cats ≥ 5 years of age in the UK, accounting for 13.6% of deaths.³

The syndrome of the clinical, biochemical and imaging abnormalities that are correlated with CKD and renal osteodystrophy is termed “chronic kidney disease-mineral and bone disorders” (CKD-MBD).⁴ Development of CKD-MBD is considered to be a consequence of reduced Glomerular Filtration Rate (GFR), and associated phosphate retention. Calcium-phosphate (Ca-P) derangements appear prevalent in cats with CKD; with the most common biochemical derangement previously documented to be hyperparathyroidism, occurring in 84% of azotemic cats.⁵ Additionally, the phosphatonin fibroblast growth factor-23 (FGF-23) has been demonstrated to increase with advancing International Renal Interest Society (IRIS) stage of CKD and to be further increased in cats with higher plasma phosphate concentrations.⁶

FGF-23 is secreted primarily by osteocytes⁷ and osteoblasts,^{8, 9} in response to hyperphosphatemia and increased plasma calcitriol concentrations.¹⁰ FGF-23 acts in the kidney to inhibit calcitriol production by reducing mRNA expression of the vitamin D synthesis enzyme (25-hydroxyvitamin D-1 α -hydroxylase) and to increase phosphaturia by down-regulating sodium-phosphorus type II co-transporters (NaPi-IIa and NaPi-IIc) in the proximal tubules.¹¹ It also acts in the parathyroid gland to decrease parathyroid hormone (PTH) production and secretion.¹²

Plasma FGF-23 concentrations in human patients are an independent predictor of progression of CKD¹³ and are independently associated with case fatality in patients starting haemodialysis for CKD.¹⁴ Additionally, very high levels of PTH are associated with increased risk of death in CKD patients on hemodialysis.¹⁵ Plasma phosphate concentration is an independent predictor of survival time,^{16, 17} and progression¹⁸ in cats with CKD, however, no previous studies have examined the relationship between plasma PTH or FGF-23 concentration and CKD progression or survival time in cats.

To explore the impact of CKD-MBD in cats the aim of this study was to examine if plasma FGF-23 or PTH concentration at diagnosis of azotemic CKD were independent predictors of progression of the disease over the following 12 months, and of survival time.

Materials and Methods

Case Selection

Records from 2 London-based first opinion practices between 7th January 1992 and 1st September 2011 were reviewed and cats ≥ 9 years of age with renal azotemia were identified. The majority of these cases had been included in a previous study.¹⁸ The date of diagnosis of renal azotemia was defined as the date that a plasma creatinine concentration >2.0 mg/dL and a concurrent urine specific gravity (USG) <1.035 were documented. The Ethics and Welfare Committee of the Royal Veterinary College had approved the study protocol and informed consent was obtained from all owners. Biochemical analyses and measurement of plasma total thyroxine were performed at an external laboratory on the date samples were obtained.¹ Urine samples collected by cystocentesis and previously stored at -80°C were used for batch analysis of urine protein-to-creatinine ratio (UPC) at an external laboratory.¹ UPC measurements from samples with urinary tract infections (identified by the presence of active urine sediment on microscopy and confirmed by bacterial culture²) were not included in any analyses.

Cases required residual stored frozen EDTA plasma samples to be available for measurement of FGF-23 and PTH. Cases were excluded if they were already being fed a renal diet at diagnosis as this can result in a reduction in plasma FGF-23 and PTH concentrations.¹⁹ Cases that demonstrated USG > 1.035 following diagnosis were excluded and those with concurrent hyperthyroidism, or that developed hyperthyroidism during follow-up. Cats were excluded if they had evidence of other concurrent disease, except systemic hypertension, or were taking medications that alter calcium or phosphate absorption or metabolism. All owners were advised to feed a commercially available renal diet after diagnosis of CKD. Other medications were prescribed as necessary on an individual case basis. No attempt was made to account for management strategies in the analyses.

Measurement of PTH and FGF-23

EDTA plasma samples previously stored at -80°C were used for batch analysis of PTH and FGF-23. The samples had been stored in 0.5 mL aliquots that had not undergone any freeze-thaw cycles in the majority of cases. In a small number of cases aliquots used to measure FGF-23 had undergone 1 freeze-thaw cycle, which has previously been shown not to significantly affect feline FGF-23 concentrations.⁶ PTH concentrations were measured using a validated total intact PTH immunoradiometric assay,³ as previously described.²⁰ Samples with a PTH concentration below the

lower limit of detection (<5.2 pg/mL)²⁰ were assigned a value of 2.6 pg/mL. Intact plasma FGF-23 concentrations were measured using a validated ELISA⁴ as previously described.⁶

Statistical Analysis

Survival times using all-cause mortality were calculated from the date of diagnosis of azotemic CKD to the date of death or euthanasia. Date of death was assigned as the 15th of the month if the exact date was unknown. Cases still alive on 1st January 2013 or lost to follow-up were censored. Survival times were compared between groups using a Mann-Whitney *U*-test.

Progression of CKD was defined as an increase of $>25\%$ in plasma creatinine during the 12 months after the date of diagnosis of azotemic CKD. Cases were classified as stable if they were followed-up for a minimum of 12 months, without an increase of $>25\%$ in plasma creatinine concentration. Cases followed-up for <12 months were censored unless they demonstrated CKD progression. Renal follow-up was defined as the time from date of diagnosis of azotemic CKD to the date of the last available plasma creatinine concentration. Blood and urine samples were routinely taken approximately every 16 weeks, or sooner if it was deemed clinically necessary.

Statistics were performed using SPSS.⁵ Results are reported as median [25th, 75th percentiles] and significance was determined as $P < .05$. Available data for the following variables obtained at the time of CKD diagnosis were included in analyses: plasma creatinine, phosphate, total calcium, PTH, FGF-23, potassium, total protein, globulin, albumin and cholesterol concentrations, packed cell volume (PCV), USG, UPC, age, presence of a urinary tract infection (UTI) and a diagnosis of hypertension (systolic blood pressure [SBP] >160 mmHg with evidence of concurrent hypertensive retinopathy or SBP >170 mmHg on 2 separate occasions).

Variables associated with survival time were examined using Cox regression models. Variables were recoded into categorical variables if they did not meet the assumption of proportional hazards, assessed by inspection of Kaplan-Meier curves. Categorical variables were assigned based on biologically relevant cutpoints where possible or otherwise by terciles. Variables that were significantly associated with survival time at the 10% level were included in a backwards conditional multivariable Cox regression model.

Variables associated with CKD progression were examined using binary logistic regression models. FGF-23 and PTH were logarithmically transformed because of their highly skewed distributions. USG values were subjected to removal of the one before the decimal point and multiplied by 1,000 to allow interpretation of odds ratio (OR). UPC values were multiplied by 10 to allow interpretation of OR. To allow the inclusion of variables of particular interest, variables significant at the 20% level in the univariable analyses were carried into a backwards conditional multivariable binary logistic regression model to determine independent predictors of progression.

No attempt was made to impute missing data or to assess interactions because of the large number of variables in the models.

Results

Survival Analysis

A total of 722 cats with a diagnosis of azotemic CKD were identified. Cases were excluded for the following reasons: a diagnosis of hyperthyroidism ($n = 233$), no sample available for measurement of FGF-23 and PTH ($n = 272$), no

urine sample taken on date of diagnosis ($n = 2$) and a diagnosis of nephrotic syndrome with atypical disease pathology ($n = 1$). This left 214 cats available for inclusion in the survival analysis. The majority ($n = 180$) were nonpedigree domestic cats, but there were 13 Persian or Persian crosses, 10 Burmese or Burmese crosses, 3 Tiffany, 2 British short-hair, 2 Chinchilla and one of each of the following: American shorthair, Ocicat, Siamese and Russian Blue cross. Ninety-nine cats were female (one entire, 98 neutered) and 115 were male (2 entire, 113 neutered). UPC measurements were performed in 174 cases, the remaining cases either had a UTI or no urine sample was available in storage.

The median age at diagnosis was 14.4 [11.0, 16.0] years. Seven cats were acquired as adults and their age was unknown. One hundred and sixty cats (74.8%) had died or were euthanized at or before the end of the follow-up period (1st January 2013) with a median survival time of 327 [109, 646] days. Fifty-four cats were either still alive at the study end point (33) or were lost to follow-up (21) and were censored from the analysis. The median follow-up time for censored cases was 658 [282, 919] days. Follow-up time was significantly longer in the censored cases compared to cases that died or were euthanized ($P < .001$).

Variables associated with survival time at the 10% level ($P < .1$) in univariable Cox regression analyses are shown in Table ¹ and were included in the multivariable model. PCV was considered as a continuous variable as it met the assumption of proportional hazards. For FGF-23 and phosphate 4 categories were used for the analysis based on biologically relevant cutpoints, for other variables 3 biologically relevant cutpoints or terciles (for age, albumin and USG) were used. A Kaplan-Meier curve of survival using plasma FGF-23 concentration at diagnosis of CKD is shown in Figure ¹. In the final multivariable model (Table ²), survival was negatively associated with plasma creatinine ($P = .002$) and FGF-23 concentrations ($P = .014$), UPC ($P < .001$) and age ($P < .001$). Survival was positively associated with PCV ($P = .004$).

Table 1

Univariable Cox regression analysis of factors associated with survival (all-cause mortality) at the 10% level ($P < .1$) at diagnosis of azotemic chronic kidney disease

Variable	n	Survival Time by Cutpoints (Days) Median [25th, 75th Percentiles]	P Value	HR	95% CI for HR
Creatinine (≤ 2.8 mg/dL)	148	490 [208, 836]	<.001		
>2.8–5.0	55	263 [42, 657]	.068	1.39	0.975–1.99
>5.0	11	20 [2, 108]	<.001	10.7	5.45–21.0
Phosphate (≤ 4.5 mg/dL)	99	462 [200, 812]	<.001		
4.5–5.0	26	708 [333, 976]	.16	0.679	0.394–1.17
5.0–6.0	37	343 [186, 875]	.55	1.14	0.737–1.77
>6.0	52	98 [18, 502]	<.001	1.99	1.36–2.92
USG (> 1.020)	67	586 [207, 840]	.020		
≤ 1.016	79	318 [54, 628]	.005	1.75	1.18–2.60
1.017–1.020	68	365 [131, 848]	.092	1.42	0.994–2.13

Variable	n	Survival Time by Cutpoints (Days) Median [25th, 75th Percentiles]	P Value	HR	95% CI for HR
UPC (<0.2)	95	544 [280, 905]	<.001		
0.2–0.4	51	246 [67, 705]	<.001	2.07	1.38–3.10
>0.4	28	126 [31, 464]	<.001	2.63	1.63–4.26
FGF23 (≤700 pg/mL)	100	577 [289, 895]	<.001		
>700–3,000	69	354 [168, 734]	.53	1.13	0.777–1.64
>3,000–10,000	24	277 [65, 632]	.003	2.08	1.29–3.36
>10,000	21	38 [10, 152]	<.001	7.40	4.35–12.6
PTH (≤17.6 pg/mL)	110	377 [158, 873]	.026		
>17.6–30	40	570 [172, 909]	.15	0.714	0.454–1.12
>30	64	261 [50, 601]	.076	1.38	0.968–1.96
PCV (%)	213	NA	<.001	0.911	0.885–0.937
Albumin (>3.21 g/dL)	71	546 [161, 880]	.010		
≤2.95	72	255 [70, 581]	.003	1.81	1.23–2.67
2.96–3.21	71	394 [211, 705]	.21	1.29	0.866–1.93
Age (≤12.7 years)	69	484 [138, 928]	<.001		
12.7–15.4	69	587 [231, 910]	.39	1.19	0.795–1.79
>15.4	69	236 [109, 506]	<.001	2.29	1.58–3.60
Normotensive cases	164	423 [126, 840]			
Diagnosis of hypertension	50	310 [181, 608]	.004	1.68	1.18–2.37

n, number of cats in group; HR, hazard ratio; CI, confidence interval; NA, not applicable.

For continuous variables, the hazard ratio represents the effect of a unit change in the predictor variable on the frequency of the outcome (death).

Figure 1

Kaplan-Meier curve for survival of cats (n = 214) based on plasma fibroblast growth factor-23 (FGF-23) concentration at diagnosis of chronic kidney disease. Cats with baseline plasma FGF-23 > 10,000 pg/mL ($P = .003$) and 3,000–10,000 pg/mL ($P < .001$), but not 700–3,000 pg/mL ($P = .53$) had significantly shorter survival times than cats with FGF-23 within the reference interval at baseline (≥ 700 pg/mL). Censored individuals are represented by ticks.

Table 2

Multivariable Cox regression analysis of factors associated with survival time (all-cause mortality) in cats at diagnosis of azotemic chronic kidney disease (n = 167)

Variable	β	SE	P Value	HR	95% CI for HR
PCV (%)	−0.060	0.021	.004	0.942	0.904–0.981
Creatinine (≤ 2.8 mg/dL)			.002		
>2.8–5.0	0.378	0.231	.10	1.46	0.927–2.30
>5.0	1.71	0.478	<.001	5.50	2.16–14.0
UPC (<0.2)			<.001		
0.2–0.4	0.789	0.221	<.001	2.20	1.43–3.39
>0.4	0.965	0.325	.003	2.62	1.39–4.96
Age (≤ 12.7 years)			<.001		
12.7–15.4	0.702	0.271	.010	2.02	1.19–3.43
>15.4	1.53	0.271	<.001	4.63	2.72–7.88
FGF23 (≤ 700 pg/mL)			.014		
>700–3,000	0.042	0.233	.86	1.04	0.660–1.65
>3,000–10,000	0.668	0.333	.045	1.95	1.02–3.74
>10,000	1.36	0.458	.003	3.91	1.59–9.58

β , coefficient; SE, standard error; HR, hazard ratio; CI, confidence interval.

Variables significant at the 10% level in univariable analyses were included in this model to determine independent predictors of survival time (all-cause mortality). For continuous variables, the hazard ratio represents the effect of a unit change in the predictor variable on the frequency of the outcome (death).

Progression Analysis

Of the 214 cases available for inclusion in this study, 148 (69.2%) cases were classified as IRIS stage 2 at diagnosis, 55 (25.7%) were stage 3, and 11 (5.1%) were stage 4. Of the stage 2 cases, 106 (71.6%) remained in stage 2 during renal follow-up, 39 (26.4%) reached stage 3, and 3 (2.0%) reached stage 4. Of the stage 3 cases, 48 (87.3%) remained in stage 3 during renal follow-up and 7 (12.7%) reached stage 4.

One hundred and nine cats (50.9%) did not demonstrate progression but were followed-up for less than 12 months and were therefore censored from the progression analysis. In the remaining 105 cats, 38 cases (36.2%) demonstrated CKD progression during the initial 12 month follow-up period and 67 cases (63.8%) did not demonstrate progression of CKD but were followed-up for a minimum of 12 months and were therefore classified as stable. Renal follow-up was 277 [144, 399] days for progressive cases, 742 [539, 1001] days for stable cases and 56 [0, 147] days for censored cases. Summary statistics for clinicopathological variables for progressive, stable and censored cases taken on the date of diagnosis of CKD are shown in Table ³.

Table 3

Summary statistics for baseline clinicopathological variables for cats with stable or progressive chronic kidney disease over the following 12 months

Variable	Stable (n = 67)	Progressive (n = 38)	Short Follow-Up (n = 109)
Creatinine (mg/dL)	2.4 [2.2, 2.8]	2.5 [2.3, 2.9]	2.5 [2.3, 3.6]
Urea (mg/dL)	47.6 [40.0, 56]	50.7 [44.8, 66.1]	53.5 [42.6, 78.7]
USG	1.020 [1.016, 1.024]	1.017 [1.014, 1.026]	1.018 [1.016, 1.021]
UPC	0.14 [0.07, 0.21]	0.20 [0.14, 0.39]	0.23 [0.14, 0.36]
Phosphate (mg/dL)	4.4 [3.6, 5.1]	4.7 [3.8, 8.0]	5.1 [4.0, 6.7]
FGF-23 (pg/mL)	504.3 [340.3, 1291.2]	1243.2 [555.6, 2066.5]	1103.8 [414.1, 3977.4]
PTH (pg/mL)	13.8 [5.8, 26.6]	19.2 [7.0, 32.7]	17.8 [7.8, 45.9]
Potassium (mEq/L)	4.0 [3.7, 4.3]	4.1 [3.8, 4.2]	4.0 [3.7, 4.4]
T. Calcium (mg/dL)	10.1 [9.76, 10.5]	9.96 [9.6, 10.4]	10.1 [9.72, 10.52]
Total Protein (g/dL)	7.83 [7.47, 8.28]	7.91 [7.53, 8.45]	7.67 [7.25, 8.27]
Globulin (g/dL)	4.66 [4.31, 5.02]	4.75 [4.32, 5.54]	4.58 [4.18, 5.26]
Albumin (g/dL)	3.14 [2.97, 3.32]	3.09 [2.81, 3.25]	3.07 [2.81, 3.24]
Cholesterol (mg/dL)	205 [158, 251]	185 [162, 250]	197 [154, 236]
PCV (%)	35 [32, 37]	32 [29, 38]	32 [27, 36]
SBP (mmHg)	142 [127, 154]	154 [130, 174]	141 [126, 160]
% with hypertension	18	39	21
% with UTI	16	8	7
Age (years)	13 [11, 15]	15 [12, 16]	15 [11, 17]
Weight	4.13 [3.54, 4.88]	3.80 [3.10, 4.60]	3.4 [2.8, 4.2]
% male	58	58	50
Follow-up (days)	742 [539, 1001]	277 [144, 399]	56 [0, 147]
Year of diagnosis	2008 [2005, 2010]	2007 [2003, 2009]	2008 [2005, 2010]

Data presented as Median [25th, 75th percentile] or prevalence (%).

Variables associated with CKD progression within 12 months at the 20% level ($P < .2$) in univariable logistic regression analyses are shown in Table ⁴. In the final multivariable model, independent predictors of CKD progression within 12 months of diagnosis included logFGF-23 and age (Table ⁵). An increase in age of 1 year was associated with an increase in the odds of progression of 21% and a 1 unit increase in logFGF-23 (or a 10-fold increase in plasma FGF-23 concentration) was associated with an increase in the odds of progression of 289%.

Table 4

Univariable binary logistic regression analysis of baseline variables associated with progression of chronic kidney disease (CKD) over 12 months at the 20% level ($P < .2$)

Variable	n	OR with 95% CI	P Value
Diagnosis of hypertension	105	2.989 [1.213, 7.365]	.017
LogFGF-23 (pg/mL)	105	2.657 [1.183, 5.967]	.018
USG $\times$ 1000	105	0.925 [0.858, 0.998]	.043
PCV (%)	105	0.925 [0.854, 1.002]	.055
UPC $\times$ 10	88	1.128 [0.982, 1.296]	.088
Albumin (g/dL)	105	0.333 [0.082, 1.353]	.12
Age (years)	97	1.122 [0.969, 1.301]	.13
Phosphate (mg/dL)	105	1.207 [0.936, 1.556]	.15
Creatinine (mg/dL)	105	1.54 [0.806, 2.93]	.19

The odds ratio indicates the increase in the odds of CKD progressing for each unit increase in the explanatory variable.

Table 5

Multivariable binary logistic regression analysis of baseline variables associated with progression of chronic kidney disease (CKD) over 12 months (n = 84)

Variable	Odds Ratio with 95% CI	P Value
Log (FGF-23 pg/mL)	3.89 [1.27, 11.9]	.017
Age (years)	1.21 [1.001, 1.45]	.049

The odds ratio indicates the increase in the odds of CKD progressing for each unit increase in the explanatory variable.

Discussion

This study examined if plasma FGF-23 or PTH concentration are associated with disease progression and survival time in cats with azotemic CKD. Plasma FGF-23 concentration, but not plasma PTH or phosphate concentrations was found to be independently associated with survival time using all-cause mortality and to be an independent predictor of progression of CKD over 12 months.

CKD-MBD is thought to cause damage to renal tissue and contribute to disease progression, although the mechanisms involved are not completely understood. Phosphate might have direct deleterious effects on the kidney by causing nephrocalcinosis,²¹ and indirect effects by initiating CKD-MBD development. Plasma phosphate concentration is independently associated with risk of death in a meta-analysis of human CKD patients,²² and to be an independent

predictor of survival time^{16, 17} in cats with CKD. In the present study, plasma phosphate was associated with survival time in univariable analyses, but was not an independent predictor of survival time when FGF-23 was also included. In a previous study, plasma phosphate concentration at CKD diagnosis was an independent predictor of CKD progression in cats,¹⁸ but this result was not replicated in the present study. This might be because of the differences in the severity of CKD in each study population, with a smaller proportion of IRIS stage 3 and 4 cases in the present study versus the previous study (25% [26/105] versus 38% [81/213]). Since phosphate retention increases as GFR declines, having a greater proportion of cases in the higher IRIS stages would have included more cats with hyperphosphatemia in the previous study. Comparison of plasma phosphate concentrations reveals that these values are very similar in both studies for the stable cases (4.4 [3.6, 5.1] versus 4.4 [3.6, 5.2]), but lower in this study for the progressive cases (4.7 [3.8, 8.0] versus 5.1 [4.0, 6.8]), which might account for why phosphate was not associated with progression in this study.

PTH was a predictor of survival time in univariable analyses in this study, but was not found to be an independent predictor of mortality in the multivariable model. In human patients, high levels of PTH are associated with an increased risk of death in CKD patients on hemodialysis;¹⁵ however, this result has not been consistent in all studies and a meta-analysis did not demonstrate a consistent association between death and serum PTH concentration in CKD patients of varying stages.²² This study examined the association between PTH and survival time or CKD progression in the cat. Plasma PTH is increased in nonazotemic cats that develop azotemia within 12 months, compared to cats that remain nonazotemic (plasma creatinine ≤ 1.6 mg/dL).²³ It is likely that PTH was increased in the cats that went on to develop azotemic CKD to help maintain normal plasma phosphate concentrations in the face of a reduced GFR. What could not be determined from this association study was whether PTH was contributing to CKD progression in these cats or acting as a biomarker for reduced renal function. PTH is considered to act as a uremic toxin because of its effects on bone remodeling and the heart and vasculature.²⁴ Whether increased PTH can directly contribute to CKD progression remains to be demonstrated, but the present study suggests that its use as a biomarker for more progressive CKD in the cat is limited. The only variable involved in Ca-P homeostasis that was independently associated with survival time and CKD progression in this study was plasma FGF-23 concentration.

Plasma FGF-23 increases early in the course of CKD, before the development of overt hyperphosphatemia,²⁵ where it acts to increase the fractional excretion of phosphate in remaining nephrons. It has been shown to be increased in cats that go on to develop azotemia within 12 months, compared to those remaining nonazotemic.²⁶ Feline plasma FGF-23 concentration is therefore likely to better reflect total body phosphate retention during mild to moderate CKD than plasma phosphate concentration, which may explain firstly, why an association between FGF-23 and CKD progression was identified when an association between phosphate and CKD progression was not and secondly, why including FGF-23 in the survival analysis resulted in the exclusion of phosphate from the multivariable model.

A clear concentration-dependent relationship between plasma FGF-23 concentration and survival time was found in this study, where highly increased FGF-23 concentrations ($>10,000$ pg/mL) were associated with an almost 4-times increase in risk of death. Additionally, a 10-fold increase in FGF-23 concentration at diagnosis was associated with a 289% increase in the odds of feline CKD progression over the next 12 months. These effects were independent of plasma creatinine concentration, suggesting that derangements in Ca-P homeostasis are important contributors to morbidity and mortality in all stages of azotemic CKD.

FGF-23 predicts the progression of CKD in human patients,^{13, 27, 28} and FGF-23 concentrations are also well established as being independently associated with case fatality in CKD patients at the start of hemodialysis,^{14, 29} and in earlier stages of CKD.^{27, 28} At present, just as has been debated for many years concerning PTH, it is unclear whether FGF-23 is a uremic toxin, or whether it is a surrogate marker for other causes of uremic toxicity. Recent evidence suggests that in humans, FGF-23 can induce left ventricular hypertrophy,³⁰ and is an independent predictor of myocardial infarction, stroke, and cardiovascular mortality in CKD patients.²⁷ However, although these findings may shed some light on how FGF-23 may act as a uremic toxin in human patients, cardiovascular incidents are not a commonly recognized cause of death or euthanasia in feline CKD patients. It has been demonstrated retrospectively that plasma FGF-23 concentrations can be reduced in feline CKD patients by feeding a phosphate restricted diet.¹⁹ Further work is now required to explore why FGF-23 is associated with CKD progression in cats and if targeted reduction of FGF-23 can improve survival time.

In this study, age was an independent predictor of survival time and of progression of CKD, with an increase of 1 year in age at diagnosis of CKD associated with a 21% increase in the risk of progression over the next 12 months. An association between age and survival time is not a novel finding,³¹ and makes sense biologically. Aging is associated with a decreased ability to repair and regenerate the kidney³² and renal histopathologic changes are observed with increasing frequency in older cats.³³ Aged rodent kidneys develop significantly more interstitial fibrosis post-reperfusion than young kidneys, despite normal aged kidneys having no interstitial fibrosis.³⁴ Therefore, it is possible that any injury to the kidney in an older cat could lead to an exaggerated repair response and a more progressive form of CKD.

Median survival time for cats in this study was 367 [147, 813] days, which is similar to other reports in the literature.^{16, 31, 35} Plasma creatinine concentration, UPC, and PCV were independent predictors of all-cause mortality in the present study. Plasma creatinine concentration was found to be associated with survival time in this, and in previous studies of cats with CKD,^{17, 31, 36} and median survival times for each IRIS stage in this study are similar to those reported previously.^{16, 37} UPC has previously been shown to be an independent predictor of survival time in cats with CKD,³¹ and in this study being classified as borderline proteinuric (UPC 0.2–0.4) or proteinuric (UPC > 0.4) under the IRIS proteinuria classification scheme⁶ was associated with a greater than 2-times increase in risk of death. A previous study found UPC to be an independent predictor of progression of CKD;¹⁸ however, UPC did not remain in the multivariable model in the present analysis. A placebo-controlled interventional study failed to demonstrate a significant increase in survival time when feline CKD patients with proteinuria were administered an angiotensin converting enzyme (ACE) inhibitor, despite a significant lowering of UPC with treatment.³⁸ The role of proteinuria in progression of CKD is therefore currently unclear.

PCV or hematocrit has previously been found to be significantly associated with survival time in univariable analyses³⁶ and found to be associated with death within 1 month for cats with CKD.¹⁷ PCV is not an independent predictor of survival time, although it has been linked to CKD progression in stage 2 cats, but not when cats in all stages were considered together.¹⁸ No attempt was made to examine the cats in IRIS stages 2 and 3 separately in the present study, because there were only 9 cases in IRIS stage 3 that demonstrated progression. Anemia of CKD is considered to be multifactorial, but in part a result of decreased erythropoietin production by the kidney.³⁹ Although few cases in this study were anemic, with a median PCV of 33% [30, 37], as PCV decreases there will be a reduction

in the oxygen carrying capacity of the blood, which in the diseased kidney might result in hypoxia. One marker of hypoxia, urinary vascular endothelial growth factor, has been associated with shorter survival times and progression in feline CKD,⁴⁰ therefore it is possible that small changes in PCV might exacerbate CKD and shorten survival time through hypoxia.

There were a number of limitations in this study. Cases might not have had a blood sample taken at the time of death or euthanasia, therefore some of the cats might have progressed without it being documented. Additionally, plasma creatinine concentration was used as a marker for GFR, which is dependent on muscle mass; therefore changes in muscle mass might have resulted in cats being identified as stable that actually had a substantial change in their GFR. Furthermore, all-cause mortality was used in the survival analysis, with a large number of cats being euthanized. There might have been circumstances in which owners elected for euthanasia for reasons other than severe retractable illness, which will have introduced some bias into the analysis.

In summary, increased FGF-23 at diagnosis of CKD is associated with an increased risk of progression of the disease over the next 12 months and associated with an increased risk of death. These findings might make it a useful biomarker for predicting poorer outcomes in cats with CKD. Further work is now required to elucidate the complex relationships between variables involved in Ca-P derangements in CKD and to establish if FGF-23 is a mediator, or simply a marker of CKD-MBD in cats.

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Blood fibroblast growth factor 23 concentration in cats with and without chronic kidney disease: a scoping review

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Abstract

Objectives

This study undertook a scoping review of research on blood fibroblast growth factor 23 (FGF-23) concentrations in healthy non-azotemic cats and cats with chronic kidney disease (CKD) to describe the volume and nature of existing literature, to determine whether published studies provide adequate evidence to support the use of FGF-23 as a biomarker in clinical practice and to identify any existing gaps in knowledge.

Methods

PRISMA Extension for Scoping Reviews guidelines were used to design and perform the scoping review. Online databases were used to identify observational and clinical studies of blood FGF-23 concentrations in healthy cats and cats with CKD published before December 2022. Study and population characteristics and descriptive data on FGF-23 concentrations were extracted.

Results

A total of 205 publications were reviewed; 17 were retained for inclusion. Most studies were retrospective. Most studies included cats with International Renal Interest Society stage 2–4 CKD, with some variation. Key concepts explored in the literature include FGF-23 concentrations by CKD stage, effect of dietary phosphate restriction on FGF-23 concentrations, relationship between FGF-23 concentrations and blood phosphorus, calcium and magnesium concentrations, and FGF-23 concentrations in cats with progressive CKD. FGF-23 concentrations tended to be higher in cats with CKD compared with healthy cats, with an overlap between healthy and CKD populations, and there was significant variation within stages of CKD.

Conclusions and relevance

FGF-23 is a biomarker of interest for the management and monitoring of phosphate overload in cats. Studies support several potential clinical applications for measuring FGF-23 concentration in practice; however, evidence is limited. Research on FGF-23 in cats with CKD would benefit from longitudinal, prospective studies that standardize CKD diagnosis and categorize cats by stage using current guidelines. Studies should include cats with early-stage, non-azotemic CKD and use commercially available assays so such results are comparable across studies.

Introduction

Renal secondary hyperparathyroidism is common in cats with chronic kidney disease (CKD) and is characterized by elevated parathyroid hormone (PTH) secondary to derangements in phosphorus, calcium and vitamin D homeostasis.^{1–3} CKD–mineral and bone disorder (CKD-MBD) is a consequence of these hormonal and mineral

derangements. In people, CKD-MBD is defined as a systemic disorder manifested by either one or a combination of the following: (1) abnormalities of calcium, phosphorus, PTH, fibroblast growth factor 23 (FGF-23) and vitamin D metabolism; (2) abnormalities in bone turnover, mineralization, volume, linear growth or strength; or (3) vascular or other soft tissue calcification.⁴ Clinical manifestations of CKD-MBD in people can include muscle weakness, bone pain, pruritus and skeletal fractures.^{5,6} Although the hormonal derangements of CKD-MBD in cats and dogs are similar to those in people, the clinical signs are less defined in veterinary medicine. In dogs^{7–11} and cats,^{8,12,13} CKD-MBD can manifest as reduced cortical bone quality, dystrophic organ and soft tissue mineralization, and osteopenia and osteodystrophy, most often reported in the skull and mandible.

A key hormone in the regulation of calcium–phosphate homeostasis is the phosphatonin peptide hormone FGF-23. FGF-23 binds to its membrane-bound receptor complexed with α -Klotho co-receptor on target cells to increase renal phosphate excretion and reduce calcitriol production in the kidney.¹⁴ Traditional understanding of the pathogenesis of CKD-MBD suggested that hyperphosphatemia and diminished renal mass resulted in reduced renal production of calcitriol and the subsequent development of renal secondary hyperparathyroidism.¹⁵ In the early 2000s, FGF-23 and its co-receptor Klotho were identified, and the binding of FGF-23 to FGF receptor/ α -Klotho complexes was shown to increase fractional excretion of phosphate in the nephron to maintain phosphate homeostasis early in CKD. Thus, blood FGF-23 concentrations increase early in people with CKD and before the development of diminished renal mass or overt hyperphosphatemia.¹⁶ An elevated blood FGF-23 concentration, therefore, has been proposed as an early potential indicator of phosphate dysregulation and trigger of CKD-MBD by inhibiting renal calcitriol synthesis, resulting in hypocalcemia and increased PTH synthesis.

FGF-23 is a biomarker of interest in veterinary medicine, and over the past decade several research groups have published studies of plasma or serum FGF-23 concentrations measured in healthy non-azotemic cats and cats diagnosed with CKD. Recently, two assays to measure serum FGF-23 concentration in cats became commercially available to veterinarians to use as a tool in the monitoring and management of phosphate overload in cats with CKD (IDEXX Laboratories; FUJIFILM VET Systems). Overall, it is unclear what kind of information is available in the literature supporting the measurement of blood FGF-23 concentrations in cats with CKD. This study undertook a scoping review of research on FGF-23 in healthy cats and cats with CKD. The objectives of the study were to conduct a systematic search of databases using published methodology to describe the volume and nature of existing literature on FGF-23 in cats with CKD, to chart and summarize FGF-23 concentrations in healthy cats and according to International Renal Interest Society (IRIS) CKD stage, to review and appraise the literature on the use of FGF-23 in feline medicine, and to chart the study methodology from existing publications to identify knowledge gaps and form recommendations for future research.

Materials and methods

Full articles published up to 31 December 2022 were evaluated for inclusion. Articles were considered published if they were available online on the journal website before 31 December 2022, even if the article did not appear in print until 2023. Abstracts without associated manuscripts and unpublished data were excluded. The search was performed using PubMed (<https://pubmed.ncbi.nlm.nih.gov>), CAB Abstracts and Google Scholar by an investigator (HTM). CAB Abstracts was searched on 17 October 2022. PubMed and Google Scholar were searched on 18 April 2023. For

Google Scholar, the first three pages of search results were included in the review. No specialized software was used. Articles identified in the search were then assessed for inclusion by reviewing their abstracts. Only observational or clinical studies that measured serum or plasma FGF-23 concentrations in the context of naturally occurring renal disease in cats were included. Articles published in languages other than English were evaluated for inclusion if they had abstracts translated into English. Article names were searched in Google Scholar, which identified one additional study containing FGF-23 data that was not otherwise identified. Authors were not directly contacted for additional articles. The screening of excluded articles was undertaken by a second reviewer (SS). For the included publications, author names, year of print publication, article title, population description, age of population under study, sample size, inclusion and exclusion criteria, and, if exact values were reported, all available descriptive statistics on FGF-23 (median, first quartile, third quartile, minimum, maximum, range and interquartile range) were extracted. Data were extracted by independent double data extraction (RM, DS) and then further evaluated by other investigators (HTM, SS).

Due to the variations in methodology and lack of clarity around the potential overlap of data points between studies, no meta-analysis or systematic review could be performed, and the authors moved forward with a scoping review. A scoping review was conducted and reported according to the Preferred Reporting Items for Systemic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines.¹⁷

Results

Study selection

Search terms used in each search engine and the number of items identified by each search term are presented in [Figure 1](#). The search resulted in 205 unique items for evaluation and the number of articles excluded is in [Figure 1](#). A complete report of all items identified from each search term and the reasons for exclusion are presented in [Table 1](#) in the supplementary material. In total, 16 articles were identified from the literature search. A Google Scholar search of individual authors identified one additional article that was not identified in any of the other searches, resulting in a total of 17 articles for inclusion. No single search engine identified all included articles (see [Table 2](#) in the supplementary material). Eight articles were identified by all three search engines, five were identified by two search engines and three were identified by only one search engine. Google Scholar identified the highest number of included articles (n = 15), including one identified in the author search, compared with 11 for PubMed and CAB Abstracts.

Figure 1

Flow diagram showing the number of publications identified and reviewed, and reasons for those excluded from the scoping review of the literature on fibroblast growth factor 23 (FGF-23) in cats with chronic kidney disease (CKD)

Study characteristics

The 17 included studies were published between 2013 and 2022. The sample type and kit used in each study and the study objectives are summarized in [Table 1](#). All studies used a sandwich ELISA to measure intact FGF-23, manufactured by one of four companies: 12 used the Human FGF-23 ELISA Kit (Kainos Laboratories); two used the Cat Fibroblast Growth Factor 23 ELISA Kit (MyBioSource); two used MedFrontier FGF23 (Minaris Medical;

MedFrontier); and one used the Feline FGF-23 ELISA Kit (Deyihengda Biotech) (Table 1). Ten of 12 studies that used the Kainos assay and one of two studies that used the MyBioSource assay measured FGF-23 in plasma and the rest used serum. Geddes et al.¹⁹ validated the use of the Kainos ELISA to measure intact plasma FGF-23 concentrations in cats, and found the assay to have acceptable precision, reproducibility and specificity for feline FGF-23. The Kainos and MedFrontier assays were reported to use the same primary anti-FGF-23 antibodies with different secondary antibodies.³⁵ Validation of the MedFrontier assay was presented in the study by Miyakawa et al.³² Information about the antibodies used in the MyBioSource and Deyihengda kits was not available. Assay validation for the Deyihengda assay was published in the study by Lin et al.²⁸ At the time of writing, there was no published validation of the MyBioSource assay but information on the precision of the assay was available from the manufacturer.³⁶ Studies comparing FGF-23 concentration measurements between manufacturers in human and feline samples have found assay-based differences that preclude direct comparison across methodologies.^{35,37} Therefore, FGF-23 concentrations were not compared across the different kit manufacturers in this study.

Table 1

Sample type, FGF-23 assay kit and study objectives for the 17 articles included in the scoping review

Study	Sample type	Assay	Study objectives
Finch et al, 2013 ¹⁸	Plasma	Kainos	Determine if FGF-23 is higher in non-azotemic cats that become azotemic within 12 months Explore the relationship between FGF-23 and PTH, and between FGF-23 and GFR
Geddes et al, 2013 ¹⁹	Plasma	Kainos	Method validation for FGF-23 measurement in cats Assess the relationship between FGF-23, PTH, creatinine and phosphorus in cats with CKD
Geddes et al, 2013 ²⁰	Plasma	Kainos	Evaluate the effect of feeding a renal diet on FGF-23 in cats with stable azotemic CKD
Geddes et al, 2015 ²¹	Plasma	Kainos	Investigate the relationship of FGF-23 and PTH concentrations at CKD diagnosis to survival, and to CKD progression over 12 months
Geddes et al, 2016 ²²	Plasma	Kainos	Evaluate whether a protein- and phosphate-restricted diet decreases FGF-23 and PTH in healthy older cats and whether it delays onset of azotemic CKD
van den Broek et al, 2017 ²³	Plasma	Kainos	Determination of risk factors for total hypercalcemia
van den Broek et al, 2018 ²⁴	Plasma	Kainos	Explore associations of Mg, FGF-23, mortality and disease progression in cats with CKD
Liao et al, 2019 ²⁵	Plasma	MyBioSource	Explore relationship between indoxyl sulfate and FGF-23 in cats with CKD

Study	Sample type	Assay	Study objectives
Sargent et al, 2019 ²⁶	Plasma	Kainos	Explore relationship between FGF-23 and SDMA in geriatric cats without azotemic CKD
Geddes et al, 2021 ²⁷	Plasma	Kainos	Describe effect of moderately phosphate-restricted diet in cats with azotemic CKD and hypercalcemia
Lin et al, 2021 ²⁸	Serum	Deyihengda	Evaluate FGF-23 in cats with different stages of CKD Evaluate effect of age on FGF-23 in healthy cats
Radloff et al, 2021 ²⁹	Serum	Kainos	Evaluate relationship between FGF-23 and aldosterone in human, feline and canine patients with CKD Test hypothesis that aldosterone drives increased FGF-23 in a mouse model of CKD
Tang et al, 2021 ³⁰	Plasma	Kainos	Identify risk factors for hypercalcemia in cats with CKD on phosphate-restricted diets Evaluate effect of hypercalcemia on CKD-MBD
van den Broek et al, 2022 ³¹	Plasma	Kainos	Describe the prevalence and incidence of hypercalcemia in cats with azotemic CKD Explore variables that identify cats with elevated risk of hypercalcemia
Miyakawa et al, 2022 ³²	Serum	MedFrontier	Investigate FGF-23 in healthy cats and cats with different stages of CKD
Miyakawa et al, 2022 ³³	Serum	MedFrontier	Evaluate whether FGF-23 concentration is associated with total and ionized Ca in cats with CKD and upper urolithiasis
Grelová et al, 2022 ³⁴	Serum	MyBioSource	Evaluate relationship between FGF-23, SDMA, creatinine, urea and phosphate in cats with CKD

Ca = calcium; CKD = chronic kidney disease; FGF-23 = fibroblast growth factor 23; GFR = glomerular filtration rate; MBD = mineral and bone disorder; Mg = magnesium; PTH = parathyroid hormone; SDMA = symmetric dimethylarginine

Information about the study design, study populations, including geographic location, definitions of health and CKD, and exclusion criteria is provided in Table 2. The included studies were conducted on cats in the UK (11/17), Japan (2/17), and Taiwan, Austria, Slovakia and China (one each). Most studies were retrospective (n = 14), of which eight were cohort studies, five were cross-sectional studies and one included both cross-sectional and cohort components. Three studies were prospective: one was a cross-sectional study; one was a double-blinded, randomized, placebo-

controlled clinical diet trial; and one was a cohort study. Seven studies recruited only cats aged 9 years or older and the remaining studies did not state age as a criterion for inclusion.

Table 2

Study type, country and time frame for sample collection, definition of CKD and inclusion/exclusion criteria for the studies of fibroblast growth factor 23 in cats

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
Finch et al, 2013 ¹⁸	P-CO	UK ^{*†}	NR	Age >9 years and healthy non-azotemic (Cr <2.0 mg/dl) at enrollment	Non-azotemic (Cr <2 mg/dl) grouped by Cr values at 12 month follow-up Group 1: Plasma Cr ≤1.58 mg/dl Group 2: Cr >1.58 mg/dl and ≤2.0 mg/dl, or Cr >2.0 mg/dl and USG >1.035 Group 3: Cr >2.0 mg/dl and USG <1.035	N/A	NR
Geddes et al, 2013 ¹⁹	R-CX	UK ^{*†}	2000–2010	Age >9 years; healthy or CKD	>9 years with normal physical examination and biochemical profile at entry and for 12-month period	Cr >2.0 mg/dl and USG <1.035 or Cr >2.0 mg/dl consecutively 2–4 weeks apart without pre-renal cause	Azotemic cats being fed a protein-and phosphate-restricted diet Concurrent disease; medications that alter Ca or phosphate absorption or metabolism; plasma TT4 >40 nmol/l, clinical signs consistent with hyperthyroidism or undergoing treatment for hyperthyroidism
Geddes et al, 2013 ²⁰	R-CO	UK ^{*†}	2000–2011	Age ≥9 years and diagnosed with renal azotemia	N/A	Cr >2.0 mg/dl and USG <1.035 or	Plasma TT4 >40 nmol/l, clinical signs consistent with hyperthyroidism or undergoing treatment

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
						Cr >2.0 mg/dl consecutively 2–4 weeks apart without pre-renal cause	for hyperthyroidism; concurrent disease; medications that alter Ca or phosphate metabolism; developed hypertension during study period; Cr concentration increased >25% during study period
Geddes et al, 2015 ²¹	R-CO	UK ^{*†}	1992–2011	Age ≥9 years and diagnosed with renal azotemia	N/A	Cr >2.0 mg/dl and USG <1.035	Fed renal diet at diagnosis; USG >1.035 after diagnosis; had or developed hyperthyroidism; concurrent disease (except hypertension); medications that alter Ca or phosphate absorption or metabolism
Geddes et al, 2016 ²²	P-CL	UK [†]	2011–2013	Age ≥9 years and ‘apparently healthy’	Normal physical examination and without reported illness. Cats with Cr >2.0 mg/dl and normal USG >1.035 were included Cats with osteoarthritis, heart murmur, dental disease or palpation of small kidneys were eligible for inclusion	N/A	Cr >2 mg/dl in conjunction with USG <1.035; long-term medical problems within past 6 months; TT4 <40 nmol/l; recurrent lower urinary tract problems; fed diet for lower urinary tract conditions; uncontrolled systemic hypertension
van den Broek et al, 2017 ²³	R-CO	UK ^{*†}	2000–2014			Cr ≥2.0 mg/dl and USG <1.035 or	Clinical signs of hyperthyroidism, TT4 >40 nmol/l, medical treatment for hyperthyroidism,

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
						Cr $\geq$ 2.0 mg/dl consecutively 2–4 weeks apart without pre-renal cause	diabetes mellitus or corticosteroid treatment
van den Broek et al, 2018 ²⁴	R-CO	UK ^{*†}	1999–2013	Age $\geq$ 9 years Healthy cats must have stored heparinized plasma sample CKD cats must have an FGF-23 concentration and stored heparinized plasma sample from time of diagnosis	‘Apparently healthy’ cats with no significant abnormalities in clinical history, examination, blood or urine testing, and no prescribed medications	Cr $\geq$ 2.0 mg/dl and USG $<$ 1.035 or Cr $\geq$ 2.0 mg/dl consecutively 2–4 weeks apart without pre-renal cause	Clinical signs of hyperthyroidism, TT4 $>$ 40 nmol/l, medical treatment for hyperthyroidism, diabetes mellitus or corticosteroid treatment
Liao et al, 2019 ²⁵	R-CO	Taiwan [‡]	NR		Lacked any clinical signs of CKD and normal CBC, serum biochemistry panel (Cr $<$ 1.6 mg/dl), urinalysis and kidney ultrasound	Cr $>$ 1.6 mg/dl and one of the following: clinical signs of renal disease, USG $<$ 1.030, persistent proteinuria (UPC $>$ 0.5), with or without abnormal renal ultrasound, for at least 3 months	Acute kidney disease or acutely worsening azotemia, hyperthyroidism, lower urinary tract disease, clinical signs of non-renal disease
Sargent et al, 2019 ²⁶	R-CX	UK ^{*†}	2011–2016	Age $>$ 9 years Healthy cats with plasma FGF-23 or SDMA concentration recorded (or sufficient stored sample to measure)	‘Apparently healthy’ cats without increased Cr	N/A	Current or historic increases in Cr; plasma TT4 $>$ 40 nmol/l or treatment for hyperthyroidism; diabetes mellitus
Geddes et al, 2021 ²⁷	R-CO	UK ^{*†}	2014–2018	Prior or new diagnosis of azotemic CKD and diagnosed with ionized	N/A	Cr $>$ 2.0 mg/dl and USG $<$ 1.035 or	Diagnosis of diabetes mellitus; corticosteroid treatment; diagnosed with hyperthyroidism

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
				hypercalcemia (>1.4 mmol/l) between January 2014 and 2018 while eating a phosphate-restricted clinical renal diet		Cr >2.0 mg/dl consecutively 2–4 weeks apart	
Lin et al, 2021 ²⁸	R-CX	China [§]	2017–2019		No clinical signs or medical evidence of previous kidney injury, renal disease or nephrotoxin exposure with normal CBC, serum biochemistry and urinalysis	Renal azotemia defined as Cr >1.6 mg/dl (>140 µmol/l) with USG <1.035 or persistent azotemia on two consecutive occasions Stage 1: inadequate urinary concentrating ability without identifiable non-renal cause, radiographic changes of kidneys and/or persistent increase in serum Cr within RI in past 3 months	Missing FGF-23, iPTH or other laboratory tests
Radloff et al, 2021 ²⁹	R-CO	Austria [¶]	NR	Age >1 year with frozen serum samples with or without CKD	Cr concentration and USG within RI	History of polyuria, polydipsia, azotemia and USG <1.035 over at least 3 months	Diagnosis of AKI, primary parathyroid disease, recent acute-to-chronic kidney disease transition, corticosteroid or angiotensin-converting enzyme inhibitor therapy

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
Tang et al, 2021 ³⁰	R-CO	UK ^{*†}	2014–2019	Age ≥9 years with azotemic CKD Plasma tCa at time of CKD diagnosis and at least two follow-up visits within the first 200 days after transition to renal diet	N/A	Cr ≥2.0 mg/dl and USG <1.035 or Cr ≥2.0 mg/dl consecutively 2–4 weeks apart without pre-renal cause	TT4 >40 nmol/l or treatment for hyperthyroidism; diabetes mellitus; receiving phosphate-restricted diet at diagnosis; corticosteroid treatment; tCa >11.8 mg/dl or iCa >5.6 mg/dl (1.4 mmol/l) before starting renal diet
van den Broek et al, 2022 ³¹	R-CO	UK ^{*†}	2012–2018	iCa available from date of diagnosis with azotemic CKD	N/A	Cr ≥2.0 mg/dl (upper limit of RI) and USG <1.035 or Cr ≥2.0 mg/dl consecutively 2–4 weeks apart	Diagnosis of hyperthyroidism or diabetes mellitus; corticosteroid treatment; being fed a renal diet, plasma TT4 >40 nmol/l
Miyakawa et al, 2022 ³³	R-CX	Japan [#]	2015–2020	Cats with stable CKD (ie, <25% change in Cr within 3 months) and presence of nephroliths or ureteroliths	N/A	Ultrasonographic renal abnormalities including small kidneys, decreased corticomedullary differentiation and irregular renal contours Renal azotemia defined as Cr >1.8 mg/dl (>159 µmol/l; upper limit of RI) and USG <1.035	Cr >2.8 mg/dl (250 µmol/l; ie, consistent with IRIS stage 3 or 4); P ≥1.50 mmol/l; diagnosis or suspicion of AKI, neoplasia or hyperthyroidism; absence of iCa concentration
Miyakawa et al, 2022 ³²	R-CX	Japan [#]	2015–2020	Age 1–8 years ;healthy or CKD	Laboratory or client-owned cats Healthy based on history, physical examination, serum biochemistry,	≥3 months of Cr >1.8 mg/dl (>159 µmol/l; upper limit of RI) with USG <1.035, UPC ≥0.4 without apparent extrarenal cause and/or	Diagnosis or suspicion of AKI, tumor or hyperthyroidism

Study	Type	Country	Dates	Eligibility requirements	Definition of healthy	Definition of CKD	Exclusion criteria
					urinalysis, abdominal ultrasound	ultrasonographic changes	
Grelová et al, 2022 ³⁴	P-CX	Slovakia [∞]	2020–2021	Age >10 years ;healthy or CKD	No history of AKI, exposure to nephrotoxins, evidence of renal disease (weight loss, PU/PD, anorexia, vomiting) or imaging abnormalities Normal serum biochemistry and SDMA within RI	At least three clinical signs (weight loss, poor hair quality, PU/PD, anorexia or radiographic kidney changes) and azotemia or persistent SDMA >14 µg/dl	Treatment for hyperthyroidism, hypertension or diabetes mellitus

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People's Dispensary for Sick Animals (PDSA)

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Beaumont Sainsbury Animal Hospital

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National Taiwan University Veterinary Hospital

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China Agricultural University Animal Teaching Hospital

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Teaching Hospital of the University of Veterinary Medicine Vienna

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University of Veterinary Medicine and Pharmacy in Košice

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Nippon Veterinary and Life Science University

AKI = acute kidney injury; Ca = calcium; CKD = chronic kidney disease; Cr = creatinine; FGF-23 = fibroblast growth factor 23; iCa = ionized calcium; IRIS = International Renal Interest Society; N/A = not applicable; NR = not reported; P = phosphate concentration; P-CL = prospective clinical trial; P-CO = prospective cohort; PTH = parathyroid hormone; PU/PD = polyuria/polydipsia; R-CO = retrospective cohort; R-CX = retrospective cross-sectional; RI =

reference interval; SDMA = symmetric dimethylarginine; tCa = total calcium; TT4 = total thyroxine concentration; UPC = urine protein:creatinine ratio; USG = urine specific gravity

Cat population characteristics

The determination of health status (healthy vs CKD) and exclusion criteria varied greatly between studies (Table 2). Most studies (n = 14) included a population of cats with a diagnosis of CKD; the remaining three studies included only healthy cats. All cats included in the studies were client-owned, except for one study³² that included laboratory cats in the healthy control population. Healthy cats were often defined as non-azotemic, non-diabetic, euthyroid cats (although two studies^{18,28} did not mention hyperthyroidism as an exclusion criterion, nor did they mention the measurement of total thyroxine concentration in the study population). One study²⁶ that measured FGF-23 concentrations in apparently healthy older cats (aged >9 years) reported that 25% of included cats had increased concentrations of symmetric dimethylarginine (SDMA; >14 µg/dl). A persistent increase in SDMA concentration is consistent with a diagnosis of early-stage CKD based on the current IRIS CKD staging guidelines.³⁸ This group of cats could not be classified by the authors of this review as either healthy or CKD since the publication did not report any information about persistence of SDMA concentrations >14 µg/dl. It is therefore plausible that this population of apparently healthy cats contained both healthy cats and cats with stage 1 or 2 CKD based on current IRIS guidelines. A second study by Finch et al¹⁸ included older non-azotemic cats (creatinine <2 mg/dl), and after 12 months grouped cats according to creatinine concentration and, in some instances, urine specific gravity (USG) (groups 1–3) (Table 2). No further detail is provided regarding the diagnosis of CKD, and it is plausible that some cats in group 1 or group 2 were healthy cats without renal dysfunction.

In the 14 studies that included cats with CKD, the definition of renal azotemia varied and included creatinine >2.0 mg/dl (7/14), >1.8 mg/dl (2/14), >1.6 mg/dl (2/14) and was not defined in two studies. The diagnosis of CKD was based on renal azotemia with inadequately concentrated urine (USG <1.030 or <1.035) or persistent azotemia without pre-renal cause in most studies (13/14). One study³⁴ diagnosed CKD based on clinical signs (including radiographic kidney changes) and either the presence of azotemia (undefined) or persistent elevation in SDMA concentration (>14 µg/dl) without consideration of USG. A few studies included abnormal renal ultrasonographic or radiographic findings (5/14), historical clinical signs of kidney disease (ie, weight loss, polyuria/polydipsia; 3/14) or urine protein:creatinine ratio measurement (one study) in determining the diagnosis of CKD. Only two studies^{26,34} mentioned measuring SDMA concentrations in their study populations, of which one study³⁴ used SDMA to diagnose CKD based on the persistence of an elevated value. The other study²⁶ that measured SDMA concentration included only apparently healthy cats and, as mentioned above, some of the cats had an elevated SDMA concentration.

All the studies from the UK drew patients from the same two clinics with overlapping study periods (Table 2). It was not possible to determine from the information available in the publications if results from individual cats were used in multiple studies or how often or how many overlapping data points may be in the data sets. The two studies performed in Japan^{32,33} likely also have overlapping data points, but this is similarly unclear. These studies that may have overlapping patient populations included cats with IRIS stage 1 and 2 CKD.

FGF-23 and CKD stage

Five studies subdivided cats by CKD stage, of which four studies provided descriptive information about FGF-23 concentrations by IRIS stage (Table 3; see Figure 1 in the supplementary material).^{19,25,28,32,33} All these studies used the creatinine concentration to stage. Four studies^{19,28,32,33} used the 2019 IRIS staging guidelines (stage 2: 1.6–2.8 mg/dl; stage 3: 2.9–5.0 mg/dl; stage 4: >5 mg/dl); however, one study¹⁹ used a creatinine concentration in the range of 2.1–2.8 mg/dl to subclassify cats as stage 2 CKD. One study did not clarify which version of IRIS staging was used.²⁵ Three studies^{28,32,33} subclassified cats as stage 1 CKD, of which two studies provided descriptive information about FGF-23 concentrations. Stage 1 CKD was defined as creatinine concentration <1.6 mg/dl (<140 µmol/l) in conjunction with either ultrasonographic renal abnormalities in two studies^{32,33} or as inadequate urine concentrating ability, radiographic renal changes and/or persistent increase in creatinine concentration within the reference interval (RI) in the past 3 months in one study.²⁸ No study used SDMA concentrations to stage cats with CKD.

Table 3.

Article characteristics and population information for studies evaluating serum or plasma concentrations of fibroblast growth factor-23 (pg/mL) in healthy cats and in cats with CKD stages 1–4

Citation	Manufacturer	Population	Sample Size	Age (Years)*	Median	Q1	Q3	IQR	Range
Geddes 2013 ¹⁹	Kainos	Healthy	44	> 9	146.0	111.0	264.0	–	–
		CKD - Stage 2	20	> 9	354.0	239.0	473.0	–	–
		CKD - Stage 3	22	> 9	1,282.0	428.0	7,658.0	–	–
		CKD - Stage 4	14	> 9	33,478.0	4,241.0	86,204.0	–	–
Geddes 2013 ²⁰	Kainos	CKD - Stage 2-3	11	≥ 9	1,126.2	492.6	2,723.5	–	–
		CKD - Stage 2-4	33	≥ 9	428.7	341.6	973.5	–	–
Geddes 2016 ²²	Kainos	Healthy	25	12.2 (11.2, 13.5)	153.4	105.1	208.6	–	–
		Healthy	28	12.1 (11.1, 13.9)	124.3	99.2	209.9	–	–
Sargent 2019 ²⁶	Kainos	Healthy - SDMA ≤14 µg/dl	107	12.2 (11.2, 14.5)	155	127	243	–	–
		Healthy - SDMA >14 µg/dl	36	14.0 (12.1, 15.3)	271	176	523	–	–
Liao 2019 ²⁵	MyBioSource	Healthy	20	4 (3.0-5.75)	121.05	–	–	71.85	

Citation	Manufacturer	Population	Sample Size	Age (Years)*	Median	Q1	Q3	IQR	Range
		CKD - Stage 2	38	10.0 (5.8-15.0)	163.1	–	–	126.9	
		CKD - Stage 3	23	12.0 (8.0-15.0)	270.8	–	–	167.5	
		CKD - Stage 4	12	8.0 (5.5-9.5)	728.3	–	–	793.5	
Miyakawa 2022 ³²	MedFrontier	Healthy	7	3.8 (2.0-8.3)	115	–	–	–	88–194
		CKD – Stage 1	7	4.2 (1.9-8.2)	284	–	–	–	119–746
		CKD – Stage 2	32	5.8 (1.3-8.9)	365	–	–	–	94–8,720
		CKD – Stage 3-4	15	6.0 (2.0-8.2)	1,378	–	–	–	182–14,978
Lin et al, 2021 ²⁸	Deyinhengda	Healthy	108	3.0 (0.3-20)	110	–	–	–	40–396
		CKD – Stage 1	34	–	189	–	–	–	86–437
		CKD – Stage 2	74	–	208	–	–	–	105–432
		CKD – Stage 3	51	–	279	–	–	–	100–552
		CKD – Stage 4	37	–	302	–	–	–	235–597

*

Age data are presented as median (range) or median (25th, 75th percentiles) in years unless otherwise noted CKD, chronic kidney disease; IQR, interquartile range; Q1, first quartile; Q3, third quartile

Table 3 shows descriptive statistics (median with measure of variability) for FGF-23 concentrations from the seven studies that provided FGF-23 values for healthy cats and/or cats with CKD of different stages. Two studies^{18,29} did not report exact FGF-23 concentrations and provided only graphical representation. Although FGF-23 concentrations are not directly comparable across methodologies, all studies show a similar trend of increasing median FGF-23 concentration with later stages of CKD. FGF-23 concentrations measured using the Kainos in healthy cats^{19,22,26} and cats with different stages of CKD¹⁸ are shown in Figure 2. When studies had overlapping sample collection dates, FGF-23 concentrations from only one study are presented to avoid the potential duplication of data. FGF-23 concentrations measured by the MedFrontier³² and Deyihengda²⁸ assays in cats with CKD of different IRIS stages are shown in Figure 3.

Figure 2

(a,b) Plasma fibroblast growth factor 23 (FGF-23) concentrations (pg/ml) for healthy controls and cats with chronic kidney disease (CKD) stages 2–4 measured using the Kainos assay. Healthy cats are represented by dark purple boxes^{19 22 26} and subgroups with CKD are represented by green (stage 2),¹⁹ light purple (stage 3)¹⁹ or blue (stage 4)¹⁹ boxes based on stage of disease. Boxes represent the interquartile range (Q1–Q3). The shaded area in panel (a) represents the area that is expanded in panel (b)

Figure 3

Serum fibroblast growth factor 23 (FGF-23) concentrations (pg/ml) measured using (a) the MedFrontier assay and (b) the Deyinhengda assay for healthy cats and cats with chronic kidney disease (CKD) stages 1–4. Healthy cats are represented by dark purple boxes^{28 32} and subgroups with CKD are represented by dark blue (stage 1),^{28 32} dark green (stage 2),^{28 32} light purple (stage 3),²⁸ light blue (stage 4)²⁸ or turquoise (stage 3–4) boxes.³² Boxes represent the minimum to maximum concentration range

Across all studies and assays, the median FGF-23 concentration was lowest in healthy cats and increased with each IRIS CKD stage (Table 3). In addition, there was wide variability in FGF-23 concentrations among within each CKD stage (Table 3; Figures 2 and 3). None of the published studies showed clear differentiation between healthy cats and cats with CKD and between CKD stages 1–4 (Table 3; Figures 2 and 3) based on FGF-23 concentrations. There was overlap in the ranges of FGF-23 concentrations between healthy cats and cats with CKD, particularly cats with stage 1 and 2 CKD, although overlap between healthy cats and cats with stage 4 CKD was reported in two publications (Figure 3).^{28,32} In Geddes et al,¹⁹ a small number of cats with stage 4 CKD had extremely high FGF-23 concentrations, with a range of 81,963 pg/ml from the first to the third quartile. No similarly dramatic FGF-23 values were reported in cats with stage 4 CKD in other studies.

Lin et al²⁸ compared FGF-23 concentrations in healthy cats subdivided by age (0–2 years, 3–6 years, 7–10 years, 11–14 years, ≥15 years) and concluded that age was not a determinant of FGF-23 concentrations. In addition, the study found no difference in FGF-23 concentrations based on sex or neuter status.

FGF-23 and phosphate overload

FGF-23 concentrations were found to positively correlate with serum phosphorus concentrations in cats with CKD in some studies^{20 25 28 32} but some studies showed no correlation.^{18 26 33 34} Geddes et al¹⁹ found that phosphorus concentration was an independent predictor of FGF-23 in cats with stage 2–4 CKD. The same study subdivided cats with CKD by stage 2 and 3 and by phosphorus concentration (stage 2: ≤4.5 mg/dl; stage 3: ≤5.0 mg/dl) and found that FGF-23 concentrations were significantly higher in cats with phosphorus concentrations above the cutoff (Table 4).

Table 4

Article characteristics and population information for a study evaluating plasma concentrations of FGF-23 (pg/ml) in cats with normophosphatemic and hyperphosphatemic CKD

Citation	Manufacturer	Population	Sample size	Median	Q1	Q3
Geddes et al, 2013 ¹⁹	Kainos	CKD stage 2a*	15	329	237	370

Citation	Manufacturer	Population	Sample size	Median	Q1	Q3
	Kainos	CKD stage 2b*	5	770	376	1912
	Kainos	CKD stage 3a*	9	435	294	1230
	Kainos	CKD stage 3b*	13	6448	887	24,499

*

Stage 2a: phosphate concentration ≤ 4.5 mg/dl; stage 2b: >4.5 mg/dl; stage 3a: ≤ 5.0 mg/dl; stage 3b: >5.0 mg/dl

CKD = chronic kidney disease; FGF-23 = fibroblast growth factor 23; Q1, first quartile; Q3, third quartile

A prospective diet trial by Geddes et al²² evaluated the effects of feeding a moderate protein (76 g/Mcal) and phosphate (1.6 g/Mcal) diet and an adult maintenance diet (protein 86 g/Mcal; phosphate 2.6 g/Mcal) to apparently healthy older cats on several markers of MBD, including FGF-23, over an 18-month period. The study found that neither FGF-23 nor phosphorus concentrations changed over time and that neither analyte was affected by diet. In a retrospective study, Geddes et al²⁰ evaluated the effect of renal diets (dietary phosphate range 0.77–1.12 g/Mcal) on phosphorus and FGF-23 concentrations over approximately 4–6 weeks in cats with stage 2–4 CKD and compared the results with those from cats with CKD without a diet change that remained on various maintenance diets. Cats with CKD fed a renal diet had a significant reduction in phosphorus and FGF-23 concentrations, unlike the comparator cats. In that study, phosphorus and FGF-23 concentrations were compared before and after a renal diet between cats that were categorized as either hyperphosphatemic (stage 2: >4.5 mg/dl; stage 3: >5.0 mg/dl; stage 4: >6.0 mg/dl) or normophosphatemic (stage 2: ≤ 4.5 mg/dl; stage 3: ≤ 5.0 mg/dl; stage 4: ≤ 6.0 mg/dl) based on CKD stage. FGF-23 concentrations decreased in both the hyperphosphatemic and normophosphatemic groups despite phosphorus concentrations decreasing in only the hyperphosphatemic cats.

FGF-23 and calcium and magnesium

Five studies evaluated the relationship between FGF-23 and calcium concentrations (Table 5). Studies reported that total calcium (Geddes et al¹⁹) and ionized calcium (Miyakawa et al³²) concentrations were independent predictors of FGF-23 concentrations in cats with CKD. Miyakawa et al³³ found that in cats with CKD and upper urolithiasis, FGF-23 concentrations were higher in those cats with ionized hypercalcemia and a normal phosphorus concentration (<4.6 mg/dl). In that study, FGF-23 concentrations positively correlated with both total and ionized calcium. A study by van den Broek et al,²³ which evaluated total calcium concentration in healthy cats and cats with stage 2–4 CKD, found no difference in FGF-23 concentrations between cats with hypercalcemia and those with normocalcemia. On the contrary, a study evaluating ionized calcium in cats with stage 2–4 CKD by van den Broek et al³¹ showed higher FGF-23 concentrations in cats with hypercalcemia compared with cats with normocalcemia.

Table 5

Article characteristics and population information for studies evaluating serum or plasma concentrations of FGF-23 (pg/ml) and calcium or magnesium in cats with CKD

Citation	Manufacturer	Population	Sample size	Age group *	Median	Q1	Q3	Range
van den Broek et al, 2017 ²³	Kainos	Healthy – normal tCa Healthy – high tCa CKD – normal tCa CKD – high tCa	57356229	12.0 (10.4, 14.0) 13.9 (10.6, 14.7) 14.6 (12.0, 16.1) 13.8 (12.0, 15.7)	164261491518	108221283342	24360410351701	——
Geddes et al, 2021 ²⁷	Kainos	CKD – high iCa at diagnosis CKD – high iCa on renal diet	77	16.9 (15.1, 17.3) 14.6 (12.8, 18.1)	22951489	1227634	25276408	—
Tang et al, 2021 ³⁵	Kainos	CKD – tCa uptrend	26	15.7 (14.0, 17.9)	599	273	862	–
	CKD – tCa non-uptrend	21	15.8 (14.5, 17.1)	562	312	1093	–	
van den Broek et al, 2022 ³¹	Kainos	CKD – normal tCa	123	15.2 (13.0, 16.9)	443	265	1015	–
	CKD – high tCa	29	15.2 (11.8, 17.1)	1608	643	2527	–	
Miyakawa et al, 2022 ³³	MedFrontier	CKD – normal iCa CKD – high iCa	26	6.7 (1.3–11.8)	245	–	–	94–637
6	7.0 (2.5–8.9)	1479	–	–	509–3888			
		CKD – normal tCa	20	6.2 (1.3–11.8)	245	—	—	94–627
		CKD – high tCa	12	7.4 (1.7–9.5)	573			125–3888

Citation	Manufacturer	Population	Sample size	Age group *	Median	Q1	Q3	Range
van den Broek et al, 2018 ²⁴	Kainos	CKD – low tMg	20	16.6 (15.0, 18.2)	4950	1931	15,893	–
	CKD – normal tMg	143	14.8 (12.0, 16.2)	637	351	1941	–	
	CKD – high tMg	11	12.0 (9.8, 15.0)	2658	684	8582	–	

*

Age is presented as median (range) or median (25th, 75th percentile)

CKD = chronic kidney disease; FGF-23 = fibroblast growth factor 23; iCa = ionized calcium; Q1 = first quartile; Q3 = third quartile; tCa = total calcium; tMg = total magnesium

Tang et al³⁰ evaluated the risk factors associated with the development of total hypercalcemia in cats with stage 2 and 3 CKD within the first 200 days of transitioning to a phosphate-restricted renal diet. No significant difference in FGF-23 concentrations at baseline (ie, before transitioning to the renal diet) was found between those cats with an uptrend in total calcium concentration and those without (Table 5). Furthermore, unlike phosphorus concentrations, baseline FGF-23 concentrations were not found to be an independent risk factor for increasing total calcium concentration after transitioning to the renal diet. van den Broek et al³¹ evaluated FGF-23 concentrations in cats with CKD based on ionized calcium concentrations at the time of diagnosis and over a 1-year follow-up period. There was no difference in FGF-23 concentrations at the time of CKD diagnosis between those cats that developed persistent ionized hypercalcemia and those cats that continued to have normocalcemia over a 1-year follow-up period. Diet was not controlled in that study; however, there was no difference in the proportion of daily food intake consisting of a renal diet between cats that became hypercalcemic and cats that remained normocalcemic over time. Geddes et al²⁷ looked at whether transitioning cats that develop ionized hypercalcemia on a renal diet to a moderate-protein, moderate-phosphate diet improved their calcium–phosphate homeostasis. The dietary change reduced ionized calcium concentrations without changing either the FGF-23 or phosphorus concentrations.

One study by van den Broek et al²⁴ investigated the relationship between total magnesium and FGF-23 concentrations and found that higher FGF-23 concentrations were independently associated with hypomagnesemia and that cats with hypomagnesemia had higher FGF-23 concentrations across all CKD stages (Table 5).

FGF-23 and development of azotemic CKD

Finch et al¹⁸ measured FGF-23 concentrations in healthy non-azotemic (creatinine <2.0 mg/dl) older cats (aged >9 years) over a 12-month period. Baseline screening FGF-23 concentrations were higher in the healthy cats that developed azotemic CKD (creatinine >2.0 mg/dl) within 12 months than in those that did not. A weak correlation

between FGF-23 and creatinine was found at baseline and at the 12-month follow-up. Sargent et al²⁶ found that FGF-23 concentrations were higher in apparently healthy cats with an SDMA concentration above the RI (>14 mg/dl) and creatinine <2.0 mg/dl compared with cats with a normal SDMA concentration (Table 3) despite no difference in phosphorus and PTH concentrations.

FGF-23 and CKD progression

Two studies^{21,25} looked at initial FGF-23 concentration at CKD diagnosis and progression of CKD based on increasing creatinine concentration (Table 6). Liao et al²⁵ classified progression as an increase of >0.5 mg/dl over 3 months or death, and Geddes et al²¹ classified progression as an increase of >25% in creatinine concentration over 12 months after the date of diagnosis of azotemic CKD. Both studies found that FGF-23 concentration was independently associated with disease progression. Specifically, Liao et al²⁵ found that FGF-23 concentrations were significantly higher in cats with stage 2 and 3 progressive CKD than in cats with non-progressive disease at the same stage. Cats with stage 1 CKD were not included in that study, and there were too few cats with stage 4 CKD enrolled to include in the evaluation of progression. Geddes et al²¹ included cats with stage 2–4 CKD and similarly found that FGF-23 concentrations were higher in cats with progressive disease than in those with stable disease.

Table 6

Article characteristics and population information for studies evaluating serum or plasma concentrations of FGF-23 (pg/ml) in cats with progressive or non-progressive CKD

Citation	Manufacturer	Population	Sample size	Age (years) *	Median	Q1	Q3
Geddes et al, 2015 ²¹	Kainos	CKD – progression	38	15 (12, 16)	1243.2	555.6	2066.5
		CKD – stable	67	13.0 (11, 15)	504.3	340.3	1291.2
Liao et al, 2019 ²⁵	MyBioSource	CKD – progression	33	11 (8–15)	245.0	201.5	491.5
		CKD – non-progression	40	8 (5–15)	165.0	121.5	255.4

*

Age is presented as median (range) or median (25th, 75th percentile)

CKD = chronic kidney disease; FGF-23 = fibroblast growth factor 23

Discussion

This scoping review describes the current literature on FGF-23 in the context of kidney disease in cats and was intended to provide evidence for veterinarians to make informed decisions about the use of measuring FGF-23 concentrations in clinical practice during the management and treatment of cats with CKD. In addition, we aimed to identify knowledge gaps and opportunities for future research. The review identified 17 studies published between 2013 and 2022. Most studies were retrospective and included cats with CKD. Most studies subcategorized cats with CKD by stage, with some variation between studies in the creatinine concentrations used to determine stage. FGF-23

concentrations tended to be higher in cats with CKD compared with healthy cats and trended higher with later IRIS CKD stages, albeit with significant variation within CKD stages and overlap between stages. These are expected findings given that FGF-23 is excreted by the kidneys and similar findings have been reported in people.^{39–41}

This review found several key concepts within the literature regarding measuring FGF-23 concentrations in cats with CKD. First, several studies evaluated the relationship between phosphate overload and FGF-23 concentrations, with some studies, but not all, showing a positive correlation between phosphorus and FGF-23 concentrations. The onset of CKD-MBD and of hyperphosphatemia and hyperparathyroidism varies between cats, especially those with early-stage CKD, and is most common in cats with late-stage disease.^{1,3} Therefore, IRIS treatment guidelines for cats with IRIS stage 1 and 2 CKD recommend monitoring blood phosphorus concentrations to identify phosphate overload and reducing phosphate intake if needed to maintain normal phosphorus concentrations in the range of 2.7–4.6 mg/dl. This recommendation is supported by a study indicating improved renal biomarkers and short-term outcomes with phosphate reduction in cats with early CKD before the onset of hyperphosphatemia.²² In this review, only one study¹⁹ explored FGF-23 as an early marker of CKD-MBD before the development of overt hyperphosphatemia in cats with stage 2 and 3 CKD. Based on the findings of that study, measuring FGF-23 concentration may help identify cats with early-stage CKD that have total body phosphate overload but have not yet developed a phosphorus concentration outside the target range. Unfortunately, most studies in this scoping review included cats with stage 2 and 3 CKD, with a paucity of research in cats with non-azotemic CKD (stage 1). Further study of cats with stage 1 CKD and early (non-azotemic) stage 2 CKD would be valuable to define the expected FGF-23 concentrations in these cats and to determine the clinical decision limits to identify early CKD-MBD.

Two commercial laboratories now offer an FGF-23 ELISA assay using serum to measure FGF-23 concentrations in practice. IDEXX Laboratories offers a feline-only FGF-23 ELISA at reference laboratories globally using the Kainos assay. Fujifilm offers a chemiluminescent FGF-23 assay at reference laboratories in Japan (CL-Jack; Minaris Medical) for dogs and cats. The CL-Jack assay uses the same antibodies as the MedFrontier assay but has been shown to yield relatively lower concentrations compared with the MedFrontier ELISA at higher FGF-23 concentrations.⁴² The IDEXX test suggests measuring an FGF-23 concentration in cats with a diagnosis of stage 1 or 2 CKD to help veterinarians decide which cats may benefit from phosphate reduction therapy. The RI for the assay is <300 pg/ml. If a cat with stage 1 or 2 CKD has an elevated FGF-23 concentration (≥ 400 pg/ml), then the commercial laboratory suggests initiating targeted therapy to reduce phosphate overload, such as a reduction in dietary phosphate or adding a phosphate binder at mealtimes. While this is a guide for veterinarians in practice, research is needed to determine if these therapeutic targets are appropriate in client-owned cats with early-stage CKD.

A second concept explored in the literature was the effect of dietary phosphorus restriction on FGF-23 concentrations. Dietary phosphate restriction followed by the addition of a phosphate binder as needed is the recommended practice in veterinary medicine to slow the progression of renal secondary hyperparathyroidism and CKD-MBD.⁴³ A variety of prescription therapeutic renal diets marketed for cats are commercially available, and they have different formulations with variation in phosphate and protein concentrations. This review included two studies that evaluated the effect of a phosphate-restricted diet on FGF-23 concentrations in healthy older cats²² and cats with stage 2–4 CKD.²⁰ The study of cats with CKD,²⁰ but not that of healthy older cats,²² showed that feeding a moderately phosphorus-restricted diet can decrease FGF-23 concentrations. Some phosphate binders have been found to reduce FGF-23 concentration in

human patients;⁴⁴ however, to date, no study has evaluated the effect of phosphate binders on FGF-23 concentrations in veterinary medicine. Whether measuring FGF-23 concentration is useful to guide the treatment of phosphate overload in cats with CKD is worthy of further investigation.

A third concept explored in the literature is the relationship between FGF-23 and calcium concentrations. Both ionized and total calcium concentrations were found to be independent predictors of FGF-23 concentrations in cats with CKD.^{18,20} Since hypercalcemia can directly increase blood FGF-23 concentrations in cats independent of glomerular filtration rate and phosphate status,³³ calcium concentrations should be evaluated in cats with a normal phosphorus concentration but elevated FGF-23 concentration before dietary phosphate restriction. In addition, recent evidence supports that feeding a phosphate-restricted diet (<1.3 mg/Mcal) may put cats with CKD at risk of developing hypercalcemia, particularly those with a phosphorus concentration within the IRIS target.^{27,30} Two studies^{30,31} included in the scoping review measured FGF-23 concentrations in cats with CKD that developed hypercalcemia and both found that FGF-23 concentrations did not differ between cats that developed hypercalcemia and those that continued to have normocalcemia on a renal diet. Thus, FGF-23 concentration does not appear to be able to predict which cats will develop hypercalcemia after the transition to a phosphate-restricted diet.

Lastly, some studies evaluated the prognostic value of FGF-23 concentrations in healthy cats and cats with CKD. In healthy geriatric cats, FGF-23 concentrations predicted the development of azotemia.¹⁸ In addition, the findings of a few studies support the potential use of FGF-23 as a biomarker of disease progression and survival in cats with CKD, similar to in people.⁴⁶⁻⁴⁸ In the two studies^{21,25} that defined progression as increasing creatinine concentration, cats with CKD that progressed had higher baseline FGF-23 concentrations, and FGF-23 was an independent predictor of disease progression to a higher IRIS stage. Whether the association between FGF-23 and disease progression in cats is causal or correlative remains unknown.

This scoping review revealed several knowledge gaps in the literature worthy of further research. First, there is still missing information about the expected range of FGF-23 concentrations in healthy cats. Based on current IRIS definitions, it is possible that some studies included cats with early-stage, non-azotemic CKD in their healthy cat population. Several studies were published before and/or included cats with data collected before the commercial availability of a feline SDMA test in 2015. Of the four studies that enrolled cats starting in 2015 or later, only one study used SDMA in the criteria to diagnose CKD. The serial measurement of SDMA concentrations could assist in identifying some cats with stage 1 CKD and better identify and stage cats with CKD with muscle atrophy. Second, little is known about the biological variability of FGF-23 concentrations in healthy cats. A recent study of cats using the MedFrontier assay showed that FGF-23 was relatively stable within a 24 h period with no significant circadian rhythm or postprandial effect, supporting that the timing of sample collection is of little importance.⁴⁹ Documentation of the degree of day-to-day variability and the biological variation of FGF-23 concentration in cats is needed to better understand the potential use of FGF-23 measurement in clinical practice. Finally, our results are up to date only as of 31 December 2022, and additional studies investigating FGF-23 in the context of renal disease may have since become available.

Studies should be undertaken prospectively, with cats diagnosed and staged with CKD using current IRIS guidelines and the collection of longitudinal data. SDMA concentrations should be measured, especially in healthy cats and in

those with muscle atrophy, to help differentiate between healthy cats and those cats with early-stage, non-azotemic CKD, as well as to appropriately stage cats with late-stage disease. The methodology of FGF-23 measurement should be standardized among studies to allow comparison of results. Consideration should be given to blood phosphorus and calcium concentrations and to study population size to ensure that a minimum number of patients are enrolled to reach statistical significance.

Conclusions

This scoping review identified and described the available literature relating to FGF-23 in cats with CKD. Most studies were retrospective and varied in their methodologies for measuring FGF-23 concentration and for categorizing cats by disease state (healthy vs CKD) and CKD stage. Although research into FGF-23 in cats with CKD is sparse, FGF-23 shows potential as a useful measurement in feline medicine to help improve the identification and understanding of CKD-MBD, and to improve monitoring and guide therapeutic choices in cats with CKD. There are some clear areas for further clinical study in the role and clinical significance of FGF-23 in early-stage CKD. There is a need to understand the timing and clinical significance of increased FGF-23 concentrations and CKD-MBD in early-stage CKD, and whether early dietary intervention can improve clinical outcomes for those cats with increased FGF-23. There is also a need for further investigation into the role of FGF-23 in identifying or predicting the progression of CKD.

Supplemental Material

Supplementary Table 1:

Complete search results and reasons for exclusion for all articles identified in the literature search.

Supplementary Table 2:

The search engines that identified the 17 articles included in the scope review.

Supplementary Figure 1:

The median fibroblast growth factor 23 concentration (pg/ml) by IRIS stage (1–4) and in healthy cats measured using one of four assays (Deyinhengda, Kainos, MedFrontier, MyBioSource).

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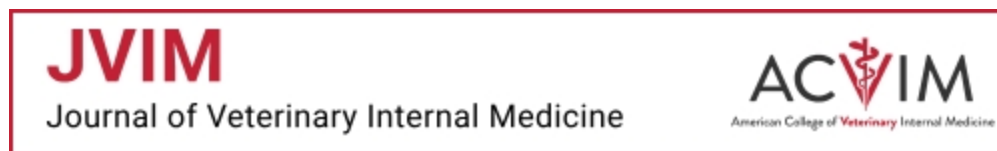
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The association of indoxyl sulfate with fibroblast growth factor - 23 in cats with chronic kidney disease

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Abstract

Background

Indoxyl sulfate (IS) has been reported not only to increase with the severity of impaired renal function, but also possibly to be a factor associated with bone abnormalities linked to fibroblast growth factor-23 (FGF-23) in humans with chronic kidney disease (CKD). It is not yet known whether this correlation between IS and FGF-23 holds true for cats with CKD.

Hypothesis

Accumulation of IS is related to FGF-23 secretion in cats with CKD.

Animals

Twenty clinically healthy cats and 73 cats with CKD cases were evaluated retrospectively.

Methods

The concentrations of IS and FGF-23 in plasma were determined by high-performance liquid chromatography and ELISA, respectively. Progression was defined as an increment of 0.5 mg/dL of serum creatinine concentration within 3 months.

Results

Plasma IS and FGF-23 concentrations were significantly increased concurrently with decreasing renal function. Higher concentration of FGF-23 was significantly associated with higher concentration of IS after adjusting for various confounding factors including creatinine and phosphate. Furthermore, the correlation between IS and phosphate was higher than that between FGF-23 and phosphate. When the renal progression group was compared with the non-progression group, both IS and FGF-23 were found to be significantly increased ($P < .05$). In addition, the area under receiver operator curve of the combination of IS and FGF-23 predicted renal progression at a level >0.9 .

Conclusions and Clinical Importance

Both FGF-23 and IS are associated with phosphate metabolism and CKD progression.

Keywords: azotemia, feline renal diseases, phosphate regulator, progression, uremic toxin

Abbreviations

AUROC

area under ROC

BUN

blood urea nitrogen

CKD

chronic kidney disease

CV

coefficient of variation

FGF-23

fibroblast growth factor-23

GFR

glomerular filtration rate

HCT

hematocrit

Hgb

hemoglobin

HPLC

high-performance liquid chromatography

IRIS

International Renal Interest Society

IS

indoxyl sulfate

ROC

receiver operating characteristics

UPC

urine protein-to-creatinine ratio

WBC

white blood cells

1. INTRODUCTION

Indoxyl sulfate (IS) is a protein-bound uremic toxin derived from dietary protein and mainly eliminated by renal excretion.[1](#) It has been reported to increase with the severity of impaired renal function in humans[2](#) and laboratory animals,[3](#) specifically dogs and cats.[4](#), [5](#) Additionally, increased IS also has been found to be related to prognosis in humans with chronic kidney disease (CKD)[6](#) and to be associated with progression in cats with CKD.[4](#) Some studies in animals have indicated that IS also plays a role in the inhibition of endothelial proliferation and osteoblast function.[7](#) These findings suggest that accumulation of IS may be factor involved in the bone abnormalities found in human CKD patients in whom it may alter calcium and phosphate concentrations and affect their regulation.

Fibroblast growth factor-23 (FGF-23), which is secreted by osteocytes and osteoblasts in response to increases serum phosphate concentration, acts with parathyroid hormone to regulate serum phosphate concentration, calcitriol, and phosphate homeostasis.[8](#) The concentration of FGF-23 also

was reported to increase with the severity of impaired renal function in humans⁹ and laboratory animals¹⁰ and results in the deposition of calcium phosphate within the body.¹¹ Studies also have shown that FGF-23 increases with the severity of CKD in both cats¹² and dogs.¹³ Furthermore, an increased FGF-23 concentration was found to be associated with progression of CKD in people⁹ and shorter survival time in cats with CKD.¹⁴

Indoxyl sulfate in humans has been reported to correlate independently with FGF-23 and to be involved in mineral metabolism.¹⁵ Chronic kidney disease is a common disease with a high prevalence in cats, and both IS and FGF-23 have been reported to be related to severity and outcome but it is not yet known whether or not an association of IS with FGF-23 exists in cats with CKD. Therefore, our aims were to: (1) elucidate whether serum FGF-23 concentrations are correlated with IS concentrations and other clinical variables across the various stages of CKD in cats and (2) evaluate the possibility of using IS and FGF-23 as predictive markers for CKD progression in cats.

2. MATERIALS AND METHODS

2.1. Patient and sample collection

Plasma and urine samples were collected from cats that attended National Taiwan University Veterinary Hospital and stored at -80°C until ELISA and high-performance liquid chromatography (HPLC). All cases that formed the control group lacked any clinical signs of CKD and also met our inclusion criteria, namely they had CBC, serum biochemistry analysis, and had blood urea nitrogen (BUN) and serum creatinine concentrations within normal limits (ie, BUN concentration <30 mg/dL and serum creatinine concentration <1.6 mg/dL). These cats also had urinalysis performed and ultrasound examination of their kidneys and these findings also were within normal limits. Cats with abnormal serum biochemistry analysis (serum creatinine concentration ≥ 1.6 mg/dL) and showing 1 of the following conditions: a history of clinical signs (eg, polyuria and polydipsia), abnormal urinalysis (urine specific gravity <1.030 or persistent proteinuria based on urine protein-to-creatinine ratio [UPC] >0.5 on 3 occasions and >2 weeks apart), with or without abnormal ultrasonographic findings (small irregular kidneys, decreased corticomedullary distinction) for at least 3 months were enrolled in the CKD group. Patients were excluded from the study if they had 1 of the following problems: acute kidney disease or acute worsening azotemia (>0.3 mg/dL increase in serum creatinine concentration within 7 days) or when they had concurrent hyperthyroidism, lower urinary tract disease, or any clinical signs of other nonrenal diseases (hepatic, cardiac, infectious, or neoplastic disease). All cats with CKD were further grouped using the International Renal Interest Society (IRIS) staging system. In accordance with

IRIS CKD staging, these patients were classified into stages 1, 2, 3, or 4. Eligible CKD patients were classified into a progression group and a non-progression group. Progression was defined as a 0.5 mg/dL increment in serum creatinine concentration or death, including those that were euthanized for progression within the 3-month period.

2.2. Measurement of IS by HPLC

Total IS concentration was determined using an HPLC system equipped with a pump (Hitachi L-7100, Hitachi, Tokyo, Japan), fluorescence detector (Hitachi F-1050, Hitachi, Tokyo, Japan) and reversed phase HPLC column (Mightysil RP-18 GP, 4.6 × 250 mm, 5 µm). Indoxyl sulfate potassium salt was dissolved in methanol as a standard solution and stored at -20°C and protected from light. One-hundred microliters of plasma were mixed with 360 µL of acetonitrile and vortexed for 30 seconds before the mixture was centrifuged at 8160*g* for 5 minutes at 4°C. The supernatant then was filtered through a 0.22 µm polyvinylidene difluoride syringe filter before HPLC was performed.

The analytical conditions were similar to those previously described.⁵ Briefly, the mobile phase consisted of sodium acetate buffer (pH 4.5) and acetonitrile (10:90, V/V). The flow rate was 1 mL/min and the injection volume was 10 µL. The excitation and emission wavelengths were set at 280 and 375 nm, respectively. Under these conditions, the retention time of IS was 2.34 ± 0.01 minutes. The limit of detection and limit of quantification were 0.05 and 0.2 mg/L, respectively. Intraday and interday assay variations were determined at both 1 and 10 mg/L. The coefficients of variation (CVs) of the intraday assays were 3.11 and 0.63% and the CVs of interday assays were 7.94 and 9.47% for 1 and 10 mg/L, respectively. The recoveries of IS at 1, 5, and 10 mg/L of spiked plasma were 89.9 ± 3.7, 85.8 ± 3.4, and 89.5 ± 0.6%, respectively.

2.3. Measurement of FGF - 23 by commercial feline FGF - 23 ELISA kit

Feline FGF-23 concentrations were measured in plasma samples using a commercial quantitative sandwich ELISA Kit (MyBioSource, San Diego, California) for the detection of intact FGF-23 by following the manufacturer's instructions. Each sample was performed in duplicate and the final concentration is presented as the mean of the 2 concentrations. All chemicals and reagents were supplied by the kit (MyBioSource), including 6 vials of standard solutions: 1000, 500, 250, 125, 62.5, and 31.2 pg/mL, sample diluent, horseradish peroxidase-conjugate reagent, washing solution, chromogen solution A, chromogen solution B, and stop solution. The sensitivity of this kit is 5.0 pg/mL, the linear detection range is 31.2-1000 pg/mL, and both the intraassay CV and interassay CV are <15% (Supporting Information S1).

2.4. Statistical analysis

Statistical analysis was performed using SPSS software (v.20). The normality of each data set was tested using the Kolmogorov-Smirnov test. Based on these results, either the Mann-Whitney *U* test or the Student's *t* test was used to assess differences between groups. Differences in categorical variables were analyzed by either the Chi-square test or Fisher's exact test. Receiver operator characteristic (ROC) curves were applied to predict renal progression and to yield the best cutoff. Kaplan-Meier curves were used to assess 90-day CKD progression. Spearman correlation coefficients, Pearson correlation coefficients, and linear regression were calculated to determine the associations between different variables. Logistic regression was applied to evaluate the correlation between variables and CKD progression and to calculate odds ratios. Multivariate logistic regression was applied to construct a covariate set for adjustment of the variables. The probability of the *F* value to be entered and probability of the *F* value to be removed during multiple regression analysis were set at $F \leq 0.05$ and $F \geq 0.1$, respectively.

3. RESULTS

Based on IRIS stage, 38 cats were in stage 2, 23 cats were in stage 3, and 12 cats were in stage 4, and these cats then were enrolled. In addition, 20 clinically healthy cats also were enrolled. Demographic findings and clinical characteristics of the cats are summarized in Table [1](#). The results showed that CKD cats were significantly older and had lower hematocrits (HCT), hemoglobin (Hgb) concentrations, and serum albumin concentrations than did the control group. Serum creatinine, phosphate, and BUN concentrations, the product of calcium and phosphate (tCa $\times$ Pi), UPC, FGF-23, and IS all were significantly increased in the advanced CKD groups. The results of the univariate and multiple regression analysis for FGF-23 as a dependent variable are shown in Table [2](#). In the simple regression analysis, FGF-23 was found to be significantly correlated with HCT, Hgb, albumin, BUN, creatinine, phosphate, sodium, chloride, and IS. However, after pooling the large number of variables together using stepwise multiple regression analysis, IS was the only independent factor associated with the FGF-23 concentration ($P = .04$). Table [3](#) shows that there were direct correlations among IS, FGF-23, and phosphate, but after adjusting for IS using partial correlation analysis, FGF-23 and phosphate did not show a significant correlation.

Table 1.

Clinical characteristics and baseline biochemical parameters of the studied patients

Variables	Control (n = 20)	CKD classification (n = 73)		
		Stage 2	Stage 3	Stage 4
Sex (male %) ^a	50%, n = 20	55.3%, n = 38	47.8%, n = 23	50%, n = 12
Age (year) ^b	4 (3-5.75), n = 20	10 (5.8-15) ^c , n = 38	12 (8-15) ^c , n = 23	8 (5.5-9.5) ^c , n = 12
BUN (mg/dL) ^b	22.5 (19-27), n = 20	28.5 (24-42) ^c , n = 38	52 (38.5-64.5) ^c , ^d , n = 23	106.5 (77.8-130.3) ^c , ^d , ^e , n = 12
Creatinine (mg/dL) ^b	1.4 (1.3-1.5), n = 20	2.2 (2-2.5) ^c , n = 38	3.7 (3.3-4.1) ^c , ^d , n = 23	6.45 (5.2-7.9) ^c , ^d , ^e , n = 12
HCT (%) ^b	41 (37.5-48.5), n = 4	36.2 (30.6-42.3) ^c , n = 30	27.2 (23.2-33.5) ^c , ^d , n = 19	24.8 (21.3-28.4) ^c , ^d , n = 12
Hemoglobin (g/L) ^b	15.9 (12-17), n = 4	12.2 (10.7-14), n = 30	10.3 (9-11.6) ^c , ^d , n = 19	8.7 (6.3-10) ^c , ^d , ^e , n = 12
WBC (/μL) ^b	6735 (5812.5-8542.5), n = 4	8165 (6400-10600), n = 30	6600 (5550-13285), n = 19	12 325 (7400-27330), n = 12
Albumin (g/dL) ^b	4 (3.3-4), n = 20	3.4 (3-3.7) ^c , n = 34	3.3 (3.1-3.5) ^c , n = 22	3 (2.9-3.3) ^c , n = 11
Ionized calcium (mmol/L) ^b	1.19 (1.16-1.23), n = 6	1.27 (1.21-1.32) ^c , n = 14	1.26 (1.2-1.3) ^c , n = 17	1.25 (1.15-1.29) ^c , n = 10
Total calcium (mg/dL) ^b	9.7 (9.2-10), n = 20	9.9 (9.8-10.9), n = 20	10.2 (8.9-11.2), n = 9	9.4 (7.9-10), n = 6
Phosphate (mg/dL) ^b	4 (4-4.9), n = 20	4 (3.8-5.2), n = 38	4.9 (4.1-7) ^c , ^d , n = 23	8 (7.1-12) ^c , ^d , ^e , n = 12
tCa × P (mg ² /dL ²) ^b	42.2 (35.6-46.5), n = 20	37.3 (32.7-46.6), n = 18	69.4 (49.6-101.2) ^c , ^d , n = 9	69.3 (52.9-141) ^c , ^d , n = 5
Sodium (mEq/L) ^b	154.5 (152-155.8), n = 20	153.9 (152-155.83), n = 38	155.1 (153-157), n = 23	155.2 (153.3-159.8), n = 12

Variables	Control (n = 20)	CKD classification (n = 73)		
		Stage 2	Stage 3	Stage 4
Potassium (mEq/L) ^b	4 (3.85-4.65), n = 20	3.9 (3.5-4), n = 38	4 (3.6-4.5), n = 23	4 (5.5), n = 12
Chloride (mEq/L) ^b	119.5 (116.3-122.8), n = 20	118 (115.7-119.3), n = 38	118 (114-121), n = 23	120.6 (116.1-128), n = 12
AST (U/L) ^b	22.5 (18.5-30.8), n = 18	34 (21-59), n = 23	33.5 (21.8-38.3), n = 14	26 (20.8-52.5), n = 8
ALT (U/L) ^b	54.5 (44-74.8), n = 20	54 (42-114.8), n = 26	69 (34-81), n = 15	44 (34.5-86), n = 9
ALP (U/L) ^b	42 (31.3-46.3), n = 20	39 (34.5-55), n = 25	32 (27-48), n = 15	27 (19.5-53), n = 9
UPC ^b	0.00 (0-0.02), n = 18	0.1 (0-0.9) ^c , n = 21	0.5 (0.167-0.8) ^c , n = 8	0.73 (0.4-1.7) ^c , n = 6
FGF-23 (pg/mL) ^b	121.05 (71.85), n = 20	163.05 (126.93), n = 38	270.8 (167.5) ^{c, d} , n = 23	728.25 (793.48) ^{c, d, e} , n = 12
IS (mg/L) ^b	1.67 (1-2), n = 20	2.91 (2-4.8) ^c , n = 38	5.8 (3.1-10) ^{c, d} , n = 23	25.7 (21-30) ^{c, d, e} , n = 12

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Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; CKD, chronic kidney disease; FGF-23, fibroblast growth factor-23; HCT, hematocrit; IS, indoxyl sulfate; tCa × P, calcium-phosphate product; UPC, urine protein-to-creatinine ratio; WBC, white blood cell.

^aChi-square test: represented as percentage (%).

^bKruskal-Wallis test: represented as the median (interquartile range).

^cAchieved significance with control group.

^dAchieved significance with stage 2 group.

^eAchieved significance with stage 3 group.

Table 2.

Simple and stepwise multiple linear regression analysis for fibroblast growth factor-23 (FGF-23) as a dependent variable

	Simple regression analysis			Stepwise multiple regression analysis		
Variables	Adjustment β	t	P_{-}^{*}	Adjustment β	t	p
Age	0.027	0.230	.82	–	–	–
SAP	0.134	0.884	.38	–	–	–
Hematocrit	–0.386	3.499	.001	–	–	–
Hemoglobin	–0.380	–3.154	.003	–	–	–
RBC	–0.316	–2.557	.01	–	–	–
Albumin	–0.339	–2.907	.005	–	–	–
BUN	0.449	4.232	<.001	–	–	–
Creatinine	0.399	3.672	<.001	–	–	–
Ionized calcium	0.020	0.153	.88	–	–	–
Phosphate	0.388	3.545	.001	–	–	–
tCa $\times$ Pi	0.331	1.921	.06	–	–	–
Sodium	0.247	2.150	.03	–	–	–
Potassium	0.104	0.879	.38	–	–	–
Chloride	0.300	2.653	.01	–	–	–
AST	–0.150	–0.995	.32	–	–	–
ALT	–0.121	–0.845	.40	–	–	–
ALP	–0.250	–1.769	.08	–	–	–
USG	0.289	1.955	.06	–	–	–
UPC	0.237	1.403	.17	–	–	–

	Simple regression analysis			Stepwise multiple regression analysis		
Variables	Adjustment β	t	P_{-}^{*}	Adjustment β	t	p
IS	0.466	4.432	<.001	0.466	2.169	.04

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Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; HCT, hematocrit; IS, indoxyl sulfate; RBC, red blood cell; SAP, systolic arterial pressure; t , t statistic; $tCa \times P$, calcium-phosphate product; UPC, urine protein-to-creatinine ratio; USG, urine specific gravity; WBC, white blood cell.

* $P < .05$ as significant.

Table 3.

Spearman correlation and partial correlation results among indoxyl sulfate (IS), fibroblast growth factor-23 (FGF-23), and phosphate

Variables	Spearman correlation analysis		Partial correlation analysis	
	P	r	P	Partial r
FGF-23 and phosphate	.008	0.310	.14	0.175
Phosphate and IS	<.001	0.530	<.001	0.462
FGF-23 and IS	<.001	0.480	.005	0.326

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The progression and non-progression groups of cats with CKD were composed of 33 and 40 cases, respectively (Table 4) and median progression was 28 days (interquartile range [IQR], 14-57 days). No cat died or was euthanized during the study. The cats with renal progression had significantly

lower HCT and WBC counts and higher BUN, phosphate, $tCa \times Pi$, UPC, FGF-23, and IS than did the non-progression group. However, serum creatinine concentration did not reach statistical significance between groups. Table 5 shows the serum creatinine, phosphate, IS, and FGF-23 concentrations of CKD cats at the different stages within the progression and non-progression groups. Because the number of stage 4 CKD cases was low, only CKD stage 2 and stage 3 groups were enrolled for evaluation of progression. Both IS and FGF-23 concentrations in the progression group remained significantly higher than in the non-progression group at the same stage. The univariate logistic regression analysis identified that higher serum FGF-23, IS, age, BUN, phosphate, $tCa \times Pi$, UPC, and lower HCT, Hgb, and albumin concentrations were significantly associated with progression of CKD (Table 6). In the multivariate logistic regression analysis, the predictive roles of IS and FGF-23 remained independently significant after adjustment for the various different models (Table 7), except for the adjustment for UPC in both the IS and FGF-23 models, as well as for $tCa \times Pi$ in the FGF-23 model. In addition, using a final model of adjustment for various covariates (BUN, creatinine, HCT, Hgb, albumin, phosphate and FGF-23, model 11a), IS was found to be an independent progression factor for CKD. However, when the model was adjusted with the same set of covariates, FGF-23 did not achieve significance when IS was added (model 11b). Moreover, the correlations between UPC and IS, as well as UPC and FGF-23, both were calculated. Only IS was correlated with UPC (r of Spearman's correlation, 0.366; $P = .05$).

Table 4.

Clinical characteristics and baseline biochemical parameters of all feline chronic kidney disease (CKD) patients, divided into the progression group and non-progression group

Variables	Progression	Non-progression	P
Sex (male %) ^a	60.6, n = 33	45, n = 40	.18
Age (year) ^b	11 (8-15), n = 33	8 (5-15), n = 40	.09
SAP (mmHg) ^b	162.5 (146-182), n = 21	150 (140-170), n = 24	.333
BUN (mg/dL) ^b	58 (38.5-74.5), n = 33	35 (25.3-51.3), n = 40	.001
Creatinine (mg/dL) ^b	3.3 (2.3-4.7), n = 33	2.6 (2-3.7), n = 40	.19
HCT (%) ^b	30 (25.6-33.5), n = 33	34.5 (25.4-41.4), n = 39	<.001
Hemoglobin (g/L) ^b	10.1 (9.1-11.9), n = 29	12.6 (9.2-14), n = 32	.06
WBC (10 ³ /μL) ^b	6700 (5850-13 130), n = 29	9510 (5755-24070), n = 32	.05
Albumin (g/dL) ^b	3.4 (3-3.7), n = 29	3.3 (3-3.5), n = 38	.06
Ionized calcium (mmol/L) ^b	1.20 (1.19-1.25), n = 17	1.27 (1.25-1.3), n = 20	.41
Total calcium (mg/dL) ^b	10.8 (10.1-11.8), n = 14	10.2 (9.3-11), n = 21	.19
Phosphate (mg/dL) ^b	5.9 (4.4-7), n = 33	4.1 (3.6-7), n = 40	.01
tCa × P (mg ² /dL ²) ^b	60.4 (48.3-80.3), n = 12	45.9 (36.2-66), n = 20	.002
Sodium (mEq/L) ^b	156.5 (152.8-160.9), n = 33	153.8 (152-155.6), n = 40	.19
Potassium (mEq/L) ^b	4 (3.7-4.4), n = 33	3.92 (3.4-4), n = 40	.67
Chloride (mEq/L) ^b	119.3 (116.4-125), n = 33	118 (115.5-120.5), n = 40	.27
AST (U/L) ^b	36 (22-62.3), n = 20	27 (20-45), n = 25	.42
ALT (U/L) ^b	73 (50-106.5), n = 21	45 (35-85.5), n = 29	.15
ALP (U/L) ^b	30 (25.5-56.7), n = 21	37 (26.5-54), n = 28	.95
UPC ^b	0.735 (0.54-1.88), n = 10	0.1 (0.1-0.7), n = 25	.02

Variables	Progression	Non-progression	<i>P</i>
FGF-23 (pg/mL) ^b	245 (201.5-491.5), n = 33	165 (121.5-255.4), n = 40	<.001
IS (mg/L) ^b	9.89 (4.2-25.4), n = 33	2.83 (2.1-5.4), n = 40	<.001

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Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; FGF-23, fibroblast growth factor-23; HCT, hematocrit; IS, indoxyl sulfate; SAP systolic arterial pressure; tCa × P, calcium-phosphate product; UPC, urine protein-to-creatinine ratio; WBC, white blood cell.

^aChi-square test: represented as percentage (%).

^bMann-Whitney test: represented as the median (interquartile range).

Table 5.

Creatinine, phosphate, indoxyl sulfate (IS), and fibroblast growth factor-23 (FGF-23) concentrations of chronic kidney disease (CKD) cats with different stage in the progression group and non-progression group

	IRIS stage	Progression group (n = 27)	Non-progression group (n = 34)	<i>P</i>^a
Creatinine (mg/dL)	2	2.25 (1.8-2.6), n = 14	2.1 (2-2.5), n = 24	.86
	3	4 (3.5-4.1), n = 13	3.5 (3.3-3.8), n = 10	.29
Phosphate (mg/dL)	2	5.1 (3.9-5.9), n = 14	4 (3.6-4.4), n = 24	.04 [*]
	3	6.6 (4.5-9.1), n = 13	4.55 (4-5), n = 10	.09
IS (mg/L)	2	5.12 (3.74-7.47), n = 14	2.24 (1.63-2.86), n = 24	<.001 [*]
	3	8.42 (5.45-15.12), n = 13	3.73 (2.82-5.87), n = 10	.008 [*]
FGF-23 (pg/mL)	2	247.3 (205.8-292.9), n = 14	128.7 (113.5-166.9), n = 24	.001 [*]
	3	354 (246.9-510.2), n = 13	229.95 (133.7-281.8), n = 10	.03 [*]

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Abbreviation: IRIS, International Renal Interest Society.

^aMann-Whitney test: represented as the median (interquartile range).

* Significant difference between progression group and non-progression group ($P < .05$).

Table 6.

Univariate logistic regression analysis for the various different variables associated with chronic kidney disease (CKD) progression in cats

Variables	OR	95% CI for OR	P
IRIS stage (1 stage)	0.449	0.156-1.289	.14
Age (year)	1.131	1.004-1.274	.04 _*
BUN (mg/dL)	1.072	1.029-1.116	.001 _*
Creatinine (mg/dL)	1.653	0.908-3.010	.1
Hematocrit (%)	0.871	0.802-0.946	.001 _*
Hemoglobin (g/L)	0.764	0.576-0.995	.05 _*
Albumin (g/dL)	0.201	0.048-0.839	.03 _*
Phosphate (mg/dL)	1.874	1.167-3.010	.009 _*
tCa × P (mg ² /dL ²)	1.090	1.017-1.169	.01 _*
UPC	3.781	1.124-12.715	.03 _*
FGF-23 (pg/mL)	1.009	1.003-1.015	.002 _*
IS (mg/L)	2.176	1.409-3.361	<.001 _*

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Abbreviations: BUN, blood urea nitrogen; CI, confidence interval; FGF-23, fibroblast growth factor-23; IRIS, International Renal Interest Society; IS, indoxyl sulfate; OR, odds ratio; tCa × P, calcium-phosphate product; UPC, urine protein-to-creatinine ratio.

* Significant correlation with CKD progression ($P < .05$).

Table 7.

Multivariate logistic regression analysis for the various different variables associated with renal progression in cats

	IS			FGF-23		
Models	Odds ratio	95% CI	P	Odds ratio	95% CI	P
Unadjusted	2.176	1.409-3.361	<.001 [*]	1.009	1.003-1.015	.002 [*]
Model 1	2.590	1.509-4.444	<.001 [*]	1.009	1.003-1.015	.004 [*]
Model 2	2.147	0.864-1.207	.001 [*]	1.008	1.002-1.014	.005 [*]
Model 3	3.455	1.046-11.405	.04 [*]	1.038	0.999-1.079	.05
Model 4	1.713	0.962-3.050	.07	1.006	0.999-1.014	.09
Model 5	2.203	1.245-3.896	.007 [*]	1.010	1.003-1.017	.006 [*]
Model 6a	1.901	1.228-2.942	.004 [*]	–	–	–
Model 6b	–	–	–	1.006	1.000-1.012	.05 [*]
Model 7a	3.81	1.404-10.33	.009 [*]	–	–	–
Model 7b				1.011	0.998-1.023	.11

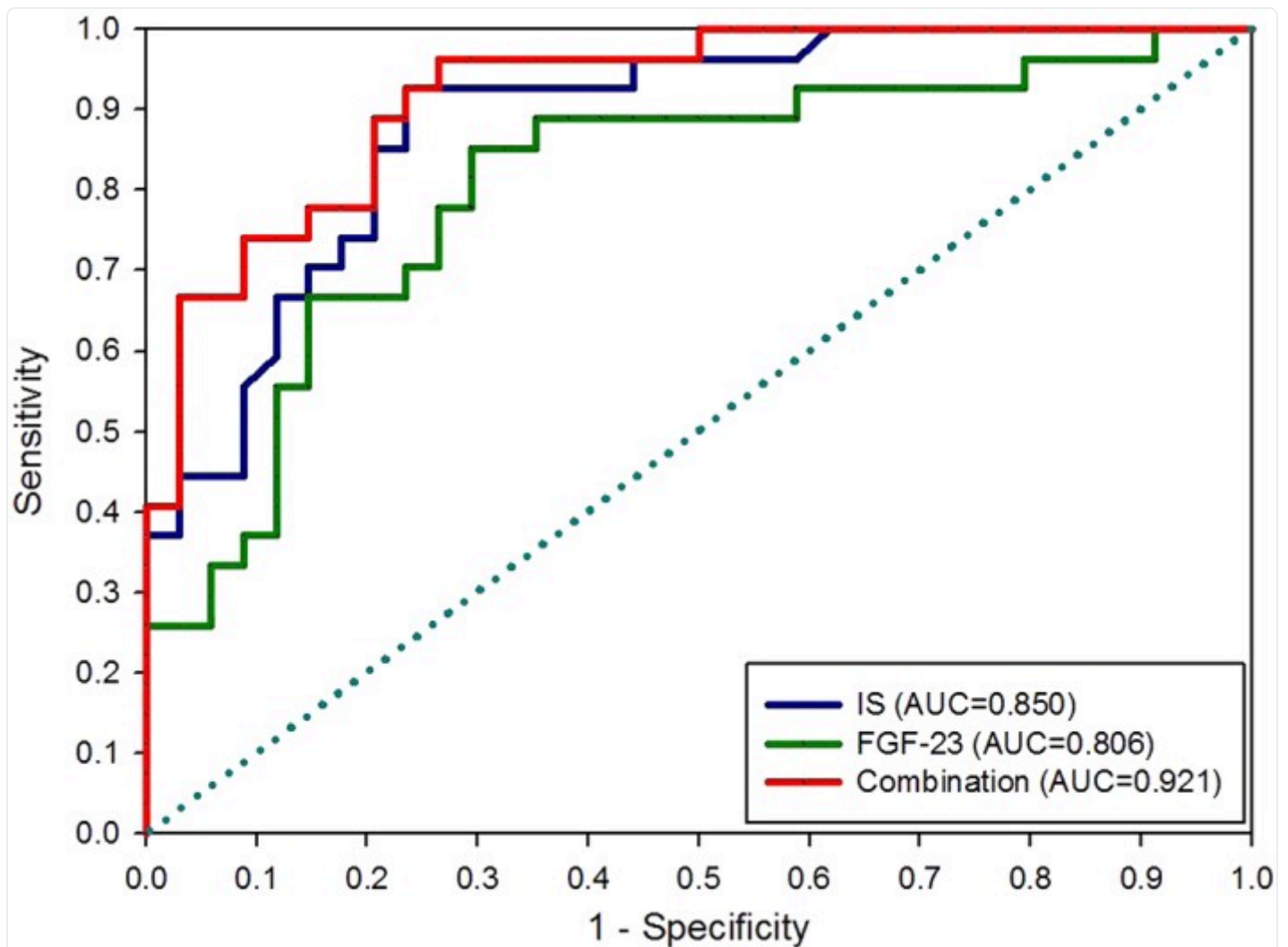
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Model 1 was adjusted for International Renal Interest Society (IRIS) stage; Model 2 was adjusted for age and gender; Model 3 was adjusted for tCa × P, calcium-phosphate product (tCa × P); Model 4 was adjusted for urine protein-to-creatinine (UPC); Model 5 was adjusted for systolic arterial pressure (SAP); Model 6a was adjusted for fibroblast growth factor-23 (FGF-23); Model 6b was adjusted for indoxyl sulfate (IS); Model 7a was adjusted for blood urea nitrogen (BUN), creatinine, hematocrit, hemoglobin, albumin, phosphate, and FGF-23. Model 7b was adjusted for BUN, creatinine, hematocrit, hemoglobin, albumin, phosphate, and IS.

* Significant correlation with CKD progression ($P < .05$).

Figure [1](#) shows the ROC curve analysis for evaluation of the predictive ability of IS, FGF-23, and a combination of both, for progression in the CKD cats. The combined predictive values (P combination) of FGF-23 and IS were calculated by multivariate logistic regression: $\log(\text{odds of progression}) = 0.642 * \text{IS} + 0.006 * \text{FGF-23}$. The following equation was derived: $P \text{ combination} = \text{IS (mg/L)} + (0.006/0.642) * \text{FGF-23 (pg/mL)}$. The area under ROC (AUC) values for the combination score, IS, and FGF-23 to predict progression were 0.921, 0.850, and 0.806, respectively. The optimal cutoff for the combination was 5.02, with a sensitivity of 96.3% and specificity of 73.5% ($P < .001$); the optimal cutoff for IS was 3.41 mg/L, with a sensitivity of 92.6% and specificity of 76.5% ($P < .001$); and the optimal cutoff for FGF-23 was 200 pg/mL, with a sensitivity of 85.2% and specificity of 70.6% ($P < .001$). Based on these cutoffs, the cats then were divided into a higher value group and a lower value group. The cats with IS concentration $>3.41 \text{ mg/L}$ ($P < .001$) and FGF-23 concentration $>200 \text{ pg/mL}$ ($P < .001$) both showed significantly faster renal progression (Figure [2A,B](#)).

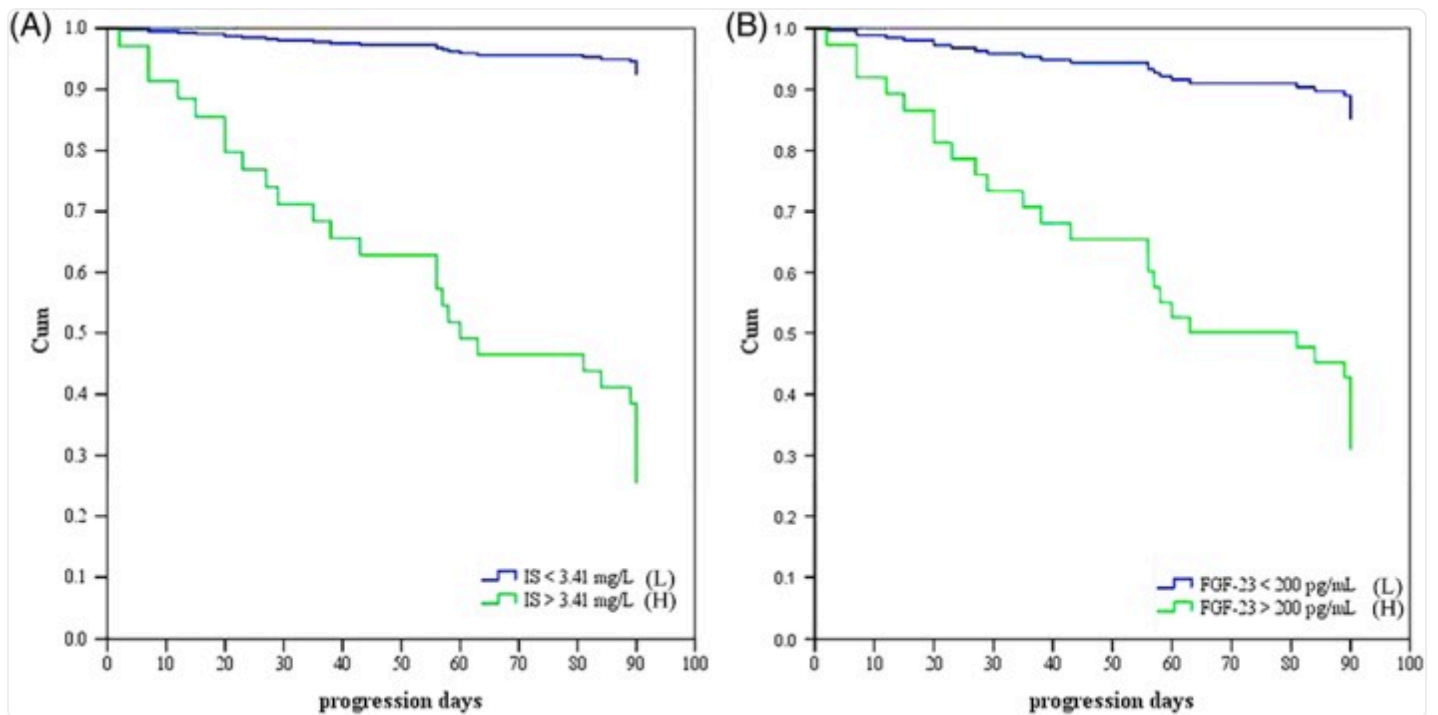
Figure 1.



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Receiver operator characteristic (ROC) curve of indoxyl sulfate (IS), fibroblast growth factor-23 (FGF-23), and combination of both ($IS + (0.006/0.642) \times FGF-23$) to predict renal progression in cats with chronic kidney disease (CKD). Area under ROC (AUC): combination (0.921) > IS (0.850) > FGF-23 (0.806)

Figure 2.



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A comparison of the Kaplan-Meier curves using the indoxyl sulfate (IS) levels (A) and fibroblast growth factor-23 (FGF-23) (B). High-value group (H): IS > 3.41 mg/L; low-value group (L): IS < 3.41 mg/L; high-value group (H): FGF-23 > 200 pg/mL; low-value group (L): FGF-23 < 200 pg/mL. Both higher levels showed significantly faster progressions. ($P < .001$, log-rank test)

4. DISCUSSION

Although both IS and FGF-23 have been reported to be biomarkers for the severity of CKD and to be predictors of progression, to the best of our knowledge, ours is the first study to investigate the relationship in cats with CKD between IS and FGF-23 and to evaluate them simultaneously along with other factors. Similar to other studies,[4](#), [14](#) IS and FGF-23 increased with the severity of CKD and were able to act as predictors of CKD progression in cats. Interestingly, multivariate logistic regression showed that IS and FGF-23 were independently associated with CKD progression after adjustment for other reported progression-related variables, such as HCT, Hgb, and phosphate.[16](#)

This result confirms the importance of IS and FGF-23 having association with the progression of CKD, and these findings are similar to previous studies in human medicine⁶ and veterinary medicine.^{4, 14} However, we also used both of these factors to construct an equation that has a higher AUC when predicting progression, and thus using the 2 factors together is a more precise measure overall.

Similar to a previous study in humans with CKD,¹⁵ our results also show that FGF-23 and IS not only show a significant correlation but also have a significant linear relationship with each other. Noticeably, after pooling together other variables that are believed to affect FGF-23 concentration, such as serum creatinine and phosphate concentrations, IS was the only variable that cannot be removed with the stepwise procedures and has significant correspondence with FGF-23. Furthermore, both plasma IS and FGF-23 concentrations are significantly correlated with serum phosphate concentration. Fibroblast growth factor-23 has been shown to increase in parallel to serum phosphate concentration in humans⁹ with CKD and cats¹² with CKD. Hyperphosphatemia is thought to be the physiological stimulus for FGF-23 secretion and the increase in FGF-23 is a response that is aimed at correcting the abnormal serum phosphate concentration; it does so by suppressing renal tubule sodium-dependent phosphate transporter expression and inhibiting the production of calcitriol.¹⁷

In our study, surprisingly, IS showed a higher correlation with phosphate than FGF-23 did. Additionally, when IS was controlled for, the significant correlation between FGF-23 and phosphate became insignificant. In contrast, IS and phosphate do maintain a significant correlation when FGF-23 is controlled for. A recent study indicated that IS is a potential factor in the induction of secondary renal hyperparathyroidism via oxidative stress and impairment of osteoblast function.⁷ Osteocytes and osteoblasts are the primary sources of FGF-23 synthesis and secretion.⁸ Moreover, 1 study also has determined that administration of IS to rats with CKD decreases renal expression of klotho, a cofactor in FGF-23 receptor interactions that enhances its phosphate-lowering function.¹⁸ These findings imply that IS may have a deleterious effect on mineral metabolism, which occurs via a disturbance of the phosphate-lowering function of FGF-23. Furthermore, our results also suggest that IS in cats with CKD also may be a potential factor affecting FGF-23 concentrations and be involved in phosphate metabolism. However, the cross talk among IS, FGF-23, and phosphate warrants further study.

In our study, the IS was significantly correlated with UPC, but the predictive power of both IS and FGF-23 regarding CKD progression was decreased to insignificant by adjusting for UPC. Similarly, proteinuria, as an important risk factor, also has been reported to be associated with CKD progression in cats.¹⁶ It is well known that IS leads to glomerular sclerosis through oxidative

stress¹⁸ and can injure the glomerular podocytes,¹⁹ which then results in defects in the glomerular filtration barrier. Eventually, excessive filtering of protein may overload the proximal tubular cells. Additionally, IS might cause damage to renal tubular cells¹ and lead to decreased reabsorption of albumin and finally to increased proteinuria. Consequently, this proteinuria might further damage tubular cells resulting in the accumulation of IS. In combination, the detrimental effects of both proteinuria and IS on renal tubules might be the reason why the predictive power of IS was affected by UPC. However, although the current design prevented us from evaluating these speculations, further studies are warranted. Higher concentrations of FGF-23 have been reported to be related to proteinuria in humans with CKD.²⁰ Oxidative stress was related to deficient expression of klotho and eventually will cause both FGF-23 resistance²¹ and proteinuria.²² Therefore, the similar effects of proteinuria and increased FGF-23 concentrations on renal tubules might be the reason that UPC is able to affect the predictive power of FGF-23. However, FGF-23 was not significantly correlated with UPC in our study (data not shown), and further investigations are needed to explore this issue.

In our study, the FGF-23 concentrations of late stage cats were different from those obtained in previous studies.^{12, 14} The ELISA kit we used was designed for the detection of feline intact FGF-23 (Supporting Information S1), and it was different from the previously reported kit, which was designed to detect human FGF-23. Although this kit was used successfully for studies in cats, whether the response level (cross reactivity) was similar for human and for feline samples remains to be confirmed by the manufacturer. One possible explanation for the difference in the response level is the sample dilution effect (in which after the response goes beyond the linear range of the assay, the sample must be further diluted) and might start to exhibit a disproportionate increase in response.²³ In addition, our study focused on the difference between the progression and non-progression groups, in which some data came from younger and less severe CKD cats that could have brought the median FGF-23 concentration lower and mostly within the assay, linear range. We believe that both sets of data are legitimate and stand on their own, given the fact that different validated ELISAs were employed. However, further studies are warranted in the future to decipher whether different ELISA kits show differences in specificity and response intensity in different animal species.

Our study had some limitations. First, because FGF-23 and IS results were derived from previously banked plasma samples, not all of the blood and urine variables could be obtained for all cases. Second, although there was no statistically significant difference, CKD stage 4 cats seemed to be younger than CKD stage 2 and 3 cats, which may be considered controversial, given that age can be a factor leading to a later stage of CKD.²⁴ Although the sample size, especially the CKD stage 4 cats, was small and may not be representative of the larger feline population, it is possible that

different kidney diseases that may compromise renal function also coexisted in these cats. Without biopsies and further examinations, the definite cause of CKD cannot be ascertained and might limit the application of our results to other types of kidney disease. Third, the expression of klotho, the cofactor of FGF-23, was not evaluated at the same time, which limits our ability to explain the mechanism of the interaction between IS and FGF-23. Fourth, serial measurements were not available for comparison of the continued change of these uremic toxins in CKD progression. Sixth, because the duration of follow-up was not fixed, the calculation of progression days might not be precise. Therefore, further studies are required to answer these questions.

5. CONCLUSIONS

In conclusion, IS is independently correlated with FGF-23, suggesting that IS may be an important factor in the regulation of mineral metabolism of cats with CKD. Not only FGF-23, but also IS play important roles in phosphate metabolism, and they are independent factors in relation to renal function. Thus, they should be considered simultaneously when examining phosphate control and mineral metabolism in cats. Both IS and FGF-23 appear to be promising renal biomarkers for the early detection of progression in cats with CKD.

CONFLICT OF INTEREST DECLARATION

Authors declare no conflict of interest.

OFF - LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

This study got IACUC approval (NTU-EL-00078) June 2016.

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

Appendix S1: Supporting Information

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Liao and Chou contributed equally to this study.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Appendix S1: Supporting Information

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